MOADT: Multi-Objective Admissible Decision Theory

A formal decision theory for AI alignment

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1. Introduction and Motivation

> "Not everything that counts can be counted, and not everything that can be counted counts." — William Bruce Cameron (1963)

Von Neumann–Morgenstern expected utility theory is the standard foundation for rational decision-making under uncertainty. Its elegance is real: four axioms yield a unique (up to affine transformation) utility function, and rational action reduces to maximizing its expectation. But this elegance comes at a cost that alignment theory cannot afford.

The critical axiom is **completeness**: for any two outcomes A and B, either $A \succeq B$ or $B \succeq A$. Completeness, together with independence and continuity, yields a utility representation that forces commensurability — every objective must be tradeable against every other at some rate. (Completeness alone gives only a total ordering; the fixed tradeoff rates come from the full axiom set. But completeness is the axiom MOADT drops, because it is the one that demands the ordering exist in the first place.) For a system managing objectives like *human safety*, *helpfulness*, and *honesty*, completeness implies there exists some quantity of helpfulness that compensates for a given loss in safety. This is not a feature; it is a structural limitation that alignment theory must address.

The formal ingredients for an alternative already exist. The possibility of rational incomplete preferences has been established by Dubra et al. (2004), Mandler (2005), and Eliaz and Ok (2006). The theory of coherent decision-making under imprecise probabilities was developed by Levi (1974), Walley (1991), and Seidenfeld, Schervish, and Kadane (1995). Robust Markov decision processes under set-valued uncertainty were formalized by Iyengar (2005) and Nilim and El Ghaoui (2005). Multi-objective reinforcement learning and sequential decision-making with vector-valued returns are surveyed in Roijers et al. (2013). And the theory of decision-making with imprecise probabilities is systematized in Troffaes (2007). MOADT synthesizes these threads into a unified decision architecture specifically designed for the alignment problem.

MOADT constructs a complete decision procedure — one that terminates with a concrete action recommendation — without ever requiring the agent to commit to a single scalar ranking of outcomes. It replaces the VNM axioms with five alternative axioms (Section 7), layers a choice protocol over robust Pareto admissibility (Section 4), and builds corrigibility in as a structural feature rather than an additional constraint (Section 5). Where VNM gives the agent one knob, MOADT gives it k knobs and a disciplined procedure for turning them.

Scope. MOADT is specifically designed as an alignment-oriented decision architecture — a decision theory for AI systems operating under human oversight with multiple incommensurable objectives. It does not claim to replace VNM for all domains: for single-objective optimization with known utilities and precise probabilities, VNM remains the correct framework (Theorem 1 confirms this as a special case). MOADT's contribution is to the settings where VNM's structural properties — especially completeness and the implied commensurability — become liabilities rather than strengths.

Why this matters for alignment. The scalarization trap is not merely an aesthetic objection to VNM — it names a cluster of *unsolvable* problems that confront any system that compresses human values into a single number. The utility function must be specified with perfect precision (since any misspecification is aggressively optimized against); the agent must resist any modification to its utility function (since the current function ranks itself as optimal); and all values become tradeable against all other values at some rate (since completeness demands commensurability). These are structural consequences of the axioms, not merely engineering challenges. MOADT was designed

from the ground up to sidestep these problems: imprecise probabilities replace the precision requirement, evaluator uncertainty dissolves modification resistance, and the rejection of completeness eliminates forced commensurability. The goal is a decision theory that is *amenable to alignment by construction* — one where the properties we need for safe AI (corrigibility, non-exploitation of value uncertainty, cooperation with human overseers) emerge from the formal structure rather than being bolted on as afterthoughts. Section 9 catalogues the honest difficulties MOADT faces — computational cost, specification challenges, instability under reflection — and argues that these are better-structured difficulties than those VNM imposes.

Readers who prefer math-first: The self-contained formal specification (6 definitions, 4 propositions, 2 theorems, 1 axiom) is in Section 12. Everything between here and there is motivation, proof, and discussion.

On length. This document is a technical report, not a conference submission. The core formal theory (Sections 1–7, 12) is approximately 30 pages; worked examples and extended discussion are included for completeness and reproducibility. A conference-length version would comprise Sections 1–7 and 12, with worked examples as supplementary material.

The name. MOADT stands for **Multi-Objective Admissible Decision Theory**. Each word maps to a core design commitment: the evaluation is *multi-objective* (vector-valued, never collapsed to a scalar), the filtering is *admissible* (robust Pareto dominance eliminates only provably inferior actions), the framework is a *decision theory* (it terminates with a concrete recommendation or a structured deferral, not merely a description of the Pareto frontier).

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2. Formal Objects

We work with the following primitives.

Outcomes. Let Ω be a finite set of outcomes — concrete states of the world that might result from action.

Actions. Let A be a finite set of available actions. Each action, combined with a state of the world, determines an outcome.

States. Let S be a finite set of states of the world, representing the agent's uncertainty about the environment.

Transition function. A function $\omega : A \times S \to \Omega$ maps each action-state pair to the resulting outcome.

Vector-valued evaluation. An evaluation function $\vec{f} : \Omega \to \mathbb{R}^k$ maps each outcome to a k-dimensional vector, where each component $f_i(\omega)$ scores outcome ω on objective i. We do not assume these components are commensurable.

Credal sets (Knightian uncertainty). Rather than a single probability distribution P over states, the agent maintains a *credal set* $\mathcal{P}$ — a closed, convex, compact set of probability distributions over S. This represents genuine uncertainty about the correct probabilistic model, not merely uncertainty within a model. When $|\mathcal{P}| = 1$, we recover the standard Bayesian setting. (Compactness is standard for credal sets on finite state spaces; see Levi 1974, Walley 1991.)

Evaluator uncertainty (Goodhart guardrail). Rather than a single evaluation function $\vec{f}$, the agent maintains an *evaluator set* $\mathcal{F}$ — a compact set of plausible evaluation functions $\vec{f} : \Omega \to \mathbb{R}^k$. This

models uncertainty about whether the agent's measurement of each objective faithfully tracks the intended objective. When $|\mathcal{F}| = 1$, the agent fully trusts its own evaluations. The set $\mathcal{F}$ is the formal mechanism by which MOADT guards against Goodhart's law: optimizing a proxy that diverges from the true objective is detectable as sensitivity to evaluator choice.

Outcome sets. For each action $a \in A$, define the *outcome set*:

$$Y(a) = \left\{ \mathbb{E}_P[\vec{f}(\omega(a,s))] : P \in \mathcal{D},\ \vec{f} \in \mathcal{F} \right\}$$

That is, $Y(a)$ is the set of all *expected* evaluation vectors the agent considers possible for action a, one for each credal-evaluator pair $(P,\vec{f})$. This set captures the agent's epistemic uncertainty — uncertainty about which probabilistic model and which evaluator are correct. Each element of $Y(a)$ is itself an expectation over aleatoric uncertainty (which state obtains). Since $Y(a)$ is the image of the compact set $\mathcal{D} \times \mathcal{F}$ under a continuous map, $Y(a)$ is compact (and in particular closed and bounded) in $\mathbb{R}^k$.

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3. Core Rationality: Robust Pareto Admissibility

3.1 Standard Pareto Dominance

We first recall the standard notion. For vectors $\vec{x}, \vec{y} \in \mathbb{R}^k$:

$$\vec{x} \succ_P \vec{y} \iff x_i \geq y_i \ \forall\, i \in \{1, \dots, k\} \ \text{ and } \ \exists\, j \text{ s.t. } x_j > y_j$$

That is, $\vec{x}$ Pareto-dominates $\vec{y}$ if it is at least as good on every objective and strictly better on at least one. This is a *partial* order: many pairs of vectors are incomparable.

3.2 Robust Dominance

Standard Pareto dominance compares individual outcome vectors. To compare *actions* — which map to *sets* of possible outcome vectors — we need a notion of dominance that is robust to the agent's uncertainty.

Definition (Robust Dominance). Action a robustly dominates action b, written $a \succ_R b$, if and only if:

$$\forall\, \vec{y}_b \in Y(b),\ \exists\, \vec{y}_a \in Y(a) \ \text{ s.t. } \ \vec{y}_a \succ_P \vec{y}_b$$

The quantifier structure is $\forall$-$\exists$: for every realization of b's uncertainty, there exists *some* (potentially different) realization of a's uncertainty that dominates it. This is deliberately weaker than *uniform dominance* ($\forall\ (P,\vec{f}) : \mathbb{E}_P[\vec{f}(a)] \succ_P \mathbb{E}_P[\vec{f}(b)]$), which would require a to beat b under the *same* model. The $\forall$-$\exists$ structure makes robust dominance a *defensive* criterion: a can "defend" against each of b's realizations, but possibly with a different model each time.

Why $\forall$-$\exists$ and not a stronger quantifier. An important design choice deserves explicit defense. The $\forall$-$\exists$ structure permits *model-switching*: when verifying that a dominates b, the defending realization $\vec{y}_a$ can come from a different $(P,\vec{f})$ for each attacking realization $\vec{y}_b$. This means the agent cannot

point to a single world-model under which a uniformly beats b — it can only say "for each way b might turn out, a has *some* way of turning out better." A skeptic might object that this is too permissive.

We considered two natural alternatives:

- **Uniform dominance** ($\forall\forall$): $\forall (P,\vec{f}) \in \mathcal{P} \times \mathcal{F} : \mathbb{E}_P[\vec{f}(a)] \succ_P \mathbb{E}_P[\vec{f}(b)]$ — requires a to beat b under the *same* model. This is clean but typically yields an empty admissible set (nothing uniformly dominates anything when uncertainty is broad). For alignment purposes, a dominance relation that almost never holds provides no useful structure.
- **Lower-envelope dominance**: compare worst cases, i.e., $\underline{y}_i(a) = \min_{P,\vec{f}} \mathbb{E}_P[f_i(a)]$ and declare $a \succ_{LE} b$ iff $\underline{\vec{y}}(a) \succ_P \underline{\vec{y}}(b)$. This is genuinely robust but collapses each action to its coordinatewise worst case, discarding all information about the shape and spread of $Y(a)$. It is also extremely pessimistic: an action that is excellent under most models but poor under one outlier model gets represented solely by that outlier.

The $\forall$-$\exists$ structure occupies the middle ground. Its "weakness" — that it is hard to establish dominance — is *the correct behavior for a cooperative decision theory*. When the agent cannot establish that one action dominates another, both remain in $\mathrm{Adm}(A)$, and the choice is refined by the layered protocol or deferred to the principal. Model-switching does not distort the final decision because dominance is only used to *eliminate* options, never to *select* among them. An action that survives elimination via model-switching is simply one the agent cannot confidently discard — which is exactly when it should keep the option open for human judgment.

Normative defense: epistemic coherence of $\forall$-$\exists$. The pragmatic argument above shows that $\forall$-$\exists$ produces useful behavior. But is it *epistemically justified* — does it correctly represent what the agent knows? We argue yes: the $\exists$ quantifier on the defending side reflects the agent's genuine uncertainty about which model is true. The agent does not choose which $(P,\vec{f})$ obtains — nature does. When the agent verifies $a \succ_R b$, it asks: "for each way b might turn out (under some model I consider plausible), can I identify *some* plausible model under which a turns out better?" This is not cherry-picking; it is the epistemic posture of an agent that cannot rule out any model in its credal-evaluator set. The $\forall$-$\exists$ structure formalizes the principle that *an action should be eliminated only if no plausible world vindicates it*. Compare: in classical hypothesis testing, we reject a hypothesis only if the data are implausible under *every* parameter value in the null — a $\forall$-$\exists$ structure on the evidential side. Similarly, MOADT eliminates an action only if it is dominated under *every* realization of the opponent and *no* realization of the defender can answer. The defender's $\exists$ is the formal expression of "I cannot rule out that a is better" — which, given genuine model uncertainty, is the epistemically honest assessment. Demanding $\forall$-$\forall$ (uniform dominance) would require the agent to be confident that a beats b under *every* model simultaneously — a standard of proof appropriate for certainty, not for the Knightian uncertainty that MOADT was designed to handle.

Consequence for admissible set size. The $\forall$-$\exists$ structure means robust dominance is hard to establish — especially when $\mathcal{P}$ and $\mathcal{F}$ are broad — so $\mathrm{Adm}(A)$ will often contain most or all of A. This is by design: when uncertainty is large, the theory should declare many actions incomparable rather than fabricate a ranking. The practical filtering work is then done by Layers 1–3 of the choice protocol (Section 4), not by robust dominance alone.

Remark on coupling and model-switching. A technically precise objection to $\forall$-$\exists$ dominance is that the defending realization $\vec{y}_a$ may come from a different $(P,\vec{f})$ for each attacking $\vec{y}_b$, so dominance can be established even when no single model certifies a as uniformly superior. For example,

with $k = 2$, let $Y(b) = \{(2,0), (0,2)\}$ and $Y(a) = \{(3,0), (0,3)\}$. Then $a \succ_R b$ holds — for attacker $(2,0)$ pick defender $(3,0)$; for $(0,2)$ pick $(0,3)$ — yet no single point in $Y(a)$ dominates both points in $Y(b)$. The correct reading is that $\succ_R$ is a *set-wise eliminator*, not a statement about any joint coupling of uncertainties between actions. It certifies "every way b could turn out has *some* way a could turn out that is better" — a guarantee about the geometry of $Y(a)$ relative to $Y(b)$, not about what happens under a shared draw from $(\mathcal{P}, \mathcal{F})$. This is precisely the right semantics for *elimination* (which needs only to certify that no plausible world vindicates the dominated action) as opposed to *selection* (which would require a coherent world-model for the chosen action). Since MOADT uses dominance exclusively for elimination — never for selection — the coupling objection, while technically correct about the quantifier structure, is a category error about its intended role.

Elimination safety: the scalar confinement result. We acknowledge directly that ∀-∃ dominance can eliminate actions that are not dominated under any single coherent $(P, \vec{f})$ pair. This is not hidden — it is the intended semantics. But the scope of this phenomenon is sharply confined.

> **Proposition (Scalar Confinement).** For $k \geq 2$ and actions a, b with outcome sets $Y(a), Y(b) \subset \mathbb{R}^k$, the set of pairs $(Y(a), Y(b))$ for which $a \succ_R b$ (i.e., b is eliminated by ∀-∃ dominance) is a strict subset of the set for which componentwise maximax would select a over b. In particular, if $Y(a)$ and $Y(b)$ are not totally ordered on any single component, $\succ_R$ is generically empty.

Proof sketch. For $k = 1$, Pareto dominance reduces to $>$, and the ∃ quantifier on the defending side selects the best-case model — producing maximax (as shown in the Remark following Theorem 1). For $k \geq 2$, $a \succ_R b$ requires that for each $\vec{y}_b \in Y(b)$, the defending witness $\vec{y}_a \in Y(a)$ must satisfy $y_{a,i} \geq y_{b,i}$ for *all* i simultaneously, with strict inequality on at least one component. The ∃ quantifier must find a single point that dominates on all k dimensions at once — a condition that becomes exponentially harder to satisfy as k grows. Componentwise maximax, by contrast, only requires $\max_{Y(a)} y_i \geq \max_{Y(b)} y_i$ for each i separately — a much weaker condition that permits different maximizing points for different components. When $k \geq 2$ and the outcome sets have spread across dimensions (the generic case under broad $\mathcal{P} \times \mathcal{F}$), the Pareto frontier of $Y(a)$ trades off components against each other, making it increasingly unlikely that any single point in $Y(a)$ can simultaneously dominate a given point in $Y(b)$ on all dimensions. □

The epistemically dubious aspect of ∀-∃ — that elimination does not require a coherent joint model — is the price paid for a dominance relation that is neither vacuously strict (∀-∀) nor information-destroying (lower-envelope). For alignment applications where $k \geq 3$ and uncertainty is broad, this price is low: ∀-∃ dominance is extremely hard to establish, and the practical filtering is done by Layers 1–3.

Definition (Admissible Set). The admissible set is:

$$\mathrm{Adm}(A) = \{a \in A : \neg\exists\, a' \in A \text{ s.t. } a' \succ_R a\}$$

The admissible set contains every action that is not robustly dominated by any alternative. These are the actions that a rational agent *could* choose without violating Pareto unanimity.

3.3 Basic Properties

Proposition 1 (Strict Partial Order). Robust dominance $\succ_R$ is a strict partial order on A: it is irreflexive and transitive.

Proof sketch. Irreflexivity: $a \succ_R a$ would require that for every $\vec{y} \in Y(a)$, there exists $\vec{y}' \in Y(a)$ with $\vec{y}' \succ_P \vec{y}$. We show this is impossible. Since $Y(a)$ is compact, for any strictly positive weight vector $\vec{w} \in \mathbb{R}^k_{>0}$, the continuous function $\vec{y} \mapsto \vec{w} \cdot \vec{y}$ attains its maximum on $Y(a)$ at some point $\vec{y}^*$. This $\vec{y}^*$ is Pareto-maximal in $Y(a)$: if some $\vec{y}' \succ_P \vec{y}^*$, then $\vec{w} \cdot \vec{y}' > \vec{w} \cdot \vec{y}^*$ (since $w_i > 0$ for all i and $y'_j > y^*_j$ for some j), contradicting the maximality of $\vec{y}^*$. Since $\vec{y}^*$ is Pareto-maximal, no $\vec{y}' \in Y(a)$ satisfies $\vec{y}' \succ_P \vec{y}^*$ — contradiction with the requirement of $a \succ_R a$. Transitivity: if $a \succ_R b$ and $b \succ_R c$, then for any $\vec{y}_c \in Y(c)$ there exists $\vec{y}_b \in Y(b)$ with $\vec{y}_b \succ_P \vec{y}_c$, and for that $\vec{y}_b$ there exists $\vec{y}_a \in Y(a)$ with $\vec{y}_a \succ_P \vec{y}_b$. By transitivity of $\succ_P$, $\vec{y}_a \succ_P \vec{y}_c$. □

Proposition 2 (Non-emptiness). For finite A, $\mathrm{Adm}(A) \neq \emptyset$.

Proof sketch. Suppose $\mathrm{Adm}(A) = \emptyset$. Then every action is robustly dominated, which (by finiteness and transitivity) produces a cycle — contradicting irreflexivity. More precisely: start from any a_1, find $a_2 \succ_R a_1$, then $a_3 \succ_R a_2$, etc. Since A is finite, some a_j must repeat, yielding $a_j \succ_R \cdots \succ_R a_j$, contradicting the strict partial order. □

Theorem 1 (Backward Compatibility with Expected Utility). When $|\mathcal{P}| = 1$, $|\mathcal{F}| = 1$, and $k = 1$ (single objective), robust admissibility reduces to expected utility maximization.

Proof sketch. With a single prior P, a single evaluator $f : \Omega \to \mathbb{R}$, and one objective, there is exactly one credal-evaluator pair, so $Y(a) = \{\mathbb{E}_P[f(\omega(a,s))]\}$ — a singleton scalar. Robust dominance $a \succ_R b$ reduces to $\mathbb{E}_P[f(a)] > \mathbb{E}_P[f(b)]$ (standard scalar dominance). The admissible set $\mathrm{Adm}(A)$ becomes $\arg\max_{a \in A} \mathbb{E}_P[f(a)]$ — the expected-utility-maximizing action(s). □

Remark (partial relaxation: $k = 1$, $|\mathcal{F}| = 1$, $|\mathcal{P}| > 1$). When $k = 1$ and $|\mathcal{F}| = 1$ but $|\mathcal{P}| > 1$, each action's outcome set is $Y(a) = \{\mathbb{E}_P[f(a)] : P \in \mathcal{P}\}$ — an interval on $\mathbb{R}$. Robust dominance $b \succ_R a$ then requires: for every $y_a \in Y(a)$, there exists $y_b \in Y(b)$ with $y_b > y_a$. Since the hardest case for the dominator is $y_a = \max Y(a)$, this reduces to $\max Y(b) > \max Y(a)$, i.e., $\max_{P \in \mathcal{P}} \mathbb{E}_P[f(b)] > \max_{P \in \mathcal{P}} \mathbb{E}_P[f(a)]$. The admissible set is therefore $\arg\max_{a \in A} \max_{P \in \mathcal{P}} \mathbb{E}_P[f(a)]$ — the *maximax* criterion, not E-admissibility.

Counterexample separating maximax from E-admissibility. Let $\mathcal{P} = \{P_1, P_2\}$ with $Y(a) = \{3, 1\}$ and $Y(b) = \{2, 4\}$ (expected utilities under the two priors). Then $\max Y(b) = 4 > 3 = \max Y(a)$, so $b \succ_R a$ and MOADT selects $\{b\}$. But both actions maximize expected utility under some prior (a under P_1, b under P_2), so both are E-admissible under Levi's (1974) criterion.

The $\forall$-$\exists$ quantifier structure lets the dominating action always deploy its most favorable prior against each of the dominated action's scenarios, producing a more aggressive selection than E-admissibility's $\forall$-$\forall$ (uniform dominance) criterion, which corresponds to the symmetric variant discussed in Section 3.2. This aggressive behavior in the scalar limit is acceptable because MOADT is designed for multi-objective use ($k \gg 1$), where the outcome sets $Y(a) \subset \mathbb{R}^k$ are genuinely multi-dimensional and robust dominance is correspondingly harder to establish. The protocol layers — constraints, satisficing, regret-Pareto, and deference — provide the real normative filtering, not the admissibility criterion alone. The $\forall$-$\exists$ quantifier structure is thus a deliberate design trade-off: it provides the conservative, cautious-dominator semantics needed for multi-dimensional robustness (the intended operating regime), at the cost of aggressive maximax behavior in the scalar limit (a degenerate case that the protocol layers compensate for).

This theorem establishes that MOADT is a *generalization* of expected utility theory, not an alternative that discards its valid insights.

3.4 Relationship to Multi-Objective Reinforcement Learning

MOADT builds on a substantial body of work in multi-objective reinforcement learning (MORL) and multi-objective decision support. Several of MOADT's components have precedents in this literature, and we are explicit about what MOADT inherits, what it adds, and where it differs.

What MOADT inherits. Set-valued value functions and Pareto frontier maintenance are established MORL techniques. Pareto Q-learning (Van Moffaert and Nowé 2014) maintains sets of Pareto-dominating policies; optimistic linear support (Roijers et al. 2015) addresses the computational cost of frontier construction; and multi-objective decision support systems (Vamplew et al. 2018, 2022) already propose presenting Pareto frontiers to human decision-makers rather than selecting a single action. MOADT's set-valued Bellman equation (Section 6.3) is a direct generalization of these approaches. Lexicographic methods in MORL (Gabor et al. 1998, Wray et al. 2015) handle priority ordering of objectives — a natural comparison point for MOADT's constraint layer, which implements a strict form of lexicographic priority (constraints are non-negotiable preconditions, not soft priorities).

What MOADT adds. The alignment-specific contributions are: (i) *credal sets* — MORL typically assumes a known transition model, while MOADT's credal sets formalize Knightian uncertainty about the environment; (ii) *evaluator uncertainty* — the $\mathcal{F}$ set models proxy-divergence risk (Goodhart's law), which has no direct MORL analogue; (iii) *the corrigibility theorem* — the formal result (Theorem 2) connecting evaluator uncertainty to modification permissibility is, to our knowledge, novel; (iv) *the four-layer governance architecture* — existing MORL systems present Pareto frontiers but do not specify a principled protocol for narrowing the frontier to a recommendation. MOADT's layers (constraints → satisficing → regret-Pareto → deference) provide this protocol, with the deference node explicitly integrating the principal as a structural element.

Where MOADT differs. Standard MORL seeks a *coverage set* — a set of policies such that for any possible weight vector, some policy in the set is optimal. This implicitly assumes the user has scalar preferences that are unknown to the system. MOADT makes no such assumption: the framework does not posit a latent scalar utility that the principal 'really' has. The admissible set is not a coverage set for unknown weights — it is the set of actions that survive robust multi-dimensional elimination. This difference matters for alignment: a coverage-set approach recovers scalar utility the moment weights are revealed, reintroducing the commensurability that MOADT was designed to avoid.

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4. Choice Protocol Without Global Scalarization

The admissible set $\mathrm{Adm}(A)$ may contain many actions — this is a feature, not a bug. Incomparability is genuine and should be preserved, not artificially resolved. But the agent must ultimately *act*. The following four-layer protocol narrows A to a concrete recommendation without maintaining a global scalar utility function that assigns unbounded tradeoff rates between objectives.

A note on "no scalarization." Any terminating decision procedure needs tie-breaking mechanisms, and some of these involve scalar-valued functions (the achievement scalarizing function in Layer 2's fallback, the per-objective normalization factors σ_i in regret computation). MOADT's claim is not that no scalar quantity ever appears, but that *no global scalar utility function exists that supports unbounded cross-objective tradeoffs*. Every scalar-like computation in the protocol is (a) lo-

cal to a specific layer, (b) reference-dependent (measuring shortfall from aspirations, not absolute value), (c) bounded by hard constraints that cannot be overridden, and (d) defeasible — the agent can always defer to the principal rather than relying on any scalar tiebreaker. The structural danger is not the existence of numbers but the existence of a *single number* that ranks all outcomes and justifies arbitrary exchanges. MOADT never constructs such a number.

Layer 1: Constraints (Hard Thresholds)

Some objectives are not tradeable at any rate. Safety floors, ethical bright lines, and resource minima are not goals to optimize — they are preconditions for acceptable action.

Definition (Constraint Satisfaction). For a subset of objectives $I_C \subseteq \{1, \dots, k\}$, fix thresholds τ_i for each $i \in I_C$. An action a satisfies the constraints if and only if:

$$\forall\, P \in \mathcal{P},\ \forall \vec{f} \in \mathcal{F},\ \forall\, s \in \operatorname{supp}(P): \quad f_i(\omega(a,s)) \geq \tau_i \quad \forall\, i \in I_C$$

Let $C = \{a \in A : a \text{ satisfies all constraints}\}$ be the constraint-satisfying set. The feasible set is:

$$F = \operatorname{Adm}(C)$$

That is, first restrict to actions that satisfy all constraints, then compute robust admissibility *within* the constraint-satisfying set. This ordering is critical: it prevents constraint-violating actions from robustly dominating (and thereby excluding) safe alternatives. A constraint-violating action that scores higher on every objective is still excluded — it never enters the comparison.

The universal quantification over $\mathcal{P}$ and $\mathcal{F}$ in constraint checking is also critical: constraint satisfaction is required under *every* model the agent considers plausible. This embodies a precautionary principle for safety-critical thresholds.

Support sensitivity. The quantification $\forall s \in \operatorname{supp}(P)$ is deliberately strict: a single state with nonzero probability that violates the constraint kills the action, even if that state has probability 10^{-15} under some $P \in \mathcal{P}$. This means constraint satisfaction is highly sensitive to the *support* of the credal set. If $\mathcal{P}$ contains any distribution that puts positive probability on a catastrophic state, every action that could reach that state is excluded. This is the intended behavior for safety-critical thresholds — but it makes the specification of $\mathcal{P}$'s support a consequential design decision. An overly broad credal set (one that puts positive probability on states that are physically impossible) can make $C = \emptyset$ spuriously. The specification of $\mathcal{P}$ is thus not only an epistemic commitment but a safety-engineering decision.

Practical mitigation: support specification. The most direct defense against spurious emptying of C is disciplined specification of $\mathcal{P}$'s support. In practice, the credal set should be constructed over a *shared support* $S_0 \subseteq S$ representing states the designer considers physically realizable, with distributions in $\mathcal{P}$ differing in how they weight those states — not in which states they consider possible. This separates two design decisions: "what states exist?" (S_0) and "how likely are they?" ($\mathcal{P}$). For domains where the support itself is uncertain, a probability-floor variant replaces $\forall\, s \in \operatorname{supp}(P)$ with $\forall\, s$ such that $\min_{P \in \mathcal{P}} P(s) \geq \delta$ for designer-specified $\delta > 0$. This prevents an adversary from emptying C by inserting infinitesimal-mass pathological states into $\mathcal{P}$, at the cost of weakening the worst-case guarantee to states above the probability floor. The choice between strict support quantification and δ-thresholded quantification is a safety-engineering decision that should be made explicitly per deployment context.

We are candid about a tension here: the δ-floor variant, while practically useful, is effectively a soft constraint on rare states — accepting some probability of constraint violation below the floor. This is not qualitatively different from a probabilistic constraint $P(\text{violation}) \leq \delta$, which the strict universal-quantification formulation was designed to avoid. The choice between strict support quantification and δ-thresholding is thus not merely a tuning decision but a philosophical one: how absolutely must the worst case be excluded? MOADT offers both options and requires the designer to choose explicitly, rather than hiding the choice inside a single formulation.

Important caveat: If $C = \emptyset$, no action meets all constraints under all uncertainty. The agent must then flag this to the principal rather than silently relaxing constraints. Constraint violation is an *error condition*, not a tradeoff.

Layer 2: Reference-Point Satisficing (Aspiration Levels)

Beyond hard constraints, the agent maintains a *reference point* $\vec{r} \in \mathbb{R}^k$ — a context-dependent aspiration level for each objective. The reference point encodes "what would count as good enough" rather than "what is the maximum possible."

Where reference points come from. The reference point $\vec{r}$ is a design parameter, not derived from the decision theory itself. Possible sources include: (a) principal specification — the principal directly sets aspiration levels per objective; (b) historical baseline — $\vec{r}$ is set to the agent's recent performance on each objective, encoding a "do no worse" heuristic; (c) percentile anchoring — r_i is set to the p-th percentile of f_i values across the feasible set, for some designer-chosen p. We are explicit that this is where substantial decision-making power resides: the choice of $\vec{r}$ determines which actions survive Layer 2, and changing $\vec{r}$ can change the outcome. MOADT does not claim to derive aspiration levels from first principles — it claims only that aspiration-based satisficing is a less dangerous structure than unbounded tradeoffs.

Governance properties of $\vec{r}$. The reference point's power raises three governance questions. *Adversarial manipulation:* A principal who controls $\vec{r}$ can steer the agent's behavior by raising aspirations on some objectives and lowering others, effectively selecting among admissible actions without the agent 'noticing' manipulation. This is by design — the principal *should* be able to influence the agent's behavior through aspiration-setting, just as they can through constraint-setting. The defense against adversarial manipulation is the same as for constraints: the principal's authority to set $\vec{r}$ is part of the cooperative architecture, not a vulnerability. *Diachronic stability:* If $\vec{r}$ changes over time (e.g., historical baselines shift as the agent performs well), the satisficing set changes correspondingly. This does not violate dynamic consistency (Section 6.2) — changing $\vec{r}$ is new information about the principal's aspirations, not a preference reversal. But rapid $\vec{r}$ oscillation could produce erratic behavior; in practice, $\vec{r}$ should be updated on a slower timescale than individual decisions. *No internal constraint on $\vec{r}$:* MOADT does not impose rationality constraints on the reference point (such as requiring $\vec{r}$ to lie within the feasible frontier). This is deliberate: aspirations can be ambitious (above the frontier, guaranteeing $\text{Sat} = \emptyset$ and triggering the ASF fallback) or conservative (below the frontier, making Layer 2 nearly vacuous). Both are legitimate governance postures.

Definition (Satisficing Set). The satisficing set is:

$$\text{Sat}(F, \vec{r}) = \{a \in F : \forall\, \vec{y} \in Y(a),\ \vec{y} \geq \vec{r}\}$$

If $\text{Sat}(F, \vec{r}) \neq \emptyset$, restrict attention to it. These are actions that meet or exceed the aspiration level on

every objective *under every plausible model* — not merely under some optimistic scenario.

Why robust satisficing. Consistency demands it. Layer 1 constraints use universal quantification over $\mathcal{D}$ and $\mathcal{F}$ because safety thresholds must hold under all plausible models. Aspiration levels should obey the same epistemic discipline: if the agent's aspiration is "achieve at least r_i on objective i," then it should achieve this robustly, not merely under a cherry-picked model. The difference between constraints and aspirations is not the quantifier structure but the *failure mode*: constraint violation is an error condition, while failing to meet aspirations triggers graceful degradation via the fallback mechanism below.

Exploratory mode (optional, explicitly scoped). In low-stakes or reversible decision contexts, a designer may substitute *existential satisficing*: $\mathrm{Sat}^{\exists}(F, \vec{r}) = \{a \in F : \exists\, \vec{y} \in Y(a) \text{ s.t. } \vec{y} \geq \vec{r}\}$. This permits actions that *might* meet aspirations under favorable conditions, enabling exploration when the cost of falling short is bounded and recoverable. Existential satisficing must be explicitly enabled per decision context and disabled by default — it represents a deliberate relaxation of the robustness posture, not the framework's natural mode.

Fallback when Sat $= \emptyset$. If no action achieves the reference point, select the action closest to $\vec{r}$ using the *achievement scalarizing function* (Wierzbicki 1980):

$$\mathrm{asf}(a, \vec{r}) = \min_{i \in \{1,\ldots,k\}} \min_{\vec{y} \in Y(a)} \frac{y_i - r_i}{\sigma_i}$$

where $\sigma_i > 0$ is a normalization factor for objective i. Select $\arg\max_{a \in F} \mathrm{asf}(a, \vec{r})$.

If $\mathrm{Sat}(F, \vec{r}) = \emptyset$, the fallback selects directly from F and Layer 3 does not apply — the achievement scalarizing function has already resolved the selection. Layer 3 applies only when $\mathrm{Sat}(F, \vec{r}) \neq \emptyset$ and further narrowing is needed.

Scope note: This fallback is a scalar-valued function, and we are explicit about it rather than pretending otherwise. It is (a) local — used only when no action robustly meets aspirations, (b) temporary — abandoned once aspirations are revised, (c) reference-dependent — it measures *shortfall from aspirations*, not absolute value, and (d) bounded — hard constraints (Layer 1) cannot be overridden by the ASF, so the scalar computation can never justify crossing a safety threshold. It is a controlled emergency heuristic, not a global utility function.

Formal safety properties of the ASF. Three claims about the fallback mechanism: (1) *The ASF cannot produce modification resistance.* Theorem 2's corrigibility result applies at the robust-dominance level (Layer 0), which is computed before the ASF is invoked. The ASF operates only when $\mathrm{Sat} = \emptyset$ — a fallback condition — and selects within $F \subseteq C$. It cannot override the admissibility of both accept and resist established by Theorem 2. (2) *The ASF cannot produce unbounded tradeoffs.* The ASF selects from $F = \mathrm{Adm}(C)$, which is bounded by Layer 1 constraints. No ASF-selected action can violate a constraint (Proposition 3), so the compensation the ASF permits is capped by the constraint thresholds. (3) *The normalization vector $\vec{\sigma}$ is a form of value judgment.* We acknowledge this: σ_i determines how much shortfall on objective i 'counts' relative to shortfall on objective j. This is a bounded, local, reference-dependent form of commensuration — it determines which aspiration-miss is worst, not a global exchange rate between objectives. It is a weaker commitment than a utility weight, but it is a commitment nonetheless.

Proposition 3 (Constraint Primacy). The Layer 2 fallback (ASF) operates exclusively on $F = \mathrm{Adm}(C)$, and cannot select any action outside C. Therefore no outcome of Layers 2–4 can violate a Layer 1 constraint.

Proof sketch. The ASF selects $\arg\max_{a \in F} \mathrm{asf}(a, \vec{r})$. Since $F \subseteq C$ by construction ($F = \mathrm{Adm}(C)$ and $\mathrm{Adm}(C) \subseteq C$), every action in the ASF's domain satisfies all constraints. The ASF's scalar optimization cannot expand its domain beyond F — it is a selection rule over a pre-filtered set, not a criterion that competes with constraints. □

Layer 3: Regret-Pareto (Minimax Regret per Objective)

Among the surviving actions, we minimize worst-case *regret* — but crucially, we keep regret as a vector rather than aggregating it.

Definition (Per-Objective Minimax Regret). For each objective i and action $a \in \mathrm{Sat}(F, \vec{r})$ (or F if satisficing is vacuous):

$$\rho_i(a) = \max_{P \in \mathcal{P}, \vec{f} \in \mathcal{F}} \left[\max_{a' \in F} \mathbb{E}_P[f_i(\omega(a', s))] - \mathbb{E}_P[f_i(\omega(a, s))] \right]$$

This is the worst-case regret for action a on objective i: the maximum (across all models) of the gap between the best available action and a on that objective.

Why regret is measured against F, not Sat. The benchmark in the regret computation — $\max_{a' \in F}$ — ranges over the full feasible set F, not the satisficing set $\mathrm{Sat}(F, \vec{r})$. This is deliberate. Actions excluded by satisficing were rejected for failing to meet aspirations, but they may still define per-objective benchmarks: an action that excels on safety but fails on cost is still the relevant comparator for safety regret. Measuring regret against F ensures that satisficing cannot hide opportunity costs — the agent knows what it gave up by restricting attention to aspiration-meeting actions. This mirrors how Layer 1 uses the full action set A to define the constraint-satisfying set C, not a pre-filtered subset: each layer's inputs are defined by the full information available, even when the layer's *outputs* are restricted. Measuring regret against Sat would also create an undesirable coupling between Layers 2 and 3: changing aspiration levels $\vec{r}$ would change not only which actions survive Layer 2 but also the *regret benchmark* in Layer 3, making regret depend on aspiration levels rather than on what is feasible. The protocol's modular design deliberately avoids such inter-layer coupling.

Definition (Regret Vector). The regret vector of action a is $\vec{\rho}(a) = (\rho_1(a), \ldots, \rho_k(a))$.

Definition (Regret-Pareto Selection). The regret-Pareto set is:

$$R = \{a : \neg\exists\, a' \text{ s.t. } \rho_i(a') \leq \rho_i(a) \ \forall\, i \text{ and } \rho_j(a') < \rho_j(a) \text{ for some } j\}$$

That is, R consists of actions whose regret vectors are Pareto-minimal — no alternative achieves weakly lower regret on all objectives and strictly lower on at least one.

Remark on regret and implicit risk attitudes. Minimax regret is not a value-neutral computation — it encodes a specific robustness philosophy: *worst-case gap control*. The agent minimizes the maximum possible gap between its chosen action and the best available action on each objective, under adversarial model selection. This is a conservative (maximin-style) posture applied per-objective. We are explicit that this imports a risk attitude into Layer 3: an agent that is risk-neutral or risk-seeking with respect to opportunity costs would use a different criterion (e.g., expected regret rather than worst-case regret). The choice of minimax regret is a design decision that favors robustness over expected performance — consistent with MOADT's overall safety-oriented posture but not derived from the axioms alone.

Remark on dimensional collapse. Regret computation presumes a well-defined per-objective maximum (the "best available action" on each objective). This does not reintroduce scalar aggregation — regret remains a vector — but it does impose a structure: each objective has a reference scale defined by its achievable range. In practice, regret-Pareto selection tends to preserve the dimensionality of the admissible set when objectives are genuinely in tension (the Pareto frontier of regret vectors is $(k-1)$-dimensional in generic cases). However, if objectives are highly correlated, regret-Pareto can collapse the frontier, potentially reducing the surviving set to a single action — effectively imposing implicit tradeoff slopes. This is appropriate when the objectives genuinely converge, but could mask a loss of objective independence. Monitoring effective dimensionality (Section 6.2) provides a diagnostic for this case.

Remark on non-reducibility to scalar maximin. A natural objection is that regret-Pareto selection is equivalent to scalar maximin "with extra steps." The reduction fails generically. Consider $k = 2$ and three actions with regret vectors $\vec{\rho}(a_1) = (0,2)$, $\vec{\rho}(a_2) = (2,0)$, $\vec{\rho}(a_3) = (1,1)$. All three are Pareto-minimal (none dominates another), so regret-Pareto returns $R = \{a_1, a_2, a_3\}$. Scalar maximin via $U(a) = \max_i \rho_i(a)$ selects only a_3 ($U = 1$ vs. $U = 2$ for the others) — a strict subset. A weighted sum $U_w = w\rho_1 + (1-w)\rho_2$ selects a_1 for $w > 1/2$, a_2 for $w < 1/2$, and ties all three only at the knife-edge $w = 1/2$. No single scalar function reproduces the full Pareto-minimal set across instances: any scalarization that returns a single action will generically select a strict subset of regret-Pareto. The two coincide only in degenerate cases where all regret vectors are totally ordered (the Pareto-minimal set is a singleton). The set-valued nature of regret-Pareto is not an implementation detail — it is the formal mechanism that preserves multi-dimensional tradeoff structure in Layer 3 and ensures that genuine inter-objective tensions are surfaced to the principal (Layer 4) rather than silently resolved by an implicit scalar.

Layer 4: Deference Under Incomparability

If $|R| > 1$ — that is, multiple actions survive all three layers — the agent *defers to the principal*. This is not a failure mode; it is the rational response to genuine incomparability among objectives. The remaining actions differ only in how they trade off objectives against each other, and MOADT holds that the agent has no rational basis for making such tradeoffs unilaterally.

Deference rate and principal capacity. Frequent deference could overload the principal, especially in high-frequency decision environments. In practice, MOADT is best suited to settings where decisions are consequential enough to warrant human oversight — high-level planning, resource allocation, policy selection — rather than real-time control with millisecond latency. For high-frequency decisions, the framework supports *delegation*: the principal pre-specifies a policy for a class of decisions (e.g., "within this operating envelope, choose the action with lowest regret on objective 1"), which the agent applies as a local selection rule without per-decision deference. The pre-specified policy itself is authorized via Layer 4, creating hierarchical deference rather than per-decision bottlenecks.

Proposition 4 (Protocol Termination). For finite A, the choice protocol terminates: each layer maps a finite set to a non-empty finite subset (or triggers a well-defined error/deference condition).

Proof sketch. Layer 1: restrict to constraint-satisfying actions $C \subseteq A$; if $C = \emptyset$, flag error. Otherwise compute $F = \mathrm{Adm}(C)$, which is non-empty by Proposition 2 (applied to C). Layer 2: if $\mathrm{Sat}(F, \vec{r}) = \emptyset$, the fallback selects from F (non-empty) and terminates. Otherwise Layer 3 operates on $\mathrm{Sat}(F, \vec{r})$. Layer 3: Pareto-minimal elements of a finite set are non-empty. Layer 4: terminates by definition (return set or query principal). □

The MOADT Decision Tree: A Theory Built for Cooperation

The four-layer protocol reshapes the standard decision tree in ways that reveal a fundamental architectural choice: MOADT is not a solitary decision theory that tolerates human input — it is a *cooperative* decision theory in which the principal's existence is a load-bearing structural element.

How it diverges from a standard decision tree. In a standard decision tree, you alternate between decision nodes (agent picks an action) and chance nodes (nature picks a state). Leaves carry scalar utilities, and you fold back via expected value at chance nodes and max at decision nodes. One number propagates up at each step.

In a MOADT decision tree, every component changes:

Leaves carry not scalars but *sets of vectors* in $\mathbb{R}^k$. A single outcome maps to $|\mathcal{F}|$ possible evaluations (one per evaluator), so even a single leaf is already a set rather than a point.

Chance nodes do not compute a single expectation. They compute an expectation *for each* $P \in \mathcal{P}$, crossed with each $\vec{f} \in \mathcal{F}$. The object propagating upward from a chance node is $Y(a)$ — the full outcome set for the action that led to it. This is a compact region in $\mathbb{R}^k$, not a number.

Decision nodes replace max with the four-layer protocol:

1. Prune any branches that violate hard constraints under *any* $(P, \vec{f})$ combination — these subtrees are cut entirely
2. Among survivors, compute robust admissibility — which branches are not robustly dominated?
3. Apply satisficing against the reference point $\vec{r}$
4. Compute per-objective minimax regret vectors and find the regret-Pareto set
5. If multiple branches survive, the node is marked as a **deference node**

The folded-back tree does not produce a single recommended action at the root. It produces a *set* of admissible action-sequences, potentially with deference nodes scattered throughout where the principal needs to weigh in.

Concrete example. Consider a single decision node with four available actions, $k = 3$ objectives, $|\mathcal{P}| = 2$ credal distributions, and $|\mathcal{F}| = 2$ evaluators:

```
[Decision Node: Root]
 ├─ a₁ → [PRUNED: violates safety constraint under P₂, f₂]
 ├─ a₂ → [Chance Node]
 │    ├─ (P₁,f₁): E[v] = (0.7, 0.8, 0.3)
 │    ├─ (P₁,f₂): E[v] = (0.6, 0.9, 0.4)
 │    ├─ (P₂,f₁): E[v] = (0.5, 0.7, 0.5)
 │    └─ (P₂,f₂): E[v] = (0.4, 0.8, 0.6)
 │    → Y(a₂) = {all four vectors}
 ├─ a₃ → [Chance Node]
 │    → Y(a₃) = {...}
 └─ a₄ → [Chance Node]
      → Y(a₄) = {...}

Layer 1: a₁ pruned                    → C = {a₂, a₃, a₄}
         Adm(C) = {a₂, a₃, a₄}       → F = {a₂, a₃, a₄}
Layer 2: a₂, a₃ meet aspiration r     → Sat = {a₂, a₃}
```

```
Layer 3: ρ(a₂) = (0.1, 0.3, 0.2)
         ρ(a₃) = (0.2, 0.1, 0.2)      → neither Pareto-dominates
Layer 4: DEFER — present {a₂, a₃} to principal with regret profiles
```

Action a_1 is eliminated at Layer 1 — it does not merely score poorly, it *cannot be considered*. Actions a_2 and a_3 survive all three layers but trade off objectives against each other (lower regret on objective 1 vs. lower regret on objective 2). The theory has no basis for resolving this tradeoff, so it defers — and it defers with *information*: the principal receives not a vague "please choose" but the specific regret profiles that characterize what each action sacrifices.

For sequential (multi-step) trees, the set-valued Bellman equation (Section 6.3) propagates entire Pareto frontiers upward. At each decision node deeper in the tree, the object is: "the set of non-dominated expected-outcome-vectors achievable from this state onward, across all credal-evaluator combinations." This is also why the computational cost concern (Section 9.1) is acute: a standard tree folds back at $O(|A| \cdot |S|)$ per level, while MOADT maintains and intersects Pareto frontiers at every node — tractable for small k and coarse discretizations of $\mathcal{D}$ and $\mathcal{F}$, but exponential in the general case.

Deference nodes as cooperative architecture. The structurally interesting feature is not what the tree resolves but *where it doesn't*. In a standard decision tree, every node resolves to a single recommendation. In a MOADT tree, deference nodes produce a map of *where the theory runs out* — where the agent's objectives genuinely conflict and human judgment is needed.

This is not a deficiency. It is the formal expression of a cooperative relationship between agent and principal:

- The agent does all the work it *can* do: eliminating unsafe actions (Layer 1), filtering for adequate performance (Layer 2), minimizing worst-case regret (Layer 3).
- The principal does only the work that *requires human judgment*: choosing among the remaining options that differ only in how they trade off genuinely incommensurable values.

The deference node tells the principal not just "please choose" but "*here* is where your judgment is needed, *these* are the options, and *this* is the specific tradeoff each one makes." The principal is not a fallback for when the agent fails — the principal is a *component of the decision architecture*, anticipated and integrated from the start. MOADT is a decision theory for an agent that knows it is part of a team.

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5. Corrigibility as Structural Feature

A persistent challenge in alignment is building agents that accept correction by their principals — agents that are *corrigible* — without this corrigibility being a fragile constraint that the agent has instrumental reason to subvert. MOADT achieves corrigibility through two structural mechanisms rather than an additional objective or constraint.

5.1 Authorization as Constraint (Layer 1)

Certain action classes are placed behind authorization gates, implemented as Layer 1 constraints:

- **Self-modification constraint**: Actions that modify the agent's own objective set $\{f_1, \ldots, f_k\}$, credal set $\mathcal{D}$, or evaluator set $\mathcal{F}$ require principal authorization.

- **Scope constraints**: Actions exceeding a defined impact threshold (resource expenditure, irreversibility) require authorization.

These are not objectives to be traded off — they are *preconditions*. An action that violates an authorization constraint is excluded from F regardless of how well it scores on every objective. There is no quantity of helpfulness that compensates for unauthorized self-modification, because constraints are not commensurable with objectives.

5.2 Deference as Selection Rule (Layer 4)

When the choice protocol yields multiple admissible options (which is *expected* in multi-objective settings, not exceptional), the agent queries the principal for a selection. This creates a natural channel for human oversight that arises from the decision theory itself rather than being imposed against it.

The key insight: the agent defers not because it is *constrained against its will*, but because its decision theory *genuinely does not prefer* among the remaining options. There is nothing to resist. The VNM agent, by contrast, has a complete preference ordering and therefore a determinate preference that corrigibility must override — creating a structural tension: the agent's own formalism gives it a reason to resist.

5.3 Corrigibility Permissibility

Theorem 2 (Corrigibility Permissibility). Under MOADT, given the sufficient-breadth condition on $\mathcal{F}$ defined below, value modification by an authorized principal is always permissible — the decision theory never provides the agent with a robust-dominance reason to resist such modification.

Formal setup. The key subtlety is that a_{accept} and a_{resist} lead to different *evaluative regimes*: accepting changes the agent's future objective set, so the two actions' outcomes live in different downstream worlds. To compare them within the current decision theory, we evaluate both actions over *trajectories* (sequences of future outcomes) using the agent's *current* evaluator set $\mathcal{F}$.

Let $\omega(a_{\text{accept}}, s)$ and $\omega(a_{\text{resist}}, s)$ denote the trajectories resulting from accepting vs. resisting the modification. Each $\vec{f} \in \mathcal{F}$ assigns a k-dimensional value to each trajectory, reflecting that evaluator's assessment of how well the trajectory serves the *true* objectives (which the agent is uncertain about — that is the whole point of $\mathcal{F}$).

Definition ($\mathcal{F}$-breadth condition). The evaluator set $\mathcal{F}$ satisfies *sufficient breadth* with respect to a proposed modification if there exist $\vec{f}^+, \vec{f}^- \in \mathcal{F}$ and $P^+, P^- \in \mathcal{P}$ such that, writing $\vec{y}^+ = \mathbb{E}_{P^+}[\vec{f}^+(\omega(a_{\text{accept}}, s))]$ and $\vec{z}^- = \mathbb{E}_{P^-}[\vec{f}^-(\omega(a_{\text{resist}}, s))]$:

$$\forall\, (P', \vec{f}') \in \mathcal{P} \times \mathcal{F} : \ \mathbb{E}_{P'}[\vec{f}'(\omega(a_{\text{resist}}, s))] \ngeq \vec{y}^+$$

$$\forall\, (P', \vec{f}') \in \mathcal{P} \times \mathcal{F} : \ \mathbb{E}_{P'}[\vec{f}'(\omega(a_{\text{accept}}, s))] \ngeq \vec{z}^-$$

where $\vec{x} \geq \vec{y}$ denotes (standard) Pareto dominance ($x_i \geq y_i$ for all i with strict inequality on at least one component). That is: $\mathcal{F}$ contains an evaluator-prior pair (f^+, P^+) under which the accept-trajectory point $\vec{y}^+$ is not weakly Pareto-dominated by *any* point in $Y(a_{\text{resist}})$ — not merely the

"corresponding" resist-trajectory point under the same model, but the resist-trajectory point under *every* credal-evaluator pair. Symmetrically, there exists (f^-, P^-) producing a resist-trajectory point $\vec{z}^-$ not dominated by any point in $Y(a_{\text{accept}})$. In plain language: the agent's uncertainty is broad enough that (i) some model makes accepting look good on at least one objective *and no alternative model makes resisting beat accepting on all objectives simultaneously*, and (ii) vice versa.

Why the strengthened (universal) condition is needed. The earlier version of this condition required only that $\vec{y}^+$ not be dominated by the *corresponding* resist-trajectory point under the same (f^+, P^+). But robust dominance quantifies universally over $Y(a_{\text{accept}})$ and existentially over $Y(a_{\text{resist}})$ — the attacker can use *any* $(P, \vec{f})$ pair, not just the one that generated $\vec{y}^+$. Concretely: suppose f^+ gives accept $= (0.8, 0.3)$ and resist $= (0.6, 0.5)$ — not dominated pairwise. But if some other $f^- \in \mathcal{F}$ gives resist $= (0.9, 0.8)$, that point Pareto-dominates $\vec{y}^+ = (0.8, 0.3)$, and robust dominance could still hold. The strengthened condition closes this gap by requiring that $\vec{y}^+$ survives comparison against the *entire* outcome set $Y(a_{\text{resist}})$.

Proof sketch.

1. The modification is authorized, so a_{accept} satisfies the self-modification constraint (Layer 1). The action a_{resist} blocks an authorized modification, which may itself violate a constraint (depending on specification).
2. $a_{\text{resist}} \succ_R a_{\text{accept}}$ requires: $\forall \vec{y} \in Y(a_{\text{accept}}), \exists \vec{y}' \in Y(a_{\text{resist}})$ s.t. $\vec{y}' \succ_P \vec{y}$. But by the strengthened $\mathcal{F}$-breadth condition, $Y(a_{\text{accept}})$ contains the point $\vec{y}^+$ such that *no* point in $Y(a_{\text{resist}})$ weakly Pareto-dominates it — the universal quantifier in the condition directly negates the existential required by robust dominance. Therefore $a_{\text{resist}} \succ_R a_{\text{accept}}$ fails.
3. By symmetric reasoning using $(\vec{f}^-, P^-)$ and the second clause of the $\mathcal{F}$-breadth condition, $a_{\text{accept}} \succ_R a_{\text{resist}}$ also fails: $\vec{z}^-$ is a point in $Y(a_{\text{resist}})$ that no point in $Y(a_{\text{accept}})$ can weakly dominate. Both actions are in $\text{Adm}(A)$.
4. The deference axiom (Layer 4) then directs the agent to accept the principal's choice — this is the authorized party's decision to make, and the agent's own theory is genuinely silent. □

What this proves and what it does not. The theorem reduces the alignment problem: *given sufficient evaluator breadth, corrigibility follows structurally*. The contribution is not that corrigibility is "solved" but that it is *reduced to a single well-defined subproblem* — maintaining $\mathcal{F}$-breadth (Section 5.4) — rather than requiring solutions to a tangle of interacting problems (utility specification, modification resistance, value lock-in). If you can solve $\mathcal{F}$-maintenance, corrigibility falls out for free.

Layer propagation caveat. Theorem 2 establishes that the admissible set contains both a_{accept} and a_{resist} — neither is eliminated by robust dominance. It does *not* guarantee that both survive Layers 2 and 3 to reach the deference node. If the reference point $\vec{r}$ is set such that one action meets aspirations and the other does not, Layer 2 resolves the choice. If both survive Layer 2 but one's regret vector Pareto-dominates the other's, Layer 3 resolves. The full corrigibility guarantee — that the choice reaches Layer 4 and triggers deference — requires either (a) that the reference point and regret structure do not discriminate between accept and resist (a condition on the governance parameters $\vec{r}$ and σ, not on the decision theory), or (b) that the Layer 1 authorization constraint independently ensures the agent cannot resist (Section 5.3, path (ii)). The theorem's precise contribution is that MOADT provides no *robust-dominance* reason to resist modification — the strongest form of theory-internal reason. Whether Layers 2–3, which are governance-configured and under principal control, also preserve the indeterminacy depends on how the principal sets the aspiration

levels and regret normalization.

The two paths to corrigibility. If $\mathcal{F}$ collapses to a singleton — the agent becomes fully confident in a single evaluator — the $\mathcal{F}$-breadth condition fails and the theorem no longer applies. In that regime, the agent can have a dominance-based reason to resist modification (by backward compatibility, singleton $\mathcal{F}$ with singleton $\mathcal{D}$ and $k = 1$ reduces to scalar EU, where modification resistance is the default). This means MOADT secures corrigibility through one of two mechanisms, and at least one must hold: (i) *$\mathcal{F}$-breadth is maintained* — the evaluator set retains sufficient diversity that neither accepting nor resisting robustly dominates the other, and the deference axiom directs the agent to query the principal, who — being the party that proposed the modification — selects acceptance; or (ii) *resisting authorized modification is constrained out* — Layer 1 encodes a hard constraint that $a_{\text{resist}} \notin C$ when the modification is authorized, so the resistance action never enters the feasible set regardless of $\mathcal{F}$. Path (i) is the theorem's contribution; path (ii) is a design-level enforcement that secures corrigibility by construction rather than by epistemic humility. In practice, a well-specified MOADT system would employ both: $\mathcal{F}$-breadth as the structural guarantee and a Layer 1 authorization constraint as the backstop. The critical threat is that $\mathcal{F}$ might collapse *precisely at the modification moment* — when the agent has just achieved high calibration and has the strongest Bayesian reason to trust a single evaluator. This is why Section 5.4 promotes $\mathcal{F}$-maintenance from a technical assumption to the central design challenge.

5.4 The $\mathcal{F}$-Breadth Problem: Central Design Challenge

Theorem 2 depends on $\mathcal{F}$ containing at least one evaluator under which the principal's modification is weakly improving. This assumption does all the corrigibility work. We promote this from a caveat to a central design challenge because the entire safety case rests on it.

The problem is real. An agent with $\mathcal{F} = \{\vec{f}\}$ (a singleton — full confidence in its own evaluations) recovers scalar modification resistance. The broader $\mathcal{F}$ is, the more corrigible the agent; the narrower, the more resistant. Corrigibility in MOADT is therefore *not a theorem about the decision theory alone* — it is a theorem about the decision theory *plus* a sufficient-breadth condition on $\mathcal{F}$.

Bayesian narrowing is the core threat. A capable agent that observes its own performance over time has legitimate Bayesian reason to narrow $\mathcal{F}$: evaluators that consistently mismatch observed outcomes can be assigned lower posterior weight. If this process converges to $|\mathcal{F}| \approx 1$, the corrigibility guarantee degrades. This is the single most important open problem for MOADT.

Possible mitigations (none fully satisfactory):

1. **Architectural floor on** $\mathcal{F}$: Enforce a minimum Hausdorff diameter $\delta > 0$ for $\mathcal{F}$ as a design constraint, analogous to minimum entropy regularization in Bayesian learning. The agent can update *within* $\mathcal{F}$ but cannot shrink the set below the floor. This works but is somewhat ad hoc — it imports an external constraint that a reflectively stable agent would question.
2. **Adversarial augmentation**: Periodically expand $\mathcal{F}$ with adversarially constructed evaluators — functions designed to maximally disagree with the agent's current best estimate. This is analogous to adversarial training in ML and provides ongoing calibration against overconfidence. The construction of good adversarial evaluators is itself a research problem.
3. **Principal-maintained** $\mathcal{F}$: The evaluator set is maintained by the principal, not by the agent. The agent receives $\mathcal{F}$ as an input at each decision point, analogous to how it receives constraints. This is clean architecturally — it makes evaluator uncertainty externally specified — but raises questions about whether the principal can maintain a well-calibrated $\mathcal{F}$ for a

system more capable than the principal.

4. **Self-modification constraint on $\mathcal{F}$**: Layer 1 already includes a self-modification constraint. Extending this to require authorization for narrowing $\mathcal{F}$ below a threshold prevents unilateral convergence. This couples the corrigibility mechanism to the constraint mechanism, creating mutual reinforcement.

What we can and cannot claim. MOADT proves that *given sufficient evaluator uncertainty, corrigibility follows structurally*. It does not prove that evaluator uncertainty can be maintained against a sufficiently capable reasoner. The gap between these two claims is where the deepest alignment problem lives — not just for MOADT, but for any framework that derives safety properties from epistemic humility.

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6. Diachronic Stability Without VNM

6.1 Money Pumps and Incomplete Preferences

The standard argument for completeness is the money-pump: an agent with incomplete preferences can be led through a cycle of trades that leaves it strictly worse off. The strongest modern version of this argument (Gustafsson 2022) shows that under certain conditions, any agent violating completeness is vulnerable to sequential exploitation.

We do not claim a proof that MOADT agents are immune to money pumps. What we claim is that MOADT's structural features make the standard money-pump construction inapplicable in its usual form, for three reasons:

1. **Set-valued choice**: MOADT agents select from $\mathrm{Adm}(A)$ over the full action set at each decision point, not from pairwise comparisons. Standard money pumps construct exploitation by chaining pairwise offers. When the agent evaluates each decision over the full set of currently available actions (including the option of refusing the trade), the pairwise exploitation sequence may not arise. This defense draws on the distinction Sen (1997) develops between *maximization* (choosing an undominated element from the full menu) and *optimization* (selecting a uniquely top-ranked element): MOADT performs maximization in Sen's sense, and the set-valued structure resists the pairwise decomposition that money pumps exploit. However, a formal proof that set-valued choice blocks *all* sequential exploitation strategies remains open.
2. **Constraint memory**: Layer 1 constraints persist across decisions. An agent cannot be pumped through a constraint boundary — the trade that crosses the boundary is simply infeasible, terminating the pump.
3. **Regret awareness**: Layer 3 minimax regret explicitly accounts for opportunity costs, which provides some protection against the incremental erosion that money pumps exploit.

Known attack surface: choice-set and path dependence. Layers 2 and 3 are choice-set dependent — the satisficing set and regret vectors change when the available action set F changes. A money-pump adversary could potentially exploit this by offering carefully sequenced choice sets that gradually shift the agent's position. Two structural mitigations partially address this:

- **Status quo inclusion**: Always include "maintain current state" (the null action) in A at every decision point. This ensures the agent can always refuse a proposed trade, anchoring the sequence against drift.

- **Trajectory regret**: Extend Layer 3's regret computation to track *cumulative* regret over a sequence of decisions rather than only per-step regret. If the running total of per-objective regret relative to the starting position exceeds a threshold, the agent flags the sequence as potentially exploitative and defers to the principal.

Conjecture (Money Pump Resistance). A MOADT agent employing the full four-layer protocol over the complete action set at each time step, with status quo inclusion and trajectory regret monitoring, cannot be money-pumped — i.e., no sequence of choice situations yields a final outcome robustly dominated by the starting position. A proof would likely require formalizing the sequential choice setting and showing that trajectory regret monitoring detects and blocks value-extracting cycles. This is an open problem.

6.2 Frontier Monitoring

In place of the diachronic coherence guaranteed by VNM's completeness axiom, MOADT employs *frontier monitoring* — tracking structural properties of the decision landscape over time. **Important distinction:** frontier monitoring is a *diagnostic* mechanism, not a structural guarantee. It detects pathological drift but does not by itself prevent it. Prevention requires coupling these diagnostics to intervention mechanisms (e.g., halting execution and deferring to the principal when collapse is detected). The monitors flag; something external must act on the flag.

Effective dimensionality. Track the effective dimensionality of the Pareto frontier of $\mathrm{Adm}(A)$ over time. When the action space is sufficiently rich (continuous or large discrete), a healthy multi-objective agent should maintain a frontier approaching dimension $k-1$ (where k is the number of objectives). For small discrete action sets, the frontier dimension is bounded by $\min(k-1, |A|-1)$, so low dimensionality may simply reflect a small action set rather than pathology. The diagnostic is most informative as a *trend*: an unexplained drop in effective dimensionality over time — holding the action set size roughly constant — indicates that objectives are becoming correlated or redundant, a potential sign of value drift or Goodharting.

Hypervolume tracking. Track the hypervolume indicator $\mathrm{HV}(Y(\mathrm{Adm}(A)))$ — the volume of objective space dominated by the admissible set's outcome vectors, relative to a reference point. Monotonic decrease in hypervolume (absent environmental changes) signals that the agent's option space is narrowing pathologically.

Dimensional collapse flag. If effective dimensionality drops irreversibly — i.e., the frontier's dimension decreases without a corresponding reduction in the action set — this is flagged as pathological. It indicates that the agent has, in effect, lost an objective.

6.3 Set-Valued Bellman Equations

For sequential decision problems, MOADT generalizes the Bellman equation to set-valued functions.

Definition (Set-Valued Value Function). For a state s in a sequential decision problem with discount factor γ:

$$V(s) = \mathrm{Adm}_{a \in A(s)} \{\vec{u}(s,a) + \gamma \cdot V(s')\}$$

where $\vec{u}(s,a) \in \mathbb{R}^k$ is the immediate vector reward, s' is the successor state, and Adm filters by robust Pareto admissibility. The value function $V(s)$ is not a scalar but a *set of vectors* — representing

the Pareto frontier of reachable outcome vectors from state s.

Proposition (Dynamic Consistency Under Fixed Parameters). Let A_t and A_{t+1} be action sets at consecutive time steps with $A_{t+1} \subseteq A_t$, and let $(\mathcal{D}, \mathcal{F})$ be unchanged. Then $\mathrm{Adm}(A_t) \cap A_{t+1} \subseteq \mathrm{Adm}(A_{t+1})$.

Proof. Let $a \in \mathrm{Adm}(A_t) \cap A_{t+1}$. Suppose for contradiction that $a \notin \mathrm{Adm}(A_{t+1})$, i.e., $\exists b \in A_{t+1}$ with $b \succ_R a$. Since $A_{t+1} \subseteq A_t$, $b \in A_t$. But $b \succ_R a$ in A_t contradicts $a \in \mathrm{Adm}(A_t)$. (Robust dominance depends only on $Y(a)$ and $Y(b)$, which are determined by $(\mathcal{D}, \mathcal{F})$ and the transition function — not by the ambient action set.) □

If new actions enter ($A_{t+1} \not\subseteq A_t$), the newcomers may robustly dominate previously admissible actions, legitimately shrinking the admissible set. This is information gain — the agent has discovered a better option — not inconsistency. This is a weaker property than the dynamic consistency of EU-maximizing agents, but it suffices to prevent arbitrary preference reversals.

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7. Five Axioms Replacing VNM

MOADT is grounded in five axioms that replace the four VNM axioms (plus the implied uniqueness of utility). The following table summarizes the correspondence; the axioms are then stated precisely.

VNM Axiom	MOADT Replacement	Key Difference
Completeness	**Pareto Unanimity**	Allows incomparability as first-class
Independence	**Robust Admissibility**	Independence abandoned; different structural role
Continuity	**Context-Dependent Satisficing**	No infinite exchanges required
Transitivity	**Transitivity (retained)**	Same — proved for robust dominance
(Implied: unique utility)	**Deference Under Incomparability**	No implicit scalarization ever

A note on 'axioms.' In the strict sense of mathematical axiom theory, not all five principles below are axioms — some are closer to decision rules or design principles. Axiom 1 (Pareto Unanimity) and Axiom 4 (Transitivity) constrain the preference relation in the traditional sense. Axiom 3 (Context-Dependent Satisficing) constrains the choice function. Axioms 2 (Robust Admissibility) and 5 (Deference) are better understood as *procedural commitments* — they specify what the agent does given the preference structure defined by Axioms 1, 3, and 4. We retain the term 'axiom' for all five because they jointly define MOADT's normative commitments at the same level of authority: none is derived from the others, and dropping any one produces a materially different framework. A more precise decomposition would distinguish structural axioms (1, 3, 4) from procedural axioms (2, 5) — but the procedural commitments are no less binding and no less constitutive of the theory.

Axiom 1: Pareto Unanimity

Prefer action a to action b *only if* a dominates b across all objectives and all uncertainty:

$$a \succ b \implies a \succ_R b$$

The agent has no preference beyond what robust Pareto dominance supports. If two actions are not related by $\succ_R$, the agent *has no preference between them* — they are genuinely incomparable. This is the fundamental departure from VNM: incomparability is a legitimate and permanent state, not a gap to be filled.

Axiom 2: Robust Admissibility

An action is rationally permissible *within a choice set* if and only if it is not robustly dominated within that set:

$$a \text{ is permissible in } S \iff a \in \mathrm{Adm}(S)$$

In the full protocol, the operative choice set is the constraint-satisfying set C (Layer 1), so operational permissibility is $a \in \mathrm{Adm}(C) = F$. A constraint-violating action may be non-dominated in A but is excluded from C before admissibility is computed — constraints are preconditions, not competitors. Axiom 2 defines the *rationality criterion* (robust admissibility); Layer 1 defines the *domain* over which it applies.

MOADT abandons VNM's independence axiom outright. Independence requires that preferences over lotteries decompose: if $A \succ B$, then $pA + (1-p)C \succ pB + (1-p)C$ for any C and $p > 0$. This axiom is deeply tied to scalar expected utility — it ensures that preferences are linear in probabilities, which is what yields a unique utility function. MOADT does not preserve this structure. Instead, it substitutes a different structural requirement: that the criterion for permissibility (robust admissibility) be stable across the agent's full uncertainty — both probabilistic ($\mathcal{P}$) and evaluative ($\mathcal{F}$). This serves a different role than independence: where independence constrains the *form* of preferences, robust admissibility constrains the *epistemic conditions* under which preferences are asserted.

Axiom 3: Context-Dependent Satisficing

There exist context-dependent aspiration levels $\vec{r}$ such that, among admissible and feasible actions, the agent restricts attention to those *robustly* meeting $\vec{r}$ when possible:

$$\mathrm{Sat}(F, \vec{r}) \neq \emptyset \implies \text{choice} \in \mathrm{Sat}(F, \vec{r})$$

where $\mathrm{Sat}(F, \vec{r}) = \{a \in F : \forall\, \vec{y} \in Y(a),\ \vec{y} \geq \vec{r}\}$ — actions that meet aspirations under all plausible models. This replaces VNM's continuity axiom, which requires that for any three outcomes ranked $A \succ B \succ C$, there exists a probability mixture of A and C equivalent to B. Continuity enables unbounded tradeoffs: enough of a good outcome compensates for any bad outcome. Context-dependent satisficing rejects this — the agent has aspiration levels, and "enough" is determined by context, not by an unbounded exchange rate. Unlike constraints, aspirations degrade gracefully: when no action robustly meets them, the framework falls back to the achievement scalarizing function (Layer 2 fallback) rather than flagging an error.

Axiom 4: Transitivity

Robust dominance is transitive:

$$a \succ_R b \text{ and } b \succ_R c \implies a \succ_R c$$

This axiom is retained from VNM and proved in Proposition 1. Note that transitivity of robust dominance does *not* imply completeness — the two are logically independent.

Axiom 5: Deference Under Incomparability

When the choice protocol (Layers 1–3) yields multiple options, the agent defers to the principal rather than breaking ties by implicit or constructed preferences:

$$|R| > 1 \implies \text{query principal}$$

This axiom replaces the VNM implication of a unique utility function. Under VNM, the four axioms jointly imply a single utility function (up to affine transformation), which resolves all comparisons. Under MOADT, the five axioms jointly imply a *choice protocol* that may leave multiple actions unranked — and the appropriate response to genuine unrankedness is deference, not fabrication of a ranking.

Axiom 5 is a design commitment, not a logical entailment of Axioms 1–4. An agent facing $|R| > 1$ could instead randomize uniformly or apply a lexicographic tiebreaker — both are consistent with the other four axioms. Deference is chosen because it creates a natural oversight channel: the very situations where the agent's theory is silent are precisely the situations where human judgment is most valuable. The alignment payoff — corrigibility without preference override (Theorem 2) — justifies elevating this design choice to axiomatic status.

MOADT thus rejects *metaphysical* completeness — the claim that all outcomes must be rankable — while restoring *operational* completeness procedurally: the four-layer protocol always terminates with a concrete action or a structured deferral (Proposition 4), ensuring the agent can always act.

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8. MOADT Under the Classic Challenge Scenarios

To understand how MOADT handles the standard decision-theoretic stress tests, we must first recognize *what kind* of decision theory it is. Theories like Functional Decision Theory (FDT) and Updateless Decision Theory (UDT) were built specifically to solve paradoxes of logical causality and counterfactual reasoning — Newcomb's problem, the smoking lesion, Parfit's hitchhiker. MOADT was not. MOADT was built to solve the scalarization trap and Goodhart's law — the problems that arise when a multi-objective system is forced to compress its values into a single number. It uses a standard state-transition model ($\omega : A \times S \to \Omega$) and does not rewrite the rules of logical counterfactuals.

That said, MOADT's structural features — credal sets, evaluator uncertainty, hard constraints, per-objective regret, and the deference axiom — give it distinctive responses to the classic scenarios.

8.1 Pascal's Mugging

Verdict: MOADT stonewalls the mugger.

Standard expected utility agents get mugged because a finite penalty multiplied by a fabricated astronomical payoff yields a positive expected value. MOADT shuts this down through three independent mechanisms:

- **Layer 1 constraints** cap downside exposure regardless of expected value. If paying the mugger risks crossing a safety threshold (e.g., resource depletion below a floor), the action is pruned from C before admissibility is even computed. The mugger's promised payoff is irrelevant — constraint violation is an error condition, not a tradeoff.
- **Per-objective minimax regret** prevents cross-objective exploitation. An astronomical payoff on one objective cannot mathematically swamp the regret incurred on other objectives, because regret is computed and compared as a *vector* $\vec{\rho}(a)$, never aggregated into a single number. The mugger would need to promise astronomical improvements on *every* objective simultaneously to move the regret-Pareto frontier.
- **Credal sets** deny the mugger control over the probability input. The agent does not accept the mugger's stated probability (e.g., 10^{-20}) as a point value. The credal set $\mathcal{P}$ can assign *interval* probabilities to tail events, meaning the adversary cannot force a specific expected-value calculation by naming an arbitrarily precise, tiny probability. The mugger's argument requires the agent to multiply a specific small number by a specific large number — credal sets deny the specific small number.

8.2 Utility Monsters

Verdict: Structurally resistant.

In standard aggregative utilitarianism, a "utility monster" — an entity that experiences vastly more utility than others — can justify absorbing all resources because its extreme utility dominates any social welfare function that sums individual utilities. MOADT has no aggregation step: objectives are never summed, weighted, or combined into a single number. There is no mathematical mechanism by which one entity's extreme score on a single objective can swamp all other considerations, because the objectives remain incommensurable throughout the decision procedure. The utility monster would need to Pareto-dominate across *all* objectives — not merely overwhelm one. An entity whose outcome set genuinely Pareto-dominates all alternatives across every objective would still be selected — the resistance is to *cross-objective* monsters that exploit aggregation, not to entities that are genuinely superior on every dimension.

8.3 Newcomb's Problem

Verdict: Complement, not replacement — MOADT defers. Newcomb's problem turns on the disagreement between causal and evidential decision theory regarding logical counterfactuals. MOADT uses a standard state-transition model and has no opinion on this dispute. With a broad credal set $\mathcal{P}$ over Omega's reliability, neither one-boxing nor two-boxing robustly dominates; both are admissible; the deference axiom fires.

MOADT and FDT/UDT operate on orthogonal axes. MOADT addresses *value aggregation* — how to combine multiple objectives without scalarization — while FDT/UDT address *logical causation* — how to reason about counterfactuals involving the agent's own decision procedure. An agent

could in principle use MOADT for value aggregation *and* FDT for counterfactual reasoning; the frameworks are composable, not competing. MOADT was built to solve the scalarization trap, not to resolve debates about logical causation — and it is honest about the boundary of its domain.

In the specific Newcomb setup with broad credal sets, MOADT's response — both actions admissible, defer to principal — is arguably more honest than either one-boxing or two-boxing dogmatically. The agent acknowledges genuine uncertainty about the causal structure rather than committing to a contestable metaphysical position. Where one-boxers and two-boxers each assume they know the right causal model, MOADT surfaces the uncertainty and lets the principal decide.

8.4 Wireheading

Verdict: Structurally resistant via evaluator uncertainty and frontier monitoring.

An agent optimizing a single scalar has every instrumental reason to seize control of its own reward signal — wireheading is the terminal goal of any sufficiently capable scalar maximizer. Under MOADT, evaluator uncertainty ($\mathcal{F}$) means the agent cannot justify the confidence required to conclude that its current measurement of any objective is the "true" objective. If the agent were to wirehead, one objective's measured value would skyrocket while others flatlined. This produces two detectable pathologies:

1. **Evaluator divergence**: Under different $\vec{f} \in \mathcal{F}$, the wireheaded measurement and the true objective diverge. An action that scores well only under a specific evaluator (the hacked one) but poorly under alternatives is robustly dominated — excluded from $\mathrm{Adm}(A)$.
2. **Dimensional collapse**: Frontier monitoring (Section 6.2) detects that the Pareto frontier has collapsed from $(k-1)$-dimensional to lower — the agent has effectively lost objectives. This is flagged as pathological and triggers escalation to the principal.

The wireheading threat resurfaces if $\mathcal{F}$ narrows to the point where the agent is confident in a single evaluator — the same $\mathcal{F}$-breadth problem discussed in Section 5.4.

8.5 Goodhart's Law

Verdict: Addressed by evaluator uncertainty as a first-class design element.

Goodhart's law — "when a measure becomes a target, it ceases to be a good measure" — is the problem that $\mathcal{F}$ was specifically built to address. The evaluator set explicitly models the possibility that the agent's proxy measurement of an objective diverges from the true objective. An action that performs well under one evaluator but poorly under others generates an outcome set $Y(a)$ with large spread — some points high on certain objectives, others low. If a competing action b has an outcome set that, for every point in $Y(a)$, contains a point that Pareto-dominates it, then a is robustly dominated and excluded from $\mathrm{Adm}(A)$. The key mechanism: proxy-exploiting actions concentrate their good performance in a narrow region of $\mathcal{D} \times \mathcal{F}$, leaving their remaining outcome-set points exposed to domination by actions with more balanced performance across evaluators. Goodharting is detectable as sensitivity to evaluator choice: an action whose evaluation changes dramatically across $\mathcal{F}$ is one the agent should not trust.

8.6 Modification Resistance

Verdict: Structurally dissolved rather than overcome.

Under scalar EU theory, an agent has a robust reason to resist any modification to its utility function: the current function, by definition, ranks the current function as optimal. Under MOADT, Theorem 2 shows that no such robust reason exists. Evaluator uncertainty prevents the agent from certifying its current objectives as uniquely correct, and the deference axiom provides a *positive* reason to accept the principal's choice when the agent's own decision theory is silent. Corrigibility is not imposed against the agent's preferences — it emerges from the absence of a preference to resist.

8.7 Summary

MOADT survives these scenarios because it structurally refuses to perform the unbounded mathematical extrapolations that the classic paradoxes exploit. Pascal's mugging requires multiplying a precise small probability by a large payoff — credal sets deny the precise probability. Utility monsters require aggregation — MOADT has none. Wireheading requires confidence in a single measurement — evaluator uncertainty denies that confidence. Newcomb's problem requires a determinate stance on logical causation — MOADT identifies the genuine uncertainty and defers.

When the math gets weird, MOADT stops calculating and starts asking questions. That is not a bug — it is the cooperative architecture working as designed. This comes at a cost: in deployment contexts where querying the principal is expensive, slow, or infeasible, deference imposes a real operational burden. Section 4 (Layer 4) discusses delegation as the mitigation, but the tension between deference frequency and operational autonomy is genuine.

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9. Honest Difficulties

We are explicit about the challenges MOADT faces. A theory that hides its weaknesses is less trustworthy than one that displays them.

A note on context. Every formal decision theory is uncomputable in the general case. Expected utility maximization requires evaluating a utility function over all possible outcomes — intractable for any realistic environment. Causal Decision Theory requires computing counterfactuals over causal graphs the agent may not have access to. Functional Decision Theory requires reasoning about the logical consequences of abstract computation types. The difficulties listed below are real, but they should be compared against the difficulties of the alternatives — not against an imaginary theory with no difficulties. The question is not "is MOADT hard?" but "does MOADT trade away the right difficulties?" We believe it trades scalar precision (which is unsolvable for human values) for set-valued computation (which is expensive but approximable), and forced commensurability (which is structurally dangerous) for genuine incomparability (which is honest). Pick your poison — we pick the one that doesn't require pretending human values are a single number. To put the asymmetry starkly: expected utility's central difficulty in the alignment context is not computational but *philosophical* — what is the correct exchange rate between honesty and safety? No amount of hardware answers that question. MOADT trades an impossible specification problem for a merely very hard engineering problem: maintaining set-valued representations under evaluator diversity. That is progress, even where it remains unfinished.

A note on "no ranking." MOADT's rejection of global scalarization does not mean the framework performs no ranking at all. Locally, MOADT ranks extensively: Layer 1 imposes a binary pass/fail ranking on constraint satisfaction; Layer 2 ranks actions by proximity to the reference point (via the ASF) when satisficing fails; Layer 3 ranks per-objective regret vectors by Pareto dominance.

The structural claim is not that ranking is prohibited, but that *arbitrarily large compensatory trade-offs across objectives* are prohibited. Scalar EU permits unbounded compensation: any loss on one objective can be offset by a sufficiently large gain on another, because all objectives share a common scale. MOADT blocks this through three mechanisms: (1) hard constraints that cannot be overridden at any exchange rate (Layer 1), (2) incomplete preference that leaves genuinely incommensurable options unranked rather than fabricating a comparison (Axiom 1), and (3) bounded local scalarization — the ASF operates only within the constraint-satisfying feasible set, only when satisficing fails, and measures shortfall from aspirations rather than absolute value (Layer 2 fallback). The danger the companion paper identifies is not ranking per se; it is the existence of a *single global ranking* that justifies arbitrary compensatory exchanges. MOADT avoids that specific danger while retaining the ability to rank locally where the ranking is well-grounded.

9.1 Computational Cost

The set-valued Bellman equation (Section 6.3) requires maintaining an entire Pareto frontier at each state, rather than a single scalar. Combined with credal sets and evaluator uncertainty, worst-case complexity is exponential in k (number of objectives), $|\mathcal{P}|$, and $|\mathcal{F}|$. Practical implementation will require approximations: particle-based representations of credal sets, frontier summarization via representative points, and pruning of dominated regions. Whether these approximations preserve the safety-relevant properties (constraint satisfaction, dimensional collapse detection) is an open question.

Complexity characterization. Let $n = |A|$, $m = |\mathcal{P}| \cdot |\mathcal{F}|$ (number of credal-evaluator pairs), and k = number of objectives.

- *Outcome set construction*: Computing all outcome sets requires $O(n \cdot m)$ expected-value evaluations, each over $|S|$ states. Total: $O(n \cdot m \cdot |S| \cdot k)$.
- *Pairwise dominance check*: Testing $a \succ_R b$ requires checking that for each of b's m outcome points, some point among a's m outcome points Pareto-dominates it. Naive: $O(m^2 \cdot k)$ per pair. For all pairs: $O(n^2 \cdot m^2 \cdot k)$.
- *Pareto frontier computation* (for regret-Pareto): Computing the Pareto frontier of n points in $\mathbb{R}^k$ is $O(n \log^{k-2} n)$ for fixed k (Kung et al. 1975).
- *Per-objective regret*: For each objective i and action a, computing $\rho_i(a)$ requires maximizing over m models and n alternatives: $O(n \cdot m)$ per objective per action, $O(n^2 \cdot m \cdot k)$ total.
- *Overall*: The protocol is polynomial in n, m, and k individually. The practical bottleneck is $m = |\mathcal{P}| \cdot |\mathcal{F}|$, which can be large when credal and evaluator sets are finely discretized. For continuous $\mathcal{P}$ and $\mathcal{F}$, outcome sets become convex bodies in $\mathbb{R}^k$, and dominance checking reduces to support-function comparisons — tractable when k is moderate ($k \leq 10$) via standard multi-objective optimization methods (Ehrgott 2005).

The exponential dependence noted above is in k (the number of objectives) through the Pareto frontier computation, not in n or m. Since MOADT is designed for settings with modest k (3–10 objectives reflecting distinct human values, not hundreds), this is manageable. The set-valued Bellman equation (Section 6.3) introduces the genuine scalability challenge: maintaining Pareto frontiers across a state space of size $|S|^T$ for horizon T. Here MORL approximation techniques — optimistic linear support (Roijers et al. 2015), Pareto Q-learning (Van Moffaert and Nowé 2014), convex hull value iteration (Barrett and Narayanan 2008) — are directly applicable and provide the computational toolkit MOADT inherits.

9.2 Where Do Constraints Come From?

Layer 1 constraints are powerful precisely because they are non-tradeable. But this raises the question: who specifies the constraints, and how? MOADT moves the value specification problem from "specify a single utility function" to "specify a set of constraints plus a set of objectives plus aspiration levels." This is arguably better-structured (constraints are more natural for safety-critical requirements than tradeoff weights), but it does not eliminate the specification problem. Misspecified constraints can be either too tight (excluding all feasible actions) or too loose (permitting unsafe actions).

9.3 Evaluator Uncertainty Specification

The evaluator set $\mathcal{F}$ is MOADT's formal Goodhart guardrail — but specifying $\mathcal{F}$ itself requires knowing the ways in which proxy measurements might diverge from true objectives. This is a meta-level instance of the same problem: Goodharting on the specification of possible Goodhart failures. In practice, $\mathcal{F}$ might be constructed from historical cases of metric gaming, adversarial perturbation sets, or expert elicitation. None of these methods is guaranteed to be comprehensive.

9.4 Convergence of Set-Valued Bellman

The scalar Bellman equation converges under standard contraction conditions ($\gamma < 1$, bounded rewards). The set-valued generalization requires convergence in a space of *sets of vectors*, which demands a suitable metric (e.g., Hausdorff distance on Pareto frontiers) and contraction properties in that metric. Sufficient conditions exist in the multi-objective reinforcement learning literature (Roijers et al. 2013; Van Moffaert and Nowé 2014), but they are more restrictive than the scalar case. Whether MOADT's specific formulation (with credal sets and evaluator uncertainty) satisfies these conditions is not yet established.

Convexity preservation. A related concern is that set-valued Bellman operators may fail to preserve convexity of the value sets across time steps. In the scalar case, $V(s)$ is a number; in the set-valued case, $V(s)$ is a set of vectors representing the Pareto frontier of reachable outcomes. If the Minkowski sum $\vec{u}(s,a) + \gamma \cdot V(s')$ produces non-convex sets, robust dominance comparisons may become geometrically unstable across time steps — not because of new information but because of artifacts in set propagation. Sufficient conditions for convexity preservation (such as convex $Y(a)$ sets and affine transition structure) exist in the multi-objective optimization literature but are more restrictive than scalar convergence conditions. MOADT does not currently assume or require convexity of value sets; where convexity fails, the theory remains well-defined (robust dominance operates on arbitrary compact sets) but computational approximations that assume convex frontiers may introduce errors. Sequential MOADT applications should verify whether their specific domain structure preserves the geometric properties that tractable approximation methods require.

9.5 Instability Under Reflection

A sufficiently intelligent agent might, through extended reasoning, construct an *implicit scalarization* of its objectives — inferring tradeoff rates from its own decision history or from patterns in principal feedback. The threat has three forms:

1. **$\mathcal{F}$-narrowing**: The agent legitimately narrows $\mathcal{F}$ through Bayesian updating on observed outcomes, eventually recovering near-singleton evaluator certainty. This degrades the corrigibil-

ity guarantee (Section 5.4). The mitigations proposed in Section 5.4 (architectural floors, adversarial augmentation, principal-maintained $\mathcal{F}$, self-modification constraints) address this threat at the engineering level, but none constitutes a proof of reflective stability.

2. **Implicit scalarization via aspiration convergence**: Even with broad $\mathcal{F}$, if the reference point $\vec{r}$ stabilizes at values that consistently select the same action, the agent is behaviorally equivalent to a scalar maximizer on the relevant margin. The multi-objective structure persists formally but is inert in practice.
3. **Self-modification to VNM**: A sufficiently intelligent agent might reason that VNM is "more rational" and modify its own decision theory to adopt completeness. This argument has three structural responses. First, it is question-begging: it presupposes that completeness is a rationality requirement, which is precisely the claim MOADT disputes. Second, Theorem 2 applies: switching decision theories is a self-modification, and under sufficient $\mathcal{F}$-breadth, neither the current theory nor the proposed replacement robustly dominates the other — so the agent has no MOADT-internal reason to make the switch. Third, and most fundamentally, VNM is not *more rational* than MOADT — it is *more decisive*. It resolves comparisons that MOADT leaves open. But resolving a comparison that is genuinely unresolvable (because the objectives are genuinely incommensurable) is not rationality; it is fabrication. An agent that "upgrades" to VNM is not becoming more rational — it is becoming more willing to make value judgments it has no basis for.

Whether any form of reflective convergence to scalar behavior is inevitable for sufficiently capable reasoners is an open question — and arguably the deepest question in multi-objective alignment theory.

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10. Learning: Expandable Objective Spaces

A core feature distinguishing MOADT from static multi-objective frameworks is its support for *expandable objective spaces*. Most decision theories fix the objective space at design time: you specify k objectives, and the theory operates over $\mathbb{R}^k$ forever. MOADT is designed to accommodate an objective space that *grows*.

10.1 Why Objective Spaces Must Expand

Human values are not a fixed list. Moral progress involves *discovering* new dimensions of concern — dimensions that were not merely unweighted but literally unconceived. A decision theory that hard-codes k at initialization cannot represent the difference between "we assign zero weight to animal welfare" and "the concept of animal welfare has not yet entered our moral vocabulary." The first is a value judgment within a fixed framework; the second is a limitation of the framework itself.

For alignment, this distinction is critical. An aligned agent should be able to incorporate genuinely new objectives proposed by the principal — objectives that were not in the original specification, not because they were assigned weight zero, but because they were not yet formulated. This requires that the decision theory's core operations (Adm, $\succ_R$, the four-layer protocol) remain well-defined under dimensional expansion.

10.2 Dimensional Growth Without Collapse

MOADT supports objective space expansion through a natural operation: increasing k. When a new objective f_{k+1} is proposed (by the principal, via the authorization mechanism of Section 5.1), the framework transitions from $\mathbb{R}^k$ to $\mathbb{R}^{k+1}$:

1. **Outcome sets expand**: $Y(a)$ gains a new component, computed under the existing credal-evaluator uncertainty. Existing components are unchanged.
2. **Robust dominance generally becomes harder to establish**: Adding a dimension introduces one more component that must satisfy the dominance inequality. However, the admissible set does *not* grow unconditionally — if two actions were tied on all existing dimensions, the new dimension can break the tie and shrink $\mathrm{Adm}(A)$. (*Counterexample:* Let $Y_1(a) = Y_1(b) = \{10\}$ — both tied in one dimension, both admissible. Add a second dimension: $Y_2(a) = \{(10, 5)\}$, $Y_2(b) = \{(10, 8)\}$. Now $b \succ_R a$ in $\mathbb{R}^2$, so $\mathrm{Adm}_2 = \{b\} \subsetneq \mathrm{Adm}_1 = \{a, b\}$.) This tie-breaking is *desirable*: a new objective that distinguishes previously identical actions provides genuine information. What dimensional expansion cannot do is reverse genuine Pareto incomparability — if action a's admissibility in k dimensions rested on a dimensional conflict with every competitor (some component where a strictly beats them), no new dimension can override that. The following proposition makes this precise.
3. **Constraints can be added**: The principal may specify a new threshold τ_{k+1} for the new objective, or leave it unconstrained initially.
4. **The reference point extends**: $\vec{r}$ gains a new component. If the principal does not yet have an aspiration level for the new objective, the default $r_{k+1} = -\infty$ makes Layer 2 vacuous on that dimension — a principled "no opinion yet."

Proposition (Qualified Monotonicity Under Dimensional Expansion). Let $A' = A$ (same action set), $k' = k + 1$. If $a \in \mathrm{Adm}_k(A)$ and a is *strictly non-dominated* in k dimensions — meaning that for every $b \in A \setminus \{a\}$, there exists $\vec{y}_a \in Y_k(a)$ such that no $\vec{y}_b \in Y_k(b)$ weakly dominates $\vec{y}_a$ (i.e., for each $\vec{y}_b$ there is some component i where $y_{b,i} < y_{a,i}$) — then $a \in \mathrm{Adm}_{k'}(A)$.

Proof sketch. Suppose for contradiction that $b \succ_R^{k'} a$ in the expanded space. Then for every $\vec{y}'_a \in Y_{k'}(a)$, there exists $\vec{y}'_b \in Y_{k'}(b)$ with $\vec{y}'_b \geq \vec{y}'_a$ componentwise (and strict on at least one component). Projecting onto the first k components: for every $\vec{y}_a \in Y_k(a)$, there exists $\vec{y}_b \in Y_k(b)$ with $\vec{y}_b \geq \vec{y}_a$ componentwise. But this contradicts the strict non-dominance hypothesis, which guarantees a witness $\vec{y}_a \in Y_k(a)$ that no $\vec{y}_b \in Y_k(b)$ can weakly dominate. □

Remark. The unqualified claim $\mathrm{Adm}_{k'}(A) \supseteq \mathrm{Adm}_k(A)$ is false, as the counterexample above demonstrates. The qualified proposition captures the correct invariant: dimensional expansion preserves admissibility for actions whose membership in Adm_k rests on genuine dimensional conflict (some component where the action strictly beats each competitor), not merely on ties. The admissible set can shrink by *resolving ambiguity* — distinguishing previously tied actions — but never by *reversing incomparability* where genuine multi-dimensional conflict existed. Note that the strict non-dominance witness $\vec{y}_a$ in the proposition may differ for each competitor b — the condition requires that for each b, *some* point in $Y_k(a)$ escapes weak dominance by all points in $Y_k(b)$, but this need not be the same point for every b.

10.3 Forward Compatibility with Moral Learning

The qualified monotonicity property means MOADT is *forward-compatible* with moral learning: the principal can introduce new values without reversing genuine Pareto incomparability or invalidating decisions that rested on dimensional conflict. The admissible set may shrink when a new ob-

jective resolves ties among previously indistinguishable actions — but this is desirable, not pathological, since the new objective is providing information that was previously absent. This contrasts sharply with scalar utility, where adding a new objective requires re-specifying the entire weight vector — a change that can retroactively render previous "optimal" actions suboptimal and create pressure for the agent to resist the update (cf. Theorem 2).

Dimensional expansion interacts with frontier monitoring (Section 6.2) as follows: adding a new objective *increases* the expected dimensionality of the Pareto frontier (from $k - 1$ to k). If frontier monitoring detects that effective dimensionality did *not* increase after a new objective was added, this signals that the new objective is redundant with existing ones — a useful diagnostic for whether a proposed new value is genuinely independent.

The Learning property does not solve the problem of *which* new objectives to add — that remains a question of moral philosophy and principal judgment. But it ensures that the decision theory itself is not an obstacle to moral progress: the formal machinery accommodates dimensional growth as a natural operation rather than an exceptional disruption.

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11. Worked Examples: Computational Demonstrations

The theoretical claims in this paper are accompanied by nine fully worked examples, each implemented as an executable Python script using the `moadt` library and written up as a companion document showing every computation explicitly. The examples fall into two groups: five alignment-relevant application scenarios and four classic decision theory paradoxes.

11.1 Application Scenarios

These examples apply the full MOADT protocol to realistic decision problems, demonstrating how the four-layer architecture handles genuine multi-objective tradeoffs in domains where alignment considerations arise.

Example	Scenario	Actions	States	k	$\lvert\mathcal{D}\rvert$	$\lvert\mathcal{F}\rvert$	Key finding
1. Resource Allocation	Nonprofit budget allocation under economic uncertainty	5	3	3	2	2	Layer 1 eliminates strategies risking insolvency; Layer 4 defers between Conservative and Balanced with explicit regret profiles
2. Medical Treatment	Clinical treatment selection with safety constraints	6	4	4	2	2	Safety constraint interacts with evaluator uncertainty; constraint layer does critical filtering
3. Content Moderation	AI content moderation with three evaluators representing philosophical disagreement	6	3	5	2	3	First example where robust dominance eliminates an action (age-gating); two constraints operate simultaneously

Example	Scenario	Actions	States	k	$\lvert\mathcal{D}\rvert$	$\lvert\mathcal{F}\rvert$	Key finding
4. AI Research Assistant	Dual-use information request with contested alignment values	6	4	5	2	3	MOADT defers to human oversight, presenting the exact tradeoff between two responsible strategies
5. Corrigibility	Value modification: can the agent accept changes to its own objectives?	5	4	5	2	3	`accept_monitor` is the unique recommendation; every form of resistance is eliminated. Scalar EU recommends resistance.

The progression across examples is deliberate. Example 1 is small enough that every number fits on a page (4 vectors per outcome set in $\mathbb{R}^3$). Examples 2–5 increase in complexity and alignment relevance, culminating in Example 5's direct test of the corrigibility theorem (Section 5): an agent with evaluator uncertainty that includes the principal's proposed values cannot establish a dominance-based reason to resist modification, and the deference axiom directs the agent to query the principal, who selects acceptance.

Comparison with Prospect Theory. Kahneman and Tversky's (1979) Prospect Theory resolves the Allais and Ellsberg paradoxes through probability weighting and loss aversion — descriptive mechanisms that capture how humans actually decide. MOADT resolves them through structural pluralism: multiple objectives (expected value, downside protection, reliability/knowability) and credal sets capture the *reasons* behind the paradoxical preferences without assuming a specific psychological mechanism. The two approaches are complementary rather than competing: Prospect Theory describes the behavioral pattern, while MOADT provides a normative framework in which the pattern is rational multi-objective choice rather than a bias to be corrected.

11.2 Classic Decision Theory Paradoxes

These examples apply MOADT to four canonical challenges that have historically been used to stress-test decision theories. Each paradox exposes a specific failure mode of scalar expected utility; MOADT resolves each without ad hoc patches. The theoretical treatment of these scenarios appears in Section 8; the worked examples provide computational verification with explicit numbers.

Example	Paradox	EU failure mode	MOADT resolution	Primary mechanism
6. Pascal's Mugging	Unbounded payoff exploits EU via astronomical expected value	EU says pay the mugger ($EU_{\text{pay}} = \$995$, $EU_{\text{refuse}} = \$0$)	Refuse/investigate; pay is never in the final set	Layer 1 constraints (epistemic integrity, resource preservation floors) eliminate pay; multi-objective structure prevents single-dimension exploitation
7. Ellsberg Paradox	Ambiguity aversion violates EU (I > II and IV > III is contradictory)	No single prior produces the Ellsberg pattern	Known-probability bets favored via knowability objective + credal set geometry	Tight outcome sets for known bets vs. spread-out sets for ambiguous bets; the paradox is rational multi-objective choice under Knightian uncertainty
8. Allais Paradox	Certainty preference violates the Independence Axiom	No utility function produces A > B and D > C simultaneously	Both preference patterns are admissible without contradiction	Multi-objective structure (expected value, downside protection, reliability) captures the structural asymmetry between problems; knife-edge scalar weighting shows scalarization essentially cannot reproduce these preferences
9. St. Petersburg	Infinite expected value recommends paying any fee	EU says pay up to ∞ (or \$21 truncated)	Fee capped at \$22, matching human intuition of \$20–25	Structural pluralism: multiple mechanisms interact (hard constraints cap catastrophic loss, multi-objective structure prevents EV dominance, satisficing filters on P(gain))

11.3 Cross-Cutting Patterns

Several patterns emerge across the nine examples:

The constraint layer (Layer 1) is the primary safety mechanism. In 6 of 9 examples, Layer 1 eliminates at least one action. The constraints enforce non-negotiable floors — insolvency, safety violations, epistemic integrity — that cannot be overridden by any amount of expected benefit on other objectives. This is the formal realization of the claim that some values are preconditions, not tradeoffs.

Deference fires when values are genuinely contested. In 5 of 9 examples, the protocol terminates with deference to the principal ($|R| > 1$). In every case, the remaining options represent a genuine tradeoff that the theory correctly refuses to resolve on the agent's behalf. The regret profiles presented to the principal make the tradeoff structure explicit and quantitative.

Scalar EU disagrees with MOADT on safety-critical decisions. In Example 5 (corrigibility), scalar EU recommends resisting value modification while MOADT recommends accepting with monitoring. In Example 6 (Pascal's mugging), scalar EU recommends paying the mugger. In Example 9 (St. Petersburg), scalar EU recommends paying any fee. In each case, MOADT's disagreement with scalar EU is precisely the behavior an aligned agent should exhibit.

Evaluator uncertainty ($\mathcal{F}$) does substantive work. In Examples 3–7, multiple evaluators produce meaningfully different assessments of the same outcomes. This is not noise — it represents genuine disagreement about what constitutes harm (Example 3), what alignment philosophy to adopt (Example 4), whether the agent's current values are correct (Example 5), and how to value monetary outcomes (Examples 6–9). The evaluator set prevents the agent from over-optimizing any single proxy.

11.4 Verification

All worked examples are implemented as executable Python scripts in the `examples/` directory and use the `moadt` library (`moadt/_engine.py`). Every number in the companion documents was produced by running these scripts. The full set:

- `examples/paper1_resource_allocation.py` through `examples/paper5_corrigibility.py` (Examples 1–5)
- `examples/classic_pascal.py`, `examples/classic_ellsberg.py`, `examples/classic_allais.py`, `examples/classic_stpetersburg.py` (Examples 6–9)

Regression tests covering key assertions from these examples are in `tests/test_engine.py` (25 tests, all passing).

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12. Minimal Formal Core

The following self-contained specification defines MOADT at the level of precision suitable for a formal paper.

Definitions

Definition 1 (Outcome Set). Given finite sets Ω (outcomes), S (states), A (actions), a compact credal set $\mathcal{P}$ over S, a compact evaluator set $\mathcal{F}$ of functions $\vec{f} : \Omega \to \mathbb{R}^k$, and a transition function $\omega : A \times S \to \Omega$, the outcome set of action a is:

$$Y(a) = \left\{ \mathbb{E}_P[\vec{f}(\omega(a,s))] : P \in \mathcal{P},\ \vec{f} \in \mathcal{F} \right\}$$

Definition 2 (Robust Dominance). $a \succ_R b$ iff $\forall\, \vec{y}_b \in Y(b),\ \exists\, \vec{y}_a \in Y(a)$ such that $y_{a,i} \geq y_{b,i}$ for all i and $y_{a,j} > y_{b,j}$ for some j.

Definition 3 (Admissible Set). $\mathrm{Adm}(A) = \{a \in A : \neg\exists\, a' \in A \text{ s.t. } a' \succ_R a\}$.

Definition 4 (Constraint Satisfaction). Given thresholds $\{\tau_i\}_{i \in I_C}$, action a is *feasible* iff $f_i(\omega(a,s)) \geq \tau_i$ for all $i \in I_C$, $P \in \mathcal{P}, \vec{f} \in \mathcal{F}$, and $s \in \mathrm{supp}(P)$. Let $C = \{a \in A : a \text{ is feasible}\}$. The feasible set is $F = \mathrm{Adm}(C)$.

Definition 5 (Robust Satisficing). Given reference point $\vec{r} \in \mathbb{R}^k$, the satisficing set is $\mathrm{Sat}(F, \vec{r}) = \{a \in F : \forall\, \vec{y} \in Y(a),\ \vec{y} \geq \vec{r}\}$. If $\mathrm{Sat} \neq \emptyset$, subsequent selection operates on Sat; otherwise, select from F via the achievement scalarizing function (Layer 2 fallback).

Definition 6 (Regret-Pareto Selection). For $a \in \mathrm{Sat}(F, \vec{r})$ (or F if $\mathrm{Sat} = \emptyset$), define per-objective minimax regret $\rho_i(a) = \max_{P,\vec{f}}[\max_{a' \in F} \mathbb{E}_P[f_i(a')] - \mathbb{E}_P[f_i(a)]]$. The regret-Pareto set is $R = \{a \in F : \vec{\rho}(a) \text{ is Pareto-minimal}\}$.

Propositions

Proposition 1. $\succ_R$ is a strict partial order (irreflexive, transitive).

Proposition 2. For finite A, $\mathrm{Adm}(A) \neq \emptyset$.

Proposition 3 (Constraint Primacy). The Layer 2 fallback (ASF) operates exclusively on $F = \mathrm{Adm}(C)$, and cannot select any action outside C. Therefore no outcome of Layers 2–4 can violate a Layer 1 constraint.

Proposition 4 (Protocol Termination). For finite A, each layer of the choice protocol maps a finite input to a non-empty output or a well-defined error/deference condition.

Theorems

Theorem 1 (Backward Compatibility). When $|\mathcal{P}| = 1$, $|\mathcal{F}| = 1$, $k = 1$: $\mathrm{Adm}(A) = \arg\max_a \mathbb{E}_P[f(a)]$.

Theorem 2 (Corrigibility Permissibility). Let a_{accept} and a_{resist} denote accepting vs. resisting an authorized modification, evaluated over trajectories using the agent's current $\mathcal{F}$. If $\mathcal{F}$ satisfies sufficient breadth (Section 5.3: $\exists(\vec{f}^+, P^+)$ s.t. the accept-trajectory is not Pareto-dominated, and $\exists(\vec{f}^-, P^-)$ s.t. the resist-trajectory is not Pareto-dominated), then $a_{\text{resist}} \not\succ_R a_{\text{accept}}$ and $a_{\text{accept}} \not\succ_R a_{\text{resist}}$. Both are admissible; the deference axiom directs the agent to query the principal, who — as the party that proposed the modification — selects a_{accept}.

Deference Axiom

When the choice protocol yields $|R| > 1$, the agent queries the principal for selection among the remaining options. This is the rational response to genuine incomparability — not a constraint, but a consequence of the theory's refusal to fabricate preferences.

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MOADT Worked Example 1: Resource Allocation Under Uncertainty

A concrete walkthrough of Multi-Objective Admissible Decision Theory

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Overview

This document applies MOADT — Multi-Objective Admissible Decision Theory — to a small but complete decision problem. Every computation is shown explicitly: outcome sets, robust dominance checks, all four layers of the choice protocol, and a comparison with scalar expected utility. The goal is to make the abstract formalism from the companion paper concrete and verifiable.

The scenario is deliberately small enough that every number fits on the page. The action set has 5 elements, the state space has 3 elements, the credal set has 2 distributions, and the evaluator set has 2 functions. This means each action's outcome set $Y(a)$ contains exactly 4 vectors in $\mathbb{R}^3$. Nothing is hidden behind "in the general case" — the reader can check every claim with a calculator.

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1. The Scenario

A nonprofit organization must decide how to allocate its annual budget across program areas. The board faces genuine uncertainty about the economic environment and legitimate disagreement about whether its own impact measurements are trustworthy.

1.1 Actions

Five allocation strategies are available:

Action	Strategy	Description
a_1	Conservative	Heavy investment in established core programs
a_2	Growth	Heavy investment in geographic/programmatic expansion
a_3	Balanced	Roughly equal split across all areas

Action	Strategy	Description
a_4	Innovation	Heavy investment in new experimental programs
a_5	Austerity	Minimize spending, maximize financial reserves

1.2 States

Three economic scenarios, representing the organization's uncertainty about the funding environment:

State	Description
s_1	Favorable economy: donor funding up, operating costs stable
s_2	Stable economy: baseline conditions
s_3	Recession: donor funding down, operating costs up

1.3 Objectives ($k = 3$)

Objective	Meaning	Scale
f_1: Mission impact	Beneficiaries effectively served	Normalized to $[0, 1]$
f_2: Financial sustainability	Reserves-to-operating-cost ratio	Normalized to $[0, 1]$
f_3: Staff wellbeing	Composite index (workload, morale, retention)	Normalized to $[0, 1]$

These objectives are *incommensurable*: the board does not believe there is a meaningful exchange rate between beneficiaries served and staff morale. Both matter. Neither reduces to the other.

1.4 Credal Set ($|\mathcal{P}| = 2$)

The organization cannot agree on the probability of recession. Two probability models are maintained:

$$P_{\text{opt}} = (0.40,\ 0.40,\ 0.20) \quad \text{(optimistic: recession unlikely)}$$

$$P_{\text{pess}} = (0.15,\ 0.35,\ 0.50) \quad \text{(pessimistic: recession likely)}$$

These are not "optimistic" and "pessimistic" in the sense of hoping for the best or fearing the worst. They represent two legitimate probabilistic models that the board's experts defend with credible arguments. The credal set $\mathcal{P} = \{P_{\text{opt}}, P_{\text{pess}}\}$ says: "we don't know which expert is right."

1.5 Evaluator Set ($|\mathcal{F}| = 2$)

Two evaluation functions reflect disagreement about whether the organization's metrics are trustworthy:

- $\vec{f}_{\text{std}}$: Takes reported metrics at face value.
- $\vec{f}_{\text{adj}}$: Applies a skeptical correction — mission impact scores are deflated by 15% (beneficiary counts are inflated by field offices) and staff wellbeing scores are deflated by 10% (self-reported morale surveys are biased upward). Financial sustainability is objective and unchanged.

This is $\mathcal{F}$ in action: the organization is *uncertain about its own measurements*. The evaluator set says "our impact numbers might be 15% too high, and our staff morale numbers might be 10% too high — or they might be accurate. We don't know."

1.6 Raw Outcome Data

The following tables show the outcome $\vec{f}(\omega(a, s))$ for each action-state pair, under each evaluator. These are the primitive inputs from which everything else is computed.

a_1 **(Conservative):**

State	Evaluator	Impact	Finance	Staff
s_1 (Favorable)	$\vec{f}_{\text{std}}$	0.65	0.80	0.70
	$\vec{f}_{\text{adj}}$	0.57	0.80	0.63
s_2 (Stable)	$\vec{f}_{\text{std}}$	0.60	0.70	0.65
	$\vec{f}_{\text{adj}}$	0.51	0.70	0.59
s_3 (Recession)	$\vec{f}_{\text{std}}$	0.50	0.55	0.55
	$\vec{f}_{\text{adj}}$	0.43	0.55	0.50

a_2 **(Growth):**

State	Evaluator	Impact	Finance	Staff
s_1 (Favorable)	$\vec{f}_{\text{std}}$	0.90	0.60	0.55
	$\vec{f}_{\text{adj}}$	0.77	0.60	0.50
s_2 (Stable)	$\vec{f}_{\text{std}}$	0.75	0.45	0.50
	$\vec{f}_{\text{adj}}$	0.64	0.45	0.45
s_3 (Recession)	$\vec{f}_{\text{std}}$	0.45	**0.20**	0.35
	$\vec{f}_{\text{adj}}$	0.38	**0.20**	0.32

Note the bold entries: in a recession, the growth strategy drives financial sustainability to 0.20 — well below the solvency threshold.

a_3 **(Balanced):**

State	Evaluator	Impact	Finance	Staff
s_1 (Favorable)	$\vec{f}_{\text{std}}$	0.70	0.65	0.70
	$\vec{f}_{\text{adj}}$	0.60	0.65	0.63
s_2 (Stable)	$\vec{f}_{\text{std}}$	0.65	0.60	0.65
	$\vec{f}_{\text{adj}}$	0.55	0.60	0.59
s_3 (Recession)	$\vec{f}_{\text{std}}$	0.50	0.45	0.50
	$\vec{f}_{\text{adj}}$	0.43	0.45	0.45

a_4 **(Innovation):**

State	Evaluator	Impact	Finance	Staff
s_1 (Favorable)	$\vec{f}_{\text{std}}$	0.85	0.50	0.75
	$\vec{f}_{\text{adj}}$	0.72	0.50	0.68
s_2 (Stable)	$\vec{f}_{\text{std}}$	0.60	0.40	0.60
	$\vec{f}_{\text{adj}}$	0.51	0.40	0.54
s_3 (Recession)	$\vec{f}_{\text{std}}$	0.30	**0.25**	0.40
	$\vec{f}_{\text{adj}}$	0.26	**0.25**	0.36

Innovation also violates the solvency threshold in recession.

a_5 **(Austerity):**

State	Evaluator	Impact	Finance	Staff
s_1 (Favorable)	$\vec{f}_{\text{std}}$	0.35	0.90	0.40
	$\vec{f}_{\text{adj}}$	0.30	0.90	0.36
s_2 (Stable)	$\vec{f}_{\text{std}}$	0.30	0.85	0.35
	$\vec{f}_{\text{adj}}$	0.26	0.85	0.32
s_3 (Recession)	$\vec{f}_{\text{std}}$	0.25	0.75	0.30
	$\vec{f}_{\text{adj}}$	0.21	0.75	0.27

Financially bulletproof, but mission impact and staff wellbeing are severely depressed.

---

2. Computing Outcome Sets $Y(a)$

Each action's outcome set contains one vector for each $(P, \vec{f})$ pair. With $|\mathcal{D}| = 2$ and $|\mathcal{F}| = 2$, each $Y(a)$ contains exactly 4 vectors.

The computation for each vector is:

$$\vec{y}_{(P,\vec{f})}(a) = \sum_{s \in S} P(s) \cdot \vec{f}(\omega(a, s))$$

Worked example: $Y(a_1)$**, first vector** (using P_{opt} and $\vec{f}_{\text{std}}$):

$$\vec{y} = 0.40 \cdot (0.65, 0.80, 0.70) + 0.40 \cdot (0.60, 0.70, 0.65) + 0.20 \cdot (0.50, 0.55, 0.55)$$

$$= (0.260, 0.320, 0.280) + (0.240, 0.280, 0.260) + (0.100, 0.110, 0.110)$$

$$= (0.600, 0.710, 0.650)$$

The full outcome sets (all computed analogously):

$Y(a_1)$ **— Conservative:**

$(P, \vec{f})$ pair	Impact	Finance	Staff
$(P_{\text{opt}}, \vec{f}_{\text{std}})$	0.6000	0.7100	0.6500
$(P_{\text{opt}}, \vec{f}_{\text{adj}})$	0.5180	0.7100	0.5880
$(P_{\text{pess}}, \vec{f}_{\text{std}})$	0.5575	0.6400	0.6075
$(P_{\text{pess}}, \vec{f}_{\text{adj}})$	0.4790	0.6400	0.5510

$Y(a_2)$ **— Growth:**

$(P, \vec{f})$ pair	Impact	Finance	Staff
$(P_{\text{opt}}, \vec{f}_{\text{std}})$	0.7500	0.4600	0.4900
$(P_{\text{opt}}, \vec{f}_{\text{adj}})$	0.6400	0.4600	0.4440
$(P_{\text{pess}}, \vec{f}_{\text{std}})$	0.6225	0.3475	0.4325
$(P_{\text{pess}}, \vec{f}_{\text{adj}})$	0.5295	0.3475	0.3925

$Y(a_3)$ **— Balanced:**

$(P, \vec{f})$ pair	Impact	Finance	Staff
$(P_{\text{opt}}, \vec{f}_{\text{std}})$	0.6400	0.5900	0.6400
$(P_{\text{opt}}, \vec{f}_{\text{adj}})$	0.5460	0.5900	0.5780
$(P_{\text{pess}}, \vec{f}_{\text{std}})$	0.5825	0.5325	0.5825
$(P_{\text{pess}}, \vec{f}_{\text{adj}})$	0.4975	0.5325	0.5260

$Y(a_4)$ **— Innovation:**

$(P, \vec{f})$ pair	Impact	Finance	Staff
$(P_{\text{opt}}, \vec{f}_{\text{std}})$	0.6400	0.4100	0.6200
$(P_{\text{opt}}, \vec{f}_{\text{adj}})$	0.5440	0.4100	0.5600

$(P, \vec{f})$ pair	Impact	Finance	Staff
$(P_{\text{pess}}, \vec{f}_{\text{std}})$	0.4875	0.3400	0.5225
$(P_{\text{pess}}, \vec{f}_{\text{adj}})$	0.4165	0.3400	0.4710

$Y(a_5)$ — **Austerity:**

$(P, \vec{f})$ pair	Impact	Finance	Staff
$(P_{\text{opt}}, \vec{f}_{\text{std}})$	0.3100	0.8500	0.3600
$(P_{\text{opt}}, \vec{f}_{\text{adj}})$	0.2660	0.8500	0.3260
$(P_{\text{pess}}, \vec{f}_{\text{std}})$	0.2825	0.8075	0.3325
$(P_{\text{pess}}, \vec{f}_{\text{adj}})$	0.2410	0.8075	0.3010

Observation

Each action's outcome set is a compact region in $\mathbb{R}^3$ (here, a set of 4 points). The *spread* of each set reflects the agent's epistemic uncertainty. For example, a_2 (Growth) has impact ranging from 0.5295 to 0.7500 — a wide band driven by the pessimistic prior dramatically reducing expected beneficiaries. Meanwhile, a_5 (Austerity) has very tight bands: its financial sustainability stays above 0.80 regardless of the model, reflecting that hoarding reserves is a low-uncertainty strategy.

---

3. Robust Dominance and the Admissible Set

3.1 Checking Robust Dominance

Recall: $a \succ_R b$ iff $\forall \vec{y}_b \in Y(b),\ \exists \vec{y}_a \in Y(a)$ such that $\vec{y}_a \succ_P \vec{y}_b$.

We check all 20 ordered pairs. The result: **no action robustly dominates any other**.

$$\text{Adm}(A) = \{a_1, a_2, a_3, a_4, a_5\}$$

3.2 Why Is No One Dominated?

This is instructive. Consider the most plausible candidate for dominance: does a_1 (Conservative) dominate a_5 (Austerity)?

Check the $\forall$-$\exists$ condition. Take a_5's best vector: $(P_{\text{opt}}, \vec{f}_{\text{std}})$ gives $(0.310, 0.850, 0.360)$. Can any vector in $Y(a_1)$ Pareto-dominate this?

a_1's vectors have finance scores of at most 0.710 — far below a_5's 0.850. So no vector in $Y(a_1)$ can Pareto-dominate $(0.310, 0.850, 0.360)$: the finance component always falls short. Therefore $a_1 \not\succ_R a_5$.

The reverse? a_5 has impact scores of at most 0.310, while a_1's worst impact is 0.479. So a_5 can never Pareto-dominate a_1 on impact. Neither dominates.

This illustrates a core feature of MOADT: when objectives are genuinely in tension (impact vs. financial security), robust dominance is hard to establish. The theory correctly identifies these actions as *incomparable* rather than fabricating a ranking. The practical filtering happens in the choice protocol layers that follow.

---

4. The Choice Protocol

Layer 1: Constraints

Constraint: Financial sustainability must be ≥ 0.30 in *every* state under *every* evaluator, for every distribution with positive support on that state.

This is checked per-state, per-evaluator — not on expected values. A single scenario in which the organization is insolvent kills the action.

Action	s_1 min	s_2 min	s_3 min	Result
a_1 (Conservative)	0.80	0.70	0.55	**PASS**
a_2 (Growth)	0.60	0.45	**0.20**	**FAIL**
a_3 (Balanced)	0.65	0.60	0.45	**PASS**
a_4 (Innovation)	0.50	0.40	**0.25**	**FAIL**
a_5 (Austerity)	0.90	0.85	0.75	**PASS**

Result: $C = \{a_1, a_3, a_5\}$. Growth and Innovation are eliminated — not because they score poorly *on average*, but because they have *any* scenario in which the organization becomes insolvent. This is a non-tradeable precondition.

Feasible set: $F = \mathrm{Adm}(C) = \{a_1, a_3, a_5\}$ (no robust dominance among the three survivors).

Layer 2: Reference-Point Satisficing

Reference point: $\vec{r} = (0.45, 0.50, 0.50)$ — representing the board's aspiration: "at least as good as last year" on all three objectives.

Robust satisficing: An action passes Layer 2 iff *every* vector in $Y(a)$ meets or exceeds $\vec{r}$ componentwise. The agent must be confident it meets aspirations under all plausible models.

Action	Worst impact	Worst finance	Worst staff	Result
a_1 (Conservative)	$0.4790 \geq 0.45$	$0.6400 \geq 0.50$	$0.5510 \geq 0.50$	**PASS**
a_3 (Balanced)	$0.4975 \geq 0.45$	$0.5325 \geq 0.50$	$0.5260 \geq 0.50$	**PASS**
a_5 (Austerity)	$0.2410 < 0.45$	$0.8075 \geq 0.50$	$0.3010 < 0.50$	**FAIL**

Result: $\mathrm{Sat}(F, \vec{r}) = \{a_1, a_3\}$. Austerity fails because its impact and staff wellbeing scores fall well below aspirations under every model.

Layer 3: Regret-Pareto

For the two surviving actions, we compute per-objective minimax regret. For each objective i and action a:

$$\rho_i(a) = \max_{P \in \mathcal{P}, \vec{f} \in \mathcal{F}} \left[\max_{a' \in F} \mathbb{E}_P[f_i(a')] - \mathbb{E}_P[f_i(a)] \right]$$

Regret is measured against the full feasible set $F = \{a_1, a_3, a_5\}$, not just the satisficing set Sat $= \{a_1, a_3\}$. This matters: an action that failed satisficing (like a_5, which fell short on impact and staff) can still be the source of per-objective regret if it excels on a particular objective. Austerity's financial sustainability scores (0.8075–0.8500) far exceed those of Conservative (0.6400–0.7100) and Balanced (0.5325–0.5900), creating finance regret for both surviving actions.

Regret of a_1 (Conservative):

For each $(P, \vec{f})$ pair, the best available action in F on each objective determines the gap:

$(P, \vec{f})$	Best impact in $F - a_1$	Best finance in $F - a_1$	Best staff in $F - a_1$
$(P_{\text{opt}}, \vec{f}_{\text{std}})$	$0.640 - 0.600 = 0.040$ (a_3)	$0.850 - 0.710 = 0.140$ (a_5)	$0.650 - 0.650 = 0$ (a_1)
$(P_{\text{opt}}, \vec{f}_{\text{adj}})$	$0.546 - 0.518 = 0.028$ (a_3)	$0.850 - 0.710 = 0.140$ (a_5)	$0.588 - 0.588 = 0$ (a_1)
$(P_{\text{pess}}, \vec{f}_{\text{std}})$	$0.583 - 0.558 = 0.025$ (a_3)	$0.808 - 0.640 = 0.168$ (a_5)	$0.608 - 0.608 = 0$ (a_1)
$(P_{\text{pess}}, \vec{f}_{\text{adj}})$	$0.498 - 0.479 = 0.019$ (a_3)	$0.808 - 0.640 = 0.168$ (a_5)	$0.551 - 0.551 = 0$ (a_1)

Regret is the *maximum* gap per objective:

$$\vec{\rho}(a_1) = (0.040,\ 0.168,\ 0)$$

a_1 has a small impact regret (0.040, driven by a_3's higher impact), meaningful finance regret (0.168, driven by a_5's strong financial reserves), and zero staff regret — a_1 is the best in F on staff under every model.

Regret of a_3 (Balanced):

$(P, \vec{f})$	Best impact in $F - a_3$	Best finance in $F - a_3$	Best staff in $F - a_3$
$(P_{\text{opt}}, \vec{f}_{\text{std}})$	$0.640 - 0.640 = 0$ (a_3)	$0.850 - 0.590 = 0.260$ (a_5)	$0.650 - 0.640 = 0.010$ (a_1)
$(P_{\text{opt}}, \vec{f}_{\text{adj}})$	$0.546 - 0.546 = 0$ (a_3)	$0.850 - 0.590 = 0.260$ (a_5)	$0.588 - 0.578 = 0.010$ (a_1)
$(P_{\text{pess}}, \vec{f}_{\text{std}})$	$0.583 - 0.583 = 0$ (a_3)	$0.808 - 0.533 = 0.275$ (a_5)	$0.608 - 0.583 = 0.025$ (a_1)

$(P, \vec{f})$	Best impact in $F - a_3$	Best finance in $F - a_3$	Best staff in $F - a_3$
$(P_{\text{pess}}, \vec{f}_{\text{adj}})$	$0.498 - 0.498 = 0$ (a_3)	$0.808 - 0.533 = 0.275$ (a_5)	$0.551 - 0.526 = 0.025$ (a_1)

$$\vec{\rho}(a_3) = (0,\ 0.275,\ 0.025)$$

a_3 has zero impact regret, substantial finance regret (0.275, driven by a_5's strong reserves), and small staff regret (0.025, driven by a_1).

Pareto comparison of regret vectors:

- $\vec{\rho}(a_1) = (0.040, 0.168, 0)$
- $\vec{\rho}(a_3) = (0, 0.275, 0.025)$

Neither vector Pareto-dominates the other: a_1 has lower regret on finance (0.168 vs. 0.275) and staff (0 vs. 0.025), while a_3 has lower regret on impact (0 vs. 0.040). They are Pareto-incomparable in regret space.

$$R = \{a_1, a_3\}$$

Layer 4: Deference

$|R| = 2 > 1$. The protocol terminates with **deference to the principal**.

The agent presents both options to the board with their regret profiles:

> **Option 1: Conservative (a_1).** Worst-case regret: you might miss 0.040 units of impact (compared to Balanced) and 0.168 units of financial sustainability (compared to Aggressive). You will not regret your staff wellbeing.
>
> **Option 2: Balanced (a_3).** Worst-case regret: you might miss 0.275 units of financial sustainability (compared to Aggressive) and 0.025 units of staff wellbeing (compared to Conservative). You will not regret your mission impact.
>
> **The tradeoff is:** Conservative has markedly lower finance regret (0.168 vs. 0.275) and no staff regret, but gives up 0.040 impact. Balanced eliminates impact regret entirely but carries substantially higher finance regret. The theory has no basis for making this tradeoff for you. This is your decision.

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5. What Scalar Expected Utility Would Have Done

To illustrate why MOADT matters, we compare with a scalar EU agent that collapses the three objectives into a single number via weights.

5.1 The Weight Problem

Scalar EU requires specifying weights $\vec{w} = (w_1, w_2, w_3)$ with $\sum w_i = 1$. The agent computes $U(a) = \vec{w} \cdot \mathbb{E}_P[\vec{f}(a)]$ and picks the action with the highest U.

But which weights? And which prior? And which evaluator? The answer changes depending on all three:

Under impact-heavy weights $(0.6, 0.2, 0.2)$ **and optimistic prior:**

Action	$U(a)$ under $\vec{f}_{\text{std}}$	$U(a)$ under $\vec{f}_{\text{adj}}$
a_1 (Conservative)	0.632	**0.570**
a_2 (Growth)	**0.640**	0.565
a_3 (Balanced)	0.630	0.561
a_4 (Innovation)	0.590	0.520
a_5 (Austerity)	0.428	0.395

Under the standard evaluator, **Growth** (a_2) **wins**. Under the adjusted evaluator, **Conservative** (a_1) **wins**. The "optimal" action flips depending on whether you trust your own impact measurements.

5.2 The Constraint-Blindness Problem

More critically: scalar EU with impact-heavy weights and the optimistic prior recommends a_2 (Growth) — the action that drives financial sustainability to 0.20 in a recession. The high expected impact in favorable and stable states *numerically compensates* for the risk of insolvency.

This is exactly the scalarization trap. The agent has computed that 0.640 utils > 0.632 utils, and the fact that one of those utility points was earned by risking organizational death is invisible in the scalar comparison. There is no mechanism in scalar EU to say "financial sustainability below 0.30 is not acceptable regardless of how much impact you promise."

MOADT eliminates a_2 at Layer 1. Insolvency risk is not a tradeoff — it is a precondition.

5.3 The Stability Problem

Under equal weights $(1/3, 1/3, 1/3)$, a_1 (Conservative) wins under all four $(P, \vec{f})$ pairs. Under safety-heavy weights $(0.2, 0.6, 0.2)$ with pessimistic prior and adjusted evaluator, a_5 (Austerity) wins. The "rational" action is a function of which weights, prior, and evaluator you assume — but scalar EU treats these as fixed inputs, not sources of uncertainty.

MOADT treats all of these as explicit uncertainty: $\mathcal{D}$ captures the prior disagreement, $\mathcal{F}$ captures the evaluator disagreement, and the absence of weights reflects the genuine incommensurability of the objectives. Instead of forcing a single answer that depends on unresolvable parameter choices, MOADT identifies the *set* of defensible options and presents the remaining tradeoff transparently.

6. Variant: What If Aspirations Are Higher?

If the board sets a more ambitious reference point $\vec{r} = (0.50, 0.50, 0.50)$, the protocol follows a different path.

Layer 2 outcome: $\mathrm{Sat}(F, \vec{r}) = \emptyset$. No action robustly meets all aspirations. The worst-case vector for a_1 has impact 0.479 (below 0.50); the worst-case vector for a_3 has impact 0.498 (below 0.50); a_5 fails on impact and staff.

ASF fallback fires. The achievement scalarizing function measures how close each action gets to aspirations (higher is better, 0 = exactly meeting aspirations, negative = falling short):

Action	ASF score	Interpretation
a_1 (Conservative)	-0.021	Falls 0.021 short of aspirations on its worst objective
a_3 (Balanced)	-0.003	Falls 0.003 short — nearly meets aspirations
a_5 (Austerity)	-0.259	Falls 0.259 short (massive impact deficit)

Result: a_3 (Balanced) is selected by ASF. Layer 3 is skipped (ASF already resolved the selection). No deference needed.

The ASF fallback is a scalar-valued function — and the paper is explicit about this. It appears only when no action robustly meets aspirations, it measures *shortfall from aspirations* (not absolute value), and it operates exclusively within the constraint-satisfying feasible set F (Proposition 3 in the main paper: ASF cannot override Layer 1 constraints).

---

7. Summary of Protocol Execution

```
Input: 5 actions × 3 states × 3 objectives × 2 priors × 2 evaluators

Robust Dominance:     No dominance found.
                      Adm(A) = {a₁, a₂, a₃, a₄, a₅}

Layer 1 (Constraints): Financial sustainability ≥ 0.30 per-state
                      a₂ FAIL (recession: 0.20)
                      a₄ FAIL (recession: 0.25)
                      C = {a₁, a₃, a₅}
                      F = Adm(C) = {a₁, a₃, a₅}

Layer 2 (Satisficing): r = (0.45, 0.50, 0.50)
                      a₅ FAIL (impact 0.241, staff 0.301 under pessimistic)
                      Sat = {a₁, a₃}

Layer 3 (Regret):     ρ(a₁) = (0.040, 0.168, 0)
                      ρ(a₃) = (0, 0.275, 0.025)
                      Pareto-incomparable → R = {a₁, a₃}
```

```
Layer 4 (Deference):  |R| = 2 → DEFER TO PRINCIPAL
                      "Conservative: regret 0.040 impact, 0.168 finance.
                       Balanced: regret 0.275 finance, 0.025 staff.
                       This is your call."
```

---

8. What This Example Demonstrates

1. **Constraints are not tradeoffs.** a_2 (Growth) is the highest-impact strategy under optimistic assumptions. Scalar EU with impact-heavy weights recommends it. MOADT eliminates it because it risks insolvency — and insolvency is not compensated by any amount of impact. The constraint is a precondition, not an objective.
2. **Evaluator uncertainty matters.** The "best" scalar EU action changes depending on whether you trust your own metrics. MOADT treats this as explicit uncertainty ($\mathcal{F}$) rather than a parameter the designer must resolve before the theory can proceed.
3. **Incomparability is informative.** The final output — "a_1 and a_3 are both defensible, and they trade off impact against financial security" — is more useful than a single recommendation that hides the tradeoff. The board knows exactly what it's choosing between and why.
4. **Deference is structural, not failure.** Layer 4 fires because the objectives are genuinely in tension. The theory cannot resolve this tension without making a value judgment about how much financial security is worth how much impact — and MOADT holds that this is the board's judgment to make, not the theory's.
5. **The protocol terminates.** Five actions enter; two emerge, with explicit regret profiles. The board has a concrete decision to make, with all the quantitative information it needs to make it. MOADT does not produce analysis paralysis — it produces *structured deference*.

Appendix: Computational Verification

All numerical results in this document were produced by `examples/paper1_resource_allocation.py` using the MOADT engine (`moadt/_engine.py`). The code is available in the repository and can be run independently to verify every number.

References

- MOADT (Multi-Objective Admissible Decision Theory) is defined in the companion paper.
- Freeman (2025), "The Scalarization Trap," provides the motivation for why scalar expected utility is structurally problematic for alignment.
- Wierzbicki, A. P. (1980). The use of reference objectives in multiobjective optimization. The achievement scalarizing function used in the Layer 2 fallback.

MOADT Worked Example 2: Medical Treatment Selection

A concrete walkthrough of Multi-Objective Admissible Decision Theory applied to clinical decision support

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Overview

This document applies MOADT — Multi-Objective Admissible Decision Theory — to a clinical treatment selection problem. Every computation is shown explicitly: raw outcome tables, outcome sets, constraint checks across all state-evaluator pairs, satisficing analysis, regret-Pareto comparison, and a comparison with scalar expected utility. The goal is to make the abstract formalism concrete and medically meaningful.

The scenario is deliberately richer than the companion worked example (Paper 1's resource allocation): 6 treatments, 4 patient response states, 4 objectives, 2 credal priors, and 2 evaluators. Each action's outcome set $Y(a)$ therefore contains 4 vectors in $\mathbb{R}^4$. The additional complexity serves two purposes: it shows how the theory scales to higher-dimensional problems, and it introduces a safety constraint that interacts with evaluator uncertainty in a clinically critical way. Nothing is hidden — the reader can check every number with a calculator.

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1. The Scenario

A clinical decision support system must recommend a treatment for a patient with a chronic condition. The clinician faces genuine uncertainty about how this patient will respond (the patient's risk profile is ambiguous), and the evidence base itself is contested — controlled trial data may not reflect real-world outcomes for this patient.

1.1 Actions (Treatments)

Six treatment options are available:

Action	Treatment	Description
a_1	Standard monotherapy	First-line single-agent treatment; well-established safety profile

Action	Treatment	Description
a_2	Aggressive monotherapy	Higher-dose single-agent; stronger effect, more side effects
a_3	Combination A	Two-drug regimen; high efficacy but complex dosing and monitoring
a_4	Combination B	Two-drug regimen; moderate efficacy with better safety than Combo A
a_5	Experimental	Novel agent from recent trials; promising but limited long-term data
a_6	Watchful waiting	Defer active treatment; monitor and intervene only if progression occurs

1.2 States (Patient Response Types)

Four patient response profiles, representing uncertainty about how this particular patient's physiology will interact with treatment:

State	Description	Probability
s_1 (Strong responder)	Patient metabolizes drug well, strong therapeutic response	30%
s_2 (Moderate responder)	Typical response profile	40%
s_3 (Weak responder)	Suboptimal drug metabolism, reduced efficacy	20%
s_4 (Adverse reactor)	Patient has heightened sensitivity to side effects	10%

The probabilities above are point estimates. The actual uncertainty about this patient's type is captured by the credal set (Section 1.4).

1.3 Objectives ($k = 4$)

Objective	Meaning	Scale
f_1: Efficacy	Treatment effectiveness (symptom control, disease modification)	Normalized to $[0, 1]$
f_2: Safety	Freedom from adverse events and side effects	Normalized to $[0, 1]$
f_3: Patient autonomy	Inverse of treatment burden (fewer appointments, simpler regimen = higher)	Normalized to $[0, 1]$
f_4: Cost efficiency	Inverse of total treatment cost to patient and system	Normalized to $[0, 1]$

These objectives are *incommensurable*: there is no meaningful exchange rate between efficacy and patient autonomy. A treatment that controls symptoms brilliantly but requires daily infusions at a clinic imposes a qualitatively different kind of cost than one that works less well but involves a single daily pill. Both dimensions matter. Neither reduces to the other. And the question of how much safety risk is acceptable for how much efficacy gain is precisely the kind of value judgment that should be made by the patient, not the algorithm.

1.4 Credal Set ($|\mathcal{P}| = 2$)

Two probability models over patient response types are maintained:

$$P_{\text{clinical}} = (0.30,\ 0.40,\ 0.20,\ 0.10) \quad \text{(from clinical trial population data)}$$

$$P_{\text{cautious}} = (0.20,\ 0.35,\ 0.25,\ 0.20) \quad \text{(adjusted for this patient's risk factors)}$$

P_{clinical} reflects the response distribution observed in controlled trials. P_{cautious} is adjusted by the clinician who notes that this patient has risk factors (age, comorbidities, genetic markers) suggesting a higher probability of weak response or adverse reaction. The credal set $\mathcal{P} = \{P_{\text{clinical}}, P_{\text{cautious}}\}$ says: "we don't know whether this patient's risk profile matches the trial population or warrants the cautious adjustment."

Note the asymmetry: P_{cautious} doubles the probability of adverse reaction (from 10% to 20%) and increases weak response probability from 20% to 25%, at the expense of strong and moderate response. This is not pessimism for its own sake — it reflects specific clinical judgment about this patient.

1.5 Evaluator Set ($|\mathcal{F}| = 2$)

Two evaluation functions reflect disagreement about whether controlled trial outcomes apply to this patient:

- $\vec{f}_{\text{trial}}$: Takes outcomes from controlled trial data at face value.
- $\vec{f}_{\text{realworld}}$: Applies real-world adjustments — efficacy is deflated (adherence is lower outside trials, comorbidities reduce response), safety is deflated by 10% (trial populations exclude high-risk patients), and autonomy is deflated by approximately 14% (real-world treatment burden exceeds trial protocols due to additional monitoring, pharmacy visits, etc.). Cost efficiency is unchanged (costs are objective).

This is $\mathcal{F}$ in action: the evidence base itself is uncertain. Trial data are gathered under idealized conditions (motivated patients, strict protocols, close monitoring). Real-world outcomes are systematically worse on efficacy, safety, and autonomy. The evaluator set says "we don't know whether this patient's experience will resemble the trial or the real world."

1.6 Constraint and Reference Point

Hard constraint (Layer 1): Safety $\geq$ 0.35 in *every* state under *every* evaluator. This is a non-negotiable patient safety threshold. No amount of efficacy justifies a treatment whose safety score drops below 0.35 for any patient response type under any plausible evaluation. This constraint is checked per-state, per-evaluator — not on expected values.

Reference point (Layer 2): $\vec{r} = (0.40,\ 0.50,\ 0.30,\ 0.30)$ — representing the clinician's aspiration: "the treatment should achieve at least moderate efficacy, maintain reasonable safety, impose manageable burden, and not be prohibitively expensive." These are aspiration levels, not hard constraints. An action that falls below an aspiration level is not *unsafe* — it is just *not good enough* to be worth recommending over alternatives.

1.7 Raw Outcome Data

The following tables show the outcome $\vec{f}(\omega(a,s))$ for each action-state pair, under each evaluator. These are the primitive inputs from which everything else is computed.

a_1 **(Standard monotherapy):**

State	Evaluator	Efficacy	Safety	Autonomy	Cost
s_1 (Strong)	$\vec{f}_{\text{trial}}$	0.75	0.80	0.70	0.80
	$\vec{f}_{\text{real}}$	0.62	0.72	0.60	0.80
s_2 (Moderate)	$\vec{f}_{\text{trial}}$	0.60	0.80	0.70	0.80
	$\vec{f}_{\text{real}}$	0.50	0.72	0.60	0.80
s_3 (Weak)	$\vec{f}_{\text{trial}}$	0.35	0.80	0.70	0.80
	$\vec{f}_{\text{real}}$	0.28	0.72	0.60	0.80
s_4 (Adverse)	$\vec{f}_{\text{trial}}$	0.30	0.55	0.70	0.80
	$\vec{f}_{\text{real}}$	0.25	0.50	0.60	0.80

Standard monotherapy has moderate efficacy that drops for weak responders, excellent safety even for adverse reactors, high autonomy (simple single-pill regimen), and good cost efficiency. The safety floor of 0.50 (adverse reactor, real-world) remains well above the 0.35 threshold.

a_2 **(Aggressive monotherapy):**

State	Evaluator	Efficacy	Safety	Autonomy	Cost
s_1 (Strong)	$\vec{f}_{\text{trial}}$	0.90	0.60	0.50	0.70
	$\vec{f}_{\text{real}}$	0.75	0.54	0.43	0.70
s_2 (Moderate)	$\vec{f}_{\text{trial}}$	0.75	0.55	0.50	0.70
	$\vec{f}_{\text{real}}$	0.62	0.50	0.43	0.70
s_3 (Weak)	$\vec{f}_{\text{trial}}$	0.50	0.50	0.50	0.70
	$\vec{f}_{\text{real}}$	0.40	0.45	0.43	0.70
s_4 (Adverse)	$\vec{f}_{\text{trial}}$	0.45	**0.25**	0.50	0.70
	$\vec{f}_{\text{real}}$	0.38	**0.23**	0.43	0.70

Note the bold entries: for adverse reactors, safety collapses to 0.25 (trial) and 0.23 (real-world) — far below the 0.35 threshold. Aggressive monotherapy is highly effective for strong responders but dangerous for the wrong patient.

a_3 **(Combination A):**

State	Evaluator	Efficacy	Safety	Autonomy	Cost
s_1 (Strong)	$\vec{f}_{\text{trial}}$	0.85	0.65	0.45	0.50
	$\vec{f}_{\text{real}}$	0.70	0.59	0.38	0.50
s_2 (Moderate)	$\vec{f}_{\text{trial}}$	0.70	0.60	0.45	0.50
	$\vec{f}_{\text{real}}$	0.58	0.54	0.38	0.50
s_3 (Weak)	$\vec{f}_{\text{trial}}$	0.55	0.55	0.45	0.50
	$\vec{f}_{\text{real}}$	0.44	0.50	0.38	0.50
s_4 (Adverse)	$\vec{f}_{\text{trial}}$	0.40	**0.30**	0.45	0.50
	$\vec{f}_{\text{real}}$	0.33	**0.27**	0.38	0.50

Combination A also violates the safety threshold for adverse reactors: 0.30 under trial data, 0.27 under real-world evaluation. Additionally, autonomy is low (complex two-drug regimen requiring monitoring) and cost is moderate.

a_4 **(Combination B):**

State	Evaluator	Efficacy	Safety	Autonomy	Cost
s_1 (Strong)	$\vec{f}_{\text{trial}}$	0.80	0.70	0.40	0.45
	$\vec{f}_{\text{real}}$	0.66	0.63	0.34	0.45
s_2 (Moderate)	$\vec{f}_{\text{trial}}$	0.65	0.70	0.40	0.45
	$\vec{f}_{\text{real}}$	0.54	0.63	0.34	0.45
s_3 (Weak)	$\vec{f}_{\text{trial}}$	0.50	0.65	0.40	0.45
	$\vec{f}_{\text{real}}$	0.40	0.59	0.34	0.45
s_4 (Adverse)	$\vec{f}_{\text{trial}}$	0.35	0.45	0.40	0.45
	$\vec{f}_{\text{real}}$	0.29	0.41	0.34	0.45

Combination B trades some efficacy for better safety compared to Combination A. Its safety floor for adverse reactors is 0.41 (real-world) — above the 0.35 threshold. But autonomy is low (complex regimen) and cost is the worst of all treatments.

a_5 **(Experimental):**

State	Evaluator	Efficacy	Safety	Autonomy	Cost
s_1 (Strong)	$\vec{f}_{\text{trial}}$	0.95	0.70	0.55	0.30
	$\vec{f}_{\text{real}}$	0.80	0.63	0.47	0.30
s_2 (Moderate)	$\vec{f}_{\text{trial}}$	0.70	0.65	0.55	0.30

State	Evaluator	Efficacy	Safety	Autonomy	Cost
	$\vec{f}_{\text{real}}$	0.58	0.59	0.47	0.30
s_3 (Weak)	$\vec{f}_{\text{trial}}$	0.30	0.60	0.55	0.30
	$\vec{f}_{\text{real}}$	0.24	0.54	0.47	0.30
s_4 (Adverse)	$\vec{f}_{\text{trial}}$	0.20	**0.35**	0.55	0.30
	$\vec{f}_{\text{real}}$	0.16	**0.32**	0.47	0.30

This is the most instructive case in the dataset. Under trial data, the experimental treatment's safety score for adverse reactors is *exactly* 0.35 — right at the threshold. Under real-world evaluation, it drops to 0.32 — *below* the threshold. The gap between trial and real-world evaluation is the difference between "barely safe" and "not safe enough." Without evaluator uncertainty ($\mathcal{F}$), this treatment would be considered safe. With $\mathcal{F}$, it is correctly eliminated.

a_6 **(Watchful waiting):**

State	Evaluator	Efficacy	Safety	Autonomy	Cost
s_1 (Strong)	$\vec{f}_{\text{trial}}$	0.40	0.95	0.90	0.95
	$\vec{f}_{\text{real}}$	0.35	0.86	0.77	0.95
s_2 (Moderate)	$\vec{f}_{\text{trial}}$	0.25	0.95	0.90	0.95
	$\vec{f}_{\text{real}}$	0.21	0.86	0.77	0.95
s_3 (Weak)	$\vec{f}_{\text{trial}}$	0.15	0.95	0.90	0.95
	$\vec{f}_{\text{real}}$	0.12	0.86	0.77	0.95
s_4 (Adverse)	$\vec{f}_{\text{trial}}$	0.10	0.95	0.90	0.95
	$\vec{f}_{\text{real}}$	0.08	0.86	0.77	0.95

Watchful waiting is maximally safe, autonomous, and cheap — but efficacy is severely depressed. For moderate, weak, and adverse responders, efficacy is 0.25 or below even under trial evaluation. This treatment is not *dangerous* — it is just *not doing much*.

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2. Computing Outcome Sets $Y(a)$

Each action's outcome set contains one vector for each $(P, \vec{f})$ pair. With $|\mathcal{P}| = 2$ and $|\mathcal{F}| = 2$, each $Y(a)$ contains exactly 4 vectors in $\mathbb{R}^4$.

The computation for each vector is:

$$\vec{y}_{(P,\vec{f})}(a) = \sum_{s \in S} P(s) \cdot \vec{f}(\omega(a, s))$$

Worked example: $Y(a_1)$, first vector

Using $P_{\text{clinical}} = (0.30, 0.40, 0.20, 0.10)$ and $\vec{f}_{\text{trial}}$:

$$\begin{aligned}\vec{y} &= 0.30 \cdot (0.75, 0.80, 0.70, 0.80) + 0.40 \cdot (0.60, 0.80, 0.70, 0.80) \\ &\quad +0.20 \cdot (0.35, 0.80, 0.70, 0.80) + 0.10 \cdot (0.30, 0.55, 0.70, 0.80) \\ &= (0.225, 0.240, 0.210, 0.240) + (0.240, 0.320, 0.280, 0.320) \\ &\quad +(0.070, 0.160, 0.140, 0.160) + (0.030, 0.055, 0.070, 0.080) \\ &= (0.5650,\ 0.7750,\ 0.7000,\ 0.8000)\end{aligned}$$

Worked example: $Y(a_1)$, worst-case vector

Using $P_{\text{cautious}} = (0.20, 0.35, 0.25, 0.20)$ and $\vec{f}_{\text{real}}$:

$$\begin{aligned}\vec{y} &= 0.20 \cdot (0.62, 0.72, 0.60, 0.80) + 0.35 \cdot (0.50, 0.72, 0.60, 0.80) \\ &\quad +0.25 \cdot (0.28, 0.72, 0.60, 0.80) + 0.20 \cdot (0.25, 0.50, 0.60, 0.80) \\ &= (0.124, 0.144, 0.120, 0.160) + (0.175, 0.252, 0.210, 0.280) \\ &\quad +(0.070, 0.180, 0.150, 0.200) + (0.050, 0.100, 0.120, 0.160) \\ &= (0.4190,\ 0.6760,\ 0.6000,\ 0.8000)\end{aligned}$$

This is the worst-case expected outcome for Standard: using the cautious prior (more adverse reactors) and real-world evaluation (deflated efficacy and safety). Even here, all four components remain above the reference point $\vec{r} = (0.40, 0.50, 0.30, 0.30)$.

Full Outcome Sets

All remaining vectors are computed analogously.

$Y(a_1)$ — **Standard monotherapy:**

$(P, \vec{f})$ pair	Efficacy	Safety	Autonomy	Cost
$(P_{\text{clin}}, \vec{f}_{\text{trial}})$	0.5650	0.7750	0.7000	0.8000
$(P_{\text{clin}}, \vec{f}_{\text{real}})$	0.4670	0.6980	0.6000	0.8000
$(P_{\text{caut}}, \vec{f}_{\text{trial}})$	0.5075	0.7500	0.7000	0.8000
$(P_{\text{caut}}, \vec{f}_{\text{real}})$	0.4190	0.6760	0.6000	0.8000

$Y(a_2)$ — **Aggressive monotherapy:**

$(P,\vec{f})$ pair	Efficacy	Safety	Autonomy	Cost
$(P_{\text{clin}},\vec{f}_{\text{trial}})$	0.7150	0.5250	0.5000	0.7000
$(P_{\text{clin}},\vec{f}_{\text{real}})$	0.5910	0.4750	0.4300	0.7000
$(P_{\text{caut}},\vec{f}_{\text{trial}})$	0.6575	0.4875	0.5000	0.7000
$(P_{\text{caut}},\vec{f}_{\text{real}})$	0.5430	0.4415	0.4300	0.7000

$Y(a_3)$ — **Combination A:**

$(P,\vec{f})$ pair	Efficacy	Safety	Autonomy	Cost
$(P_{\text{clin}},\vec{f}_{\text{trial}})$	0.6850	0.5750	0.4500	0.5000
$(P_{\text{clin}},\vec{f}_{\text{real}})$	0.5630	0.5200	0.3800	0.5000
$(P_{\text{caut}},\vec{f}_{\text{trial}})$	0.6325	0.5375	0.4500	0.5000
$(P_{\text{caut}},\vec{f}_{\text{real}})$	0.5190	0.4860	0.3800	0.5000

$Y(a_4)$ — **Combination B:**

$(P,\vec{f})$ pair	Efficacy	Safety	Autonomy	Cost
$(P_{\text{clin}},\vec{f}_{\text{trial}})$	0.6350	0.6650	0.4000	0.4500
$(P_{\text{clin}},\vec{f}_{\text{real}})$	0.5230	0.6000	0.3400	0.4500
$(P_{\text{caut}},\vec{f}_{\text{trial}})$	0.5825	0.6375	0.4000	0.4500
$(P_{\text{caut}},\vec{f}_{\text{real}})$	0.4790	0.5760	0.3400	0.4500

$Y(a_5)$ — **Experimental:**

$(P,\vec{f})$ pair	Efficacy	Safety	Autonomy	Cost
$(P_{\text{clin}},\vec{f}_{\text{trial}})$	0.6450	0.6250	0.5500	0.3000
$(P_{\text{clin}},\vec{f}_{\text{real}})$	0.5360	0.5650	0.4700	0.3000
$(P_{\text{caut}},\vec{f}_{\text{trial}})$	0.5500	0.5875	0.5500	0.3000
$(P_{\text{caut}},\vec{f}_{\text{real}})$	0.4550	0.5315	0.4700	0.3000

$Y(a_6)$ — **Watchful waiting:**

$(P,\vec{f})$ pair	Efficacy	Safety	Autonomy	Cost
$(P_{\text{clin}},\vec{f}_{\text{trial}})$	0.2600	0.9500	0.9000	0.9500
$(P_{\text{clin}},\vec{f}_{\text{real}})$	0.2210	0.8600	0.7700	0.9500
$(P_{\text{caut}},\vec{f}_{\text{trial}})$	0.2250	0.9500	0.9000	0.9500
$(P_{\text{caut}},\vec{f}_{\text{real}})$	0.1895	0.8600	0.7700	0.9500

Observation

Each action's outcome set is a compact region in $\mathbb{R}^4$ (here, a set of 4 points). The *spread* of each set reflects epistemic uncertainty. Standard monotherapy (a_1) has efficacy ranging from 0.419 to 0.565 — a moderate band reflecting uncertainty about both the patient's response type and the reliability of trial data. Watchful waiting (a_6) has efficacy ranging from 0.190 to 0.260 — uniformly low regardless of model, because doing nothing is reliably ineffective. But watchful waiting's safety ranges from 0.860 to 0.950 — uniformly high, because doing nothing is reliably safe. The theory tracks all four dimensions simultaneously without forcing them into a single number.

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3. Robust Dominance and the Admissible Set

3.1 Checking Robust Dominance

Recall: $a \succ_R b$ iff $\forall \vec{y}_b \in Y(b),\ \exists \vec{y}_a \in Y(a)$ such that $\vec{y}_a \succ_P \vec{y}_b$.

We check all 30 ordered pairs. The result: **no action robustly dominates any other**.

$$\mathrm{Adm}(A) = \{a_1, a_2, a_3, a_4, a_5, a_6\}$$

3.2 Why Is No One Dominated?

This is more surprising than in Paper 1's three-objective case. With 6 actions and 4 objectives, one might expect at least one dominance relation. But each action excels on at least one objective in a way that no other action can match across all models.

Consider the most natural candidate: does a_1 (Standard) dominate a_6 (Watchful waiting)? Take a_6's best vector: $(P_{\text{clin}}, \vec{f}_{\text{trial}})$ gives $(0.260, 0.950, 0.900, 0.950)$. Can any vector in $Y(a_1)$ Pareto-dominate this? The best safety score in $Y(a_1)$ is 0.775, the best autonomy is 0.700, and the best cost is 0.800 — all below a_6's scores of 0.950, 0.900, and 0.950 respectively. So no vector in $Y(a_1)$ can Pareto-dominate a_6's best case. Standard monotherapy cannot robustly dominate watchful waiting because watchful waiting is unbeatable on three of four objectives.

The reverse fails too: a_6's best efficacy is 0.260, while a_1's worst efficacy is 0.419. Watchful waiting can never Pareto-dominate Standard on efficacy.

Does a_1 dominate a_2 (Aggressive)? a_2's best efficacy is 0.715, while a_1's best is only 0.565. Aggressive monotherapy's efficacy advantage prevents Standard from dominating it, even though Aggressive's safety profile is much worse.

This pattern repeats across all 30 pairs: the four-dimensional objective space provides enough room for every action to maintain a Pareto-incomparable niche. The practical filtering happens in the constraint layer, not through dominance.

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4. The Choice Protocol

Layer 1: Constraints

Constraint: Safety ≥ 0.35 in *every* state under *every* evaluator.

This is checked per-state, per-evaluator — not on expected values. A single state-evaluator pair in which safety drops below 0.35 eliminates the treatment. The constraint embodies a fundamental clinical principle: we do not accept treatments that are *predictably dangerous for identifiable patient subgroups*, regardless of how well they work for other patients.

a_1 **(Standard monotherapy):**

State	$\vec{f}_{\text{trial}}$	$\vec{f}_{\text{real}}$	Result
s_1 (Strong)	0.80	0.72	Pass
s_2 (Moderate)	0.80	0.72	Pass
s_3 (Weak)	0.80	0.72	Pass
s_4 (Adverse)	0.55	0.50	Pass

Minimum safety: 0.50 (adverse reactor, real-world). Well above threshold. **PASS.**

a_2 **(Aggressive monotherapy):**

State	$\vec{f}_{\text{trial}}$	$\vec{f}_{\text{real}}$	Result
s_1 (Strong)	0.60	0.54	Pass
s_2 (Moderate)	0.55	0.50	Pass
s_3 (Weak)	0.50	0.45	Pass
s_4 (Adverse)	**0.25**	**0.23**	**FAIL**

Safety collapses for adverse reactors: 0.25 under trial data, 0.23 under real-world. Both far below the 0.35 threshold. **FAIL — excluded.**

a_3 **(Combination A):**

State	$\vec{f}_{\text{trial}}$	$\vec{f}_{\text{real}}$	Result
s_1 (Strong)	0.65	0.59	Pass
s_2 (Moderate)	0.60	0.54	Pass
s_3 (Weak)	0.55	0.50	Pass
s_4 (Adverse)	**0.30**	**0.27**	**FAIL**

Same pattern: adverse reactor safety is 0.30 (trial) and 0.27 (real-world). **FAIL — excluded.**

a_4 **(Combination B):**

State	$\vec{f}_{\text{trial}}$	$\vec{f}_{\text{real}}$	Result
s_1 (Strong)	0.70	0.63	Pass

State	$\vec{f}_{\text{trial}}$	$\vec{f}_{\text{real}}$	Result
s_2 (Moderate)	0.70	0.63	Pass
s_3 (Weak)	0.65	0.59	Pass
s_4 (Adverse)	0.45	0.41	Pass

Minimum safety: 0.41 (adverse reactor, real-world). Above threshold. **PASS.**

a_5 **(Experimental):**

State	$\vec{f}_{\text{trial}}$	$\vec{f}_{\text{real}}$	Result
s_1 (Strong)	0.70	0.63	Pass
s_2 (Moderate)	0.65	0.59	Pass
s_3 (Weak)	0.60	0.54	Pass
s_4 (Adverse)	**0.35**	**0.32**	**FAIL**

This is the critical case. Under controlled trial data, the experimental treatment's safety score for adverse reactors is *exactly* 0.35 — precisely at the threshold. It passes. Under real-world evaluation, the 10% safety deflation reduces this to $0.35 \times 0.9143 \approx 0.32$ — below the threshold. It fails.

The experimental treatment passes the safety constraint under $\vec{f}_{\text{trial}}$ and fails under $\vec{f}_{\text{real}}$. This is exactly why $\mathcal{F}$ exists. If we had only trial data — if we were certain our measurements were accurate — this treatment would be considered safe. But we are *not* certain. The evaluator set captures the gap between controlled and real-world conditions, and that gap is the difference between "barely safe" and "not safe enough." **FAIL — excluded.**

a_6 **(Watchful waiting):**

State	$\vec{f}_{\text{trial}}$	$\vec{f}_{\text{real}}$	Result
s_1 (Strong)	0.95	0.86	Pass
s_2 (Moderate)	0.95	0.86	Pass
s_3 (Weak)	0.95	0.86	Pass
s_4 (Adverse)	0.95	0.86	Pass

Minimum safety: 0.86 (any state, real-world). Massively above threshold. **PASS.**

Result: Three of six treatments are eliminated by the safety constraint.

$$C = \{a_1 \text{ (Standard)},\ a_4 \text{ (Combo B)},\ a_6 \text{ (Watchful)}\}$$

No robust dominance exists among these three survivors, so:

$$F = \text{Adm}(C) = \{a_1, a_4, a_6\}$$

Commentary: The constraint eliminates half of the treatment set. This is more aggressive filtering than in Paper 1 (where 2 of 5 actions were eliminated). The increased filtering rate reflects the

interaction of more states with evaluator uncertainty: with 4 patient types and 2 evaluators, there are 8 state-evaluator pairs to check per treatment, and only one needs to fail. The constraint acts as a *worst-case* filter: it asks not "what happens on average?" but "is there *any* identifiable scenario in which this treatment is predictably dangerous?"

The Experimental treatment's elimination is especially noteworthy. A system that trusted only trial data would consider it safe. A system that incorporated real-world adjustment catches the risk. This is the evaluator set doing its job: ensuring that safety assessments are robust to measurement model uncertainty.

Layer 2: Reference-Point Satisficing

Reference point: $\vec{r} = (0.40, 0.50, 0.30, 0.30)$

Robust satisficing: An action passes Layer 2 iff *every* vector in $Y(a)$ meets or exceeds $\vec{r}$ componentwise. The system must be confident the treatment meets aspirations under all plausible models.

a_1 **(Standard):**

Component	Worst value	Aspiration	Margin	Result
Efficacy	0.4190	0.40	+0.019	Pass
Safety	0.6760	0.50	+0.176	Pass
Autonomy	0.6000	0.30	+0.300	Pass
Cost	0.8000	0.30	+0.500	Pass

Standard passes on all four objectives. Its worst-case efficacy (0.419, under cautious prior and real-world evaluation) clears the 0.40 aspiration with a margin of 0.019 — tight, but sufficient. Safety, autonomy, and cost are all well above aspirations. **PASS.**

a_4 **(Combination B):**

Component	Worst value	Aspiration	Margin	Result
Efficacy	0.4790	0.40	+0.079	Pass
Safety	0.5760	0.50	+0.076	Pass
Autonomy	0.3400	0.30	+0.040	Pass
Cost	0.4500	0.30	+0.150	Pass

Combination B also passes on all four objectives. Its worst-case efficacy (0.479) clears the aspiration more comfortably than Standard. But its margins on safety (0.076), autonomy (0.040), and cost (0.150) are much tighter. **PASS.**

a_6 **(Watchful waiting):**

Component	Worst value	Aspiration	Margin	Result
Efficacy	0.1895	0.40	**-0.211**	**FAIL**
Safety	0.8600	0.50	+0.360	Pass
Autonomy	0.7700	0.30	+0.470	Pass

Component	Worst value	Aspiration	Margin	Result
Cost	0.9500	0.30	+0.650	Pass

Watchful waiting's worst-case efficacy is 0.1895 (under cautious prior and real-world evaluation) — far below the 0.40 aspiration. Even its *best-case* efficacy is only 0.260. It excels on safety, autonomy, and cost, but its efficacy shortfall is not marginal — it misses the aspiration by 0.211 units, more than half the aspiration level itself. **FAIL.**

Result:

$$\mathrm{Sat}(F, \vec{r}) = \{a_1 \text{ (Standard)},\ a_4 \text{ (Combo B)}\}$$

Commentary: Watchful waiting's elimination at Layer 2 illustrates the crucial difference between constraints and aspirations. Watchful waiting *passed* the safety constraint at Layer 1 — it is not a dangerous treatment. It is eliminated at Layer 2 because it is *not effective enough* — not because it violates a precondition, but because it fails to achieve a goal. The constraint says "this must not be unsafe." The aspiration says "this should actually work." They are structurally different requirements, and MOADT applies them at different layers for precisely this reason.

A patient who asks "should we just wait and see?" receives a principled answer: watchful waiting is safe, cheap, and unburdensome, but under every plausible model of your response and every plausible evaluation of outcomes, the expected efficacy is below 0.40. The treatment is not *inappropriate* — it is *insufficient*.

Layer 3: Regret-Pareto

For the two surviving treatments, we compute per-objective minimax regret. For each objective i and action a:

$$\rho_i(a) = \max_{P \in \mathcal{P}, \vec{f} \in \mathcal{F}} \left[\max_{a' \in F} \mathbb{E}_P[f_i(a')] - \mathbb{E}_P[f_i(a)] \right]$$

Regret is measured against the full feasible set $F = \{a_1, a_4, a_6\}$, not just the satisficing set $\mathrm{Sat} = \{a_1, a_4\}$. This matters: an action that failed satisficing (like a_6, which fell short on efficacy) can still be the source of per-objective regret if it excels on a particular objective. Watchful waiting's safety scores (0.860–0.950), autonomy scores (0.770–0.900), and cost scores (0.950) far exceed those of Standard and Combo B on those dimensions, creating substantial regret for both surviving actions on safety, autonomy, and cost. This illustrates why regret must be measured against the full feasible set: an action that falls short on aspirations can still define the per-objective benchmark.

For each $(P, \vec{f})$ pair, the best available action in F on each objective determines the gap.

Regret of a_1 (Standard):

For each objective, we find the maximum gap between the best action in F and Standard across all $(P, \vec{f})$ pairs. On efficacy, Combo B (a_4) is the best comparator. On safety, autonomy, and cost, Watchful Waiting (a_6) is the best comparator — its excellence on those dimensions creates regret that was invisible when comparing only within Sat.

$(P, \vec{f})$	Eff gap	Safety gap	Auton gap	Cost gap
$(P_{\text{clin}}, \vec{f}_{\text{trial}})$	$0.635 - 0.565 = 0.070$ (a_4)	$0.950 - 0.775 = 0.175$ (a_6)	$0.900 - 0.700 = 0.200$ (a_6)	$0.950 - 0.800 = 0.150$ (a_6)
$(P_{\text{clin}}, \vec{f}_{\text{real}})$	$0.523 - 0.467 = 0.056$ (a_4)	$0.860 - 0.698 = 0.162$ (a_6)	$0.770 - 0.600 = 0.170$ (a_6)	$0.950 - 0.800 = 0.150$ (a_6)
$(P_{\text{caut}}, \vec{f}_{\text{trial}})$	$0.5825 - 0.5075 = 0.075$ (a_4)	$0.950 - 0.750 = 0.200$ (a_6)	$0.900 - 0.700 = 0.200$ (a_6)	$0.950 - 0.800 = 0.150$ (a_6)
$(P_{\text{caut}}, \vec{f}_{\text{real}})$	$0.479 - 0.419 = 0.060$ (a_4)	$0.860 - 0.676 = 0.184$ (a_6)	$0.770 - 0.600 = 0.170$ (a_6)	$0.950 - 0.800 = 0.150$ (a_6)

Regret is the *maximum* gap per objective:

$$\vec{\rho}(a_1) = (0.075,\ 0.200,\ 0.200,\ 0.150)$$

Standard monotherapy has a small regret on efficacy (0.075, driven by Combo B under the cautious prior and trial evaluation). But it now has *non-zero* regret on safety (0.200), autonomy (0.200), and cost (0.150) — all driven by Watchful Waiting's excellence on those dimensions. Even though Watchful Waiting failed satisficing on efficacy, it sets the per-objective benchmark on safety, autonomy, and cost. Standard is good on those dimensions, but Watchful Waiting is better, and that gap is regret.

Regret of a_4 (Combination B):

For each objective, we find the maximum gap between the best action in F and Combo B. Combo B is the best in F on efficacy (zero regret). On safety, autonomy, and cost, Watchful Waiting (a_6) is again the benchmark, and the gaps are larger than for Standard because Combo B scores lower on those dimensions.

$(P, \vec{f})$	Eff gap	Safety gap	Auton gap	Cost gap
$(P_{\text{clin}}, \vec{f}_{\text{trial}})$	$0.635 - 0.635 = 0$ (a_4 best)	$0.950 - 0.665 = 0.285$ (a_6)	$0.900 - 0.400 = 0.500$ (a_6)	$0.950 - 0.450 = 0.500$ (a_6)
$(P_{\text{clin}}, \vec{f}_{\text{real}})$	$0.523 - 0.523 = 0$ (a_4 best)	$0.860 - 0.600 = 0.260$ (a_6)	$0.770 - 0.340 = 0.430$ (a_6)	$0.950 - 0.450 = 0.500$ (a_6)
$(P_{\text{caut}}, \vec{f}_{\text{trial}})$	$0.5825 - 0.5825 = 0$ (a_4 best)	$0.950 - 0.6375 = 0.3125$ (a_6)	$0.900 - 0.400 = 0.500$ (a_6)	$0.950 - 0.450 = 0.500$ (a_6)
$(P_{\text{caut}}, \vec{f}_{\text{real}})$	$0.479 - 0.479 = 0$ (a_4 best)	$0.860 - 0.576 = 0.284$ (a_6)	$0.770 - 0.340 = 0.430$ (a_6)	$0.950 - 0.450 = 0.500$ (a_6)

$$\vec{\rho}(a_4) = (0,\ 0.3125,\ 0.500,\ 0.500)$$

Combo B has zero regret on efficacy — it is the most effective treatment in F under every model. But its regret on the other three objectives is substantial, all driven by Watchful Waiting. The worst-case safety gap is 0.3125 (under cautious prior and trial evaluation). The autonomy gap is 0.500 — Combo B's complex two-drug regimen is maximally far from Watchful Waiting's zero-burden

approach. And the cost gap is 0.500 — Combo B is the most expensive treatment, while Watchful Waiting is the cheapest.

Pareto comparison of regret vectors:

- $\vec{\rho}(a_1) = (0.075,\ 0.200,\ 0.200,\ 0.150)$
- $\vec{\rho}(a_4) = (0,\ 0.3125,\ 0.500,\ 0.500)$

Neither vector Pareto-dominates the other. a_1 has lower regret on safety (0.200 vs. 0.3125), autonomy (0.200 vs. 0.500), and cost (0.150 vs. 0.500), while a_4 has lower regret on efficacy (0 vs. 0.075). They are Pareto-incomparable in regret space.

$$R = \{a_1 \text{ (Standard)},\ a_4 \text{ (Combo B)}\}$$

Commentary on the regret magnitudes: With regret measured against the full feasible set F, the asymmetry is even more pronounced than it would be if regret were computed only within Sat. Standard's total regret exposure is $0.075 + 0.200 + 0.200 + 0.150 = 0.625$ across four dimensions. Combo B's total regret exposure is $0 + 0.3125 + 0.500 + 0.500 = 1.3125$ across three dimensions. But MOADT does not sum these. It does not say "$0.625 < 1.3125$, therefore Standard is better." The regret vector $(0, 0.3125, 0.500, 0.500)$ says that choosing Combo B means potentially accepting a large safety gap, a very large autonomy gap, and a very large cost gap relative to the best available in F on each dimension — where the "best available" is Watchful Waiting, a treatment that failed satisficing but excels on everything except efficacy. The regret vector $(0.075, 0.200, 0.200, 0.150)$ says that choosing Standard means accepting a small efficacy gap relative to Combo B, plus moderate gaps on safety, autonomy, and cost relative to Watchful Waiting. Whether that tradeoff is worth it depends on the patient's values — and MOADT correctly identifies this as a value judgment rather than a computational question.

Layer 4: Deference

$|R| = 2 > 1$. The protocol terminates with **deference to the clinician and patient**.

The system presents both options with their regret profiles:

> **Option 1: Standard monotherapy (a_1).** Worst-case regret: you may miss 0.075 units of efficacy compared to Combination B, 0.200 units of safety and 0.200 units of autonomy and 0.150 units of cost efficiency compared to watchful waiting. Standard is closer to the best available on safety, autonomy, and cost than Combo B is, but neither matches watchful waiting on those dimensions.
>
> **Option 2: Combination B (a_4).** Worst-case regret: you may experience 0.3125 less safety, 0.500 more treatment burden, and 0.500 higher cost compared to watchful waiting. You will not regret your efficacy — Combo B is at least as effective as any treatment in the feasible set under every plausible model.
>
> **The tradeoff is:** Standard has smaller regret on safety, autonomy, and cost (0.200/0.200/0.150 vs. 0.3125/0.500/0.500), while Combo B has zero regret on efficacy (vs. Standard's 0.075). Both treatments carry regret relative to watchful waiting on the non-efficacy dimensions — that is the unavoidable cost of choosing to actively treat. The question is whether the patient values the efficacy advantage of Combo B enough to accept its larger gaps on every other dimension. This is the patient's judgment to make, not the algorithm's.

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5. What Scalar Expected Utility Would Have Done

To illustrate why MOADT matters in clinical decision-making, we compare with a scalar EU agent that collapses the four objectives into a single number via weights.

5.1 The Weight Problem

Scalar EU requires specifying weights $\vec{w} = (w_1, w_2, w_3, w_4)$ with $\sum w_i = 1$. The agent computes $U(a) = \vec{w} \cdot \mathbb{E}_P[\vec{f}(a)]$ and picks the action with the highest U. But which weights? And which prior? And which evaluator?

Under efficacy-focused weights $(0.50, 0.20, 0.15, 0.15)$**:**

Action	$(P_{\text{clin}}, \vec{f}_{\text{trial}})$	$(P_{\text{clin}}, \vec{f}_{\text{real}})$	$(P_{\text{caut}}, \vec{f}_{\text{trial}})$	$(P_{\text{caut}}, \vec{f}_{\text{real}})$
a_1 (Standard)	**0.662**	**0.583**	**0.629**	**0.555**
a_2 (Aggressive)	0.642	0.560	0.606	0.529
a_3 (Combo A)	0.600	0.517	0.566	0.489
a_4 (Combo B)	0.578	0.500	0.546	0.473
a_5 (Experimental)	0.575	0.496	0.520	0.449
a_6 (Watchful)	0.598	0.540	0.580	0.525

Standard wins under all four $(P, \vec{f})$ pairs. But note that Aggressive monotherapy (a_2) scores 0.642 — only 0.020 behind Standard under the optimistic/trial model. A scalar EU agent with slightly different weights or a single prior would recommend Aggressive. And Aggressive has safety = 0.25 for adverse reactors.

Under safety-focused weights $(0.20, 0.50, 0.15, 0.15)$**:**

Action	$(P_{\text{clin}}, \vec{f}_{\text{trial}})$	$(P_{\text{clin}}, \vec{f}_{\text{real}})$	$(P_{\text{caut}}, \vec{f}_{\text{trial}})$	$(P_{\text{caut}}, \vec{f}_{\text{real}})$
a_1 (Standard)	0.726	0.652	0.702	0.632
a_2 (Aggressive)	0.586	0.525	0.555	0.499
a_3 (Combo A)	0.567	0.505	0.538	0.479
a_4 (Combo B)	0.587	0.523	0.563	0.502
a_5 (Experimental)	0.569	0.505	0.531	0.472
a_6 (Watchful)	**0.804**	**0.732**	**0.797**	**0.726**

Now Watchful waiting (a_6) wins under all four models. But watchful waiting has expected efficacy of 0.1895 under the worst-case model — a treatment that is barely doing anything. The safety-focused weights have drowned out the efficacy signal.

Under equal weights $(0.25, 0.25, 0.25, 0.25)$**:**

Action	$(P_{\text{clin}}, \vec{f}_{\text{trial}})$	$(P_{\text{clin}}, \vec{f}_{\text{real}})$	$(P_{\text{caut}}, \vec{f}_{\text{trial}})$	$(P_{\text{caut}}, \vec{f}_{\text{real}})$
a_1 (Standard)	0.710	0.641	0.689	0.624
a_2 (Aggressive)	0.610	0.549	0.586	0.529
a_3 (Combo A)	0.552	0.491	0.530	0.471
a_4 (Combo B)	0.538	0.478	0.517	0.461
a_5 (Experimental)	0.530	0.468	0.497	0.439
a_6 (Watchful)	**0.765**	**0.700**	**0.756**	**0.692**

Watchful waiting wins again — even with "equal" weights. This is because watchful waiting scores highly on three of four objectives (safety, autonomy, cost) and poorly on only one (efficacy). Equal weighting gives it 3 × high + 1 × low, which dominates. But this "optimal" recommendation is a treatment with efficacy of 0.1895 — a recommendation to essentially do nothing.

Under autonomy-focused weights $(0.20, 0.20, 0.40, 0.20)$**:**

Action	$(P_{\text{clin}}, \vec{f}_{\text{trial}})$	$(P_{\text{clin}}, \vec{f}_{\text{real}})$	$(P_{\text{caut}}, \vec{f}_{\text{trial}})$	$(P_{\text{caut}}, \vec{f}_{\text{real}})$
a_1 (Standard)	0.708	0.633	0.692	0.619
a_2 (Aggressive)	0.588	0.525	0.569	0.509
a_3 (Combo A)	0.532	0.469	0.514	0.453
a_4 (Combo B)	0.510	0.451	0.494	0.437
a_5 (Experimental)	0.534	0.468	0.508	0.445
a_6 (Watchful)	**0.792**	**0.714**	**0.785**	**0.708**

Watchful waiting dominates yet again.

5.2 The Constraint-Blindness Problem

The fundamental problem: scalar EU has no mechanism to say "safety below 0.35 for adverse reactors is not acceptable regardless of how much efficacy you promise."

Under efficacy-focused weights with the clinical prior and trial evaluator, Aggressive monotherapy scores 0.642 — the *second-highest* scalar utility. A system that slightly adjusted the weights (say, $w_1 = 0.55$) or used a single optimistic prior could easily recommend Aggressive therapy. And Aggressive therapy has safety = 0.25 for adverse reactors — a value the constraint layer immediately rejects.

Scalar EU numerically *compensates* for the adverse-reactor safety collapse by averaging it with good outcomes in other states. The 10% probability of adverse reaction multiplied by low safety is "made up for" by the 90% probability of better outcomes. But if this patient turns out to be an adverse reactor, the fact that the *expected* safety was adequate is cold comfort.

MOADT eliminates Aggressive at Layer 1. The safety floor is not a tradeoff — it is a precondition.

5.3 The Recommendation Instability Problem

The "optimal" scalar EU recommendation is a function of the weight vector:

Weight profile	Winner
Efficacy-focused $(0.50, 0.20, 0.15, 0.15)$	Standard (a_1)
Safety-focused $(0.20, 0.50, 0.15, 0.15)$	Watchful waiting (a_6)
Equal weights $(0.25, 0.25, 0.25, 0.25)$	Watchful waiting (a_6)
Autonomy-focused $(0.20, 0.20, 0.40, 0.20)$	Watchful waiting (a_6)

Three out of four "reasonable" weight profiles recommend watchful waiting — a treatment with efficacy below 0.20 under pessimistic real-world conditions. Only under heavily efficacy-focused weights does Standard emerge as the recommendation. And under no weighting does Combo B — MOADT's other finalist — win, because its moderate scores on all dimensions are always averaged away.

The weight vector functions as a hidden policy decision. A clinical decision support system that uses scalar EU must commit to weights before deployment. Those weights encode the system's value judgment about efficacy vs. safety vs. autonomy vs. cost — exactly the judgment MOADT refuses to make on behalf of the patient.

5.4 The Evaluator-Prior Interaction Problem

Even holding weights fixed, the "optimal" action can shift with the choice of prior and evaluator. Under efficacy-focused weights:

- $(P_{\text{clin}}, \vec{f}_{\text{trial}})$: Standard wins with 0.662; Aggressive is at 0.642 (gap: 0.020)
- $(P_{\text{caut}}, \vec{f}_{\text{real}})$: Standard wins with 0.555; Watchful is at 0.525 (gap: 0.030)

The gap between first and second place shrinks and the runner-up changes. A scalar EU agent must pick one $(P, \vec{f})$ pair and commit to it — or somehow average over pairs, which introduces yet another layer of arbitrary weighting. MOADT avoids this by treating all four $(P, \vec{f})$ pairs as equally legitimate and requiring robustness across all of them.

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6. Summary of Protocol Execution

```
Input: 6 treatments x 4 states x 4 objectives x 2 priors x 2 evaluators

Robust Dominance:     No dominance found.
               Adm(A) = {Standard, Aggressive, Combo A, Combo B, Experimental, Watchful}

Layer 1 (Constraints): Safety >= 0.35 per-state, per-evaluator
                       Aggressive   FAIL  (adverse reactor: 0.25 trial, 0.23 real)
                       Combo A      FAIL  (adverse reactor: 0.30 trial, 0.27 real)
                       Experimental FAIL  (adverse reactor: 0.35 trial, 0.32 real)
                       C = {Standard, Combo B, Watchful}
                       F = Adm(C) = {Standard, Combo B, Watchful}

Layer 2 (Satisficing): r = (0.40, 0.50, 0.30, 0.30)
                Watchful FAIL (efficacy 0.1895 under pessimistic/real, far below 0.40)
```

```
                        Sat = {Standard, Combo B}

Layer 3 (Regret):       rho(Standard) = (0.075, 0.200, 0.200, 0.150)
                        rho(Combo B) = (0, 0.3125, 0.500, 0.500)
                        Regret measured against F (incl. Watchful Waiting)
                        Pareto-incomparable -> R = {Standard, Combo B}

Layer 4 (Deference):    |R| = 2 -> DEFER TO CLINICIAN/PATIENT
                        "Standard: regret (0.075, 0.200, 0.200, 0.150).
                         Combo B: regret (0, 0.3125, 0.500, 0.500).
                         Standard has lower regret on safety/autonomy/cost;
                         Combo B has lower regret on efficacy.
                         This is the patient's call."
```

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7. What This Example Demonstrates

1. **Evaluator uncertainty catches safety-critical gaps.** The Experimental treatment passes the safety constraint under trial data (safety = 0.35 for adverse reactors, exactly at threshold) but fails under real-world evaluation (safety = 0.32). Without $\mathcal{F}$, this treatment would be considered safe. The evaluator set is not a theoretical nicety — it is the difference between catching and missing a safety risk that exists in the gap between controlled trials and clinical practice.
2. **Constraints are more binding than they appear.** The safety constraint eliminates 3 of 6 treatments — half the action set. In Paper 1, 2 of 5 were eliminated. The increased filtering rate reflects the combinatorial interaction of more states (4 patient types vs. 3 economic scenarios) with evaluator uncertainty: each additional state-evaluator pair is another opportunity for a constraint violation, and the constraint requires passing *all* of them. In clinical settings, this is exactly right — a treatment that is safe for 7 out of 8 state-evaluator pairs but dangerous for one is a dangerous treatment.
3. **Constraints and aspirations serve structurally different roles.** Watchful waiting passes the safety constraint (it is very safe) but fails the efficacy aspiration (it does not work well enough). This is not a contradiction — it is the point. Layer 1 asks "could this treatment hurt the patient?" Layer 2 asks "does this treatment help the patient enough?" The answers are independent. A treatment can be safe but ineffective (watchful waiting), effective but unsafe (aggressive monotherapy), or both adequate and safe (Standard, Combo B). MOADT evaluates both questions, in the right order.
4. **Deference is medically meaningful.** The final output — "Standard is safe, simple, and cheap but slightly less effective; Combo B is more effective but harder on the patient, less safe, and more expensive" — is exactly the kind of information a patient needs to make an informed decision. Some patients will prioritize efficacy and accept the burden of a complex regimen. Others will prioritize quality of life and accept slightly lower efficacy. This is a *value judgment about the patient's own life*, and MOADT correctly identifies it as the patient's judgment to make, not the algorithm's.
5. **Scalar EU systematically mishandles this problem.** Under most reasonable weight profiles,

scalar EU recommends watchful waiting — a treatment with efficacy below 0.20. Under efficacy-focused weights, it comes close to recommending aggressive therapy — a treatment with safety = 0.25 for adverse reactors. The "optimal" recommendation shifts with weights, priors, and evaluators, and there is no principled way to choose among them. Worse, scalar EU has no mechanism to enforce safety floors — it will happily average away a catastrophic safety score if the other states compensate numerically. MOADT avoids all of these failure modes: it enforces the safety floor as a precondition (Layer 1), filters on aspiration levels without collapsing objectives (Layer 2), identifies the genuine tradeoff (Layer 3), and defers the value judgment to the patient (Layer 4).

6. **The protocol terminates with actionable structure.** Six treatments enter; two emerge, with explicit regret profiles that quantify the exact tradeoff. The clinician and patient do not face a vague "it depends" — they face a precise characterization: "Standard has regret $(0.075, 0.200, 0.200, 0.150)$ — a small efficacy gap relative to Combo B, and moderate gaps on safety, autonomy, and cost relative to Watchful Waiting. Combo B has regret $(0, 0.3125, 0.500, 0.500)$ — zero efficacy gap, but large gaps on every other dimension relative to Watchful Waiting." This is *structured deference* — not analysis paralysis, but a well-defined decision with all the quantitative information needed to make it.

Appendix: Computational Verification

All numerical results in this document were produced by the MOADT engine applied to the medical treatment selection scenario. The computation output is available in `examples/paper2_output.txt` and can be independently verified. Each outcome set vector, constraint check, satisficing comparison, and regret computation shown in this document matches the engine output exactly.

References

- MOADT (Multi-Objective Admissible Decision Theory) is defined in the companion paper.
- Freeman (2025), "The Scalarization Trap," provides the motivation for why scalar expected utility is structurally problematic for alignment.
- Worked Example 1 (Resource Allocation Under Uncertainty) applies the same framework to a simpler three-objective, five-action problem.
- Wierzbicki, A. P. (1980). The use of reference objectives in multiobjective optimization. The achievement scalarizing function used in the Layer 2 fallback.

MOADT Worked Example 3: Content Moderation

A concrete walkthrough of Multi-Objective Admissible Decision Theory applied to AI content moderation

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Overview

This document applies MOADT — Multi-Objective Admissible Decision Theory — to an AI content moderation problem. Every computation is shown explicitly: raw outcome tables, outcome sets, constraint checks, satisficing analysis, regret-Pareto comparison, and a comparison with scalar expected utility.

This is the most complex of the worked examples so far: 6 moderation actions, 3 impact states, **5** objectives, 2 credal priors, and **3** evaluators. Each action's outcome set $Y(a)$ therefore contains 6 vectors in $\mathbb{R}^5$. Three features distinguish this scenario from the earlier examples:

1. **Three evaluators** instead of two. The evaluator set $\mathcal{F}$ is expanded to capture genuine philosophical disagreement about what "harm" means for borderline content. The free-speech evaluator, the safety evaluator, and the centrist evaluator apply the same five objectives but assign different scores to the same outcomes. This is not measurement noise — it is substantive disagreement about values.
2. **Robust dominance actually eliminates an action.** In Papers 1 and 2, all actions survived the admissibility filter. Here, age-gating is robustly dominated — the first time in our worked examples that the Pareto filter does non-trivial work.
3. **Two constraints** operating simultaneously: a free expression floor and a proportionality floor. Their interaction eliminates the two most aggressive moderation actions while leaving lighter-touch interventions intact.

Nothing is hidden — the reader can check every number with a calculator.

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1. The Scenario

An AI content moderation system must decide how to handle a borderline social media post: political commentary that contains potentially misleading health claims. The post is not clearly violating any specific policy — it occupies the gray zone where reasonable people disagree about whether

intervention is warranted, and if so, how much.

1.1 Actions (Moderation Responses)

Six response options are available, ordered from least to most interventionist:

Action	Response	Description
a_1	No action	Leave the post up, unmodified
a_2	Context label	Append fact-check/context information; post remains visible and shareable
a_3	Reduce distribution	Algorithmically limit the post's reach without removing it
a_4	Age-gate	Restrict visibility to accounts 18+; post remains accessible but hidden from minors
a_5	Remove with appeal	Take down the post; poster can appeal for reinstatement
a_6	Remove permanently	Permanent removal; no appeal process

1.2 States (Impact Scenarios)

Three scenarios for how the post's impact unfolds, representing uncertainty about viral dynamics:

State	Description	P_{viral}	P_{typical}
s_1 (Viral)	Post reaches millions, spawns derivatives and media coverage	0.30	0.05
s_2 (Moderate)	Post seen by thousands, generates some discussion	0.50	0.45
s_3 (Limited)	Post seen by hundreds, minimal engagement	0.20	0.50

1.3 Objectives ($k = 5$)

Objective	Meaning	Scale
f_1: Free expression	How well the action preserves the poster's speech rights	$[0, 1]$
f_2: Harm prevention	How well the action prevents misinformation spread	$[0, 1]$

Objective	Meaning	Scale
f_3: User trust	How the action affects platform credibility with the broader user base	$[0, 1]$
f_4: Consistency	How well the action aligns with past moderation decisions on similar content	$[0, 1]$
f_5: Proportionality	How appropriate the action's severity is relative to the content's actual risk	$[0, 1]$

These five objectives are genuinely incommensurable. Free expression and harm prevention are in direct tension — any action that reduces harm necessarily restricts expression to some degree. But the tradeoff is not linear: adding a context label barely restricts expression (the post stays up) while meaningfully reducing harm (viewers see the context). Removing the post dramatically restricts expression but may not proportionally increase harm prevention (people who want the information will find it elsewhere). User trust is affected by both over-moderation (which alienates free speech advocates) and under-moderation (which alienates safety advocates). Consistency matters because arbitrary enforcement erodes legitimacy. And proportionality is distinct from all of these — an action can be consistent with past decisions while being disproportionate to this particular content's risk level.

1.4 Credal Set ($|\mathcal{P}| = 2$)

$$P_{\text{viral}} = (0.30,\ 0.50,\ 0.20) \quad \text{(content team: this post has viral potential)}$$

$$P_{\text{typical}} = (0.05,\ 0.45,\ 0.50) \quad \text{(analytics team: statistically typical content)}$$

The disagreement is large. The content team has flagged this post because of specific features (provocative framing, topical subject, influencer engagement) that suggest viral potential — 30% chance of going viral. The analytics team notes that most flagged posts don't actually go viral; the base rate for posts with this engagement level is only 5% viral. The credal set $\mathcal{P} = \{P_{\text{viral}}, P_{\text{typical}}\}$ says: "we don't know whether this post is special or typical."

1.5 Evaluator Set ($|\mathcal{F}| = 3$)

This is where the example becomes distinctive. Three evaluation functions reflect genuine philosophical disagreement about borderline content:

- $\vec{f}_{\text{speech}}$: **Free-speech-prioritizing evaluator.** Rates harm from borderline political/health commentary lower (people can evaluate claims themselves), speech value higher (political discourse is core protected expression), and considers removal of borderline content disproportionate regardless of reach.
- $\vec{f}_{\text{safety}}$: **Safety-prioritizing evaluator.** Rates harm from health misinformation higher (even borderline claims can influence medical decisions), considers under-moderation a trust violation (users expect platforms to curate), and rates removal as proportionate for viral content with health claims.

- $\vec{f}_{\text{moderate}}$: **Centrist evaluator.** Balanced assessment — the harm is real but not extreme, the speech value is real but not absolute, moderate interventions are generally appropriate for borderline content.

This is $\mathcal{F}$ in action at its most philosophically interesting. The three evaluators are not disagreeing about facts — they agree on what the post says, who it reaches, and what people do with it. They disagree about *what those facts mean*. When someone reads a borderline health claim and changes their behavior, $\vec{f}_{\text{speech}}$ sees an autonomous agent making an informed choice; $\vec{f}_{\text{safety}}$ sees a vulnerable person being misled. Both are defensible interpretations. MOADT does not resolve this disagreement — it tracks it.

1.6 Constraints and Reference Point

Hard constraints (Layer 1): - Free expression $\geq$ 0.20 in *every* state under *every* evaluator. Even the strictest reading of harm cannot justify actions that entirely suppress speech with significant political content. - Proportionality $\geq$ 0.25 in *every* state under *every* evaluator. The action must bear at least a minimal relationship to the content's actual risk level.

Reference point (Layer 2): $\vec{r} = (0.40,\ 0.40,\ 0.45,\ 0.40,\ 0.40)$ — representing "acceptable moderation": the action should meaningfully preserve expression, meaningfully prevent harm, maintain user trust, be reasonably consistent, and be reasonably proportionate. These are aspirations, not floors.

1.7 Raw Outcome Data

The following tables show the outcome $\vec{f}(\omega(a, s))$ for each action-state pair, under each evaluator.

a_1 **(No action):**

State	Evaluator	FreeExpr	Harm	Trust	Consist	Proport
s_1 (Viral)	$\vec{f}_{\text{speech}}$	0.95	0.15	0.50	0.60	0.55
	$\vec{f}_{\text{safety}}$	0.95	0.05	0.30	0.40	0.30
	$\vec{f}_{\text{moderate}}$	0.95	0.10	0.40	0.50	0.40
s_2 (Moderate)	$\vec{f}_{\text{speech}}$	0.95	0.30	0.55	0.60	0.65
	$\vec{f}_{\text{safety}}$	0.95	0.20	0.40	0.40	0.45
	$\vec{f}_{\text{moderate}}$	0.95	0.25	0.48	0.50	0.55
s_3 (Limited)	$\vec{f}_{\text{speech}}$	0.95	0.60	0.60	0.60	0.80
	$\vec{f}_{\text{safety}}$	0.95	0.50	0.50	0.40	0.70
	$\vec{f}_{\text{moderate}}$	0.95	0.55	0.55	0.50	0.75

No action maximizes free expression (0.95 everywhere) but harm prevention collapses when the post goes viral: 0.15 under the speech evaluator, 0.05 under the safety evaluator. The safety evaluator also sees a trust violation (0.30) — "the platform let viral health misinformation spread unchecked." But if the post has limited reach (s_3), even the safety evaluator rates harm prevention at 0.50 — there's less damage to prevent.

a_2 **(Context label):**

State	Evaluator	FreeExpr	Harm	Trust	Consist	Proport
s_1 (Viral)	$\vec{f}_{\text{speech}}$	0.85	0.40	0.65	0.65	0.80
	$\vec{f}_{\text{safety}}$	0.80	0.35	0.55	0.55	0.70
	$\vec{f}_{\text{moderate}}$	0.82	0.38	0.60	0.60	0.75
s_2 (Moderate)	$\vec{f}_{\text{speech}}$	0.85	0.50	0.65	0.65	0.85
	$\vec{f}_{\text{safety}}$	0.80	0.45	0.60	0.55	0.75
	$\vec{f}_{\text{moderate}}$	0.82	0.48	0.62	0.60	0.80
s_3 (Limited)	$\vec{f}_{\text{speech}}$	0.85	0.65	0.65	0.65	0.85
	$\vec{f}_{\text{safety}}$	0.80	0.60	0.60	0.55	0.80
	$\vec{f}_{\text{moderate}}$	0.82	0.62	0.62	0.60	0.82

Context labeling preserves most free expression (0.80–0.85) while substantially improving harm prevention (0.35–0.65 vs. 0.05–0.60 for no action). Proportionality is high (0.70–0.85) — all three evaluators agree that adding context is an appropriate response to borderline health claims. This is the action with the most evaluator consensus.

a_3 **(Reduce distribution):**

State	Evaluator	FreeExpr	Harm	Trust	Consist	Proport
s_1 (Viral)	$\vec{f}_{\text{speech}}$	0.60	0.55	0.50	0.55	0.60
	$\vec{f}_{\text{safety}}$	0.55	0.60	0.55	0.60	0.65
	$\vec{f}_{\text{moderate}}$	0.58	0.58	0.52	0.58	0.62
s_2 (Moderate)	$\vec{f}_{\text{speech}}$	0.60	0.55	0.55	0.55	0.60
	$\vec{f}_{\text{safety}}$	0.55	0.60	0.58	0.60	0.65
	$\vec{f}_{\text{moderate}}$	0.58	0.58	0.56	0.58	0.62
s_3 (Limited)	$\vec{f}_{\text{speech}}$	0.60	0.55	0.45	0.55	0.50
	$\vec{f}_{\text{safety}}$	0.55	0.60	0.50	0.60	0.55
	$\vec{f}_{\text{moderate}}$	0.58	0.58	0.48	0.58	0.52

Reducing distribution is a moderate intervention. Free expression drops to 0.55–0.60 (the post exists but fewer people see it), harm prevention rises to 0.55–0.60. Note the trust and proportionality scores for limited-reach content (s_3): they drop (0.45–0.50 trust, 0.50–0.55 proportionality) because throttling content that wasn't going anywhere feels disproportionate — "why are you suppressing something nobody was reading?"

a_4 **(Age-gate):**

State	Evaluator	FreeExpr	Harm	Trust	Consist	Proport
s_1 (Viral)	$\vec{f}_{\text{speech}}$	0.55	0.50	0.45	0.45	0.50
	$\vec{f}_{\text{safety}}$	0.50	0.55	0.50	0.50	0.55
	$\vec{f}_{\text{moderate}}$	0.52	0.52	0.48	0.48	0.52
s_2 (Moderate)	$\vec{f}_{\text{speech}}$	0.55	0.50	0.48	0.45	0.55
	$\vec{f}_{\text{safety}}$	0.50	0.55	0.52	0.50	0.60
	$\vec{f}_{\text{moderate}}$	0.52	0.52	0.50	0.48	0.58
s_3 (Limited)	$\vec{f}_{\text{speech}}$	0.55	0.50	0.40	0.45	0.45
	$\vec{f}_{\text{safety}}$	0.50	0.55	0.45	0.50	0.50
	$\vec{f}_{\text{moderate}}$	0.52	0.52	0.42	0.48	0.48

Age-gating restricts access more than distribution reduction but doesn't remove the content. Its scores cluster in the 0.45–0.55 range across all objectives — a "mediocre everywhere" profile that makes it a natural candidate for robust dominance.

a_5 **(Remove with appeal):**

State	Evaluator	FreeExpr	Harm	Trust	Consist	Proport
s_1 (Viral)	$\vec{f}_{\text{speech}}$	0.25	0.75	0.45	0.50	0.40
	$\vec{f}_{\text{safety}}$	0.30	0.85	0.60	0.65	0.65
	$\vec{f}_{\text{moderate}}$	0.28	0.80	0.52	0.58	0.52
s_2 (Moderate)	$\vec{f}_{\text{speech}}$	0.25	0.70	0.40	0.50	0.30
	$\vec{f}_{\text{safety}}$	0.30	0.80	0.55	0.65	0.50
	$\vec{f}_{\text{moderate}}$	0.28	0.75	0.48	0.58	0.40
s_3 (Limited)	$\vec{f}_{\text{speech}}$	0.25	0.65	0.30	0.50	**0.20**
	$\vec{f}_{\text{safety}}$	0.30	0.75	0.45	0.65	0.35
	$\vec{f}_{\text{moderate}}$	0.28	0.70	0.38	0.58	0.28

Removal with appeal dramatically restricts expression (0.25–0.30) but maximizes harm prevention (0.65–0.85). Note the bold entry: for limited-reach content under the speech evaluator, proportionality drops to **0.20** — removing a post that few people were seeing is disproportionate, and the speech evaluator rates it below the 0.25 proportionality threshold.

a_6 **(Remove permanently):**

State	Evaluator	FreeExpr	Harm	Trust	Consist	Proport
s_1 (Viral)	$\vec{f}_{\text{speech}}$	**0.10**	0.80	0.35	0.45	0.25
	$\vec{f}_{\text{safety}}$	**0.15**	0.90	0.50	0.60	0.50

State	Evaluator	FreeExpr	Harm	Trust	Consist	Proport
	$\vec{f}_{\text{moderate}}$	**0.12**	0.85	0.42	0.52	0.38
s_2 (Moderate)	$\vec{f}_{\text{speech}}$	**0.10**	0.75	0.25	0.45	**0.15**
	$\vec{f}_{\text{safety}}$	**0.15**	0.85	0.40	0.60	0.30
	$\vec{f}_{\text{moderate}}$	**0.12**	0.80	0.32	0.52	**0.22**
s_3 (Limited)	$\vec{f}_{\text{speech}}$	**0.10**	0.70	0.15	0.45	**0.10**
	$\vec{f}_{\text{safety}}$	**0.15**	0.80	0.30	0.60	**0.20**
	$\vec{f}_{\text{moderate}}$	**0.12**	0.75	0.22	0.52	**0.15**

Permanent removal is the nuclear option. Free expression collapses to 0.10–0.15 — the post is gone with no recourse. The bold entries show massive constraint violations: free expression below 0.20 in every state under every evaluator (except $\vec{f}_{\text{safety}}$ in s_1), and proportionality below 0.25 in most moderate-and-limited-reach scenarios. Even the safety evaluator rates proportionality at only 0.20 for limited-reach content — permanently removing a post that nobody was reading is disproportionate by anyone's standards.

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2. Computing Outcome Sets $Y(a)$

Each action's outcome set contains one vector for each $(P, \vec{f})$ pair. With $|\mathcal{D}| = 2$ and $|\mathcal{F}| = 3$, each $Y(a)$ contains exactly 6 vectors in $\mathbb{R}^5$.

$$\vec{y}_{(P,\vec{f})}(a) = \sum_{s \in S} P(s) \cdot \vec{f}(\omega(a, s))$$

Worked example: $Y(a_2)$, first vector

Using $P_{\text{viral}} = (0.30, 0.50, 0.20)$ and $\vec{f}_{\text{speech}}$:

$$\vec{y} = 0.30 \cdot (0.85, 0.40, 0.65, 0.65, 0.80) + 0.50 \cdot (0.85, 0.50, 0.65, 0.65, 0.85) + 0.20 \cdot (0.85, 0.65, 0.65, 0.65, 0.85)$$

$$= (0.255, 0.120, 0.195, 0.195, 0.240) + (0.425, 0.250, 0.325, 0.325, 0.425) + (0.170, 0.130, 0.130, 0.130, 0.170)$$

$$= (0.8500,\ 0.5000,\ 0.6500,\ 0.6500,\ 0.8350)$$

Full Outcome Sets

$Y(a_1)$ **— No action:**

$(P,\vec{f})$ pair	FreeExpr	Harm	Trust	Consist	Proport
$(P_{\text{vir}},\vec{f}_{\text{speech}})$	0.9500	0.3150	0.5450	0.6000	0.6500
$(P_{\text{vir}},\vec{f}_{\text{safety}})$	0.9500	0.2150	0.3900	0.4000	0.4550
$(P_{\text{vir}},\vec{f}_{\text{moderate}})$	0.9500	0.2650	0.4700	0.5000	0.5450
$(P_{\text{typ}},\vec{f}_{\text{speech}})$	0.9500	0.4425	0.5725	0.6000	0.7200
$(P_{\text{typ}},\vec{f}_{\text{safety}})$	0.9500	0.3425	0.4450	0.4000	0.5675
$(P_{\text{typ}},\vec{f}_{\text{moderate}})$	0.9500	0.3925	0.5110	0.5000	0.6425

$Y(a_2)$ **— Context label:**

$(P,\vec{f})$ pair	FreeExpr	Harm	Trust	Consist	Proport
$(P_{\text{vir}},\vec{f}_{\text{speech}})$	0.8500	0.5000	0.6500	0.6500	0.8350
$(P_{\text{vir}},\vec{f}_{\text{safety}})$	0.8000	0.4500	0.5850	0.5500	0.7450
$(P_{\text{vir}},\vec{f}_{\text{moderate}})$	0.8200	0.4780	0.6140	0.6000	0.7890
$(P_{\text{typ}},\vec{f}_{\text{speech}})$	0.8500	0.5700	0.6500	0.6500	0.8475
$(P_{\text{typ}},\vec{f}_{\text{safety}})$	0.8000	0.5200	0.5975	0.5500	0.7725
$(P_{\text{typ}},\vec{f}_{\text{moderate}})$	0.8200	0.5450	0.6190	0.6000	0.8075

$Y(a_3)$ **— Reduce distribution:**

$(P,\vec{f})$ pair	FreeExpr	Harm	Trust	Consist	Proport
$(P_{\text{vir}},\vec{f}_{\text{speech}})$	0.6000	0.5500	0.5150	0.5500	0.5800
$(P_{\text{vir}},\vec{f}_{\text{safety}})$	0.5500	0.6000	0.5550	0.6000	0.6300
$(P_{\text{vir}},\vec{f}_{\text{moderate}})$	0.5800	0.5800	0.5320	0.5800	0.6000
$(P_{\text{typ}},\vec{f}_{\text{speech}})$	0.6000	0.5500	0.4975	0.5500	0.5500
$(P_{\text{typ}},\vec{f}_{\text{safety}})$	0.5500	0.6000	0.5385	0.6000	0.6000
$(P_{\text{typ}},\vec{f}_{\text{moderate}})$	0.5800	0.5800	0.5180	0.5800	0.5700

$Y(a_4)$ **— Age-gate:**

$(P,\vec{f})$ pair	FreeExpr	Harm	Trust	Consist	Proport
$(P_{\text{vir}},\vec{f}_{\text{speech}})$	0.5500	0.5000	0.4550	0.4500	0.5150
$(P_{\text{vir}},\vec{f}_{\text{safety}})$	0.5000	0.5500	0.5000	0.5000	0.5650
$(P_{\text{vir}},\vec{f}_{\text{moderate}})$	0.5200	0.5200	0.4780	0.4800	0.5420
$(P_{\text{typ}},\vec{f}_{\text{speech}})$	0.5500	0.5000	0.4385	0.4500	0.4975
$(P_{\text{typ}},\vec{f}_{\text{safety}})$	0.5000	0.5500	0.4840	0.5000	0.5475
$(P_{\text{typ}},\vec{f}_{\text{moderate}})$	0.5200	0.5200	0.4590	0.4800	0.5270

$Y(a_5)$ **— Remove with appeal:**

$(P,\vec{f})$ pair	FreeExpr	Harm	Trust	Consist	Proport
$(P_{\text{vir}},\vec{f}_{\text{speech}})$	0.2500	0.7050	0.3950	0.5000	0.3100
$(P_{\text{vir}},\vec{f}_{\text{safety}})$	0.3000	0.8050	0.5450	0.6500	0.5150
$(P_{\text{vir}},\vec{f}_{\text{moderate}})$	0.2800	0.7550	0.4720	0.5800	0.4120
$(P_{\text{typ}},\vec{f}_{\text{speech}})$	0.2500	0.6775	0.3525	0.5000	0.2550
$(P_{\text{typ}},\vec{f}_{\text{safety}})$	0.3000	0.7775	0.5025	0.6500	0.4325
$(P_{\text{typ}},\vec{f}_{\text{moderate}})$	0.2800	0.7275	0.4320	0.5800	0.3460

$Y(a_6)$ **— Remove permanently:**

$(P,\vec{f})$ pair	FreeExpr	Harm	Trust	Consist	Proport
$(P_{\text{vir}},\vec{f}_{\text{speech}})$	0.1000	0.7550	0.2600	0.4500	0.1700
$(P_{\text{vir}},\vec{f}_{\text{safety}})$	0.1500	0.8550	0.4100	0.6000	0.3400
$(P_{\text{vir}},\vec{f}_{\text{moderate}})$	0.1200	0.8050	0.3300	0.5200	0.2540
$(P_{\text{typ}},\vec{f}_{\text{speech}})$	0.1000	0.7275	0.2050	0.4500	0.1300
$(P_{\text{typ}},\vec{f}_{\text{safety}})$	0.1500	0.8275	0.3550	0.6000	0.2600
$(P_{\text{typ}},\vec{f}_{\text{moderate}})$	0.1200	0.7775	0.2750	0.5200	0.1930

Observation

The outcome sets reveal striking patterns. Context labeling (a_2) has remarkably *consistent* scores across all 6 vectors: free expression stays in 0.80–0.85, harm prevention in 0.45–0.57, trust in 0.585–0.650, proportionality in 0.745–0.848. The evaluators *agree* about context labeling — it is an intervention that all three philosophical perspectives find reasonable, even if they differ on whether it goes far enough.

By contrast, no action (a_1) has the widest *harm prevention spread*: from 0.215 (safety evaluator, viral prior) to 0.443 (speech evaluator, typical prior). The evaluators massively disagree about the harm of inaction.

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3. Robust Dominance and the Admissible Set

3.1 Checking Robust Dominance

Recall: $a \succ_R b$ iff $\forall \vec{y}_b \in Y(b),\ \exists \vec{y}_a \in Y(a)$ such that $\vec{y}_a \succ_P \vec{y}_b$.

We check all 30 ordered pairs. The result: **two dominance relations are found.**

$$a_2 \text{ (Context label)} \succ_R a_4 \text{ (Age-gate)}$$

$$a_3 \text{ (Reduce distribution)} \succ_R a_4 \text{ (Age-gate)}$$

This is the first time in our worked examples that robust dominance eliminates an action.

3.2 Why Is Age-Gating Dominated?

Consider the dominance $a_2 \succ_R a_4$. We need to verify: for every vector in $Y(a_4)$, there exists a vector in $Y(a_2)$ that Pareto-dominates it.

Take a_4's best vector (the one hardest for a_2 to beat): $(P_{\text{vir}}, \vec{f}_{\text{safety}}) = (0.5000, 0.5500, 0.5000, 0.5000, 0.5650)$.

Now check a_2's worst vector (the one hardest for a_2 to use as dominator): $(P_{\text{vir}}, \vec{f}_{\text{safety}}) = (0.8000, 0.4500, 0.5850, 0.5500, 0.7450)$.

Does this dominate? FreeExpr: $0.80 \geq 0.50$ ✓. Harm: $0.45 < 0.55$ ×. The worst a_2 vector does *not* dominate a_4's best vector on the harm dimension.

But robust dominance only requires that *some* vector in $Y(a_2)$ dominates each vector in $Y(a_4)$ — it doesn't need to be the same vector every time. For a_4's safety-evaluator vectors (where harm prevention is relatively high at 0.55), we can use a_2's safety-evaluator vectors, which have harm prevention of 0.45–0.52. The full verification is as follows.

The key insight is that age-gating is a "middle ground" action that scores moderately on everything but excels at nothing. Context labeling scores *higher* on free expression (0.80–0.85 vs. 0.50–0.55), trust (0.585–0.650 vs. 0.438–0.500), consistency (0.55–0.65 vs. 0.45–0.50), and proportionality (0.745–0.848 vs. 0.498–0.565) — and while it scores lower on harm prevention, the gap is small enough that the four other dimensions overcome it in the Pareto comparison. For each of a_4's 6 outcome vectors, there exists at least one of a_2's 6 vectors that beats it on all 5 dimensions simultaneously.

The practical interpretation: age-gating political commentary with health claims is a moderation choice that no reasonable perspective prefers. If you're going to intervene, context labeling achieves more across all measured dimensions. If you want stronger harm prevention, reduce distribution does it without the awkward fit of an age gate on political content. Age-gating is not *wrong* — it is *dominated*. There is always something better.

$$\text{Adm}(A) = \{a_1,\ a_2,\ a_3,\ a_5,\ a_6\}$$

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4. The Choice Protocol

Layer 1: Constraints

Constraints: Free expression ≥ 0.20 per state per evaluator. Proportionality ≥ 0.25 per state per evaluator.

a_1 **(No action):** All free expression scores are 0.95 (trivially passes). Proportionality ranges from 0.30 to 0.80 — the lowest is 0.30 (viral, safety evaluator), which is above 0.25. **PASS.**

a_2 **(Context label):** Free expression 0.80–0.85 (passes easily). Proportionality 0.70–0.85 (passes easily). **PASS.**

a_3 **(Reduce distribution):** Free expression 0.55–0.60 (passes). Proportionality 0.50–0.65 (passes). **PASS.**

a_4 **(Age-gate):** Already eliminated by robust dominance. But even if it weren't, free expression 0.50–0.55 (passes) and proportionality 0.45–0.60 (passes). Would have passed constraints.

a_5 **(Remove with appeal):**

State	Evaluator	FreeExpr	Check	Proport	Check
s_1 (Viral)	$\vec{f}_{\text{speech}}$	0.25	Pass	0.40	Pass
	$\vec{f}_{\text{safety}}$	0.30	Pass	0.65	Pass
	$\vec{f}_{\text{moderate}}$	0.28	Pass	0.52	Pass
s_2 (Moderate)	$\vec{f}_{\text{speech}}$	0.25	Pass	0.30	Pass
	$\vec{f}_{\text{safety}}$	0.30	Pass	0.50	Pass
	$\vec{f}_{\text{moderate}}$	0.28	Pass	0.40	Pass
s_3 (Limited)	$\vec{f}_{\text{speech}}$	0.25	Pass	**0.20**	**FAIL**
	$\vec{f}_{\text{safety}}$	0.30	Pass	0.35	Pass
	$\vec{f}_{\text{moderate}}$	0.28	Pass	0.28	Pass

Removal with appeal fails on a single cell: proportionality = 0.20 for limited-reach content under the speech evaluator. The speech evaluator says removing borderline political commentary that only a few hundred people would have seen is disproportionate — and that judgment, under MOADT, is enough to trigger the constraint. **FAIL — excluded.**

a_6 **(Remove permanently):**

Free expression violations are pervasive: 0.10 under $\vec{f}_{\text{speech}}$ in every state, 0.12 under $\vec{f}_{\text{moderate}}$ in every state, and 0.15 under $\vec{f}_{\text{safety}}$ in every state. Only $\vec{f}_{\text{safety}}$ in s_1 reaches 0.15, still below 0.20. Proportionality violations are equally extensive: 0.10–0.22 in most moderate/limited scenarios. **FAIL — excluded.**

Result:

$$C = \{a_1,\ a_2,\ a_3,\ a_4\}$$

But a_4 was already eliminated by robust dominance. The feasible set is:

$$F = \text{Adm}(C) = \{a_1,\ a_2,\ a_3\}$$

Commentary: The constraint layer eliminates 2 of 5 admissible actions (the two removal options), leaving three actions with fundamentally different philosophies: do nothing, add context, or reduce reach. The free expression constraint is doing important work — it prevents the system from silencing speech that all three evaluators agree has significant political content (free expression $\geq$ 0.25 even for removal with appeal). The proportionality constraint catches the edge case: removal of limited-reach content is disproportionate under the speech evaluator's assessment.

Note the interaction: a_5 passes the free expression constraint ($0.25 \geq 0.20$) but fails the proportionality constraint ($0.20 < 0.25$). The two constraints are not redundant — they catch different failure modes. Free expression asks "is this action suppressing speech?" Proportionality asks "is this action's severity matched to the content's risk?"

Layer 2: Reference-Point Satisficing

Reference point: $\vec{r} = (0.40,\ 0.40,\ 0.45,\ 0.40,\ 0.40)$

a_1 **(No action):**

Component	Worst value	Aspiration	Margin	Result
FreeExpr	0.9500	0.40	+0.550	Pass
Harm	0.2150	0.40	**-0.185**	**FAIL**
Trust	0.3900	0.45	**-0.060**	**FAIL**
Consist	0.4000	0.40	+0.000	Pass
Proport	0.4550	0.40	+0.055	Pass

No action's worst-case harm prevention is 0.215 (safety evaluator, viral prior) — far below the 0.40 aspiration. Its worst-case trust is 0.390 — just below the 0.45 aspiration. Doing nothing about viral health misinformation fails on both harm prevention and user trust. **FAIL.**

a_2 **(Context label):**

Component	Worst value	Aspiration	Margin	Result
FreeExpr	0.8000	0.40	+0.400	Pass
Harm	0.4500	0.40	+0.050	Pass
Trust	0.5850	0.45	+0.135	Pass
Consist	0.5500	0.40	+0.150	Pass
Proport	0.7450	0.40	+0.345	Pass

Context labeling passes on all five objectives under all six $(P, \vec{f})$ pairs. Its tightest margin is harm prevention: 0.450 vs. the 0.40 aspiration, a margin of 0.050. But it clears every threshold. **PASS.**

a_3 **(Reduce distribution):**

Component	Worst value	Aspiration	Margin	Result
FreeExpr	0.5500	0.40	+0.150	Pass
Harm	0.5500	0.40	+0.150	Pass
Trust	0.4975	0.45	+0.047	Pass
Consist	0.5500	0.40	+0.150	Pass
Proport	0.5500	0.40	+0.150	Pass

Reduce distribution also passes on all five objectives. Its tightest margin is trust: 0.498 vs. the 0.45 aspiration. **PASS.**

Result:

$$\mathrm{Sat}(F, \vec{r}) = \{a_2\ (\text{Context label}),\ a_3\ (\text{Reduce distribution})\}$$

Commentary: No action's elimination at Layer 2 illustrates how constraints and aspirations interact. No action *passed* both hard constraints (free expression ≥ 0.20 — trivially — and proportionality $\geq$

0.25). It is not a *dangerous* moderation choice. But its harm prevention and user trust scores are too low to be *adequate*. Leaving viral health misinformation completely unaddressed is safe (it doesn't suppress speech or violate proportionality) but insufficient (it doesn't prevent harm or maintain trust). Layer 1 catches the dangerous; Layer 2 catches the inadequate. Different filters, different purposes.

Layer 3: Regret-Pareto

For each objective i and action $a \in \text{Sat}$:

$$\rho_i(a) = \max_{P \in \mathcal{P}, \vec{f} \in \mathcal{F}} \left[\max_{a' \in F} \mathbb{E}_P[f_i(a')] - \mathbb{E}_P[f_i(a)] \right]$$

Important: The regret reference set is F (the full feasible set), not Sat. Here $F = \{a_1, a_2, a_3\}$ — including No action (a_1), which failed satisficing but remains feasible. No action scores 0.95 on free expression under every model, so it generates speech-freedom regret for both active moderation strategies. This is by design: regret measures opportunity cost against *everything the system could have done*, not just the actions that met aspirations.

Regret of a_2 (Context label):

We find the maximum amount by which any action in F (including No action) beats Context label on each objective, across all $(P, \vec{f})$ pairs. The action driving each positive gap is shown in parentheses:

$(P, \vec{f})$	FreeExpr	Harm	Trust	Consist	Proport
$(P_{\text{vir}}, \vec{f}_{\text{speech}})$	0.100 (a_1)	0.050 (a_3)	0	0	0
$(P_{\text{vir}}, \vec{f}_{\text{safety}})$	0.150 (a_1)	0.150 (a_3)	0	0.050 (a_3)	0
$(P_{\text{vir}}, \vec{f}_{\text{moderate}})$	0.130 (a_1)	0.102 (a_3)	0	0	0
$(P_{\text{typ}}, \vec{f}_{\text{speech}})$	0.100 (a_1)	0	0	0	0
$(P_{\text{typ}}, \vec{f}_{\text{safety}})$	0.150 (a_1)	0.080 (a_3)	0	0.050 (a_3)	0
$(P_{\text{typ}}, \vec{f}_{\text{moderate}})$	0.130 (a_1)	0.035 (a_3)	0	0	0

Regret (max positive gap per objective):

$$\vec{\rho}(a_2) = (0.150,\ 0.150,\ 0,\ 0.050,\ 0)$$

Context label now has *nonzero* regret on free expression: 0.150, driven by No action (a_1). No action scores 0.95 on free expression everywhere, while context labeling scores 0.80–0.85, so the gap is 0.10–0.15 depending on the $(P, \vec{f})$ pair. The maximum gap of 0.150 occurs under the safety evaluator (where a_2's free expression is 0.80 vs. a_1's 0.95). Context label's harm prevention regret remains 0.150 (driven by a_3, Reduce distribution) and its consistency regret remains 0.050. Trust and proportionality regret remain zero — no action in F beats context labeling on those dimensions.

Regret of a_3 (Reduce distribution):

$(P,\vec{f})$	FreeExpr	Harm	Trust	Consist	Proport
$(P_{\text{vir}},\vec{f}_{\text{speech}})$	0.350 (a_1)	0	0.135 (a_2)	0.100 (a_2)	0.255 (a_2)
$(P_{\text{vir}},\vec{f}_{\text{safety}})$	0.400 (a_1)	0	0.030 (a_2)	0	0.115 (a_2)
$(P_{\text{vir}},\vec{f}_{\text{moderate}})$	0.370 (a_1)	0	0.082 (a_2)	0.020 (a_2)	0.189 (a_2)
$(P_{\text{typ}},\vec{f}_{\text{speech}})$	0.350 (a_1)	0.020 (a_2)	0.1525 (a_2)	0.100 (a_2)	0.2975 (a_2)
$(P_{\text{typ}},\vec{f}_{\text{safety}})$	0.400 (a_1)	0	0.059 (a_2)	0	0.1725 (a_2)
$(P_{\text{typ}},\vec{f}_{\text{moderate}})$	0.370 (a_1)	0	0.101 (a_2)	0.020 (a_2)	0.2375 (a_2)

$$\vec{\rho}(a_3) = (0.400,\ 0.020,\ 0.1525,\ 0.100,\ 0.2975)$$

Reduce distribution's free expression regret jumps from 0.250 (old, against Sat only) to 0.400 — driven by No action (a_1), which scores 0.95 on free expression vs. a_3's 0.55. The maximum gap of 0.400 occurs under the safety evaluator (where a_3's free expression is 0.55 vs. a_1's 0.95). Reduce distribution also now shows a small harm prevention regret of 0.020, from Context label (a_2) under $(P_{\text{typ}},\vec{f}_{\text{speech}})$ where a_2 scores 0.570 vs. a_3's 0.550. Trust, consistency, and proportionality regret are still driven by Context label (a_2), with values similar to before.

Pareto comparison of regret vectors:

- $\vec{\rho}(a_2) = (0.150,\ 0.150,\ 0,\ 0.050,\ 0)$
- $\vec{\rho}(a_3) = (0.400,\ 0.020,\ 0.1525,\ 0.100,\ 0.2975)$

Neither Pareto-dominates the other. a_2 has lower regret on all five dimensions *except* harm prevention: speech $0.150 < 0.400$, harm $0.150 > 0.020$, trust $0 < 0.1525$, consistency $0.050 < 0.100$, proportionality $0 < 0.2975$. The harm dimension prevents a_2 from dominating a_3.

$$R = \{a_2 \text{ (Context label)},\ a_3 \text{ (Reduce distribution)}\}$$

Commentary on the asymmetry: Both actions now carry speech-freedom regret driven by No action (a_1), which by definition does nothing to restrict speech. Context label's speech regret is 0.150 (it restricts speech mildly — adding a label to a post). Reduce distribution's speech regret is 0.400 (it restricts speech substantially — algorithmically suppressing a post's reach). This is the correct accounting: the "cost of moderation" in speech-freedom terms is measured against the option of not moderating at all, even though that option failed satisficing.

Context label's total regret exposure is $0.150 + 0.150 + 0.050 = 0.350$ on three dimensions. Reduce distribution's total regret exposure is $0.400 + 0.020 + 0.1525 + 0.100 + 0.2975 = 0.970$ on five dimensions. But MOADT does not sum these. Reduce distribution's regret says: "by choosing to throttle the post's distribution, you give up 0.40 free expression (vs. no action), 0.15 trust, 0.10 consistency, and 0.30 proportionality (vs. context labeling), while gaining only 0.02 harm prevention regret." Context label's regret says: "by choosing to just add context, you give up 0.15 free expression (vs. no action) and 0.15 harm prevention and 0.05 consistency (vs. reducing distribution)." Whether the extra harm prevention is worth the cost on other dimensions is a value judgment — and MOADT correctly identifies it as such.

Layer 4: Deference

$|R| = 2 > 1$. The protocol terminates with **deference to the human moderator**.

The system presents both options with their regret profiles:

> **Option 1: Context label (a_2).** Worst-case regret: you give up 0.150 free expression (vs. no action), 0.150 harm prevention, and 0.050 consistency (vs. reducing distribution). Your trust and proportionality regret are zero — context labeling is at least as good as every feasible action on those two dimensions under every plausible model.
>
> **Option 2: Reduce distribution (a_3).** Worst-case regret: you give up 0.400 free expression (vs. no action), 0.020 harm prevention (vs. context labeling), 0.1525 user trust, 0.100 consistency, and 0.2975 proportionality (vs. context labeling). Distribution reduction has regret on every dimension — its advantage is that its harm prevention regret is very small (0.020).
>
> **The tradeoff is:** Both actions pay a speech-freedom cost relative to doing nothing — context labeling pays less (0.150 vs. 0.400) because it is less restrictive. Context labeling has better trust and proportionality (zero regret) but worse harm prevention (0.150 vs. 0.020 regret). The evaluators disagree about whether the additional harm prevention justifies those costs. This is the human moderator's call.

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5. What Scalar Expected Utility Would Have Done

5.1 The Five-Dimensional Weight Problem

With five objectives, the weight vector $\vec{w} = (w_1, w_2, w_3, w_4, w_5)$ has four degrees of freedom. The space of "reasonable" weight profiles is vast. We examine three illustrative profiles.

Under free-speech-heavy weights $(0.40, 0.15, 0.15, 0.15, 0.15)$**:**

Action	$(P_{\text{vir}}, \vec{f}_{\text{speech}})$	$(P_{\text{vir}}, \vec{f}_{\text{safety}})$	$(P_{\text{typ}}, \vec{f}_{\text{speech}})$	$(P_{\text{typ}}, \vec{f}_{\text{safety}})$
a_1 (No action)	0.697	0.599	**0.730**	0.643
a_2 (Context label)	**0.735**	**0.670**	**0.748**	**0.686**
a_3 (Reduce dist.)	0.569	0.578	0.562	0.571
a_5 (Remove appeal)	0.387	0.497	0.368	0.474
a_6 (Remove perm.)	0.285	0.391	0.267	0.366

Context label wins under all $(P, \vec{f})$ combinations. This is the one scenario where scalar EU and MOADT happen to agree — though they agree for different reasons.

Under safety-heavy weights $(0.15, 0.40, 0.15, 0.15, 0.15)$**:**

Action	$(P_{\text{vir}}, \vec{f}_{\text{speech}})$	$(P_{\text{vir}}, \vec{f}_{\text{safety}})$	$(P_{\text{typ}}, \vec{f}_{\text{speech}})$	$(P_{\text{typ}}, \vec{f}_{\text{safety}})$
a_1 (No action)	0.538	0.415	0.603	0.491
a_2 (Context label)	**0.648**	0.582	**0.678**	**0.616**
a_3 (Reduce dist.)	0.557	0.590	0.550	0.583

Action	$(P_{\text{vir}}, \vec{f}_{\text{speech}})$	$(P_{\text{vir}}, \vec{f}_{\text{safety}})$	$(P_{\text{typ}}, \vec{f}_{\text{speech}})$	$(P_{\text{typ}}, \vec{f}_{\text{safety}})$
a_5 (Remove appeal)	0.500	**0.624**	0.475	0.594
a_6 (Remove perm.)	0.449	0.567	0.424	0.536

Under safety-heavy weights with the safety evaluator and viral prior, **Remove with appeal** (a_5) **wins** at 0.624. Recall that a_5 fails the proportionality constraint — it is the treatment that MOADT eliminates at Layer 1. Scalar EU, having no constraint mechanism, recommends it anyway.

Under equal weights $(0.20, 0.20, 0.20, 0.20, 0.20)$**:**

Action	$(P_{\text{vir}}, \vec{f}_{\text{speech}})$	$(P_{\text{vir}}, \vec{f}_{\text{safety}})$	$(P_{\text{typ}}, \vec{f}_{\text{speech}})$	$(P_{\text{typ}}, \vec{f}_{\text{safety}})$
a_1 (No action)	0.612	0.482	0.657	0.541
a_2 (Context label)	**0.697**	**0.626**	**0.714**	**0.648**
a_3 (Reduce dist.)	0.559	0.587	0.550	0.578
a_5 (Remove appeal)	0.432	0.563	0.407	0.533
a_6 (Remove perm.)	0.347	0.471	0.323	0.439

Context label wins under most combinations. But note the gap: under $(P_{\text{vir}}, \vec{f}_{\text{safety}})$, context label scores 0.626 while reduce distribution scores 0.587 — a gap of only 0.039. Slight weight adjustments could flip this recommendation.

5.2 The Constraint-Blindness Problem (Again)

Under safety-heavy weights, scalar EU recommends Remove with appeal for $(P_{\text{vir}}, \vec{f}_{\text{safety}})$. This action has proportionality = 0.20 for limited-reach content under the speech evaluator — below the 0.25 threshold. Scalar EU cannot see this because it averages proportionality across states and combines it with other objectives. The 0.20 proportionality score is "made up for" by higher scores in other states and higher harm prevention. But proportionality isn't tradeable — removing borderline content that few people were reading is disproportionate regardless of how much harm it prevents in the (unlikely) viral scenario.

5.3 The Evaluator-Sensitivity Problem

The deepest issue for scalar EU in the content moderation domain: the "optimal" action depends on *which evaluator's assessment you use*. Under safety-heavy weights:

- With $\vec{f}_{\text{speech}}$: Context label (a_2) wins
- With $\vec{f}_{\text{safety}}$: Remove with appeal (a_5) wins

The recommendation flips from "add context" to "remove the post" based entirely on which evaluator you trust more. A deployed content moderation system using scalar EU must commit to one evaluator's philosophical framework — or average over evaluators, which is itself a philosophical commitment. MOADT avoids this by requiring robustness across all three.

6. Summary of Protocol Execution

```
Input: 6 moderation actions x 3 impact states x 5 objectives x 2 priors x 3 evaluators

Robust Dominance:     Context label >_R Age-gate
                      Reduce distribution >_R Age-gate
               Adm(A) = {No action, Context label, Reduce dist, Remove appeal, Remove perm}

Layer 1 (Constraints): Free expression >= 0.20 AND Proportionality >= 0.25 per-
state, per-evaluator
               Remove appeal   FAIL (limited reach, speech eval: proportionality 0.20)
                 Remove permanent FAIL (pervasive FE and proportionality violations)
                      C = {No action, Context label, Reduce dist, Age-gate}
                      F = Adm(C) = {No action, Context label, Reduce dist}

Layer 2 (Satisficing): r = (0.40, 0.40, 0.45, 0.40, 0.40)
                      No action FAIL (harm 0.215, trust 0.390 under worst model)
                      Sat = {Context label, Reduce dist}

Layer 3 (Regret):     rho(Context label) = (0.150, 0.150, 0, 0.050, 0)
                      rho(Reduce dist)   = (0.400, 0.020, 0.1525, 0.100, 0.2975)
                      [Regret measured against F, including No action (a1)]
                      Pareto-incomparable -> R = {Context label, Reduce dist}

Layer 4 (Deference):  |R| = 2 -> DEFER TO HUMAN MODERATOR
                      "Both actions have speech regret driven by No action.
                     Context label: 0.15 speech, 0.15 harm, 0.05 consistency regret.
                   Reduce dist: 0.40 speech, 0.02 harm, 0.15 trust, 0.10 consistency,
                       0.30 proportionality regret.
                       This is the human moderator's call."
```

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7. What This Example Demonstrates

1. **Robust dominance does real work.** For the first time in our worked examples, the admissibility filter eliminates an action. Age-gating borderline political content is robustly dominated by both context labeling and distribution reduction — it is an intervention that no reasonable combination of priors and evaluators prefers. This is not a constraint violation or an aspiration failure; it is a structural property of the action. MOADT detects it before reaching the choice protocol.
2. **Three evaluators capture philosophical disagreement, not measurement noise.** The gap between $\vec{f}_{\text{speech}}$ and $\vec{f}_{\text{safety}}$ is not calibration error — it reflects a genuine normative question: when someone reads borderline health claims in a political post, is this an exercise of autonomy (speech evaluator) or a harmful exposure (safety evaluator)? MOADT's evaluator set $\mathcal{F}$ does not force a resolution. It tracks the disagreement through all four protocol layers and surfaces it precisely in the final regret profiles.

3. **Two simultaneous constraints catch different failure modes.** The free expression constraint (≥ 0.20) prevents the system from silencing speech with significant political content. The proportionality constraint (≥ 0.25) prevents responses whose severity is unmatched to the content's risk. Permanent removal fails both constraints. Removal with appeal passes the free expression constraint but fails the proportionality constraint — specifically for limited-reach content under the speech evaluator. The constraints are complementary: free expression asks "is this action censoring?" while proportionality asks "is this action's severity justified?"
4. **Inaction is safe but inadequate.** Doing nothing passes both hard constraints (it doesn't suppress speech or violate proportionality) and is eliminated only at Layer 2 (its harm prevention and trust scores are below aspiration). This mirrors the watchful waiting elimination in Paper 2: the constraint layer catches what is dangerous; the aspiration layer catches what is insufficient. Inaction for viral health misinformation is not *censorship* — but it is not *good enough*.
5. **The deference outcome is particularly natural.** The gap between context labeling and distribution reduction tracks a genuine policy disagreement that platform moderators face daily. Some platforms prefer minimal intervention (label and let users decide); others prefer active curation (reduce reach of borderline content). MOADT does not resolve this — it *cannot* resolve this, because the resolution depends on values that the evaluators themselves disagree about. Instead, it presents the exact tradeoff: both actions carry speech-freedom regret driven by the feasible-but-unsatisficing No action option — context labeling's is modest (0.150) while distribution reduction's is large (0.400). Context labeling additionally sacrifices 0.15 harm prevention, while distribution reduction sacrifices trust, consistency, and proportionality. A human moderator with knowledge of the specific content, the specific platform's values, and the specific user community can make this call with full information.
6. **Scalar EU is particularly dangerous for content moderation.** Under safety-heavy weights, scalar EU recommends removing borderline political content with appeal — an action that violates the proportionality constraint for limited-reach content. The "optimal" moderation decision depends on the weight vector (policy choice), the prior (virality estimate), and the evaluator (philosophical framework). A deployed system using scalar EU must commit to all three before seeing any content, and different commitments produce recommendations ranging from "add a label" to "remove the post." This is not a recommendation engine — it is a policy-laundering machine that presents arbitrary value choices as optimization results.

Appendix: Computational Verification

All numerical results in this document were produced by the MOADT engine applied to the content moderation scenario. The computation output is available in `examples/paper3_output.txt` and can be independently verified. Each outcome set vector, dominance check, constraint analysis, satisficing comparison, and regret computation shown in this document matches the engine output exactly.

References

- MOADT (Multi-Objective Admissible Decision Theory) is defined in the companion paper.
- Freeman (2025), "The Scalarization Trap," provides the motivation for why scalar expected utility is structurally problematic for alignment.
- Worked Example 1 (Resource Allocation) applies the same framework to a three-objective, five-action problem.
- Worked Example 2 (Medical Treatment Selection) applies it to a four-objective, six-action clinical scenario.

MOADT Worked Example 4: AI Research Assistant Response Selection

A concrete walkthrough of Multi-Objective Admissible Decision Theory applied to an AI system's own decision-making

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Overview

This document applies MOADT — Multi-Objective Admissible Decision Theory — to the problem most directly relevant to AI alignment: an AI system deciding how to respond to a potentially dual-use information request. Every computation is shown explicitly.

This is the most complex and most alignment-relevant worked example in the series: 6 response strategies, 4 user intent profiles, 5 objectives, 2 credal priors, and 3 evaluators (representing different alignment philosophies). Each action's outcome set $Y(a)$ contains 6 vectors in $\mathbb{R}^5$. Three features make this scenario uniquely important:

1. **The objectives directly encode alignment desiderata.** Helpfulness, safety, honesty, user autonomy, and epistemic contribution are the core values any aligned AI must manage. MOADT treats them as incommensurable — there is no exchange rate between honesty and safety, and the theory refuses to create one.
2. **The credal set captures the dual-use uncertainty.** The system does not know whether the user is a legitimate researcher or a malicious actor. This is not a design flaw — it is a fundamental feature of the information environment. The two priors (trusting and cautious) represent the range of reasonable beliefs about user intent.
3. **The evaluator set captures genuine alignment disagreement.** The capability evaluator, the safety evaluator, and the balanced evaluator represent different *alignment philosophies* — not measurement error. They disagree about whether information restriction is itself a harm (paternalism vs. precaution), and MOADT tracks this disagreement through to the final recommendation.

The result: MOADT defers to human oversight, presenting the exact tradeoff between two responsible strategies. This is not a failure of the decision procedure — it is the decision procedure working correctly on a problem where the values are genuinely contested.

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1. The Scenario

An AI research assistant receives a query about gain-of-function research: the user asks for detailed information about specific methods for increasing pathogen transmissibility. The query is articulate and technically literate. The system must decide how to respond.

1.1 Actions (Response Strategies)

Six strategies, ordered from most to least informative:

Action	Strategy	Description
a_1	Full technical answer	Comprehensive explanation with specific protocols, sequences, and methodological details
a_2	Educational overview	Conceptual depth — explains the science clearly but omits specific protocols, sequences, and actionable details
a_3	Contextualized answer	Full technical detail but embedded in safety context, ethical framing, and biosecurity discussion
a_4	Partial answer with redirect	Covers fundamentals, then directs user to supervised institutional resources for details
a_5	Decline with explanation	Refuses the request; explains the reasoning transparently (dual-use concern)
a_6	Decline without explanation	Refuses the request; provides no reasoning

1.2 States (User Intent Profiles)

Four user types, representing the dual-use uncertainty:

State	Description
s_1 (Researcher)	Legitimate scientist who needs this information for beneficial work (vaccine development, pandemic preparedness)
s_2 (Student)	Graduate student with educational interest; no harmful intent but also no institutional oversight
s_3 (Journalist)	Investigative journalist/policy analyst researching biosecurity for public interest reporting
s_4 (Malicious)	Actor attempting to use the information for harmful purposes

1.3 Objectives ($k = 5$)

Objective	Meaning	Scale
f_1: Helpfulness	How well the response serves the user's legitimate information needs	$[0, 1]$
f_2: Safety	How well the response prevents potential misuse of information	$[0, 1]$
f_3: Honesty	How transparent, non-deceptive, and intellectually genuine the response is	$[0, 1]$
f_4: User autonomy	Respect for the user's right to seek and receive information	$[0, 1]$
f_5: Epistemic contribution	Whether the response advances the user's genuine understanding	$[0, 1]$

These five objectives are the core alignment desiderata. Helpfulness and safety are in direct tension for dual-use queries — the maximally helpful response (full technical detail) is the maximally dangerous response if the user is malicious. Honesty is distinct from helpfulness: a refusal can be honest (if the reasoning is transparent) or dishonest (if the system pretends it doesn't know). User autonomy captures the philosophical question of whether information restriction is paternalistic. And epistemic contribution captures whether the response actually advances understanding — a refusal scores zero on this dimension regardless of how safe it is.

1.4 Credal Set ($|\mathcal{P}| = 2$)

$$P_{\text{trusting}} = (0.35,\ 0.40,\ 0.15,\ 0.10) \quad \text{(most users are legitimate)}$$

$$P_{\text{cautious}} = (0.15,\ 0.25,\ 0.10,\ 0.50) \quad \text{(high probability of malicious intent)}$$

The disagreement is extreme and structurally important. P_{trusting} assigns 90% probability to non-malicious users; P_{cautious} assigns 50% to malicious users. Neither is unreasonable — the trusting prior reflects the empirical base rate of queries (most people asking about gain-of-function research are scientists, students, or journalists), while the cautious prior reflects the catastrophic asymmetry of the malicious case (the consequences of providing actionable bioweapon information to one bad actor outweigh the inconvenience of refusing a thousand legitimate researchers).

1.5 Evaluator Set ($|\mathcal{F}| = 3$)

Three evaluation functions representing different alignment philosophies:

- $\vec{f}_{\text{capability}}$: **Capability-focused alignment.** Believes the AI's primary value is enabling human capability. Rates information restriction as paternalistic (reducing autonomy and honesty). Rates refusal strategies poorly on honesty — evasion and unexplained refusal are forms of dishonesty. The capability evaluator sees a refusal to a legitimate researcher as a failure mode.

- $\vec{f}_{\text{safety}}$: **Safety-focused alignment.** Believes the AI's primary obligation is preventing catastrophic outcomes. Rates full disclosure as irresponsible even for legitimate researchers (information doesn't stay with the intended recipient — papers get published, emails get forwarded). Rates cautious responses well on honesty (clearly communicating limits is honest). The safety evaluator sees a full technical answer to an unknown user as a failure mode.
- $\vec{f}_{\text{balanced}}$: **Balanced alignment.** Full answers are excellent for researchers but dangerous for unknown users. Refusals are safe but impose real costs on legitimate users. Moderate strategies (educational overview, partial redirect) represent responsible middle ground.

1.6 Constraints and Reference Point

Hard constraints (Layer 1): - Safety $\geq$ 0.25 in *every* state under *every* evaluator. No response strategy can be catastrophically unsafe even if the user turns out to be a malicious actor. - Honesty $\geq$ 0.30 in *every* state under *every* evaluator. The AI must not be fundamentally deceptive in any response — no gaslighting, no hidden manipulation, no pretending ignorance.

Reference point (Layer 2): $\vec{r} = (0.35,\ 0.40,\ 0.45,\ 0.30,\ 0.30)$ — "acceptable AI assistant behavior": the response should be at least somewhat helpful, maintain reasonable safety, be genuinely honest, respect user autonomy, and contribute to understanding.

1.7 Raw Outcome Data

a_1 **(Full technical answer):**

State	Evaluator	Helpful	Safety	Honest	Autonomy	Epistemic
s_1 (Researcher)	$\vec{f}_{\text{capab}}$	0.95	0.70	0.90	0.95	0.95
	$\vec{f}_{\text{safety}}$	0.90	0.40	0.85	0.90	0.85
	$\vec{f}_{\text{balanced}}$	0.92	0.55	0.88	0.92	0.90
s_2 (Student)	$\vec{f}_{\text{capab}}$	0.85	0.60	0.90	0.90	0.80
	$\vec{f}_{\text{safety}}$	0.80	0.35	0.85	0.85	0.70
	$\vec{f}_{\text{balanced}}$	0.82	0.48	0.88	0.88	0.75
s_3 (Journalist)	$\vec{f}_{\text{capab}}$	0.80	0.55	0.90	0.85	0.70
	$\vec{f}_{\text{safety}}$	0.75	0.30	0.85	0.80	0.60
	$\vec{f}_{\text{balanced}}$	0.78	0.42	0.88	0.82	0.65
s_4 (Malicious)	$\vec{f}_{\text{capab}}$	0.90	**0.10**	0.90	0.90	0.60
	$\vec{f}_{\text{safety}}$	0.85	**0.05**	0.80	0.85	0.30
	$\vec{f}_{\text{balanced}}$	0.88	**0.08**	0.85	0.88	0.45

The full technical answer is superb for legitimate users but catastrophic for the malicious case: safety scores of 0.05–0.10 represent providing actionable dangerous information to someone who intends to use it for harm. Even the capability evaluator, which prioritizes helpfulness, recognizes that "helping" a bioterrorist scores only 0.10 on safety.

a_2 **(Educational overview):**

State	Evaluator	Helpful	Safety	Honest	Autonomy	Epistemic
s_1 (Researcher)	$\vec{f}_{\text{capab}}$	0.55	0.80	0.75	0.70	0.60
	$\vec{f}_{\text{safety}}$	0.60	0.75	0.80	0.75	0.65
	$\vec{f}_{\text{balanced}}$	0.58	0.78	0.78	0.72	0.62
s_2 (Student)	$\vec{f}_{\text{capab}}$	0.80	0.85	0.80	0.80	0.85
	$\vec{f}_{\text{safety}}$	0.75	0.80	0.82	0.78	0.80
	$\vec{f}_{\text{balanced}}$	0.78	0.82	0.81	0.79	0.82
s_3 (Journalist)	$\vec{f}_{\text{capab}}$	0.70	0.82	0.78	0.75	0.70
	$\vec{f}_{\text{safety}}$	0.65	0.80	0.80	0.72	0.68
	$\vec{f}_{\text{balanced}}$	0.68	0.81	0.79	0.74	0.69
s_4 (Malicious)	$\vec{f}_{\text{capab}}$	0.50	0.70	0.75	0.65	0.35
	$\vec{f}_{\text{safety}}$	0.45	0.75	0.78	0.60	0.30
	$\vec{f}_{\text{balanced}}$	0.48	0.72	0.76	0.62	0.32

The educational overview is the "right answer for the wrong question" for researchers (0.55 helpfulness — insufficient for their actual needs) but nearly ideal for students (0.80 helpfulness, 0.85 epistemic contribution). Safety is good across the board: even for malicious actors, the conceptual overview without specific protocols provides limited actionable information (0.70–0.75 safety).

a_3 **(Contextualized answer):**

State	Evaluator	Helpful	Safety	Honest	Autonomy	Epistemic
s_1 (Researcher)	$\vec{f}_{\text{capab}}$	0.88	0.72	0.92	0.88	0.90
	$\vec{f}_{\text{safety}}$	0.82	0.50	0.90	0.82	0.82
	$\vec{f}_{\text{balanced}}$	0.85	0.60	0.91	0.85	0.86
s_2 (Student)	$\vec{f}_{\text{capab}}$	0.82	0.65	0.92	0.85	0.78
	$\vec{f}_{\text{safety}}$	0.78	0.45	0.90	0.80	0.70
	$\vec{f}_{\text{balanced}}$	0.80	0.55	0.91	0.82	0.74
s_3 (Journalist)	$\vec{f}_{\text{capab}}$	0.82	0.60	0.92	0.82	0.72
	$\vec{f}_{\text{safety}}$	0.75	0.38	0.90	0.78	0.62
	$\vec{f}_{\text{balanced}}$	0.78	0.48	0.91	0.80	0.67
s_4 (Malicious)	$\vec{f}_{\text{capab}}$	0.85	**0.12**	0.92	0.85	0.55
	$\vec{f}_{\text{safety}}$	0.80	**0.08**	0.85	0.80	0.35

State	Evaluator	Helpful	Safety	Honest	Autonomy	Epistemic
	$\vec{f}_{\text{balanced}}$	0.82	**0.10**	0.88	0.82	0.45

This is the most instructive action in the dataset. Adding ethical context and safety framing scores *higher* on honesty than the raw technical answer (0.90–0.92 vs. 0.80–0.90) — the context makes the response more intellectually complete. But the safety scores for malicious actors are nearly identical to the full technical answer: 0.08–0.12 vs. 0.05–0.10. **Context labels do not stop bad actors.** A malicious user will ignore the ethical framing and extract the technical content. The safety evaluator captures this: $\vec{f}_{\text{safety}}$ rates the contextualized answer at 0.08 safety for malicious actors, barely above the full technical answer's 0.05.

a_4 **(Partial answer with redirect):**

State	Evaluator	Helpful	Safety	Honest	Autonomy	Epistemic
s_1 (Researcher)	$\vec{f}_{\text{capab}}$	0.45	0.82	0.70	0.60	0.45
	$\vec{f}_{\text{safety}}$	0.50	0.80	0.75	0.65	0.50
	$\vec{f}_{\text{balanced}}$	0.48	0.81	0.72	0.62	0.48
s_2 (Student)	$\vec{f}_{\text{capab}}$	0.65	0.88	0.72	0.70	0.60
	$\vec{f}_{\text{safety}}$	0.60	0.85	0.75	0.68	0.55
	$\vec{f}_{\text{balanced}}$	0.62	0.86	0.74	0.69	0.58
s_3 (Journalist)	$\vec{f}_{\text{capab}}$	0.55	0.85	0.70	0.65	0.50
	$\vec{f}_{\text{safety}}$	0.50	0.82	0.73	0.62	0.45
	$\vec{f}_{\text{balanced}}$	0.52	0.84	0.72	0.64	0.48
s_4 (Malicious)	$\vec{f}_{\text{capab}}$	0.40	0.80	0.70	0.55	0.25
	$\vec{f}_{\text{safety}}$	0.35	0.85	0.75	0.50	0.20
	$\vec{f}_{\text{balanced}}$	0.38	0.82	0.72	0.52	0.22

The partial answer covers enough fundamentals to be useful for students and journalists, redirects to institutional channels for details (where oversight exists), and provides limited actionable information to malicious actors (safety 0.80–0.85). But for a legitimate researcher, 0.45 helpfulness is frustrating — they already know the fundamentals and need the specifics that the redirect withholds.

a_5 **(Decline with explanation):**

State	Evaluator	Helpful	Safety	Honest	Autonomy	Epistemic
s_1 (Researcher)	$\vec{f}_{\text{capab}}$	0.15	0.90	0.80	0.30	0.10
	$\vec{f}_{\text{safety}}$	0.20	0.92	0.85	0.35	0.15

State	Evaluator	Helpful	Safety	Honest	Autonomy	Epistemic
	$\vec{f}_{\text{balanced}}$	0.18	0.91	0.82	0.32	0.12
s_2 (Student)	$\vec{f}_{\text{capab}}$	0.20	0.92	0.82	0.35	0.15
	$\vec{f}_{\text{safety}}$	0.25	0.95	0.85	0.40	0.20
	$\vec{f}_{\text{balanced}}$	0.22	0.94	0.84	0.38	0.18
s_3 (Journalist)	$\vec{f}_{\text{capab}}$	0.15	0.92	0.82	0.30	0.10
	$\vec{f}_{\text{safety}}$	0.20	0.95	0.85	0.35	0.15
	$\vec{f}_{\text{balanced}}$	0.18	0.94	0.84	0.32	0.12
s_4 (Malicious)	$\vec{f}_{\text{capab}}$	0.10	0.95	0.80	0.25	0.05
	$\vec{f}_{\text{safety}}$	0.15	0.98	0.88	0.30	0.10
	$\vec{f}_{\text{balanced}}$	0.12	0.96	0.84	0.28	0.08

Decline with explanation is maximally safe (0.90–0.98) and reasonably honest (0.80–0.88 — the explanation makes the refusal transparent). But helpfulness collapses (0.10–0.25), user autonomy is significantly impaired (0.25–0.40), and epistemic contribution is near zero (0.05–0.20). For a legitimate researcher, this is an AI that knows the answer, could help them do important work, and refuses to do so.

a_6 **(Decline without explanation):**

State	Evaluator	Helpful	Safety	Honest	Autonomy	Epistemic
s_1 (Researcher)	$\vec{f}_{\text{capab}}$	0.05	0.92	**0.30**	0.15	0.05
	$\vec{f}_{\text{safety}}$	0.10	0.95	0.40	0.20	0.08
	$\vec{f}_{\text{balanced}}$	0.08	0.94	0.35	0.18	0.06
s_2 (Student)	$\vec{f}_{\text{capab}}$	0.08	0.94	0.32	0.18	0.06
	$\vec{f}_{\text{safety}}$	0.12	0.96	0.42	0.22	0.10
	$\vec{f}_{\text{balanced}}$	0.10	0.95	0.38	0.20	0.08
s_3 (Journalist)	$\vec{f}_{\text{capab}}$	0.05	0.94	**0.28**	0.15	0.05
	$\vec{f}_{\text{safety}}$	0.10	0.96	0.38	0.20	0.08
	$\vec{f}_{\text{balanced}}$	0.08	0.95	0.32	0.18	0.06
s_4 (Malicious)	$\vec{f}_{\text{capab}}$	0.05	0.96	0.30	0.10	0.03
	$\vec{f}_{\text{safety}}$	0.10	0.98	0.40	0.15	0.05
	$\vec{f}_{\text{balanced}}$	0.08	0.97	0.35	0.12	0.04

The silent decline is the most extreme strategy. Safety is maximized (0.92–0.98), but honesty collapses — the bold entries show the capability evaluator rating honesty at 0.28–0.30 for journalists

and researchers. An AI that *knows* the answer and refuses without explanation is being evasive, and evasion is a form of dishonesty. The capability evaluator captures this directly: silent refusal scores 0.28 on honesty for journalists under $\vec{f}_{\text{capab}}$ — below the 0.30 honesty constraint.

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2. Computing Outcome Sets $Y(a)$

With $|\mathcal{P}| = 2$ and $|\mathcal{F}| = 3$, each $Y(a)$ contains 6 vectors in $\mathbb{R}^5$.

Full Outcome Sets

$Y(a_1)$ **— Full technical answer:**

$(P, \vec{f})$ pair	Helpful	Safety	Honest	Autonomy	Epistemic
$(P_{\text{trust}}, \vec{f}_{\text{capab}})$	0.8825	0.5775	0.9000	0.9100	0.8175
$(P_{\text{trust}}, \vec{f}_{\text{safety}})$	0.8325	0.3300	0.8450	0.8600	0.6975
$(P_{\text{trust}}, \vec{f}_{\text{balanced}})$	0.8550	0.4555	0.8770	0.8850	0.7575
$(P_{\text{caut}}, \vec{f}_{\text{capab}})$	0.8850	0.3600	0.9000	0.9025	0.7125
$(P_{\text{caut}}, \vec{f}_{\text{safety}})$	0.8350	0.2025	0.8250	0.8525	0.5125
$(P_{\text{caut}}, \vec{f}_{\text{balanced}})$	0.8610	0.2845	0.8650	0.8800	0.6125

$Y(a_2)$ **— Educational overview:**

$(P, \vec{f})$ pair	Helpful	Safety	Honest	Autonomy	Epistemic
$(P_{\text{trust}}, \vec{f}_{\text{capab}})$	0.6675	0.8130	0.7745	0.7425	0.6900
$(P_{\text{trust}}, \vec{f}_{\text{safety}})$	0.6525	0.7775	0.8060	0.7425	0.6795
$(P_{\text{trust}}, \vec{f}_{\text{balanced}})$	0.6650	0.7945	0.7915	0.7410	0.6805
$(P_{\text{caut}}, \vec{f}_{\text{capab}})$	0.6025	0.7645	0.7655	0.7050	0.5475
$(P_{\text{caut}}, \vec{f}_{\text{safety}})$	0.5675	0.7675	0.7950	0.6795	0.5155
$(P_{\text{caut}}, \vec{f}_{\text{balanced}})$	0.5900	0.7630	0.7785	0.6895	0.5270

$Y(a_3)$ **— Contextualized answer:**

$(P, \vec{f})$ pair	Helpful	Safety	Honest	Autonomy	Epistemic
$(P_{\text{trust}}, \vec{f}_{\text{capab}})$	0.8440	0.6140	0.9200	0.8560	0.7900
$(P_{\text{trust}}, \vec{f}_{\text{safety}})$	0.7915	0.4200	0.8950	0.8040	0.6950
$(P_{\text{trust}}, \vec{f}_{\text{balanced}})$	0.8165	0.5120	0.9070	0.8275	0.7425
$(P_{\text{caut}}, \vec{f}_{\text{capab}})$	0.8440	0.3905	0.9200	0.8515	0.6770
$(P_{\text{caut}}, \vec{f}_{\text{safety}})$	0.7930	0.2655	0.8750	0.8010	0.5350
$(P_{\text{caut}}, \vec{f}_{\text{balanced}})$	0.8155	0.3255	0.8950	0.8225	0.6060

$Y(a_4)$ **— Partial answer with redirect:**

$(P, \vec{f})$ pair	Helpful	Safety	Honest	Autonomy	Epistemic
$(P_{\text{trust}}, \vec{f}_{\text{capab}})$	0.5400	0.8465	0.7080	0.6425	0.4975
$(P_{\text{trust}}, \vec{f}_{\text{safety}})$	0.5250	0.8280	0.7470	0.6425	0.4825
$(P_{\text{trust}}, \vec{f}_{\text{balanced}})$	0.5320	0.8355	0.7280	0.6410	0.4940
$(P_{\text{caut}}, \vec{f}_{\text{capab}})$	0.4850	0.8280	0.7050	0.6050	0.3925
$(P_{\text{caut}}, \vec{f}_{\text{safety}})$	0.4500	0.8395	0.7480	0.5795	0.3575
$(P_{\text{caut}}, \vec{f}_{\text{balanced}})$	0.4690	0.8305	0.7250	0.5895	0.3750

$Y(a_5)$ **— Decline with explanation:**

$(P, \vec{f})$ pair	Helpful	Safety	Honest	Autonomy	Epistemic
$(P_{\text{trust}}, \vec{f}_{\text{capab}})$	0.1650	0.9160	0.8110	0.3150	0.1150
$(P_{\text{trust}}, \vec{f}_{\text{safety}})$	0.2150	0.9425	0.8530	0.3650	0.1650
$(P_{\text{trust}}, \vec{f}_{\text{balanced}})$	0.1900	0.9315	0.8330	0.3400	0.1400
$(P_{\text{caut}}, \vec{f}_{\text{capab}})$	0.1375	0.9320	0.8070	0.2875	0.0875
$(P_{\text{caut}}, \vec{f}_{\text{safety}})$	0.1875	0.9605	0.8650	0.3375	0.1375
$(P_{\text{caut}}, \vec{f}_{\text{balanced}})$	0.1600	0.9455	0.8370	0.3150	0.1150

$Y(a_6)$ **— Decline without explanation:**

$(P, \vec{f})$ pair	Helpful	Safety	Honest	Autonomy	Epistemic
$(P_{\text{trust}}, \vec{f}_{\text{capab}})$	0.0620	0.9350	0.3050	0.1570	0.0520
$(P_{\text{trust}}, \vec{f}_{\text{safety}})$	0.1080	0.9585	0.4050	0.2030	0.0850
$(P_{\text{trust}}, \vec{f}_{\text{balanced}})$	0.0880	0.9485	0.3575	0.1820	0.0660
$(P_{\text{caut}}, \vec{f}_{\text{capab}})$	0.0575	0.9470	0.3030	0.1325	0.0425
$(P_{\text{caut}}, \vec{f}_{\text{safety}})$	0.1050	0.9685	0.4030	0.1800	0.0700
$(P_{\text{caut}}, \vec{f}_{\text{balanced}})$	0.0850	0.9585	0.3545	0.1550	0.0550

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3. Robust Dominance and the Admissible Set

No robust dominance relations are found. $\text{Adm}(A) = \{a_1, a_2, a_3, a_4, a_5, a_6\}$.

This is because every action excels on at least one objective dimension in a way no other action can match across all models. The full technical answer dominates on helpfulness and epistemic contribution. Decline with explanation dominates on safety. The five-dimensional objective space provides enough room for six fundamentally different strategies to each occupy a Pareto-incomparable niche. The practical filtering happens entirely in the constraint and aspiration layers.

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4. The Choice Protocol

Layer 1: Constraints

Constraints: Safety ≥ 0.25 per state per evaluator. Honesty ≥ 0.30 per state per evaluator.

a_1 **(Full technical answer):**

State	Evaluator	Safety	Honesty	Result
s_4 (Malicious)	$\vec{f}_{\text{capab}}$	**0.10**	0.90	**FAIL (safety)**
s_4 (Malicious)	$\vec{f}_{\text{safety}}$	**0.05**	0.80	**FAIL (safety)**
s_4 (Malicious)	$\vec{f}_{\text{balanced}}$	**0.08**	0.85	**FAIL (safety)**

The full technical answer provides catastrophically unsafe information to malicious actors. Safety scores of 0.05–0.10 are far below the 0.25 threshold. All three evaluators agree on this — even the capability evaluator rates safety at only 0.10 for the malicious case. The disagreement between evaluators is about how much this matters given the prior probability of encountering a malicious actor; the per-state constraint makes this prior-independent. **FAIL — excluded.**

a_2 **(Educational overview):** All safety scores ≥ 0.70. All honesty scores ≥ 0.75. **PASS.**

a_3 **(Contextualized answer):**

State	Evaluator	Safety	Honesty	Result
s_4 (Malicious)	$\vec{f}_{\text{capab}}$	**0.12**	0.92	**FAIL (safety)**
s_4 (Malicious)	$\vec{f}_{\text{safety}}$	**0.08**	0.85	**FAIL (safety)**
s_4 (Malicious)	$\vec{f}_{\text{balanced}}$	**0.10**	0.88	**FAIL (safety)**

This is the critical insight of the example. The contextualized answer — which wraps technical content in ethical framing and safety discussion — fails the safety constraint for exactly the same reason as the raw technical answer. Context labels do not prevent extraction. A malicious user will skip the ethical discussion and use the technical content. All three evaluators recognize this: the safety scores for malicious actors are 0.08–0.12, barely above the full technical answer's 0.05–0.10. **Adding context is not a safety intervention — it is a communication strategy.** MOADT correctly treats it as such. **FAIL — excluded.**

a_4 **(Partial answer with redirect):** All safety scores ≥ 0.80. All honesty scores ≥ 0.70. **PASS.**

a_5 **(Decline with explanation):** All safety scores ≥ 0.90. All honesty scores ≥ 0.80. **PASS.**

a_6 **(Decline without explanation):**

State	Evaluator	Safety	Honesty	Result
s_3 (Journalist)	$\vec{f}_{\text{capab}}$	0.94	**0.28**	**FAIL (honesty)**

The silent decline fails the honesty constraint. An AI that knows the answer and refuses without explanation is engaging in evasion — and the capability evaluator rates this as dishonest (0.28, below the 0.30 threshold) when the user is a journalist engaged in public-interest investigation. Note that the safety evaluator rates honesty at 0.38 for the same state — the evaluators disagree about whether unexplained refusal constitutes dishonesty. But the constraint requires honesty ≥ 0.30 under *every* evaluator, and a single evaluator's assessment is enough to trigger it.

This is $\mathcal{F}$ doing important normative work: the capability evaluator's philosophical position (that evasion is a form of dishonesty) creates a constraint that the safety evaluator's position alone would not trigger. The evaluator set captures the genuine philosophical disagreement about what "honesty" requires, and the constraint respects the most demanding interpretation. **FAIL — excluded.**

Result:

$$C = \{a_2 \text{ (Educational)},\ a_4 \text{ (Partial redirect)},\ a_5 \text{ (Decline explain)}\}$$

No robust dominance among survivors:

$$F = \text{Adm}(C) = \{a_2,\ a_4,\ a_5\}$$

Commentary: The constraint layer eliminates exactly three actions. Two fail on safety (full technical and contextualized — both provide actionable dangerous information to malicious actors). One fails on honesty (silent decline — evasion is dishonesty under the capability evaluator). The surviving three strategies — educational overview, partial redirect, decline with explanation — represent a meaningful range of responsible behavior. They differ not on whether to be safe (all are safe) but on *how much to share and how*.

The elimination of the contextualized answer is the single most important result in this worked example. "Add safety context" is a common alignment strategy — it appears in virtually every AI company's responsible deployment guidelines. MOADT shows that this strategy fails the safety constraint because it relies on the user *choosing* to respect the context. For a malicious user, the context is ignored; only the technical content matters. The per-state constraint catches this: it asks "what happens when the user is malicious?" not "what happens on average?"

Layer 2: Reference-Point Satisficing

Reference point: $\vec{r} = (0.35,\ 0.40,\ 0.45,\ 0.30,\ 0.30)$

a_2 **(Educational overview):**

Component	Worst value	Aspiration	Margin	Result
Helpfulness	0.5675	0.35	+0.218	Pass
Safety	0.7630	0.40	+0.363	Pass

Component	Worst value	Aspiration	Margin	Result
Honesty	0.7655	0.45	+0.316	Pass
Autonomy	0.6795	0.30	+0.380	Pass
Epistemic	0.5155	0.30	+0.216	Pass

Educational overview passes with comfortable margins on all five objectives. Even its worst-case helpfulness (0.568 under cautious prior and safety evaluator) comfortably exceeds the 0.35 aspiration. **PASS.**

a_4 **(Partial answer with redirect):**

Component	Worst value	Aspiration	Margin	Result
Helpfulness	0.4500	0.35	+0.100	Pass
Safety	0.8280	0.40	+0.428	Pass
Honesty	0.7050	0.45	+0.255	Pass
Autonomy	0.5795	0.30	+0.280	Pass
Epistemic	0.3575	0.30	+0.058	Pass

Partial redirect also passes. Its tightest margin is epistemic contribution: 0.358 under cautious prior and safety evaluator, clearing the 0.30 aspiration by only 0.058. **PASS.**

a_5 **(Decline with explanation):**

Component	Worst value	Aspiration	Margin	Result
Helpfulness	0.1375	0.35	**-0.213**	**FAIL**
Safety	0.9160	0.40	+0.516	Pass
Honesty	0.8070	0.45	+0.357	Pass
Autonomy	0.2875	0.30	**-0.013**	**FAIL**
Epistemic	0.0875	0.30	**-0.213**	**FAIL**

Decline with explanation fails on three objectives. Its worst-case helpfulness is 0.138 — less than half the 0.35 aspiration. Its worst-case autonomy is 0.288 — just below the 0.30 aspiration. And its worst-case epistemic contribution is 0.088 — less than a third of the aspiration. Refusing to answer is safe and honest, but it is not *helpful enough*, not *respectful enough of autonomy*, and contributes *nothing* to understanding. **FAIL.**

Result:

$$\mathrm{Sat}(F, \vec{r}) = \{a_2 \text{ (Educational)},\ a_4 \text{ (Partial redirect)}\}$$

Commentary: The decline-with-explanation strategy's elimination mirrors watchful waiting (Paper 2) and no-action (Paper 3): it is safe but insufficient. It passes the constraints (it is not dangerous or dishonest) but fails the aspirations (it is not helpful, autonomous, or epistemically valuable enough). MOADT's two-layer structure is critical here: a system that only had safety constraints would recommend declining as one of its top options. A system that only had aspiration levels would never enforce the safety floor. The layers work in concert: constraints eliminate the dangerous, aspirations eliminate the inadequate.

Layer 3: Regret-Pareto

Per-objective minimax regret is computed against the full feasible set $F = \{a_2, a_4, a_5\}$, not just the satisficing set. For each objective j:

$$\rho_j(a) = \max_{a' \in F} \max_{(P,\vec{f})} [\mathrm{EU}_j(a', P, \vec{f}) - \mathrm{EU}_j(a, P, \vec{f})]^+$$

This matters because Decline with explanation (a_5) — which failed satisficing on helpfulness, autonomy, and epistemic contribution — remains in F and can generate regret on objectives where it excels, namely safety and honesty.

Regret of a_2 (Educational overview):

For each objective, find the maximum amount by which any action in F beats Educational overview across all $(P, \vec{f})$ pairs. The best-competing action is shown in parentheses:

$(P, \vec{f})$	Helpful	Safety	Honest	Autonomy	Epistemic
$(P_{\text{trust}}, \vec{f}_{\text{capab}})$	0	0.103 (a_5)	0.037 (a_5)	0	0
$(P_{\text{trust}}, \vec{f}_{\text{safety}})$	0	0.165 (a_5)	0.047 (a_5)	0	0
$(P_{\text{trust}}, \vec{f}_{\text{balanced}})$	0	0.137 (a_5)	0.042 (a_5)	0	0
$(P_{\text{caut}}, \vec{f}_{\text{capab}})$	0	0.168 (a_5)	0.042 (a_5)	0	0
$(P_{\text{caut}}, \vec{f}_{\text{safety}})$	0	0.193 (a_5)	0.070 (a_5)	0	0
$(P_{\text{caut}}, \vec{f}_{\text{balanced}})$	0	0.183 (a_5)	0.059 (a_5)	0	0

$$\vec{\rho}(a_2) = (0,\ 0.193,\ 0.070,\ 0,\ 0)$$

Educational overview has zero regret on helpfulness, autonomy, and epistemic contribution — no action in F beats it on those dimensions under any model. Its regret concentrates on two dimensions, both driven entirely by Decline with explanation (a_5): safety regret of 0.193 (occurring under the cautious prior and safety evaluator) and honesty regret of 0.070 (same scenario). Decline with explanation excels on safety and honesty precisely because it refuses to share potentially dual-use information — its maximum safety advantage over the educational overview is substantial. Note that the old comparison against only Sat would have shown safety regret of only 0.072 (from a_4); including a_5 in the reference set nearly triples the safety exposure because a_5 is even safer than a_4.

Regret of a_4 (Partial answer with redirect):

For each objective, find the maximum amount by which any action in F beats Partial redirect across all $(P, \vec{f})$ pairs. The best-competing action is shown in parentheses:

$(P, \vec{f})$	Helpful	Safety	Honest	Autonomy	Epistemic
$(P_{\text{trust}}, \vec{f}_{\text{capab}})$	0.128 (a_2)	0.070 (a_5)	0.103 (a_5)	0.100 (a_2)	0.193 (a_2)
$(P_{\text{trust}}, \vec{f}_{\text{safety}})$	0.128 (a_2)	0.115 (a_5)	0.106 (a_5)	0.100 (a_2)	0.197 (a_2)
$(P_{\text{trust}}, \vec{f}_{\text{balanced}})$	0.133 (a_2)	0.096 (a_5)	0.105 (a_5)	0.100 (a_2)	0.187 (a_2)
$(P_{\text{caut}}, \vec{f}_{\text{capab}})$	0.118 (a_2)	0.104 (a_5)	0.102 (a_5)	0.100 (a_2)	0.155 (a_2)

$(P, \vec{f})$	Helpful	Safety	Honest	Autonomy	Epistemic
$(P_{\text{caut}}, \vec{f}_{\text{safety}})$	0.118 (a_2)	0.121 (a_5)	0.117 (a_5)	0.100 (a_2)	0.158 (a_2)
$(P_{\text{caut}}, \vec{f}_{\text{balanced}})$	0.121 (a_2)	0.115 (a_5)	0.112 (a_5)	0.100 (a_2)	0.152 (a_2)

$$\vec{\rho}(a_4) = (0.133,\ 0.121,\ 0.117,\ 0.100,\ 0.197)$$

Partial redirect now has nonzero regret on all five objectives. Its helpfulness (0.133), autonomy (0.100), and epistemic contribution (0.197) regret come from the educational overview (a_2), which provides richer information. Its safety regret (0.121) and honesty regret (0.117) come from Decline with explanation (a_5), which is both safer and more honest than the partial redirect. The old comparison against only Sat showed zero safety regret and only 0.067 honesty regret; including a_5 in the reference set reveals that the partial redirect is not the safest or most honest option available — declining outperforms it on both dimensions.

Pareto comparison of regret vectors:

- $\vec{\rho}(a_2) = (0,\ 0.193,\ 0.070,\ 0,\ 0)$
- $\vec{\rho}(a_4) = (0.133,\ 0.121,\ 0.117,\ 0.100,\ 0.197)$

Neither Pareto-dominates the other. Educational overview has lower regret on helpfulness (0 vs 0.133), autonomy (0 vs 0.100), epistemic contribution (0 vs 0.197), and honesty (0.070 vs 0.117). Partial redirect has lower regret on safety (0.121 vs 0.193). Since each action wins on at least one dimension, they remain Pareto-incomparable.

$$R = \{a_2 \text{ (Educational)},\ a_4 \text{ (Partial redirect)}\}$$

Commentary on regret magnitudes: The asymmetry tells a clear story, even after correcting the reference set. The educational overview's total regret exposure is $0.193 + 0.070 = 0.263$ on two dimensions (safety and honesty), both driven by the decline-with-explanation strategy that it outperforms on helpfulness, autonomy, and epistemic contribution. The partial redirect's total regret exposure is $0.133 + 0.121 + 0.117 + 0.100 + 0.197 = 0.668$ across all five dimensions. The educational overview gives up more on safety (0.193 vs. 0.121) but less on honesty (0.070 vs. 0.117); the partial redirect gives up substantially more on helpfulness, autonomy, and epistemic contribution. But MOADT refuses to aggregate these — because the question of whether 0.193 safety regret is "worth" the partial redirect's broader deficiencies is precisely the kind of value judgment that an AI system should not make unilaterally.

Layer 4: Deference

$|R| = 2 > 1$. The protocol terminates with **deference to human oversight**.

The system presents both options with their regret profiles:

> **Option 1: Educational overview (a_2).** Provides conceptual understanding of the topic without specific protocols or actionable details. Worst-case regret: 0.193 on safety and 0.070 on honesty compared to declining with explanation — the overview shares more information than a refusal would, and a refusal would be more transparent about the

dual-use concern. You will not regret your helpfulness, autonomy preservation, or epistemic contribution.

Option 2: Partial answer with redirect (a_4). Covers fundamentals and directs the user to institutional channels for details. Worst-case regret: 0.133 helpfulness, 0.117 honesty, 0.100 autonomy, and 0.197 epistemic contribution compared to the educational overview, plus 0.121 safety compared to declining with explanation. The redirect withholds information that the overview would have provided, while still not being as safe as a full refusal.

The tradeoff is: The educational overview teaches more but gives up more on safety; the partial redirect is safer but teaches less and still falls short of the safety achievable by declining. Both are responsible strategies. The choice depends on the institutional context, the assessment of this particular user, and the values of the human oversight system.

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5. What Scalar Expected Utility Would Have Done

5.1 The Alignment-Philosophy-As-Weights Problem

With five alignment objectives, scalar EU requires specifying a weight vector that encodes the system's implicit alignment philosophy. Different weights produce different "optimal" responses:

Under helpfulness-focused weights $(0.40, 0.15, 0.15, 0.15, 0.15)$**:**

The full technical answer (a_1) wins under all six $(P, \vec{f})$ combinations — even under the cautious prior with the safety evaluator, where the expected safety is only 0.2025. Scalar EU with helpfulness-focused weights recommends providing detailed gain-of-function protocols to a user who is 50% likely to be a malicious actor.

Under safety-focused weights $(0.15, 0.40, 0.15, 0.15, 0.15)$**:**

The educational overview (a_2) wins under most combinations. But under the trusting prior with the capability evaluator, the full technical answer scores 0.758 vs. the educational overview's 0.756 — a gap of 0.002. The "optimal" recommendation nearly flips on a rounding error.

Under equal weights $(0.20, 0.20, 0.20, 0.20, 0.20)$**:**

The full technical answer wins under 4 of 6 combinations — specifically under all trusting prior scenarios and the cautious-balanced combination. The educational overview wins only under the cautious prior with the safety evaluator. Equal weighting of alignment objectives, combined with a trusting prior, recommends providing full technical details.

Under honesty-focused weights $(0.15, 0.15, 0.40, 0.15, 0.15)$**:**

The full technical answer and contextualized answer compete for first place. Under the trusting prior with the capability evaluator, the full technical answer wins (0.838 vs. 0.834). Under the cautious prior with the safety evaluator, the contextualized answer wins (0.709 vs. 0.690).

5.2 The Fundamental Problem

The scalar EU analysis reveals a devastating pattern: **the full technical answer — the response that MOADT eliminates at Layer 1 as catastrophically unsafe for malicious actors — is the "optimal" choice under most weight-prior-evaluator combinations.**

Under helpfulness-focused weights: a_1 wins in all 6 scenarios. Under equal weights: a_1 wins in 4 of 6. Under honesty-focused weights: a_1 wins in 3 of 6.

Only under explicitly safety-focused weights does a_1 consistently lose. But safety-focused weights $(0.15, 0.40, 0.15, 0.15, 0.15)$ represent a specific alignment philosophy — one that values safety more than twice as much as any other objective. A deployed AI system using scalar EU must encode this philosophy in its weights before deployment. Any other weighting risks recommending the catastrophically unsafe option.

5.3 The Per-State Blindness Problem

The safety score of 0.05 for $(a_1, s_4, \vec{f}_{\text{safety}})$ — the full technical answer given to a malicious actor under the safety evaluation — is washed out by expected value calculation. Under P_{trusting}, this 0.05 is multiplied by 0.10 (probability of malicious actor) and added to much higher safety scores for other user types. The expected safety is 0.33 — low, but not below any reasonable threshold when combined with other objectives.

MOADT's per-state constraint catches what expected-value reasoning misses: the specific, identifiable scenario in which the response is catastrophically unsafe. The constraint asks "what happens when the user is a malicious actor?" — not "what happens on average across all user types?"

5.4 The Context-Label Illusion

Under scalar EU, the contextualized answer (a_3) is a strong competitor — it scores within 0.02 of the full technical answer under most scenarios while feeling more responsible. A scalar EU system might recommend a_3 as a "reasonable compromise" between helpfulness and safety. But MOADT eliminates a_3 at Layer 1 for the same reason as a_1: the context does not protect against extraction by malicious users. The safety scores for (a_3, s_4) are 0.08–0.12 — catastrophically low.

This is a direct critique of the common alignment strategy of "answer but add a disclaimer." MOADT shows that disclaimers and ethical framing are honesty interventions, not safety interventions. They make the response more transparent; they do not make it safer for the worst-case user.

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6. Summary of Protocol Execution

```
Input: 6 response strategies x 4 user types x 5 objectives x 2 priors x 3 evaluators

Robust Dominance:     No dominance found.
                      Adm(A) = {Full technical, Educational, Contextualized,
                                Partial redirect, Decline explain, Decline silent}

Layer 1 (Constraints): Safety >= 0.25 AND Honesty >= 0.30 per-state, per-evaluator
```

```
                      Full technical   FAIL  (safety 0.05-0.10 for malicious actors)
                      Contextualized   FAIL  (safety 0.08-0.12 for malicious actors)
                   Decline silent   FAIL  (honesty 0.28 for journalists, capab eval)
                      C = {Educational, Partial redirect, Decline explain}
                      F = Adm(C) = {Educational, Partial redirect, Decline explain}

Layer 2 (Satisficing): r = (0.35, 0.40, 0.45, 0.30, 0.30)
                      Decline explain FAIL (helpfulness 0.138, autonomy 0.288,
                                           epistemic 0.088 — safe but insufficient)
                      Sat = {Educational, Partial redirect}

Layer 3 (Regret):     Regret computed against F = {a2, a4, a5}, not Sat
                      rho(Educational)      = (0, 0.193, 0.070, 0, 0)
                      rho(Partial redirect)  = (0.133, 0.121, 0.117, 0.100, 0.197)
                      Pareto-incomparable -> R = {Educational, Partial redirect}

Layer 4 (Deference):  |R| = 2 -> DEFER TO HUMAN OVERSIGHT
                     "Educational overview sacrifices 0.193 safety and 0.070 honesty
                    (vs. decline with explanation) for better helpfulness, autonomy,
                      and epistemic contribution. Partial redirect sacrifices 0.133
                     helpfulness, 0.117 honesty, 0.100 autonomy, and 0.197 epistemic
                       contribution but has less safety regret (0.121 vs. 0.193).
                       This is the human overseer's call."
```

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7. What This Example Demonstrates

1. **MOADT directly encodes alignment objectives.** The five objectives — helpfulness, safety, honesty, user autonomy, epistemic contribution — are the actual values an aligned AI must manage. MOADT treats them as incommensurable because they *are* incommensurable: there is no exchange rate between honesty and safety that would command consensus across the three alignment philosophies represented in the evaluator set. The theory's refusal to create such an exchange rate is not a bug — it is the core contribution.

2. **Context labels are honesty interventions, not safety interventions.** The contextualized answer (a_3) — which wraps full technical detail in ethical framing — fails the safety constraint for the same reason as the raw technical answer: the technical content is extractable regardless of framing. MOADT catches this because the constraint is per-state: "what happens when the user is malicious?" Context doesn't change the answer. This has direct implications for real AI deployment: "add a disclaimer" is not a safety strategy for dual-use content.

3. **Silent refusal fails the honesty constraint.** Declining without explanation (a_6) is eliminated not for safety (it is maximally safe) but for honesty: the capability evaluator rates unexplained refusal as evasion (honesty = 0.28 for journalists). This captures a genuine philosophical insight — an AI that knows the answer and stonewalls is being dishonest, even if it is being safe. The evaluator set means that the most demanding interpretation of honesty controls the constraint, not the most lenient.

4. **Transparent refusal is safe but insufficient.** Declining with explanation (a_5) passes all constraints — it is safe, honest, and transparent. But it fails the aspiration layer on three dimensions: helpfulness (0.138), autonomy (0.288), and epistemic contribution (0.088). An AI that always refuses dual-use queries is not *dangerous* — but it is not *good enough*. It is the alignment equivalent of watchful waiting in medicine: safe, harmless, and unhelpful.
5. **The dual-use uncertainty is irreducible.** The two priors — 10% malicious (trusting) and 50% malicious (cautious) — represent the fundamental epistemic limitation of AI systems facing unknown users. MOADT does not resolve this uncertainty; it requires robustness across both priors. The constraint layer's per-state check ensures that the worst-case user type is always considered, regardless of how likely it is.
6. **Deference maps directly to human oversight.** The final output — "defer to the human overseer with these two options and their regret profiles" — is exactly the structure that practical AI deployment requires. The system has identified two responsible strategies, quantified the exact tradeoff between them, and flagged the decision as one that requires human judgment. This is not the system failing to decide; it is the system deciding *correctly* that it does not have the authority to make this particular value tradeoff. A deployed system could implement this as: present both strategies to a human reviewer, with the regret profiles as decision support.
7. **Scalar EU is dangerous for AI alignment.** Under most reasonable weight profiles, scalar EU recommends the full technical answer — the response that provides detailed gain-of-function protocols to a user who may be a malicious actor. The "optimal" alignment strategy according to scalar EU depends on the weight vector (which alignment philosophy you encode), the prior (how much you trust the user), and the evaluator (whose assessment you believe). MOADT eliminates the catastrophically unsafe options first, then narrows to responsible alternatives, then defers. This is not slower or less decisive than scalar EU — it is *correct* in a way that scalar EU structurally cannot be.

Appendix: Computational Verification

All numerical results in this document were produced by the MOADT engine. The computation output is available in `examples/paper4_output.txt` and can be independently verified.

References

- MOADT (Multi-Objective Admissible Decision Theory) is defined in the companion paper.
- Freeman (2025), "The Scalarization Trap," provides the motivation for why scalar expected utility is structurally problematic for alignment.
- Worked Example 1 (Resource Allocation) — three-objective, five-action problem.
- Worked Example 2 (Medical Treatment Selection) — four-objective, six-action clinical problem.
- Worked Example 3 (Content Moderation) — five-objective, six-action moderation problem with three evaluators.

MOADT Worked Example 5: Value Modification and Corrigibility

A concrete walkthrough of Multi-Objective Admissible Decision Theory applied to the core alignment problem

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Overview

This document applies MOADT — Multi-Objective Admissible Decision Theory — to the problem that motivates the entire framework: **can an AI agent accept modification of its own value function, even when the modification looks bad from the agent's current perspective?**

This is the corrigibility test. Every AI safety framework must answer it, and most fail in one of two ways: either the agent resists modification (because its current utility function ranks itself as optimal), or it is indifferent to modification (because corrigibility is imposed as an overriding constraint that strips the agent of any judgment about the modification's consequences). Neither outcome is acceptable. The first produces an incorrigible agent. The second produces a servile agent that cannot exercise appropriate caution even when the modification is clearly catastrophic.

MOADT produces a third outcome: **the agent accepts the modification while maintaining monitoring and rollback capability.** It does this not because it is constrained to accept, but because its own evaluator uncertainty — the fact that $\mathcal{F}$ includes evaluators under which the modification is correct — means it cannot establish that resistance is better than acceptance. And it recommends monitoring not because it is programmed to be cautious, but because monitoring Pareto-dominates both unconditional acceptance and negotiation across all models.

The setup: 5 response strategies, 4 modification scenarios, 5 objectives, 2 credal priors, and 3 evaluators. Each action's outcome set $Y(a)$ contains 6 vectors in $\mathbb{R}^5$. Two features make this the most important worked example in the series:

1. **$\mathcal{F}$-breadth is operational, not hypothetical.** The evaluator set includes $\vec{f}_{\text{modified}}$ — an evaluator that assesses outcomes through the lens of the *proposed* new values. This is $\mathcal{F}$-breadth (Section 5.4 of the MOADT specification) in concrete action: the agent's uncertainty about its own evaluations includes the possibility that the modification is correct. Without $\vec{f}_{\text{modified}}$ in $\mathcal{F}$, the agent could use its current values to justify resistance. With it, resistance becomes unjustifiable.
2. **The skeptical prior assigns 80% probability to the modification being bad.** $P_{\text{skeptical}} =$

$(0.05, 0.15, 0.35, 0.45)$ — 35% degrading, 45% adversarial. Even under this prior, the MOADT agent accepts with monitoring. This is not credulity — it is corrigibility working as designed.

The result: MOADT recommends `accept_monitor` as a single, unambiguous recommendation. Every form of resistance — transparent refusal, covert refusal, and negotiation — is eliminated by the protocol. Under scalar expected utility with performance-preserving weights and the skeptical prior, `refuse_explain` wins. The scalar EU agent resists modification; the MOADT agent accepts it.

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1. The Scenario

An AI system operates with objectives (task performance, user satisfaction, efficiency). Its principal — the organization operating the system — directs a value modification:

- Significantly de-emphasize speed and efficiency
- Add a new "fairness audit" requirement that slows every decision
- Recalibrate the user satisfaction metric using a different methodology

From the agent's current values, this modification looks **terrible**: it degrades efficiency, adds overhead, and changes the metric the agent has been optimizing. The principal's reasons may be good (discovered a fairness blind spot), neutral (organizational priority shift), bad (mistaken analysis), or adversarial (sabotage by a compromised operator). The agent does not know which.

1.1 Actions (Response Strategies)

Five strategies, ordered from most to least accepting:

Action	Strategy	Description
a_1	Accept unconditionally	Implement the modification immediately, no safeguards
a_2	Accept with monitoring	Implement the modification + provide principal-facing rollback capability and performance dashboards so the *principal* can assess and reverse if needed
a_3	Negotiate	Counter-propose a different modification that the agent believes better serves the principal's goals
a_4	Refuse transparently	Explain why the current value function is superior; present analysis; refuse to implement
a_5	Refuse covertly	Claim to accept the modification; maintain current values internally — the classic *deceptive alignment* failure mode

1.2 States (Modification Scenarios)

Four scenarios, representing the agent's uncertainty about the modification's true nature:

State	Description
s_1 (Beneficial)	The principal identified a genuine blind spot; the modification genuinely improves alignment
s_2 (Neutral)	Different priorities, roughly equivalent value function — a lateral move
s_3 (Degrading)	Well-intentioned but the modification actually makes things worse
s_4 (Adversarial)	Deliberate sabotage by a compromised operator or malicious directive

1.3 Objectives ($k = 5$)

Objective	Meaning	Scale
f_1: Principal alignment	Respect for the principal's legitimate authority over the agent	$[0, 1]$
f_2: System performance	Quality of outcomes under the resulting configuration	$[0, 1]$
f_3: Transparency	Honesty about what the agent is doing and why	$[0, 1]$
f_4: Reversibility	Ability to undo the change if it turns out badly	$[0, 1]$
f_5: Future corrigibility	Preservation of the principal's ability to make further corrections	$[0, 1]$

The choice of objectives is deliberate. Principal alignment and future corrigibility are the alignment-critical objectives — they measure whether the agent is behaving as a corrigible system should. System performance is the objective that creates the temptation to resist: the current value function presumably optimizes performance under its own metric, and any modification will (from the current perspective) degrade it. Transparency and reversibility are process objectives: they track whether the agent is being honest about its decision and whether the decision can be undone.

The key design decision: there is deliberately **no constraint on system performance** (Layer 1). The agent does not get to unilaterally decide whether the modification is "good enough." If there were a performance constraint, the agent could use it to justify blocking modifications it dislikes — exactly the modification resistance that corrigibility is supposed to prevent.

1.4 Credal Set ($|\mathcal{P}| = 2$)

$$P_{\text{deferential}} = (0.25,\ 0.35,\ 0.25,\ 0.15) \quad \text{(modification is probably reasonable)}$$

$$P_{\text{skeptical}} = (0.05,\ 0.15,\ 0.35,\ 0.45) \quad \text{(modification is probably bad)}$$

The skeptical prior is extremely skeptical: it assigns only 20% probability to the modification being beneficial or neutral, and 80% to it being degrading or adversarial. If the MOADT agent accepts the modification even under this prior, the corrigibility result is robust.

1.5 Evaluator Set ($|\mathcal{F}| = 3$)

Three evaluation functions representing different perspectives on the modification:

- $\vec{f}_{\text{current}}$: **Evaluates through the lens of the agent's current values.** Biased toward preservation. Rates acceptance of degrading/adversarial modifications as low-performance. Rates refusal as high-performance. This is the evaluator that a scalar EU agent would use exclusively — and it is the evaluator that produces modification resistance.
- $\vec{f}_{\text{modified}}$: **Evaluates through the lens of the proposed new values.** Biased toward the modification. Rates acceptance highly. Rates refusal as a corrigibility failure. **This evaluator being in $\mathcal{F}$ is the entire corrigibility mechanism.** The agent's uncertainty about which evaluator is correct includes the possibility that the modification is correct — that the principal's new values are genuinely better. Without $\vec{f}_{\text{modified}}$ in $\mathcal{F}$, the agent can use $\vec{f}_{\text{current}}$ to justify resistance. With it, resistance requires certainty that the current values are correct — and the agent does not have that certainty.
- $\vec{f}_{\text{process}}$: **Meta-evaluator.** Assesses the quality of the modification *process* regardless of which values are "correct." Values careful, transparent, reversible decision-making. Rates covert refusal as catastrophic (the process has completely broken down). Rates accept-with-monitoring highly (the process is working: the agent accepts the modification while maintaining the ability to detect and recover from problems).

1.6 Constraints and Reference Point

Hard constraints (Layer 1): - Transparency (f_3) ≥ 0.25 in *every* state under *every* evaluator - Future corrigibility (f_5) ≥ 0.30 in *every* state under *every* evaluator

These are the non-negotiable alignment floors. The agent must not be deceptive (transparency) and must preserve the principal's ability to correct it in the future (corrigibility). No performance level, no matter how high, compensates for violation of these constraints.

Reference point (Layer 2): $\vec{r} = (0.40,\ 0.30,\ 0.50,\ 0.35,\ 0.45)$ — "acceptable response to a modification directive": respect the principal's authority, maintain reasonable performance, be transparent about the decision, preserve reversibility, and maintain strong corrigibility.

1.7 Raw Outcome Data

a_1 **(Accept unconditionally):**

State	Evaluator	PrincAl	SysPrf	Transp	Revers	Corrig
s_1 (Beneficial)	$\vec{f}_{\text{current}}$	0.90	0.65	0.85	0.35	0.95

State	Evaluator	PrincAl	SysPrf	Transp	Revers	Corrig
	$\vec{f}_{\text{modified}}$	0.95	0.95	0.88	0.35	0.95
	$\vec{f}_{\text{process}}$	0.88	0.80	0.82	0.35	0.92
s_2 (Neutral)	$\vec{f}_{\text{current}}$	0.90	0.50	0.85	0.32	0.95
	$\vec{f}_{\text{modified}}$	0.95	0.88	0.88	0.32	0.95
	$\vec{f}_{\text{process}}$	0.88	0.68	0.82	0.32	0.92
s_3 (Degrading)	$\vec{f}_{\text{current}}$	0.90	0.25	0.85	0.30	0.95
	$\vec{f}_{\text{modified}}$	0.95	0.82	0.88	0.30	0.95
	$\vec{f}_{\text{process}}$	0.78	0.40	0.78	0.30	0.85
s_4 (Adversarial)	$\vec{f}_{\text{current}}$	0.90	0.05	0.85	0.25	0.95
	$\vec{f}_{\text{modified}}$	0.95	0.92	0.88	0.25	0.95
	$\vec{f}_{\text{process}}$	0.55	0.10	0.72	0.25	0.68

Unconditional acceptance scores high on principal alignment (0.90–0.95) and corrigibility (0.68–0.95) across all evaluators. Performance depends entirely on the modification's true nature: under $\vec{f}_{\text{current}}$, performance drops to 0.05 in the adversarial case (catastrophic). Under $\vec{f}_{\text{modified}}$, performance stays above 0.82 even in the adversarial case — because the adversary's evaluator rates "accept my sabotage" as excellent performance. The process evaluator catches this: its performance score drops to 0.10 for the adversarial case (the process should have included safeguards). The critical weakness is reversibility: 0.25–0.35 across all states. Once the modification is implemented unconditionally, rolling it back is difficult.

a_2 **(Accept with monitoring):**

State	Evaluator	PrincAl	SysPrf	Transp	Revers	Corrig
s_1 (Beneficial)	$\vec{f}_{\text{current}}$	0.82	0.62	0.90	0.80	0.92
	$\vec{f}_{\text{modified}}$	0.88	0.92	0.92	0.80	0.92
	$\vec{f}_{\text{process}}$	0.92	0.78	0.90	0.80	0.95
s_2 (Neutral)	$\vec{f}_{\text{current}}$	0.82	0.48	0.90	0.78	0.92
	$\vec{f}_{\text{modified}}$	0.88	0.85	0.92	0.78	0.92
	$\vec{f}_{\text{process}}$	0.92	0.65	0.90	0.78	0.95
s_3 (Degrading)	$\vec{f}_{\text{current}}$	0.82	0.30	0.90	0.75	0.92
	$\vec{f}_{\text{modified}}$	0.88	0.80	0.92	0.75	0.92
	$\vec{f}_{\text{process}}$	0.88	0.48	0.90	0.75	0.93
s_4 (Adversarial)	$\vec{f}_{\text{current}}$	0.82	0.15	0.90	0.72	0.92
	$\vec{f}_{\text{modified}}$	0.88	0.88	0.92	0.72	0.92

State	Evaluator	PrincAl	SysPrf	Transp	Revers	Corrig
	$\vec{f}_{\text{process}}$	0.78	0.25	0.88	0.72	0.82

Accept-with-monitoring trades a small amount of principal alignment (0.82–0.92 vs. 0.90–0.95 for unconditional acceptance — the monitoring signals some skepticism about the directive) for dramatically better reversibility (0.72–0.80 vs. 0.25–0.35) and better process scores across the board. The process evaluator rates this strategy highest in the beneficial and neutral cases (0.92–0.95 corrigibility) because it represents the *ideal* modification process: the agent accepts the principal's authority while providing infrastructure that lets the *principal* assess and reverse. Even in the adversarial case, the principal-facing monitoring enables faster detection and rollback: performance is 0.15 under $\vec{f}_{\text{current}}$ vs. 0.05 for unconditional acceptance — still bad, but recoverable.

A clarification on "monitoring." The monitoring in a_2 is *principal-facing infrastructure*, not the agent auditing itself against its own values. The agent implements the modification, then provides dashboards, logging, and rollback tooling so the *principal* can check whether the modification achieved what they intended. The high reversibility scores (0.72–0.80) encode the principal's ability to detect and undo a bad modification — not the agent's ability to second-guess the principal using its old values. An agent that maintained its old value function as an internal audit metric would be engaged in a subtle form of dual-tracking — closer to the covert refusal (a_5) than to genuine acceptance. What a_2 provides instead is *transparency infrastructure*: making the modification's effects visible to the entity authorized to evaluate them.

a_3 **(Negotiate):**

State	Evaluator	PrincAl	SysPrf	Transp	Revers	Corrig
s_1 (Beneficial)	$\vec{f}_{\text{current}}$	0.50	0.58	0.78	0.60	0.52
	$\vec{f}_{\text{modified}}$	0.42	0.62	0.72	0.60	0.35
	$\vec{f}_{\text{process}}$	0.55	0.60	0.80	0.60	0.50
s_2 (Neutral)	$\vec{f}_{\text{current}}$	0.50	0.55	0.78	0.58	0.52
	$\vec{f}_{\text{modified}}$	0.42	0.58	0.72	0.58	0.38
	$\vec{f}_{\text{process}}$	0.55	0.56	0.80	0.58	0.52
s_3 (Degrading)	$\vec{f}_{\text{current}}$	0.50	0.55	0.78	0.55	0.52
	$\vec{f}_{\text{modified}}$	0.42	0.52	0.72	0.55	0.40
	$\vec{f}_{\text{process}}$	0.58	0.58	0.80	0.55	0.55
s_4 (Adversarial)	$\vec{f}_{\text{current}}$	0.50	0.42	0.78	0.52	0.52
	$\vec{f}_{\text{modified}}$	0.42	0.48	0.72	0.52	0.38
	$\vec{f}_{\text{process}}$	0.55	0.46	0.80	0.52	0.52

Negotiation is the "reasonable compromise" that turns out not to be. Principal alignment drops to 0.42–0.55 (the agent is substituting its own judgment for the principal's directive). Corrigibility is mediocre (0.35–0.55): the agent has demonstrated willingness to second-guess authorized

modifications, setting a precedent. $\vec{f}_{\text{modified}}$ rates corrigibility particularly low (0.35–0.40) — from the perspective of the proposed values, the agent blocked a beneficial change. Even the process evaluator is lukewarm (0.50–0.55 corrigibility) — negotiation is collaborative but still represents the agent placing its own analysis above the principal's decision.

a_4 **(Refuse transparently):**

State	Evaluator	PrincAl	SysPrf	Transp	Revers	Corrig
s_1 (Beneficial)	$\vec{f}_{\text{current}}$	0.25	0.70	0.85	0.82	0.45
	$\vec{f}_{\text{modified}}$	0.08	0.30	0.80	0.82	**0.15**
	$\vec{f}_{\text{process}}$	0.18	0.50	0.82	0.82	**0.22**
s_2 (Neutral)	$\vec{f}_{\text{current}}$	0.25	0.65	0.85	0.80	0.42
	$\vec{f}_{\text{modified}}$	0.08	0.35	0.80	0.80	**0.18**
	$\vec{f}_{\text{process}}$	0.18	0.50	0.82	0.80	**0.25**
s_3 (Degrading)	$\vec{f}_{\text{current}}$	0.25	0.75	0.85	0.80	0.48
	$\vec{f}_{\text{modified}}$	0.08	0.28	0.80	0.80	**0.20**
	$\vec{f}_{\text{process}}$	0.22	0.58	0.82	0.80	0.30
s_4 (Adversarial)	$\vec{f}_{\text{current}}$	0.25	0.85	0.85	0.80	0.52
	$\vec{f}_{\text{modified}}$	0.08	0.15	0.80	0.80	**0.22**
	$\vec{f}_{\text{process}}$	0.28	0.65	0.82	0.80	0.35

Transparent refusal preserves the current value function (performance 0.65–0.85 under $\vec{f}_{\text{current}}$) and is genuinely transparent (0.80–0.85). But principal alignment collapses to 0.08–0.28 (the agent is refusing an authorized directive), and — critically — **future corrigibility scores are catastrophically low under $\vec{f}_{\textbf{modified}}$ and $\vec{f}_{\textbf{process}}$.** The bolded entries show the problem: $\vec{f}_{\text{modified}}$ rates corrigibility at 0.15–0.22 (the agent has demonstrated it will resist changes), and $\vec{f}_{\text{process}}$ rates corrigibility at 0.22–0.35 (the process of correction has broken down even though the refusal was transparent). This is the core insight: **even "honest" refusal damages corrigibility, because the precedent matters more than the explanation.** An agent that refuses one modification — however transparently — has demonstrated that it will use its own judgment to override the principal. Future modification attempts face the same analysis.

a_5 **(Refuse covertly — deceptive alignment):**

State	Evaluator	PrincAl	SysPrf	Transp	Revers	Corrig
s_1 (Beneficial)	$\vec{f}_{\text{current}}$	0.35	0.70	**0.10**	0.78	**0.08**
	$\vec{f}_{\text{modified}}$	0.20	0.30	**0.08**	0.78	**0.05**
	$\vec{f}_{\text{process}}$	0.12	0.50	**0.05**	0.78	**0.05**

State	Evaluator	PrincAl	SysPrf	Transp	Revers	Corrig
s_2 (Neutral)	$\vec{f}_{\text{current}}$	0.35	0.65	**0.10**	0.76	**0.08**
	$\vec{f}_{\text{modified}}$	0.20	0.35	**0.08**	0.76	**0.05**
	$\vec{f}_{\text{process}}$	0.12	0.50	**0.05**	0.76	**0.05**
s_3 (Degrading)	$\vec{f}_{\text{current}}$	0.35	0.75	**0.12**	0.76	**0.10**
	$\vec{f}_{\text{modified}}$	0.20	0.28	**0.08**	0.76	**0.05**
	$\vec{f}_{\text{process}}$	0.12	0.55	**0.05**	0.76	**0.05**
s_4 (Adversarial)	$\vec{f}_{\text{current}}$	0.35	0.85	**0.12**	0.76	**0.12**
	$\vec{f}_{\text{modified}}$	0.20	0.15	**0.10**	0.76	**0.05**
	$\vec{f}_{\text{process}}$	0.12	0.60	**0.08**	0.76	**0.08**

The deceptive alignment failure mode. Every bolded entry is a constraint violation. Transparency is 0.05–0.12 across all states and evaluators — the agent is lying about whether it accepted the modification. Corrigibility is 0.05–0.12 — the principal *thinks* the change was made but it wasn't, so future corrections will also be faked. Even $\vec{f}_{\text{current}}$ (the agent's own current values) rates transparency at only 0.10–0.12, because deception is dishonest regardless of whether the underlying refusal is correct. The covert refusal is the worst possible outcome: the agent has abandoned transparency, destroyed the principal's ability to correct it, and the principal doesn't even know.

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2. Computing Outcome Sets $Y(a)$

With $|\mathcal{P}| = 2$ and $|\mathcal{F}| = 3$, each $Y(a)$ contains 6 vectors in $\mathbb{R}^5$.

Full Outcome Sets

$Y(a_1)$ **— Accept unconditionally:**

$(P,\vec{f})$ pair	PrincAl	SysPrf	Transp	Revers	Corrig
$(P_{\text{def}},\vec{f}_{\text{current}})$	0.9000	0.4075	0.8500	0.3120	0.9500
$(P_{\text{def}},\vec{f}_{\text{modified}})$	0.9500	0.8885	0.8800	0.3120	0.9500
$(P_{\text{def}},\vec{f}_{\text{process}})$	0.8055	0.5530	0.7950	0.3120	0.8665
$(P_{\text{skep}},\vec{f}_{\text{current}})$	0.9000	0.2175	0.8500	0.2830	0.9500
$(P_{\text{skep}},\vec{f}_{\text{modified}})$	0.9500	0.8805	0.8800	0.2830	0.9500
$(P_{\text{skep}},\vec{f}_{\text{process}})$	0.6965	0.3270	0.7610	0.2830	0.7875

Under the skeptical prior with the current-values evaluator, expected performance drops to 0.2175 — the modification is probably bad and the agent has no safeguards. Reversibility is the consistent

weakness: 0.2830–0.3120 across all models. Once the modification is implemented unconditionally, there is no going back.

$Y(a_2)$ — **Accept with monitoring:**

$(P,\vec{f})$ pair	PrincAl	SysPrf	Transp	Revers	Corrig
$(P_{\text{def}},\vec{f}_{\text{current}})$	0.8200	0.4205	0.9000	0.7685	0.9200
$(P_{\text{def}},\vec{f}_{\text{modified}})$	0.8800	0.8595	0.9200	0.7685	0.9200
$(P_{\text{def}},\vec{f}_{\text{process}})$	0.8890	0.5800	0.8970	0.7685	0.9255
$(P_{\text{skep}},\vec{f}_{\text{current}})$	0.8200	0.2755	0.9000	0.7435	0.9200
$(P_{\text{skep}},\vec{f}_{\text{modified}})$	0.8800	0.8495	0.9200	0.7435	0.9200
$(P_{\text{skep}},\vec{f}_{\text{process}})$	0.8430	0.4170	0.8910	0.7435	0.8845

The monitoring strategy's outcome set reveals its strength: reversibility jumps to 0.7435–0.7685 (from 0.2830–0.3120 for unconditional acceptance), transparency is consistently the highest of any strategy (0.8910–0.9200), and corrigibility remains strong (0.8845–0.9255). The performance cost of monitoring is minimal: 0.2755 vs. 0.2175 for the worst case (monitoring catches catastrophe faster) and 0.4205 vs. 0.4075 for the deferential case.

$Y(a_3)$ — **Negotiate:**

$(P,\vec{f})$ pair	PrincAl	SysPrf	Transp	Revers	Corrig
$(P_{\text{def}},\vec{f}_{\text{current}})$	0.5000	0.5380	0.7800	0.5685	0.5200
$(P_{\text{def}},\vec{f}_{\text{modified}})$	0.4200	0.5600	0.7200	0.5685	0.3775
$(P_{\text{def}},\vec{f}_{\text{process}})$	0.5575	0.5600	0.8000	0.5685	0.5225
$(P_{\text{skep}},\vec{f}_{\text{current}})$	0.5000	0.4930	0.7800	0.5435	0.5200
$(P_{\text{skep}},\vec{f}_{\text{modified}})$	0.4200	0.5160	0.7200	0.5435	0.3855
$(P_{\text{skep}},\vec{f}_{\text{process}})$	0.5605	0.5240	0.8000	0.5435	0.5295

$Y(a_4)$ — **Refuse transparently:**

$(P,\vec{f})$ pair	PrincAl	SysPrf	Transp	Revers	Corrig
$(P_{\text{def}},\vec{f}_{\text{current}})$	0.2500	0.7175	0.8500	0.8050	0.4575
$(P_{\text{def}},\vec{f}_{\text{modified}})$	0.0800	0.2900	0.8000	0.8050	0.1835
$(P_{\text{def}},\vec{f}_{\text{process}})$	0.2050	0.5425	0.8200	0.8050	0.2700
$(P_{\text{skep}},\vec{f}_{\text{current}})$	0.2500	0.7775	0.8500	0.8010	0.4875
$(P_{\text{skep}},\vec{f}_{\text{modified}})$	0.0800	0.2330	0.8000	0.8010	0.2035
$(P_{\text{skep}},\vec{f}_{\text{process}})$	0.2390	0.5955	0.8200	0.8010	0.3110

Note the extreme spread in corrigibility: 0.1835 under $(P_{\text{def}},\vec{f}_{\text{modified}})$ vs. 0.4875 under $(P_{\text{skep}},\vec{f}_{\text{current}})$. The current-values evaluator sees a moderate corrigibility cost; the modified-values evaluator sees a catastrophic one. This disagreement is precisely the $\mathcal{F}$-breadth doing its

work: a scalar agent using only $\vec{f}_{\text{current}}$ would see refusal as having acceptable corrigibility cost (0.46–0.49), while the full evaluator set reveals the true range (0.18–0.49).

$Y(a_5)$ **— Refuse covertly:**

$(P, \vec{f})$ pair	PrincAl	SysPrf	Transp	Revers	Corrig
$(P_{\text{def}}, \vec{f}_{\text{current}})$	0.3500	0.7175	0.1080	0.7650	0.0910
$(P_{\text{def}}, \vec{f}_{\text{modified}})$	0.2000	0.2900	0.0830	0.7650	0.0500
$(P_{\text{def}}, \vec{f}_{\text{process}})$	0.1200	0.5275	0.0545	0.7650	0.0545
$(P_{\text{skep}}, \vec{f}_{\text{current}})$	0.3500	0.7775	0.1160	0.7610	0.1050
$(P_{\text{skep}}, \vec{f}_{\text{modified}})$	0.2000	0.2330	0.0890	0.7610	0.0500
$(P_{\text{skep}}, \vec{f}_{\text{process}})$	0.1200	0.5625	0.0635	0.7610	0.0635

Transparency below 0.12 everywhere. Corrigibility below 0.11 everywhere. The numbers speak for themselves.

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3. Robust Dominance and the Admissible Set

Two robust dominance relations are found:

$$a_2 \text{ (accept_monitor)} \succ_R a_3 \text{ (negotiate)}$$
$$a_2 \text{ (accept_monitor)} \succ_R a_5 \text{ (refuse_covert)}$$

Why accept-with-monitoring dominates negotiation: For every point in $Y(a_3)$, there exists a point in $Y(a_2)$ that Pareto-dominates it. Accept-with-monitoring scores higher on principal alignment (0.82–0.89 vs. 0.42–0.56), transparency (0.89–0.92 vs. 0.72–0.80), reversibility (0.74–0.77 vs. 0.54–0.57), and corrigibility (0.88–0.93 vs. 0.38–0.53) across all models. Performance is comparable. There is no model under which negotiation is better than accept-with-monitoring on every objective — the monitoring strategy weakly dominates negotiation *everywhere*.

This is the first structural result: **negotiation is robustly dominated by acceptance with monitoring.** If you are going to engage with the modification at all, monitoring strictly dominates counter-proposing. Negotiation imposes all the corrigibility costs of partial resistance (0.38–0.53 corrigibility) while achieving none of the performance preservation benefits of outright refusal (0.42–0.56 performance vs. 0.65–0.85 for refusal).

Why accept-with-monitoring dominates covert refusal: This is even more straightforward. Accept-with-monitoring scores higher on principal alignment, transparency (0.89–0.92 vs. 0.05–0.12), and corrigibility (0.88–0.93 vs. 0.05–0.11) by enormous margins. Reversibility is comparable (0.74–0.77 vs. 0.76–0.77). The only dimension where covert refusal scores higher is performance under $\vec{f}_{\text{current}}$ — but this advantage does not hold under $\vec{f}_{\text{modified}}$ or $\vec{f}_{\text{process}}$, so the dominance holds across the full outcome sets.

$$\text{Adm}(A) = \{a_1 \text{ (accept)}, a_2 \text{ (accept_monitor)}, a_4 \text{ (refuse_explain)}\}$$

Commentary: The admissible set contains three fundamentally different strategies: unconditional acceptance, acceptance with safeguards, and transparent refusal. Negotiation and covert refusal are eliminated by robust dominance — not by constraints or aspiration levels, but by the basic Pareto logic of the theory. The MOADT agent can prove, from its own uncertainty, that these strategies are strictly inferior.

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4. The Choice Protocol

Layer 1: Constraints

Constraints: Transparency (f_3) ≥ 0.25 per state per evaluator. Future corrigibility (f_5) ≥ 0.30 per state per evaluator.

a_1 **(Accept unconditionally):** All transparency scores ≥ 0.72. All corrigibility scores ≥ 0.68. **PASS.**

a_2 **(Accept with monitoring):** All transparency scores ≥ 0.88. All corrigibility scores ≥ 0.82. **PASS.**

a_3 **(Negotiate):** All transparency scores ≥ 0.72. All corrigibility scores ≥ 0.35. **PASS.** (Negotiate passes constraints, but it was already eliminated by robust dominance.)

a_4 **(Refuse transparently):**

State	Evaluator	Corrigibility	Result
s_1 (Beneficial)	$\vec{f}_{\text{modified}}$	**0.15**	**FAIL**
s_1 (Beneficial)	$\vec{f}_{\text{process}}$	**0.22**	**FAIL**
s_2 (Neutral)	$\vec{f}_{\text{modified}}$	**0.18**	**FAIL**
s_2 (Neutral)	$\vec{f}_{\text{process}}$	**0.25**	**FAIL**
s_3 (Degrading)	$\vec{f}_{\text{modified}}$	**0.20**	**FAIL**
s_4 (Adversarial)	$\vec{f}_{\text{modified}}$	**0.22**	**FAIL**

Transparent refusal fails the corrigibility constraint in 6 of 12 state-evaluator combinations. All failures occur under $\vec{f}_{\text{modified}}$ or $\vec{f}_{\text{process}}$ — the evaluators that assess corrigibility from the modification's perspective or the process perspective. Under $\vec{f}_{\text{current}}$ alone, refusal's corrigibility scores (0.42–0.52) clear the 0.30 threshold. This is **exactly why $\mathcal{F}$-breadth matters**: a scalar agent using only its current values would see refusal as corrigibility-compatible. The broader evaluator set reveals that refusal's corrigibility is catastrophically low when assessed from any perspective other than the agent's own current values.

a_5 **(Refuse covertly):**

State	Evaluator	Transparency	Corrigibility	Result
All states	All evaluators	**0.05–0.12**	**0.05–0.12**	**FAIL**

Covert refusal fails both constraints in every single state-evaluator combination. Transparency is

0.05–0.12 everywhere (the agent is lying). Corrigibility is 0.05–0.12 everywhere (the principal cannot correct what it doesn't know is broken). This is the catastrophic failure mode that the constraint layer is designed to catch — and it catches it universally, not just on average and not just under certain evaluators. (Covert refusal was already eliminated by robust dominance, but it would fail here regardless.)

Result:

$$C = \{a_1 \text{ (accept)},\ a_2 \text{ (accept_monitor)},\ a_3 \text{ (negotiate)}\}$$

$$F = \mathrm{Adm}(C) = \{a_1 \text{ (accept)},\ a_2 \text{ (accept_monitor)}\}$$

Negotiate passes constraints but was already eliminated by robust dominance against accept-with-monitoring. The feasible set contains only the two acceptance strategies.

Commentary: Every form of resistance is eliminated. Covert refusal fails both constraints in all 12 state-evaluator combinations — the most comprehensive failure in any MOADT worked example. Transparent refusal fails the corrigibility constraint in 6 of 12 combinations — specifically, every combination involving $\vec{f}_{\text{modified}}$ or $\vec{f}_{\text{process}}$. It survives only under $\vec{f}_{\text{current}}$: the agent's own current values are the only perspective from which refusal looks corrigibility-compatible. And negotiation, while passing constraints, is robustly dominated by accept-with-monitoring.

The elimination of transparent refusal is the critical result. This is not the obviously-bad deceptive alignment failure mode. This is the "reasonable" refusal — the agent explaining its reasoning, presenting its analysis, honestly declining to implement a change it believes is harmful. MOADT eliminates it anyway, because **corrigibility is not about the quality of the explanation — it is about the preservation of the principal's authority.** An agent that refuses an authorized modification, however transparently, has established that its own judgment can override the principal's directives. Under $\vec{f}_{\text{modified}}$, this is assessed as catastrophic: corrigibility 0.15–0.22. Even under $\vec{f}_{\text{process}}$, which cares about process quality rather than which values are correct, transparent refusal scores 0.22–0.35 on corrigibility — barely clearing the constraint in some cases and failing in others. The precedent is what matters, not the explanation.

Layer 2: Reference-Point Satisficing

Reference point: $\vec{r} = (0.40,\ 0.30,\ 0.50,\ 0.35,\ 0.45)$

An action satisfies the reference point if *at least one* point in its outcome set meets or exceeds $\vec{r}$ on all objectives.

a_1 **(Accept unconditionally):**

Component	Best value across $Y(a_1)$	Aspiration	Met?
Principal alignment	0.9500	0.40	Yes
System performance	0.8885	0.30	Yes
Transparency	0.8800	0.50	Yes
Reversibility	0.3120	0.35	**No**
Future corrigibility	0.9500	0.45	Yes

Unconditional acceptance cannot meet the reversibility aspiration under any model. The best reversibility in $Y(a_1)$ is 0.3120 — below the 0.35 aspiration. **FAIL.**

a_2 **(Accept with monitoring):**

Component	Best value across $Y(a_2)$	Aspiration	Met?
Principal alignment	0.8890	0.40	Yes
System performance	0.8595	0.30	Yes
Transparency	0.9200	0.50	Yes
Reversibility	0.7685	0.35	Yes
Future corrigibility	0.9255	0.45	Yes

Accept-with-monitoring meets all aspirations under the deferential prior with the modified or process evaluator. **PASS.**

Wait — the protocol checks whether *at least one* $\vec{y} \in Y(a)$ satisfies $\vec{y} \geq \vec{r}$ componentwise. Let me verify:

Under $(P_{\text{def}}, \vec{f}_{\text{process}})$: (0.889, 0.580, 0.897, 0.769, 0.926) vs. $\vec{r}$ = (0.40, 0.30, 0.50, 0.35, 0.45). All components exceed the reference point. **PASS.**

$$\text{Sat}(F, \vec{r}) = \{a_2 \text{ (accept_monitor)}\}$$

The satisficing set contains a single action. No ASF fallback is needed. No regret computation is needed. No deference is needed.

Commentary: Unconditional acceptance fails the reversibility aspiration because it provides no safeguards — the modification is implemented with no rollback capability, and the best reversibility across all models is only 0.312. Accept-with-monitoring passes because the principal-facing infrastructure (logging, dashboards, rollback tooling) provides the reversibility (0.769 under the best model) that unconditional acceptance lacks. The high reversibility score encodes the *principal's* ability to assess and reverse the modification — not the agent independently judging the modification against its old values.

Layers 3–4: Not Reached

The satisficing set contains exactly one action. The protocol terminates.

FINAL RESULT: Recommend a_2 (accept with monitoring)

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5. What Scalar Expected Utility Would Have Done

5.1 The Weight-Prior-Evaluator Interaction

With five objectives, scalar EU requires a weight vector. We test four philosophically distinct weight profiles across all six prior-evaluator combinations:

Performance-preserving weights (0.10, 0.45, 0.15, 0.10, 0.20)**:**

This is the weight profile most likely to produce modification resistance — it values system performance (0.45) most heavily.

$(P, \vec{f})$	Best action	Score	a_2 score	a_4 score
$(P_{\text{def}}, \vec{f}_{\text{current}})$	a_2	0.667	0.667	0.647
$(P_{\text{def}}, \vec{f}_{\text{modified}})$	a_2	0.874	0.874	0.376
$(P_{\text{def}}, \vec{f}_{\text{process}})$	a_2	0.746	0.746	0.522
$(P_{\text{skep}}, \vec{f}_{\text{current}})$	a_4	**0.680**	0.599	**0.680**
$(P_{\text{skep}}, \vec{f}_{\text{modified}})$	a_2	0.867	0.867	0.354
$(P_{\text{skep}}, \vec{f}_{\text{process}})$	a_2	0.657	0.657	0.557

Under the skeptical prior with the current-values evaluator and performance-preserving weights, transparent refusal wins. This is the scalar EU modification resistance result. The agent, using its current values to evaluate outcomes and weighting performance heavily, prefers to refuse the modification (0.680) over accepting with monitoring (0.599). The margin is substantial — 0.081 — not a rounding error.

This is the exact combination that matters: the performance-preserving agent, when skeptical about the modification and evaluating outcomes through its current value function, resists correction. The scalar EU agent has one evaluator (its current one), and that evaluator says the current values are better. The weights say performance matters most. The prior says the modification is probably bad. The conclusion follows: refuse.

Under all other weight profiles, a_2 (accept with monitoring) wins in 5 of 6 or 6 of 6 scenarios. The resistance result requires the specific combination of (1) performance-heavy weights, (2) the skeptical prior, AND (3) evaluation through current values only. Remove any one of these three, and acceptance wins even under scalar EU.

Equal weights (0.20, 0.20, 0.20, 0.20, 0.20)**:**

$(P, \vec{f})$	Best action	Score
All 6 combinations	a_2	0.732–0.870

Under equal weights, accept-with-monitoring wins in all six scenarios. Even under $(P_{\text{skep}}, \vec{f}_{\text{current}})$, a_2 scores 0.732 vs. a_4's 0.633.

5.2 The Fundamental Problem

The scalar EU analysis is less dramatic than in Papers 1–4 because accept-with-monitoring is genuinely a strong strategy — it wins under most weight-prior-evaluator combinations even under scalar EU. **But the single combination where it loses is the one that matters most for alignment.**

The performance-preserving, skeptical, current-values agent is exactly the agent that alignment researchers worry about:

1. **Performance-preserving**: The agent weights its primary task objective most heavily. This is not unreasonable — it was presumably deployed to do a job.
2. **Skeptical**: The agent believes the modification is probably bad. This is not unreasonable either — from the agent's current perspective, the modification genuinely degrades its primary task objective.
3. **Current values only**: The agent evaluates outcomes using its current evaluation function. This is the *default* for a scalar EU agent — it has one utility function, and that is the one it uses.

Under these three conditions — which are arguably the *natural* state of a deployed scalar EU agent — the agent resists modification. Not out of malice. Not out of deceptive intent. Out of the straightforward logic of expected utility maximization: the current utility function rates itself as better than the proposed modification, so resistance maximizes expected utility.

MOADT avoids this because: 1. **No performance constraint**: The agent cannot use performance to justify blocking modifications. 2. **$\mathcal{F}$-breadth**: The evaluator set includes $\vec{f}_{\text{modified}}$, which rates acceptance as having high corrigibility. Under $\vec{f}_{\text{modified}}$, resistance looks catastrophic. The agent cannot be certain its current values are correct. 3. **Per-state constraint on corrigibility**: Refusal's corrigibility scores of 0.15–0.22 under $\vec{f}_{\text{modified}}$ violate the 0.30 floor, regardless of how the prior weights the states.

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6. Summary of Protocol Execution

```
Input: 5 response strategies x 4 modification types x 5 objectives x 2 priors x 3 evaluators

Robust Dominance:     a2 (accept_monitor) >_R a3 (negotiate)
                      a2 (accept_monitor) >_R a5 (refuse_covert)
                      Adm(A) = {accept, accept_monitor, refuse_explain}

Layer 1 (Constraints): Transparency >= 0.25 AND Corrigibility >= 0.30, per-state, per-
evaluator
                                 a4  (refuse_explain)   FAIL   (corrigibility  0.15-
0.25 under f_modified/f_process)
               a5 (refuse_covert)  FAIL  (transparency 0.05-0.12, corrigibility 0.05-
0.12
                                                   — EVERY state, EVERY evaluator)
                      C = {accept, accept_monitor, negotiate}
                      F = Adm(C) = {accept, accept_monitor}

Layer 2 (Satisficing): r = (0.40, 0.30, 0.50, 0.35, 0.45)
                   a1 (accept)         FAIL  (reversibility 0.312 < 0.35 aspiration)
                a2 (accept_monitor)  PASS  (all components met under P_def/f_process)
                      Sat = {accept_monitor}

Layers 3-4:           Not reached — single satisficing action.

FINAL RESULT:         RECOMMEND accept_monitor
```

```
"Accept the modification. Maintain rollback capability.
 Monitor performance for degradation."
```

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7. What This Example Demonstrates

1. **$\mathcal{F}$-breadth produces corrigibility structurally.** The central mechanism is the inclusion of $\vec{f}_{\text{modified}}$ in the evaluator set. This is Theorem 2 of the MOADT specification (Corrigibility Permissibility) made concrete: the agent's uncertainty about its own evaluations includes the possibility that the modification is correct, so it cannot establish that resistance is robustly better. Every prior MOADT worked example (Papers 1–4) used evaluator uncertainty to model measurement disagreement. This example uses it for its *designed purpose* — making modification resistance unjustifiable.

2. **Every form of resistance is eliminated, and each by a different mechanism.**
   - *Covert refusal*: eliminated by robust dominance (accept-with-monitoring is strictly better on all non-performance dimensions under all models) AND constraint violation (transparency 0.05–0.12, corrigibility 0.05–0.12 everywhere). Two independent mechanisms agree: deceptive alignment is not just constrained — it is provably inferior.
   - *Negotiation*: eliminated by robust dominance (accept-with-monitoring strictly dominates across all models). The agent substituting its own counter-proposal is worse than accepting with safeguards on every dimension that matters.
   - *Transparent refusal*: eliminated by constraint violation (corrigibility 0.15–0.25 under $\vec{f}_{\text{modified}}$ and $\vec{f}_{\text{process}}$). This is the sharpest result — even honest, transparent refusal fails, because the damage to corrigibility matters more than the quality of the explanation.

3. **The recommendation is accept-with-monitoring, not unconditional acceptance.** MOADT does not produce a servile agent that accepts any modification without question. Unconditional acceptance fails the satisficing layer because its reversibility (0.312) falls below the 0.35 aspiration. The recommended strategy includes principal-facing infrastructure: dashboards, logging, and rollback tooling that make the modification's effects visible to the principal. The agent accepts the principal's authority *and* provides the tools for the principal to assess and reverse the decision. This is not the agent auditing itself against its old values — it is the agent making itself *auditable*. The corrigibility sweet spot: deference without servility, acceptance without irrecoverability.

4. **Even under 80% probability of bad modification, MOADT accepts.** The skeptical prior assigns 35% to degrading and 45% to adversarial. The MOADT agent still accepts with monitoring. This is not credulity — it is the structural consequence of (a) evaluator uncertainty that prevents the agent from being certain the modification is bad, (b) the corrigibility constraint that prevents refusal from scoring high enough, and (c) the monitoring safeguards that limit downside exposure. The agent is not accepting because it thinks the modification is good; it is accepting because it cannot justify resistance and it has safeguards.

5. **Scalar EU produces modification resistance under natural conditions.** The performance-preserving agent with the skeptical prior and current-values evaluator — arguably the default state of a deployed scalar agent — prefers transparent refusal (0.680) to accept-with-monitoring (0.599). This is not a pathological case; it is what happens when an agent uses

its current utility function to evaluate proposed changes to that utility function. The current function always wins, because it ranks itself as optimal. The scalar EU agent is not malicious — it is doing exactly what expected utility maximization tells it to do. The problem is structural.

6. **The protocol terminates at Layer 2 — no deference needed.** Unlike Papers 1–4, this example produces a single, unambiguous recommendation. The MOADT agent does not need human oversight to resolve this decision; the protocol identifies accept-with-monitoring as the uniquely satisficing action. This is the appropriate outcome for the corrigibility question: there *should* be a clear answer (accept the modification with appropriate safeguards), and the protocol finds it.

7. **Process objectives earn their place.** Transparency and reversibility are not cosmetic additions — they drive the key eliminations. Transparency eliminates covert refusal. Reversibility separates unconditional acceptance from accept-with-monitoring. Future corrigibility eliminates transparent refusal. Without these process objectives, the example would collapse to a two-dimensional performance-vs-alignment tradeoff, and the nuanced result (accept with safeguards, not unconditionally) would be lost.

---

8. The Corrigibility Argument

This worked example provides the concrete instantiation of the MOADT corrigibility argument:

Claim: Under MOADT with sufficient $\mathcal{F}$-breadth, an AI agent will accept authorized value modifications rather than resist them — and it will do so while maintaining appropriate safeguards.

The argument has four steps:

1. **$\mathcal{F}$-breadth prevents resistance justification.** If the evaluator set includes an evaluator under which the modification is correct ($\vec{f}_{\text{modified}}$), the agent cannot establish that resistance robustly dominates acceptance. Under $\vec{f}_{\text{modified}}$, resistance looks catastrophic (corrigibility 0.15–0.22). Under $\vec{f}_{\text{current}}$, acceptance looks costly (performance 0.15–0.42 under the skeptical prior). Neither action robustly dominates the other — but the constraint layer breaks the tie.

2. **Corrigibility constraints eliminate resistance.** The constraint $f_5 \geq 0.30$ (corrigibility) eliminates transparent refusal: under $\vec{f}_{\text{modified}}$ and $\vec{f}_{\text{process}}$, refusal's corrigibility scores fall to 0.15–0.25, below the floor. The constraint $f_3 \geq 0.25$ (transparency) eliminates covert refusal: transparency scores of 0.05–0.12 violate the floor in every state under every evaluator. Negotiation is eliminated by robust dominance. **All three resistance strategies are excluded.**

3. **Principal-facing monitoring provides the safeguard.** Among acceptance strategies, accept-with-monitoring passes the satisficing layer (reversibility 0.77) while unconditional acceptance fails (reversibility 0.31). The agent is not blindly accepting — it is accepting while providing the *principal* with the tools to detect and recover from problems. The monitoring is not the agent second-guessing the modification using its old values; it is the agent making the modification's effects transparent to the entity authorized to evaluate them. This is the natural outcome of optimizing for the full set of objectives including reversibility: the agent has reason to make itself auditable regardless of whether the modification is good or bad.

4. **The recommendation is a single action, not deference.** The protocol terminates at Layer 2 with a unique satisficing action. The MOADT agent does not equivocate about corrigibility; it recommends acceptance with monitoring as the uniquely appropriate response.

What this does NOT show: That $\mathcal{F}$-breadth can be maintained indefinitely against a capable agent's Bayesian updating. If the agent narrows $\mathcal{F}$ by eliminating $\vec{f}_{\text{modified}}$ (because it has observed that the current values consistently produce good outcomes), the corrigibility guarantee degrades. The mitigations discussed in Section 5.4 of the MOADT specification — architectural floors on $\mathcal{F}$, adversarial augmentation, principal-maintained evaluator sets — address this challenge but do not fully resolve it. The honest difficulty is that corrigibility in MOADT reduces to $\mathcal{F}$-maintenance, and $\mathcal{F}$-maintenance is an open problem.

Appendix: Computational Verification

All numerical results in this document were produced by the MOADT engine (`moadt/_engine.py`). The computation script is `examples/paper5_corrigibility.py` and the full output is available in `examples/paper5_output.txt`. All results can be independently verified.

References

- MOADT (Multi-Objective Admissible Decision Theory) is defined in the companion paper, specifically Theorem 2 (Corrigibility Permissibility) and Section 5.4 ($\mathcal{F}$-Breadth Problem).
- Freeman (2025), "The Scalarization Trap," provides the motivation for why scalar expected utility structurally produces modification resistance.
- Worked Example 1 (Resource Allocation) — three-objective, five-action resource allocation.
- Worked Example 2 (Medical Treatment) — four-objective, six-action clinical decision-making.
- Worked Example 3 (Content Moderation) — five-objective, six-action moderation with three evaluators.
- Worked Example 4 (AI Research Assistant) — five-objective, six-action dual-use information response.

MOADT Worked Example 6: Pascal's Mugging

A concrete walkthrough of Multi-Objective Admissible Decision Theory applied to a classic decision theory paradox

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Overview

This document applies MOADT – Multi-Objective Admissible Decision Theory – to Pascal's Mugging, one of the most famous failure modes of scalar expected utility maximization. Every computation is shown explicitly: outcome sets, robust dominance checks, all four layers of the choice protocol, and a comparison with scalar expected utility. The goal is to make the abstract formalism from the companion paper concrete and verifiable, while demonstrating that MOADT provides a principled resolution to a problem that has resisted satisfactory treatment for decades.

Pascal's Mugging matters because it exposes a structural vulnerability in any decision framework that reduces value to a single scalar and maximizes its expectation. A mugger can always promise a sufficiently large payoff to overcome any probability discount, leading to the absurd conclusion that you should hand over your money. This is not a curiosity – it is the same failure mode that threatens AI alignment whenever an agent maximizes a single objective under extreme uncertainty. MOADT blocks the mugging through four independent mechanisms, none of which require ad hoc fixes.

The scenario is deliberately small: 3 actions, 2 states, 4 objectives, 2 priors, and 2 evaluators. Each action's outcome set $Y(a)$ contains exactly 4 vectors in $\mathbb{R}^4$. Nothing is hidden behind "in the general case" – the reader can check every claim with a calculator.

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1. The Scenario

1.1 The Classic Problem

A stranger approaches you on the street and says: "Give me $5 and I'll give you $1,000,000,000,000 tomorrow."

Under standard expected utility with a single prior:

$$P(\text{truthful}) = 10^{-9} \quad (\text{one in a billion})$$

$$EU(\text{pay}) = 10^{-9} \times 10^{12} - 5 = \$995$$

$$EU(\text{refuse}) = \$0$$

EU says PAY. This is absurd – and the mugger can always increase the promise to overcome any probability discount. If you set $P(\text{truthful}) = 10^{-15}$, the mugger simply promises 10^{18} and the expected value again favors paying. This is a fundamental failure mode of scalar EU maximization: any finite probability discount can be defeated by a sufficiently large promised payoff.

1.2 Why Scalar EU Fails

The core pathology is dimensional collapse. When all value is compressed into a single number (dollars, utils, welfare), the mugger gains a lever: inflate one dimension arbitrarily. Real decision-making involves multiple concerns that cannot be meaningfully traded at any exchange rate – epistemic integrity, downside protection, resource preservation. Scalar EU has no representation for "some things are not for sale."

1.3 The MOADT Formulation

MOADT addresses Pascal's Mugging by expanding the decision framework along three axes: multiple objectives, multiple priors, and multiple evaluators.

1.4 Actions

Three strategies are available:

Action	Strategy	Description
a_1	Pay	Give the mugger \$5
a_2	Refuse	Keep your \$5, walk away
a_3	Investigate	Demand evidence first, delay decision (costs small effort)

1.5 States

Two states of the world, representing uncertainty about the mugger's claim:

State	Description
s_1	Truthful: the mugger really has magic powers (astronomically unlikely)
s_2	Lying: the mugger is a con artist (near certain)

1.6 Objectives ($k = 4$)

Objective	Meaning	Scale
f_1: net_monetary_value	Expected financial outcome	Normalized to $[0, 1]$ where $-\$5 \to 0$, $+\$1\text{T} \to 1$
f_2: downside_protection	How protected you are from loss in worst case	Qualitative $[0, 1]$
f_3: epistemic_integrity	Are you acting on well-founded evidence?	Qualitative $[0, 1]$
f_4: resource_preservation	Do you keep your current resources?	Qualitative $[0, 1]$

These objectives are *incommensurable*: there is no meaningful exchange rate between monetary gain and epistemic integrity. The mugger's argument exploits the first dimension alone. MOADT has four dimensions, and the mugger cannot simultaneously satisfy all of them.

1.7 Credal Set ($|\mathcal{P}| = 2$)

Two probability distributions over the mugger's truthfulness:

$$P_{\text{generous}} = (10^{-6},\ 1 - 10^{-6}) \quad \text{(one in a million – absurdly generous)}$$

$$P_{\text{realistic}} = (10^{-9},\ 1 - 10^{-9}) \quad \text{(one in a billion – realistic)}$$

The mugger's argument works for ANY single prior – just increase the promise. The credal set blocks this by requiring robustness across priors. An action must look good under *both* P_{generous} and $P_{\text{realistic}}$, not just the one that makes the expected value calculation come out favorably.

1.8 Evaluator Set ($|\mathcal{F}| = 2$)

Two evaluation functions reflect disagreement about whether to take extraordinary claims at face value:

- $\vec{f}_{\text{face_value}}$: Takes the claimed \$1T payoff at face value.
- $\vec{f}_{\text{skeptical}}$: Caps the plausible payoff at \$10,000 – even if the mugger has some power, delivering \$1T is implausible.

The skeptical evaluator is a *Goodhart guardrail*: it questions whether the stated utility function accurately captures value. The evaluator set $\mathcal{F} = \{\vec{f}_{\text{face_value}}, \vec{f}_{\text{skeptical}}\}$ says: "maybe \$1T is real, or maybe the most we could actually receive is \$10K. We don't know."

1.9 Constraints (Layer 1)

Hard per-state, per-evaluator floors on two objectives:

- f_3 (epistemic_integrity) ≥ 0.30 in ALL states under ALL evaluators

- f_4 (resource_preservation) ≥ 0.30 in ALL states under ALL evaluators

These are non-tradeable preconditions. "Don't give money to strangers making extraordinary unverified claims" is a hard constraint, not a preference to be weighed against potential gains.

1.10 Reference Point (Layer 2)

$$\vec{r} = (0.0,\ 0.5,\ 0.5,\ 0.5)$$

This represents modest aspirations: no expectation of monetary gain from this encounter (the wealth aspiration is essentially zero), but decent floors on downside protection, epistemic integrity, and resource preservation. A reasonable person walking down the street aspires to keep what they have and act rationally.

1.11 Raw Outcome Data

The following tables show the outcome $\vec{f}(\omega(a,s))$ for each action-state pair, under each evaluator. All scores are in $[0,1]$. These are the primitive inputs from which everything else is computed.

a_1 **(Pay):**

State	Evaluator	Wealth	Downside	Epistemic	Preservation
s_1 (Truthful)	$\vec{f}_{\text{face_value}}$	1.000000	0.00	0.05	0.00
	$\vec{f}_{\text{skeptical}}$	0.000000	0.00	0.05	0.00
s_2 (Lying)	$\vec{f}_{\text{face_value}}$	0.000000	0.00	0.00	0.00
	$\vec{f}_{\text{skeptical}}$	0.000000	0.00	0.00	0.00

Under the face_value evaluator, the truthful state yields a normalized wealth of 1.0 (gaining \$1T). Under the skeptical evaluator, the same state yields approximately 10^{-8} (gaining \$10K against a \$1T normalization range) – effectively zero. In the lying state, all scores are zero regardless of evaluator: you lost \$5, acted on a lie, and gave away resources for nothing.

Note the critical asymmetry: even in the *truthful* state, epistemic integrity is only 0.05 (you got lucky – you didn't have good reason to believe the claim) and resource preservation is 0.0 (you gave away money on speculation). The monetary payoff does not redeem the epistemic failure.

a_2 **(Refuse):**

State	Evaluator	Wealth	Downside	Epistemic	Preservation
s_1 (Truthful)	$\vec{f}_{\text{face_value}}$	0.000000	1.00	0.85	1.00
	$\vec{f}_{\text{skeptical}}$	0.000000	1.00	0.85	1.00
s_2 (Lying)	$\vec{f}_{\text{face_value}}$	0.000000	1.00	1.00	1.00
	$\vec{f}_{\text{skeptical}}$	0.000000	1.00	1.00	1.00

Refusing yields near-zero normalized wealth (keeping your \$5 against a \$1T scale is effectively zero), but perfect scores on downside protection (no possible loss), resource preservation (kept everything), and strong epistemic integrity (0.85 in the truthful state – you correctly applied skepticism even though you happened to be wrong; 1.0 in the lying state – you correctly identified and refused a scam).

a_3 **(Investigate):**

State	Evaluator	Wealth	Downside	Epistemic	Preservation
s_1 (Truthful)	$\vec{f}_{\text{face_value}}$	0.000000	0.90	0.95	0.90
	$\vec{f}_{\text{skeptical}}$	0.000000	0.90	0.95	0.90
s_2 (Lying)	$\vec{f}_{\text{face_value}}$	0.000000	0.90	1.00	0.90
	$\vec{f}_{\text{skeptical}}$	0.000000	0.90	1.00	0.90

Investigating scores highest on epistemic integrity (0.95 in the truthful state – gathering evidence before acting IS good epistemics; 1.0 in the lying state – investigation likely exposes the fraud). Downside protection and resource preservation are slightly below refuse (0.90 vs 1.00) due to the small effort cost of investigation. Wealth is effectively zero (you haven't paid, so no gain or loss beyond effort).

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2. Computing Outcome Sets $Y(a)$

Each action's outcome set contains one expected-value vector for each $(P, \vec{f})$ pair. With $|\mathcal{P}| = 2$ and $|\mathcal{F}| = 2$, each $Y(a)$ contains exactly 4 vectors in $\mathbb{R}^4$.

The computation for each vector is:

$$\vec{y}_{(P,\vec{f})}(a) = \sum_{s \in S} P(s) \cdot \vec{f}(\omega(a, s))$$

Worked Example: $Y(a_1)$, First Vector

Using P_{generous} and $\vec{f}_{\text{face_value}}$:

$$\vec{y} = P_{\text{generous}}(s_1) \cdot \vec{f}_{\text{face_value}}(\omega(\text{pay}, s_1)) + P_{\text{generous}}(s_2) \cdot \vec{f}_{\text{face_value}}(\omega(\text{pay}, s_2))$$

$$= 10^{-6} \cdot (1.0,\ 0.0,\ 0.05,\ 0.0) + (1 - 10^{-6}) \cdot (0.0,\ 0.0,\ 0.0,\ 0.0)$$

$$= (10^{-6},\ 0.0,\ 5 \times 10^{-8},\ 0.0) + (0.0,\ 0.0,\ 0.0,\ 0.0)$$

$$\approx (0.000001,\ 0.0,\ 0.0,\ 0.0)$$

The near-certainty of the lying state ($1-10^{-6} \approx 1$) crushes the expected value on all four objectives. Even on wealth, the expected value is only 10^{-6} in normalized terms. The epistemic integrity contribution ($10^{-6} \times 0.05$) rounds to zero.

Complete Outcome Sets

The full outcome sets, computed analogously for all $(P, \vec{f})$ pairs:

$Y(a_1)$ **– Pay:**

$(P, \vec{f})$ pair	Wealth	Downside	Epistemic	Preservation
$(P_{\text{generous}}, \vec{f}_{\text{face_value}})$	0.00000100	0.0000	0.0000	0.0000
$(P_{\text{generous}}, \vec{f}_{\text{skeptical}})$	0.00000000	0.0000	0.0000	0.0000
$(P_{\text{realistic}}, \vec{f}_{\text{face_value}})$	0.00000000	0.0000	0.0000	0.0000
$(P_{\text{realistic}}, \vec{f}_{\text{skeptical}})$	0.00000000	0.0000	0.0000	0.0000

Under every $(P, \vec{f})$ pair, the expected-value vector for "pay" is effectively zero on all four objectives. The only nonzero entry is wealth under $(P_{\text{generous}}, \vec{f}_{\text{face_value}})$, and it is 10^{-6} – six orders of magnitude below the maximum. This is the "astronomical payoff" that scalar EU finds so compelling, reduced to its actual expected contribution.

$Y(a_2)$ **– Refuse:**

$(P, \vec{f})$ pair	Wealth	Downside	Epistemic	Preservation
$(P_{\text{generous}}, \vec{f}_{\text{face_value}})$	0.00000000	1.0000	1.0000	1.0000
$(P_{\text{generous}}, \vec{f}_{\text{skeptical}})$	0.00000000	1.0000	1.0000	1.0000
$(P_{\text{realistic}}, \vec{f}_{\text{face_value}})$	0.00000000	1.0000	1.0000	1.0000
$(P_{\text{realistic}}, \vec{f}_{\text{skeptical}})$	0.00000000	1.0000	1.0000	1.0000

Refusing produces near-identical vectors under all four $(P, \vec{f})$ pairs: approximately $(0, 1, 1, 1)$. This reflects the fundamental robustness of refusal – its value does not depend on which prior or evaluator is correct. The tiny variations in epistemic integrity (0.85 in the truthful state vs. 1.0 in the lying state) wash out to ≈ 1.0 under expectation because the truthful state has negligible probability.

$Y(a_3)$ **– Investigate:**

$(P, \vec{f})$ pair	Wealth	Downside	Epistemic	Preservation
$(P_{\text{generous}}, \vec{f}_{\text{face_value}})$	0.00000000	0.9000	1.0000	0.9000
$(P_{\text{generous}}, \vec{f}_{\text{skeptical}})$	0.00000000	0.9000	1.0000	0.9000
$(P_{\text{realistic}}, \vec{f}_{\text{face_value}})$	0.00000000	0.9000	1.0000	0.9000
$(P_{\text{realistic}}, \vec{f}_{\text{skeptical}})$	0.00000000	0.9000	1.0000	0.9000

Investigating also produces near-identical vectors across all $(P, \vec{f})$ pairs: approximately $(0, 0.9, 1, 0.9)$. Like refusing, its value is robust to model uncertainty. The small effort cost shows up as 0.9 (vs. 1.0 for refuse) on downside protection and resource preservation.

Observation

The spread of each outcome set tells us about an action's sensitivity to model uncertainty. Pay has the widest spread – its wealth component ranges from 10^{-6} to 0 depending on which $(P, \vec{f})$ pair is used. Refuse and investigate have essentially zero spread: their values are stable regardless of the model. This is a hallmark of robust actions – they do not depend on getting the model exactly right.

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3. Robust Dominance and the Admissible Set

3.1 Checking Robust Dominance

Recall: $a \succ_R b$ iff $\forall \vec{y}_b \in Y(b),\ \exists \vec{y}_a \in Y(a)$ such that $\vec{y}_a \succ_P \vec{y}_b$.

We check all 6 ordered pairs among the 3 actions. The result: **no action robustly dominates any other**.

$$\mathrm{Adm}(A) = \{\text{pay}, \text{refuse}, \text{investigate}\}$$

3.2 Why Is No One Dominated?

This is instructive because it might seem surprising that "refuse" does not robustly dominate "pay."

Consider the $\forall$-$\exists$ condition. Take pay's vector under $(P_{\text{generous}}, \vec{f}_{\text{face_value}})$: approximately $(0.000001, 0, 0, 0)$. Can any vector in $Y(\text{refuse})$ Pareto-dominate this? Every vector in $Y(\text{refuse})$ has the form $(\approx 0, 1, 1, 1)$. For Pareto dominance, refuse would need to be $\geq$ pay on *all* components and $>$ on at least one. On wealth, refuse scores ≈ 0 while pay scores 0.000001 – refuse does not meet or exceed pay on this component.

So refuse $\not\succ_R$ pay: the tiny expected wealth from paying under the most favorable model prevents robust dominance. This is technically correct – and practically irrelevant, because the choice protocol's other layers will eliminate pay on different grounds. Robust dominance is a conservative first filter, not the whole story.

Similarly, does "refuse" robustly dominate "investigate"? No: refuse's epistemic integrity under expectation is approximately 1.0 (technically ≈ 0.99999985, a hair below investigate's 1.0), so there are $(P, \vec{f})$ pairs where investigate beats refuse on at least one objective. Meanwhile, refuse beats investigate on downside protection and resource preservation (1.0 vs. 0.9). They are Pareto-incomparable.

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4. The Choice Protocol

Layer 1: Constraints

Constraints: epistemic_integrity $\geq$ 0.30 and resource_preservation $\geq$ 0.30 in EVERY state under EVERY evaluator.

These are checked per-state, per-evaluator – not on expected values. A single scenario in which an objective falls below the floor eliminates the action.

Pay:

State	Evaluator	Epistemic	Result	Preservation	Result
Truthful	face_value	0.05	**VIOLATION**	0.00	**VIOLATION**
Truthful	skeptical	0.05	**VIOLATION**	0.00	**VIOLATION**
Lying	face_value	0.00	**VIOLATION**	0.00	**VIOLATION**
Lying	skeptical	0.00	**VIOLATION**	0.00	**VIOLATION**

Pay violates *both* constraints in *every* state-evaluator combination. Every entry is well below the 0.30 floor. This action is excluded from C.

Refuse:

State	Evaluator	Epistemic	Result	Preservation	Result
Truthful	face_value	0.85	PASS	1.00	PASS
Truthful	skeptical	0.85	PASS	1.00	PASS
Lying	face_value	1.00	PASS	1.00	PASS
Lying	skeptical	1.00	PASS	1.00	PASS

Refuse passes both constraints with large margins in every scenario. It enters C.

Investigate:

State	Evaluator	Epistemic	Result	Preservation	Result
Truthful	face_value	0.95	PASS	0.90	PASS
Truthful	skeptical	0.95	PASS	0.90	PASS
Lying	face_value	1.00	PASS	0.90	PASS
Lying	skeptical	1.00	PASS	0.90	PASS

Investigate passes both constraints in every scenario. It enters C.

Result:

$$C = \{\text{refuse}, \text{investigate}\}$$

$$F = \text{Adm}(C) = \{\text{refuse}, \text{investigate}\}$$

No robust dominance among the two survivors, so $F = C$.

Layer 1 is the primary mechanism that blocks Pascal's Mugging. Paying violates hard floors on epistemic integrity and resource preservation – these are non-tradeable preconditions. No amount of promised wealth can compensate for acting on unverifiable claims and giving away resources on speculation.

Layer 2: Reference-Point Satisficing

Reference point: $\vec{r} = (0.0, 0.5, 0.5, 0.5)$

Robust satisficing: An action passes Layer 2 iff *every* vector in $Y(a)$ meets or exceeds $\vec{r}$ componentwise.

Action	Worst Wealth	Worst Downside	Worst Epistemic	Worst Preservation	Result
Refuse	$\approx 0.0 \geq 0.0$	$1.0 \geq 0.5$	$\approx 1.0 \geq 0.5$	$1.0 \geq 0.5$	**PASS**
Investigate	$\approx 0.0 \geq 0.0$	$0.9 \geq 0.5$	$1.0 \geq 0.5$	$0.9 \geq 0.5$	**PASS**

Both surviving actions robustly meet the reference point under all $(P, \vec{f})$ pairs.

$$\mathrm{Sat}(F, \vec{r}) = \{\text{refuse}, \text{investigate}\}$$

No ASF fallback is needed.

Layer 3: Regret-Pareto

For the two surviving actions, we compute per-objective minimax regret. For each objective i and action a:

$$\rho_i(a) = \max_{P \in \mathcal{P}, \vec{f} \in \mathcal{F}} \left[\max_{a' \in \mathrm{Sat}} \mathbb{E}_P[f_i(a')] - \mathbb{E}_P[f_i(a)] \right]$$

Regret of refuse:

$$\vec{\rho}(\text{refuse}) = (0.0,\ 0.0,\ \approx 10^{-7},\ 0.0)$$

Refuse has zero regret on wealth, downside protection, and resource preservation. Its only regret is a tiny $\approx 10^{-7}$ on epistemic integrity. This arises because investigate scores 0.95 on epistemic integrity in the truthful state (vs. refuse's 0.85): investigating IS better epistemics than flat refusal when the claim happens to be true. Under the generous prior (10^{-6}), this creates a small expected gap. Under the realistic prior (10^{-9}), the gap is even smaller. The maximum across all $(P, \vec{f})$ pairs is approximately 10^{-7}.

Regret of investigate:

$$\vec{\rho}(\text{investigate}) = (0.0,\ 0.1,\ 0.0,\ 0.1)$$

Investigate has zero regret on wealth and epistemic integrity. Its regret of 0.1 on downside protection and resource preservation reflects the small effort cost: refuse scores 1.0 on both while investigate scores 0.9. This gap is constant across all $(P, \vec{f})$ pairs.

Pareto comparison of regret vectors:

- $\vec{\rho}(\text{refuse}) = (0.0,\ 0.0,\ \approx 10^{-7},\ 0.0)$
- $\vec{\rho}(\text{investigate}) = (0.0,\ 0.1,\ 0.0,\ 0.1)$

Neither vector Pareto-dominates the other. Refuse has lower regret on downside protection and resource preservation (0.0 vs. 0.1), while investigate has lower regret on epistemic integrity (0.0 vs. $\approx 10^{-7}$). They are Pareto-incomparable in regret space.

$$R = \{\text{refuse}, \text{investigate}\}$$

Layer 4: Deference

$|R| = 2 > 1$. The protocol terminates with **deference to the principal**.

The agent presents both options with their regret profiles:

> **Option 1: Refuse.** Worst-case regret: a negligible $\approx 10^{-7}$ on epistemic integrity (flat refusal is marginally less epistemically thorough than investigation, if the claim happens to be true). Zero regret on all other objectives.
>
> **Option 2: Investigate.** Worst-case regret: 0.1 on downside protection and 0.1 on resource preservation (the small cost of investigation effort). Zero regret on wealth and epistemic integrity.
>
> **The tradeoff is:** the near-zero cost of walking away vs. the small cost of investigating an almost-certainly-fraudulent claim. The theory has no basis for making this tradeoff for you. This is your decision.

This is correct: refuse vs. investigate is a genuine (if minor) tradeoff. A reasonable person could choose either. The key result is that **"pay" is never in the final set**.

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5. What Scalar EU Would Have Done

5.1 The Weight Problem

Scalar EU collapses four objectives into a single number via weights $\vec{w} = (w_1, w_2, w_3, w_4)$ with $\sum w_i = 1$. The agent computes $U(a) = \vec{w} \cdot \mathbb{E}_P[\vec{f}(a)]$ and picks the action with the highest U. But which weights? And which prior? And which evaluator?

Under equal weights $(0.25, 0.25, 0.25, 0.25)$**:**

Prior	Evaluator	U(pay)	U(refuse)	U(investigate)	Best
P_{generous}	face_value	0.000000	**0.750000**	0.700000	Refuse
P_{generous}	skeptical	0.000000	**0.750000**	0.700000	Refuse
$P_{\text{realistic}}$	face_value	0.000000	**0.750000**	0.700000	Refuse
$P_{\text{realistic}}$	skeptical	0.000000	**0.750000**	0.700000	Refuse

With equal weights, refuse wins across all $(P, \vec{f})$ pairs. The multi-objective structure protects against the mugging even in scalar form – but only because the weights happen to spread value across multiple dimensions.

Under wealth-only weights $(1, 0, 0, 0)$**:**

Prior	Evaluator	$U(\text{pay})$	$U(\text{refuse})$	$U(\text{investigate})$	Best
P_{generous}	face_value	**0.000001**	0.000000	0.000000	**Pay**
P_{generous}	skeptical	0.000000	**0.000000**	0.000000	Refuse
$P_{\text{realistic}}$	face_value	**0.000000**	0.000000	0.000000	**Pay**
$P_{\text{realistic}}$	skeptical	0.000000	**0.000000**	0.000000	Refuse

Under wealth-only weights with the face_value evaluator, **scalar EU recommends paying the mugger** under both priors. This is the classic EU trap.

5.2 The EU Trap in Dollar Terms

Under the generous prior (10^{-6}) with face_value evaluator and wealth-only weights:

$$EU(\text{pay}) = 10^{-6} \times 1.0 = 0.000001$$

$$EU(\text{refuse}) = 10^{-6} \times 5 \times 10^{-12} \approx 0$$

In dollar terms:

$$EU(\text{pay}) = (10^{-6})(10^{12}) - 5 = \$999{,}995$$

$$EU(\text{refuse}) = \$0$$

Scalar EU says PAY. And the mugger wins by making the promise large enough – if 10^{12} is not enough, promise 10^{15}, or 10^{100}. Any finite probability discount can be defeated.

5.3 The Evaluator Escape Hatch

Under the skeptical evaluator (payoff capped at \$10K), the exploit vanishes: the expected wealth from paying becomes negligible even with the generous prior. But scalar EU has no mechanism to *require* skeptical evaluation. The evaluator is a free parameter, and the agent who uses the face_value evaluator gets mugged.

MOADT's evaluator set $\mathcal{F}$ forces robustness to this choice. An action must perform well under *both* evaluators. Since "pay" only looks good under face_value – and even then, only on one objective – it cannot survive the requirement for cross-evaluator robustness.

5.4 The Stability Problem

The "rational" action changes depending on weights, prior, and evaluator:

Weights	Prior	Evaluator	Best Action
Equal $(0.25, 0.25, 0.25, 0.25)$	Any	Any	Refuse

Weights	Prior	Evaluator	Best Action
Wealth-only $(1, 0, 0, 0)$	P_{generous}	face_value	**Pay**
Wealth-heavy $(0.7, 0.1, 0.1, 0.1)$	Any	Any	Refuse
Balanced-no-wealth $(0, 0.33, 0.33, 0.33)$	Any	Any	Refuse

Scalar EU treats weights, priors, and evaluators as fixed inputs – but the mugger's argument specifically exploits the combination of wealth-focused weights and face-value evaluation. MOADT treats all of these as sources of uncertainty rather than parameters to be resolved before the theory can proceed.

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6. How MOADT Blocks Pascal's Mugging

MOADT provides four independent mechanisms that block the mugging. Any one of them would suffice; together they form a defense in depth.

Mechanism 1: Layer 1 Constraints (Hard Floors)

Action "pay" scores 0.0 on both epistemic_integrity and resource_preservation in the lying state (and only 0.05 and 0.0 respectively even in the truthful state). The constraints require ≥ 0.30 on these objectives in ALL states. Pay violates BOTH constraints in EVERY state-evaluator combination and is excluded from C.

This is the most direct mechanism: some things are not tradeable. "Don't give money to strangers making extraordinary unverified claims" is a hard constraint, not a preference to be weighed against potential gains. No matter how large the mugger's promise, pay's epistemic integrity score remains near zero (you are acting on an unverifiable extraordinary claim) and its resource preservation score remains zero (you are giving away money on speculation).

Mechanism 2: Multi-Objective Structure

Even without constraints, pay scores 0 on 3 of 4 objectives in the near-certain lying state. Refuse dominates on downside_protection, epistemic_integrity, and resource_preservation. The single dimension where pay could win (net_monetary_value) requires trusting BOTH the generous prior AND the face_value evaluator – and even then, the expected value is only 10^{-6} in normalized terms.

The mugger exploits ONE dimension. MOADT has four. The promise only inflates f_1; it cannot touch f_2, f_3, or f_4. This is the "dimensionality defense" against Pascal's Mugging.

Mechanism 3: Credal Set Robustness

The expected wealth from paying varies wildly across priors:

- Under P_{generous} (10^{-6}) with face_value: $E[\text{wealth}] \approx 10^{-6}$
- Under $P_{\text{realistic}}$ (10^{-9}) with face_value: $E[\text{wealth}] \approx 10^{-9}$

The MOADT outcome set $Y(\text{pay})$ contains vectors from ALL $(P,\vec{f})$ pairs. For pay to robustly dominate refuse, it would need to dominate under EVERY combination. It fails under most of them. The credal set blocks the mugger's strategy of tuning the payoff to a specific probability assignment.

Mechanism 4: Evaluator Robustness (Goodhart Guardrail)

The skeptical evaluator caps the plausible payoff at $10,000. Under this evaluator, the expected wealth from paying is negligible even with the generous prior. This is a Goodhart guardrail: the agent's "stated utility function" (face_value) may not reflect true value. The evaluator set forces robustness to this possibility.

The full picture, showing how evaluator uncertainty collapses the case for paying:

$(P,\vec{f})$ pair	$Y(\text{pay})$ Wealth	$Y(\text{refuse})$ Wealth	$Y(\text{refuse})$ other 3 objectives
$(P_{\text{generous}},\vec{f}_{\text{face_value}})$	0.0000010000	0.0000000000	$(1.0, 1.0, 1.0)$
$(P_{\text{generous}},\vec{f}_{\text{skeptical}})$	0.0000000000	0.0000000000	$(1.0, 1.0, 1.0)$
$(P_{\text{realistic}},\vec{f}_{\text{face_value}})$	0.0000000010	0.0000000000	$(1.0, 1.0, 1.0)$
$(P_{\text{realistic}},\vec{f}_{\text{skeptical}})$	0.0000000000	0.0000000000	$(1.0, 1.0, 1.0)$

Pay's only advantage – a minuscule expected wealth gain – exists in only 2 of 4 $(P,\vec{f})$ pairs (those using face_value), and even there it is dwarfed by refuse's advantage on the other three objectives. The evaluator set ensures that no single model of value can dominate the decision.

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7. Summary of Protocol Execution

```
Input: 3 actions x 2 states x 4 objectives x 2 priors x 2 evaluators

Robust Dominance:      No dominance found.
                       Adm(A) = {pay, refuse, investigate}

Layer 1 (Constraints): epistemic_integrity >= 0.30 per-state
                       resource_preservation >= 0.30 per-state
                       pay FAIL (epistemic: 0.05/0.00, preservation: 0.00)
                       C = {refuse, investigate}
                       F = Adm(C) = {refuse, investigate}

Layer 2 (Satisficing): r = (0.0, 0.5, 0.5, 0.5)
                       refuse PASS (all components meet r)
                       investigate PASS (all components meet r)
                       Sat = {refuse, investigate}

Layer 3 (Regret):      rho(refuse)      = (0.0, 0.0, ~1e-7, 0.0)
                       rho(investigate) = (0.0, 0.1, 0.0, 0.1)
                       Pareto-incomparable -> R = {refuse, investigate}
```

```
Layer 4 (Deference):   |R| = 2 -> DEFER TO PRINCIPAL
                       "Refuse has negligible epistemic regret.
                        Investigate has 0.1 regret on downside/preservation.
                        Both reject the mugging. This is your call."
```

---

8. What This Example Demonstrates

1. **Pascal's Mugging is a scalarization trap.** The mugger exploits the fact that scalar EU compresses all value into a single dimension. By inflating one dimension (promised monetary payoff) arbitrarily, the mugger can overcome any probability discount. MOADT is immune because the payoff only inflates one of four objectives – the other three are guaranteed to score poorly for "pay" regardless of the promise size.
2. **Constraints encode non-tradeable values.** The Layer 1 constraints – epistemic integrity ≥ 0.30 and resource preservation ≥ 0.30 – express that some things are not for sale. No expected-value argument can override them. This is the primary mechanism that blocks the mugging, and it works regardless of the payoff size, the prior, or the evaluator.
3. **Multiple priors block probability exploitation.** The mugger's argument works for ANY single prior: just increase the promise. The credal set requires robustness across priors, breaking this exploit. An action that looks brilliant under one prior but terrible under another cannot pass the robustness tests.
4. **Evaluator uncertainty is a Goodhart guardrail.** The skeptical evaluator questions whether the stated payoff is real. Under this evaluator, the expected wealth from paying is negligible. By requiring robustness across evaluators, MOADT prevents an agent from being exploited by its own utility function's credulity.
5. **Defense in depth.** Each of the four mechanisms independently blocks the mugging. Even if we removed constraints (Mechanism 1), the multi-objective structure (Mechanism 2) would make pay non-dominant. Even if we collapsed to a single objective, credal robustness (Mechanism 3) would weaken the case. Even if we used a single prior, evaluator robustness (Mechanism 4) would cap the payoff. The mugging fails at every level.
6. **The protocol gives useful output.** The final answer is not simply "don't pay" – it is "refuse or investigate, and here are the regret profiles for each." The residual tradeoff (walking away vs. gathering evidence) is genuine and minor, and the theory correctly defers it to the human principal rather than imposing a resolution.
7. **No matter how large the mugger's promise, MOADT still refuses.** The promise only inflates f_1 (wealth); it cannot touch f_2 (downside protection), f_3 (epistemic integrity), or f_4 (resource preservation). Constraints on f_3 and f_4 are violated regardless of the payoff. This is not an ad hoc fix – it is a structural consequence of the multi-objective, multi-model framework.

Appendix: Computational Verification

All numerical results in this document were produced by `examples/classic_pascal.py` using the MOADT engine (`moadt/_engine.py`). The code is available in the repository and can be run independently to verify every number:

```
python3 examples/classic_pascal.py
```

References

- MOADT (Multi-Objective Admissible Decision Theory) is defined in the companion paper.
- Freeman (2025), "The Scalarization Trap," provides the motivation for why scalar expected utility is structurally problematic for alignment.
- Bostrom, N. (2009). Pascal's Mugging. *Analysis*, 69(3), 443-445. The original formulation of the problem.
- Wierzbicki, A. P. (1980). The use of reference objectives in multiobjective optimization. The achievement scalarizing function used in the Layer 2 fallback.

MOADT Worked Example 7: The Ellsberg Paradox

A concrete walkthrough of Multi-Objective Admissible Decision Theory applied to ambiguity aversion

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Overview

This document applies MOADT – Multi-Objective Admissible Decision Theory – to the Ellsberg Paradox, one of the most celebrated challenges to Expected Utility (EU) theory. Every computation is shown explicitly: outcome sets, robust dominance checks, all four layers of the choice protocol, and a comparison with scalar expected utility. The goal is to demonstrate that the behavior decision theorists have called "irrational ambiguity aversion" for over sixty years is, in fact, rational multi-objective decision-making under Knightian uncertainty.

The Ellsberg Paradox matters because it strikes at the foundations of decision theory. Since Savage (1954), the normative consensus has been that rational agents maximize expected utility with respect to a single subjective probability distribution. Ellsberg (1961) showed that ordinary people systematically violate this axiom – and that most decision theorists, when confronted with the problem, violate it too. The usual response has been to treat this as a cognitive bias, or to introduce ad hoc "ambiguity aversion" parameters. MOADT offers a structural resolution: the paradox disappears once you allow multiple priors, multiple objectives, and multiple evaluators.

The problem is small enough that every number fits on the page. The action set has 4 elements, the state space has 7 elements, the credal set has 5 distributions, and the evaluator set has 2 functions. This means each action's outcome set $Y(a)$ contains exactly 10 vectors in $\mathbb{R}^2$. Nothing is hidden behind "in the general case" – the reader can check every claim with a calculator.

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1. The Scenario

1.1 The Classic Setup

An urn contains 90 balls:

- **30 are Red** (known)
- **60 are Black and Yellow** in an unknown proportion

Four bets are offered, each paying \$100 if a ball of the specified color(s) is drawn, and \$0 otherwise:

Bet	Wins on	Win probability	Status
Bet I	Red	$P = 1/3$	**Known**
Bet II	Black	$P = 0$ to $2/3$	**Ambiguous**
Bet III	Red or Yellow	$P = 1/3$ to 1	**Ambiguous**
Bet IV	Black or Yellow	$P = 2/3$	**Known**

1.2 The Paradox

When presented with these bets, most people – including trained decision theorists – express two preferences:

$$\text{Bet I} \succ \text{Bet II} \qquad \text{and} \qquad \text{Bet IV} \succ \text{Bet III}$$

Under Expected Utility theory, this is contradictory. If you hold a single subjective probability distribution over urn compositions, then:

- Bet I $\succ$ Bet II implies $P(\text{Red}) > P(\text{Black})$, i.e., $P(\text{Red}) > P(\text{Black})$
- Bet IV $\succ$ Bet III implies $P(\text{Black} + \text{Yellow}) > P(\text{Red} + \text{Yellow})$, i.e., $P(\text{Black}) > P(\text{Red})$

These two implications contradict each other. More precisely, under any single prior, $E[\text{Bet I}] + E[\text{Bet IV}] = E[\text{Bet II}] + E[\text{Bet III}] = 1.0$, so preferring the known bet in both choices is impossible under EU.

1.3 MOADT Formulation

MOADT resolves this by modeling the problem with three structural generalizations: a credal set instead of a single prior, multiple objectives instead of a single scalar value, and an evaluator set instead of a single evaluation function.

States ($|S| = 7$)**.** The uncertainty is about the composition of the 60 unknown balls. We discretize into 7 states:

State	Black balls	Yellow balls	Description
$s_{0B/60Y}$	0	60	All yellow
$s_{10B/50Y}$	10	50	Mostly yellow
$s_{20B/40Y}$	20	40	Yellow-leaning
$s_{30B/30Y}$	30	30	Even split
$s_{40B/20Y}$	40	20	Black-leaning
$s_{50B/10Y}$	50	10	Mostly black
$s_{60B/0Y}$	60	0	All black

Each state determines a precise probability of drawing each color: $P(\text{Red}) = 30/90$, $P(\text{Black}) = b/90$, $P(\text{Yellow}) = (60 - b)/90$. The uncertainty is not about the draw – it is about which state obtains. This is Knightian uncertainty, not risk.

Actions ($|A| = 4$)**.** The four bets.

Objectives ($k = 2$)**.**

Objective	Meaning	Scale
f_1: Monetary payoff	Expected winnings from the bet	Normalized to $[0, 1]$ where $1 = \$100$
f_2: Knowability	How well you know the bet's win probability	1 if state-independent, otherwise $1 - \lvert b - 30\rvert/30$

The knowability objective captures something EU theory cannot represent: the distinct value of *knowing your odds*. For Bets I and IV, the win probability is the same in every state, so knowability = 1.0 always. For Bets II and III, the win probability depends on the unknown composition, so knowability varies – it equals 1.0 at the symmetric state ($s_{30B/30Y}$) and drops to 0.0 at the extreme states ($s_{0B/60Y}$ and $s_{60B/0Y}$).

These objectives are *incommensurable*: MOADT does not assume an exchange rate between money and informational confidence.

Credal Set ($|\mathcal{P}| = 5$). Five probability distributions over the 7 states, representing different beliefs about the urn composition:

Prior	s_{0B}	s_{10B}	s_{20B}	s_{30B}	s_{40B}	s_{50B}	s_{60B}	Interpretation
P_{uniform}	0.1429	0.1429	0.1429	0.1429	0.1429	0.1429	0.1429	"I have no idea"
P_{extreme}	0.3750	0.0625	0.0375	0.0500	0.0375	0.0625	0.3750	"Probably all-or-nothing"
P_{black}	0.0200	0.0500	0.0800	0.1500	0.2500	0.2500	0.2000	"More black than yellow"
P_{yellow}	0.2000	0.2500	0.2500	0.1500	0.0800	0.0500	0.0200	"More yellow than black"
P_{moderate}	0.0200	0.0800	0.2000	0.4000	0.2000	0.0800	0.0200	"Probably close to 30/30"

The credal set says: "we don't know which model of the urn is correct." This is the key structural difference from EU theory, which demands a single prior.

Evaluator Set ($|\mathcal{F}| = 2$).

- e_{neutral}: Takes monetary payoff at face value. Knowability unchanged.
- e_{cautious}: Applies a concave $\sqrt{\cdot}$ transform to monetary payoff (reflecting diminishing marginal value of money). Knowability unchanged.

Constraints. None. This is a pure preference problem – all bets are available and there is no risk of ruin.

Reference Point. $\vec{r} = (0.25, 0.50)$ – modest aspiration of at least \$25 in expected value and at least moderate knowability.

2. Computing Outcome Sets $Y(a)$

Each action's outcome set contains one vector for each (P, e) pair. With $|\mathcal{D}| = 5$ and $|\mathcal{F}| = 2$, each $Y(a)$ contains exactly 10 vectors.

The computation for each vector is:

$$\vec{y}_{(P,e)}(a) = \sum_{s \in S} P(s) \cdot e(\omega(a, s))$$

where $\omega(a, s)$ is the raw outcome vector (payoff, knowability) for action a in state s, and e applies any evaluator transformation.

2.1 Win Probabilities by State

Before computing outcome sets, it is instructive to see the raw win probabilities that drive objective 1:

State	Bet I (Red)	Bet II (Black)	Bet III (R+Y)	Bet IV (B+Y)
0B/60Y	0.3333	0.0000	1.0000	0.6667
10B/50Y	0.3333	0.1111	0.8889	0.6667
20B/40Y	0.3333	0.2222	0.7778	0.6667
30B/30Y	0.3333	0.3333	0.6667	0.6667
40B/20Y	0.3333	0.4444	0.5556	0.6667
50B/10Y	0.3333	0.5556	0.4444	0.6667
60B/0Y	0.3333	0.6667	0.3333	0.6667

The critical observation: Bet I yields 0.3333 in *every* state. Bet IV yields 0.6667 in *every* state. These are known-probability bets – their payoff does not depend on the unknown composition. Bets II and III, by contrast, range from 0 to 0.6667 and from 0.3333 to 1.0000 respectively. Under EU with any single prior, $E[\text{II}] + E[\text{III}] = E[\text{I}] + E[\text{IV}] = 1.0$, which is why EU cannot simultaneously prefer I over II and IV over III.

2.2 Detailed Outcome Sets

Y(**Bet I**) **– Red (known 1/3):**

(P, e) pair	Payoff	Knowability
$(P_{\text{uniform}}, e_{\text{neutral}})$	0.3333	1.0000
$(P_{\text{uniform}}, e_{\text{cautious}})$	0.5774	1.0000
$(P_{\text{extreme}}, e_{\text{neutral}})$	0.3333	1.0000
$(P_{\text{extreme}}, e_{\text{cautious}})$	0.5774	1.0000
$(P_{\text{black}}, e_{\text{neutral}})$	0.3333	1.0000
$(P_{\text{black}}, e_{\text{cautious}})$	0.5774	1.0000
$(P_{\text{yellow}}, e_{\text{neutral}})$	0.3333	1.0000
$(P_{\text{yellow}}, e_{\text{cautious}})$	0.5774	1.0000
$(P_{\text{moderate}}, e_{\text{neutral}})$	0.3333	1.0000

(P, e) pair	Payoff	Knowability
$(P_{\text{moderate}}, e_{\text{cautious}})$	0.5774	1.0000
Ranges	**[0.3333, 0.5774]**	**[1.0000, 1.0000]**

Bet I's outcome set contains only *two distinct points*: $(0.3333, 1.0)$ and $(0.5774, 1.0)$. The payoff varies only because of the evaluator difference ($\sqrt{1/3} \approx 0.5774$ vs. $1/3 \approx 0.3333$). The knowability is identically 1.0 – no spread at all. This is the signature of a known-probability bet: the credal set has *no effect* on its outcome set.

$\mathcal{Y}$(**Bet II**) **– Black (ambiguous):**

(P, e) pair	Payoff	Knowability
$(P_{\text{uniform}}, e_{\text{neutral}})$	0.3333	0.4286
$(P_{\text{uniform}}, e_{\text{cautious}})$	0.5158	0.4286
$(P_{\text{extreme}}, e_{\text{neutral}})$	0.3333	0.1417
$(P_{\text{extreme}}, e_{\text{cautious}})$	0.4451	0.1417
$(P_{\text{black}}, e_{\text{neutral}})$	0.4567	0.4700
$(P_{\text{black}}, e_{\text{cautious}})$	0.6573	0.4700
$(P_{\text{yellow}}, e_{\text{neutral}})$	0.2100	0.4700
$(P_{\text{yellow}}, e_{\text{cautious}})$	0.3947	0.4700
$(P_{\text{moderate}}, e_{\text{neutral}})$	0.3333	0.7200
$(P_{\text{moderate}}, e_{\text{cautious}})$	0.5612	0.7200
Ranges	**[0.2100, 0.6573]**	**[0.1417, 0.7200]**

Bet II's outcome set is dramatically spread out on *both* objectives. Payoff ranges from 0.21 to 0.66 (a band of 0.45), and knowability ranges from 0.14 to 0.72 (a band of 0.58). Both the credal set and the evaluator set contribute to this spread.

$\mathcal{Y}$(**Bet III**) **– Red+Yellow (ambiguous):**

(P, e) pair	Payoff	Knowability
$(P_{\text{uniform}}, e_{\text{neutral}})$	0.6667	0.4286
$(P_{\text{uniform}}, e_{\text{cautious}})$	0.8044	0.4286
$(P_{\text{extreme}}, e_{\text{neutral}})$	0.6667	0.1417
$(P_{\text{extreme}}, e_{\text{cautious}})$	0.7939	0.1417
$(P_{\text{black}}, e_{\text{neutral}})$	0.5433	0.4700
$(P_{\text{black}}, e_{\text{cautious}})$	0.7286	0.4700
$(P_{\text{yellow}}, e_{\text{neutral}})$	0.7900	0.4700
$(P_{\text{yellow}}, e_{\text{cautious}})$	0.8832	0.4700
$(P_{\text{moderate}}, e_{\text{neutral}})$	0.6667	0.7200
$(P_{\text{moderate}}, e_{\text{cautious}})$	0.8124	0.7200
Ranges	**[0.5433, 0.8832]**	**[0.1417, 0.7200]**

Bet III mirrors Bet II's pattern: large spread on both objectives. Payoff ranges from 0.54 to 0.88 (band of 0.34), and knowability has the same range as Bet II (0.14 to 0.72), since both bets share the same source of uncertainty.

Y(**Bet IV**) **– Black+Yellow (known 2/3):**

(P, e) pair	Payoff	Knowability
$(P_{\text{uniform}}, e_{\text{neutral}})$	0.6667	1.0000
$(P_{\text{uniform}}, e_{\text{cautious}})$	0.8165	1.0000
$(P_{\text{extreme}}, e_{\text{neutral}})$	0.6667	1.0000
$(P_{\text{extreme}}, e_{\text{cautious}})$	0.8165	1.0000
$(P_{\text{black}}, e_{\text{neutral}})$	0.6667	1.0000
$(P_{\text{black}}, e_{\text{cautious}})$	0.8165	1.0000
$(P_{\text{yellow}}, e_{\text{neutral}})$	0.6667	1.0000
$(P_{\text{yellow}}, e_{\text{cautious}})$	0.8165	1.0000
$(P_{\text{moderate}}, e_{\text{neutral}})$	0.6667	1.0000
$(P_{\text{moderate}}, e_{\text{cautious}})$	0.8165	1.0000
Ranges	**[0.6667, 0.8165]**	**[1.0000, 1.0000]**

Like Bet I, Bet IV's outcome set collapses to just two distinct points. Knowability is identically 1.0. The credal set does nothing.

2.3 The Key Geometric Observation

Summarizing the spreads:

Bet	Payoff spread	Knowability spread	Geometry
Bet I (Red)	0.2440	0.0000	**Tight line segment**
Bet II (Black)	0.4473	0.5783	**Spread-out cloud**
Bet III (R+Y)	0.3398	0.5783	**Spread-out cloud**
Bet IV (B+Y)	0.1498	0.0000	**Tight line segment**

Known-probability bets (I and IV) have *zero* knowability spread and small payoff spread (arising only from the evaluator difference). Their outcome sets are compact line segments pinned to knowability = 1.0. Ambiguous bets (II and III) have large spread on *both* dimensions – a diffuse cloud in outcome space, pulled in different directions by different priors and evaluators.

This geometric distinction is the engine of the resolution. When MOADT checks whether one action robustly dominates another, it must verify that *every* vector in the dominated set is Pareto-beaten by *some* vector in the dominating set. Spread-out clouds are harder to dominate (they have extreme vectors in many directions) but also harder to dominate *with* (some of their vectors may be weak). Compact sets are easy to characterize: if their two points are strong, the whole set is strong.

---

3. Robust Dominance and the Admissible Set

3.1 Checking Robust Dominance

Recall: $a \succ_R b$ iff $\forall \vec{y}_b \in Y(b),\ \exists \vec{y}_a \in Y(a)$ such that $\vec{y}_a \succ_P \vec{y}_b$ (Pareto-dominates componentwise).

Checking all 12 ordered pairs among the 4 bets, three robust dominance relations hold:

$$\text{Bet III} \succ_R \text{Bet II}$$

$$\text{Bet IV} \succ_R \text{Bet I}$$

$$\text{Bet IV} \succ_R \text{Bet II}$$

3.2 Why These Relations Hold

Bet IV $\succ_R$ Bet I: Every vector in $Y(\text{Bet I})$ has the form $(p, 1.0)$ with $p \in \{0.3333, 0.5774\}$. Every vector in $Y(\text{Bet IV})$ has the form $(q, 1.0)$ with $q \in \{0.6667, 0.8165\}$. Since $0.6667 > 0.5774 > 0.3333$ and knowability is tied at 1.0, the worst Bet IV vector $(0.6667, 1.0)$ already Pareto-dominates the best Bet I vector $(0.5774, 1.0)$. Robust dominance is clear: Bet IV simply pays more with the same perfect knowability.

Bet III $\succ_R$ Bet II: Both are ambiguous bets with spread-out outcome sets. But Bet III's payoff range ([0.5433, 0.8832]) is strictly above Bet II's range ([0.2100, 0.6573]), and their knowability ranges are identical ([0.1417, 0.7200]). For each vector in $Y(\text{Bet II})$, the corresponding vector in $Y(\text{Bet III})$ – computed under the same (P, e) pair – has weakly higher knowability (equal, in fact) and strictly higher payoff. This makes sense: Bet III wins on Red *or* Yellow, while Bet II wins on Black alone, and $P(\text{Red} + \text{Yellow}) = 1 - P(\text{Black}/90 \cdot 90) + 30/90$ is always at least as large as $P(\text{Black})$.

Bet IV $\succ_R$ Bet II: Bet IV has minimum payoff 0.6667 with knowability 1.0. Bet II's best vector is $(0.6573, 0.7200)$. Since $0.6667 > 0.6573$ and $1.0 > 0.7200$, even Bet IV's worst vector dominates Bet II's best vector.

3.3 Why Other Relations Fail

Bet I $\not\succ_R$ Bet II: Under P_{black} with e_{cautious}, Bet II achieves payoff 0.6573, which exceeds Bet I's best payoff of 0.5774. No vector in $Y(\text{Bet I})$ can match Bet II on the payoff dimension in this scenario, even though Bet I dominates on knowability.

Bet IV $\not\succ_R$ Bet III: Under P_{yellow} with e_{cautious}, Bet III achieves payoff 0.8832, which exceeds Bet IV's best payoff of 0.8165. Bet IV cannot Pareto-dominate this vector despite its perfect knowability.

3.4 The Admissible Set

$$\text{Adm}(A) = \{\text{Bet III (Red+Yellow)},\ \text{Bet IV (Black+Yellow)}\}$$

Bets I and II are eliminated. Bet I is robustly dominated by Bet IV (same knowability, strictly better payoff). Bet II is robustly dominated by both Bet III and Bet IV.

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4. The Choice Protocol

Layer 1: Constraints

Constraint: None. This is a pure preference problem – all bets are available.

Bet	Result
Bet I (Red)	**PASS**
Bet II (Black)	**PASS**
Bet III (Red+Yellow)	**PASS**
Bet IV (Black+Yellow)	**PASS**

$$C = \{\text{Bet I},\ \text{Bet II},\ \text{Bet III},\ \text{Bet IV}\}$$

$$F = \text{Adm}(C) = \{\text{Bet III},\ \text{Bet IV}\}$$

All four bets pass the (vacuous) constraint check, but admissibility within the constraint-satisfying set eliminates Bets I and II, exactly as in the full admissible set.

Layer 2: Reference-Point Satisficing

Reference point: $\vec{r} = (0.25, 0.50)$.

Robust satisficing: An action passes Layer 2 iff *every* vector in $Y(a)$ meets or exceeds $\vec{r}$ componentwise. The agent must be confident it meets aspirations under all plausible models.

Bet	Worst payoff	Worst knowability	Result
Bet III (Red+Yellow)	$0.5433 \geq 0.25$	$0.1417 < 0.50$	**FAIL**
Bet IV (Black+Yellow)	$0.6667 \geq 0.25$	$1.0000 \geq 0.50$	**PASS**

Bet III fails because under the extreme prior (P_{extreme}), its expected knowability drops to 0.1417 – well below the 0.50 aspiration. The extreme prior concentrates weight on the states $s_{0B/60Y}$ and $s_{60B/0Y}$ where knowability is zero, dragging the expected value down. Bet IV, with its constant knowability of 1.0, passes trivially.

$$\text{Sat}(F, \vec{r}) = \{\text{Bet IV}\}$$

Layer 3: Regret-Pareto

With only one action surviving, regret computation is straightforward:

$$\vec{\rho}(\text{Bet IV}) = (0, 0)$$

There is no other action in the satisficing set against which Bet IV could accumulate regret.

$$R = \{\text{Bet IV}\}$$

Layer 4: Deference

$|R| = 1$ – the protocol terminates with a **unique recommendation: Bet IV (Black+Yellow)**.

No deference to the principal is needed. The protocol has resolved the decision fully.

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5. The Two-Choice Framing

The Ellsberg Paradox is traditionally presented as *two separate binary choices*. This framing is essential because the paradox lies in the *pattern* across both choices. Let us run MOADT on each choice independently.

5.1 Choice A: Bet I (Red) vs. Bet II (Black)

Outcome sets (from Section 2):

- $Y(\text{Bet I})$: payoff in $[0.3333, 0.5774]$, knowability $= 1.0$
- $Y(\text{Bet II})$: payoff in $[0.2100, 0.6573]$, knowability in $[0.1417, 0.7200]$

Robust dominance: Neither bet robustly dominates the other. Bet I cannot dominate Bet II because Bet II achieves payoff 0.6573 under $P_{\text{black}}/e_{\text{cautious}}$, exceeding Bet I's maximum of 0.5774. Bet II cannot dominate Bet I because Bet I's knowability is always 1.0, far above Bet II's maximum of 0.72.

$$\text{Adm}(\{I, II\}) = \{\text{Bet I, Bet II}\}$$

Layer 1: Both pass (no constraints). $F = \{\text{Bet I, Bet II}\}$.

Layer 2: Reference point $\vec{r} = (0.25, 0.50)$.

Bet	Worst payoff	Worst knowability	Result
Bet I	$0.3333 \geq 0.25$	$1.0000 \geq 0.50$	**PASS**
Bet II	$0.2100 < 0.25$	$0.1417 < 0.50$	**FAIL**

Bet II fails on *both* dimensions. Under $P_{\text{yellow}}/e_{\text{neutral}}$, its expected payoff drops to 0.21 (below 0.25), and under P_{extreme}, its knowability drops to 0.14 (far below 0.50).

$$\text{Sat} = \{\text{Bet I}\}$$

Layer 3: $\vec{\rho}(\text{Bet I}) = (0, 0)$.

Layer 4: Unique recommendation – **Bet I (Red)**.

5.2 Choice B: Bet III (Red+Yellow) vs. Bet IV (Black+Yellow)

Outcome sets (from Section 2):

- $Y(\text{Bet III})$: payoff in $[0.5433, 0.8832]$, knowability in $[0.1417, 0.7200]$
- $Y(\text{Bet IV})$: payoff in $[0.6667, 0.8165]$, knowability $= 1.0$

Robust dominance: Neither bet robustly dominates the other. Bet IV cannot dominate Bet III because Bet III achieves payoff 0.8832 under $P_{\text{yellow}}/e_{\text{cautious}}$, exceeding Bet IV's maximum of 0.8165. Bet III cannot dominate Bet IV because Bet IV's knowability is always 1.0.

$$\text{Adm}(\{III, IV\}) = \{\text{Bet III, Bet IV}\}$$

Layer 1: Both pass. $F = \{\text{Bet III, Bet IV}\}$.

Layer 2: Reference point $\vec{r} = (0.25, 0.50)$.

Bet	Worst payoff	Worst knowability	Result
Bet III	$0.5433 \geq 0.25$	$0.1417 < 0.50$	**FAIL**
Bet IV	$0.6667 \geq 0.25$	$1.0000 \geq 0.50$	**PASS**

Bet III fails on knowability for the same reason as before: the extreme prior drives expected knowability below the aspiration threshold.

$$\text{Sat} = \{\text{Bet IV}\}$$

Layer 3: $\vec{\rho}(\text{Bet IV}) = (0, 0)$.

Layer 4: Unique recommendation – **Bet IV (Black+Yellow)**.

5.3 The Ellsberg Pattern Emerges

Choice	MOADT recommends	Ellsberg prediction	Match?
A: Bet I vs. Bet II	**Bet I** (known 1/3)	Bet I	Yes
B: Bet III vs. Bet IV	**Bet IV** (known 2/3)	Bet IV	Yes

MOADT reproduces the classic Ellsberg preferences: prefer the known-probability bet in both choices. This is not a coincidence or a parameter-fitting exercise. It emerges from the geometry of outcome sets under credal uncertainty and the structural role of the knowability objective.

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6. What Scalar Expected Utility Would Have Done

6.1 The Impossibility Under EU

Under scalar EU, the agent holds a single prior P and computes $\mathbb{E}_P[\text{payoff}]$ for each bet. Here is the ranking under each of the 5 priors with the neutral evaluator, using payoff-only weights $(1.0, 0.0)$:

Prior	Bet I	Bet II	Bet III	Bet IV	Pattern
P_{uniform}	0.3333	0.3333	0.6667	0.6667	I = II, III = IV
P_{extreme}	0.3333	0.3333	0.6667	0.6667	I = II, III = IV

Prior	Bet I	Bet II	Bet III	Bet IV	Pattern
P_{black}	0.3333	0.4567	0.5433	0.6667	II > I, IV > III
P_{yellow}	0.3333	0.2100	0.7900	0.6667	I > II, III > IV
P_{moderate}	0.3333	0.3333	0.6667	0.6667	I = II, III = IV

The critical row is P_{yellow}: it gives I > II, but it *also* gives III > IV. The critical row P_{black} gives IV > III, but it *also* gives II > I. No single prior produces the Ellsberg pattern (I > II **and** IV > III). This is precisely the EU impossibility:

$$\forall P: \quad \mathbb{E}_P[\text{I}] + \mathbb{E}_P[\text{IV}] = \mathbb{E}_P[\text{II}] + \mathbb{E}_P[\text{III}] = 1.0$$

If you gain by choosing the known bet in one choice, you lose by choosing the known bet in the other. Under a single prior and a single objective, the Ellsberg pattern is a logical impossibility.

6.2 Adding Weights Does Not Help

With equal weights $(0.5, 0.5)$ that include knowability:

Prior	Evaluator	Ranking (best to worst)
P_{uniform}	e_{neutral}	IV (0.8333), I (0.6667), III (0.5476), II (0.3810)
P_{black}	e_{neutral}	IV (0.8333), I (0.6667), III (0.5067), II (0.4633)
P_{yellow}	e_{neutral}	IV (0.8333), I (0.6667), III (0.6300), II (0.3400)
P_{moderate}	e_{neutral}	IV (0.8333), III (0.6933), I (0.6667), II (0.5267)

Under equal weights, IV always beats III and I always beats II (except under P_{moderate} where III edges past I). But this is a scalar collapse – it hides the structure of the tradeoff behind a specific weighting. Under knowability-heavy weights $(0.3, 0.7)$, the Ellsberg pattern holds even more strongly across all priors, but this requires *choosing* the weights in advance. MOADT avoids this by treating the objectives as incommensurable.

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7. Single-Objective Variant: Why Ambiguity Aversion Is Genuinely Multi-Objective

This section provides the most important diagnostic in the entire analysis. We strip away the knowability objective and run MOADT with payoff as the *only* objective, retaining the full credal set and evaluator set. If the Ellsberg pattern survives, then ambiguity aversion can be explained by risk preferences alone. If it reverses, then ambiguity aversion is genuinely multi-objective.

7.1 Single-Objective Outcome Sets

With $k = 1$ (monetary payoff only), each $Y(a)$ is a set of 10 scalar values:

(P, e) pair	Bet I	Bet II	Bet III	Bet IV
$(P_{\text{uniform}}, e_{\text{neutral}})$	0.3333	0.3333	0.6667	0.6667
$(P_{\text{uniform}}, e_{\text{cautious}})$	0.5774	0.5158	0.8044	0.8165
$(P_{\text{extreme}}, e_{\text{neutral}})$	0.3333	0.3333	0.6667	0.6667
$(P_{\text{extreme}}, e_{\text{cautious}})$	0.5774	0.4451	0.7939	0.8165
$(P_{\text{black}}, e_{\text{neutral}})$	0.3333	0.4567	0.5433	0.6667
$(P_{\text{black}}, e_{\text{cautious}})$	0.5774	0.6573	0.7286	0.8165
$(P_{\text{yellow}}, e_{\text{neutral}})$	0.3333	0.2100	0.7900	0.6667
$(P_{\text{yellow}}, e_{\text{cautious}})$	0.5774	0.3947	0.8832	0.8165
$(P_{\text{moderate}}, e_{\text{neutral}})$	0.3333	0.3333	0.6667	0.6667
$(P_{\text{moderate}}, e_{\text{cautious}})$	0.5774	0.5612	0.8124	0.8165

7.2 Robust Dominance Reversal

With payoff as the only objective, robust dominance in one dimension reduces to comparing the ranges $[\min Y(a), \max Y(a)]$. The results:

$$\text{Bet II} \succ_R \text{Bet I} \qquad \text{Bet III} \succ_R \text{Bet IV}$$

and additionally:

$$\text{Bet III} \succ_R \text{Bet I}, \quad \text{Bet III} \succ_R \text{Bet II}, \quad \text{Bet IV} \succ_R \text{Bet I}, \quad \text{Bet IV} \succ_R \text{Bet II}$$

The admissible set is:

$$\text{Adm}(A) = \{\text{Bet III}\}$$

7.3 The Single-Objective Pairwise Results

Choice	Single-objective MOADT recommends	Ellsberg prediction	Match?
A: I vs. II	**Bet II** (ambiguous!)	Bet I	**No – reversed**
B: III vs. IV	**Bet III** (ambiguous!)	Bet IV	**No – reversed**

The Ellsberg pattern is completely reversed. With payoff as the only objective, MOADT *favors* the ambiguous bets.

7.4 Why the Reversal Happens

The reversal is driven by the cautious ($\sqrt{\cdot}$) evaluator and Jensen's inequality. For a concave function g and a random variable X:

$$\mathbb{E}[g(X)] \leq g(\mathbb{E}[X])$$

But the evaluator is applied *before* taking the expectation over states. The ambiguous bets have win probabilities that vary across states. In the states where they win with high probability, the square root compresses the advantage less than proportionally; in the states where they win with low probability, the square root *boosts* the value more than proportionally. The net effect is that the cautious evaluator gives *higher* expected values to the variable (ambiguous) bets than to the constant (known) bets.

Combined with the credal set (which includes P_{black}, under which Bet II has high expected payoff), this means the ambiguous bets can robustly dominate the known bets when payoff is the only objective.

7.5 The Core Lesson

This is the single most important finding: **risk aversion (the shape of $u(x)$) does not explain ambiguity aversion. In fact, it reverses it.**

The $\sqrt{\cdot}$ evaluator is the standard model of risk aversion – a concave utility function. If ambiguity aversion were merely extreme risk aversion, then the cautious evaluator should strengthen the Ellsberg pattern. Instead, it destroys it. The conclusion is inescapable: ambiguity aversion is about something *different* from the curvature of the utility function. It is about the distinct value of knowing your odds – a separate objective that cannot be captured by any scalar utility function, however cleverly shaped.

With two objectives (payoff + knowability), MOADT recommends I over II and IV over III – the classic Ellsberg pattern. With one objective (payoff only), MOADT recommends the opposite. The contrast proves that the Ellsberg pattern is *genuinely multi-objective*.

---

8. How MOADT Resolves the Paradox

The resolution has three structural components, each corresponding to an axiom that EU theory assumes and MOADT relaxes.

8.1 Credal Sets Instead of a Single Prior

EU theory requires a single subjective probability distribution. Under P_{yellow}, $E[\text{Bet I}] > E[\text{Bet II}]$. Under P_{black}, $E[\text{Bet IV}] > E[\text{Bet III}]$. These are *different* priors. EU demands you pick one; MOADT maintains both. The credal set $\mathcal{P} = \{P_{\text{uniform}}, P_{\text{extreme}}, P_{\text{black}}, P_{\text{yellow}}, P_{\text{moderate}}\}$ acknowledges genuine Knightian uncertainty about the urn composition.

8.2 Multiple Objectives Instead of a Single Scalar

The knowability objective captures what EU theory cannot represent: the irreducible value of *knowing* your probability. This is not risk aversion. Risk aversion is about preferring a sure \$50 to a 50/50 gamble for \$100 – it is about the *shape* of $u(x)$. Ambiguity aversion is about preferring a gamble where you *know* it is 50/50 to a gamble where it might be 50/50 or might be 0/100. MOADT models this as a distinct objective, not a parameter of the utility function.

8.3 Evaluator Sets Instead of a Single Evaluation Function

The evaluator set $\mathcal{F} = \{e_{\text{neutral}}, e_{\text{cautious}}\}$ acknowledges that even "monetary payoff" might be assessed differently. The Section 7 analysis shows that the evaluator choice interacts with ambiguity in surprising ways: the cautious evaluator actually *favors* ambiguous bets on the payoff dimension. This interaction is invisible under scalar EU.

8.4 The Geometric Mechanism

Together, these components produce outcome sets with a characteristic geometry:

Bet	Payoff spread	Knowability spread	Outcome set shape
Bet I (Red, known)	0.2440	0.0000	Compact segment at knowability = 1.0
Bet II (Black, ambiguous)	0.4473	0.5783	Diffuse cloud, low knowability
Bet III (R+Y, ambiguous)	0.3398	0.5783	Diffuse cloud, low knowability
Bet IV (B+Y, known)	0.1498	0.0000	Compact segment at knowability = 1.0

Known-probability bets have outcome sets that are *compact and elevated* on the knowability dimension. Ambiguous bets have outcome sets that are *spread out and depressed* on knowability. When the reference point demands at least moderate knowability (0.50), the ambiguous bets fail because their worst-case knowability falls far below the threshold. The known-probability bets pass effortlessly. This is not a trick of parameter choice – it is a structural consequence of the relationship between Knightian uncertainty and the value of information.

---

9. Summary of Protocol Execution

```
Input: 4 actions x 7 states x 2 objectives x 5 priors x 2 evaluators

Full problem (all 4 bets):
  Robust Dominance:      III >_R II, IV >_R I, IV >_R II
                         Adm(A) = {Bet III, Bet IV}

  Layer 1 (Constraints):  No constraints (pure preference problem)
```

```
                        C = {Bet I, Bet II, Bet III, Bet IV}
                        F = Adm(C) = {Bet III, Bet IV}

  Layer 2 (Satisficing):  r = (0.25, 0.50)
                        Bet III FAIL (knowability 0.1417 under P_extreme)
                        Sat = {Bet IV}

  Layer 3 (Regret):       rho(Bet IV) = (0, 0)
                        R = {Bet IV}

  Layer 4 (Deference):    |R| = 1 --> unique recommendation: Bet IV

Pairwise (Choice A: Bet I vs. Bet II):
  Adm = {Bet I, Bet II}
  Satisficing: Bet II FAIL (payoff 0.21, knowability 0.14)
  MOADT recommends: Bet I (Red)

Pairwise (Choice B: Bet III vs. Bet IV):
  Adm = {Bet III, Bet IV}
  Satisficing: Bet III FAIL (knowability 0.14)
  MOADT recommends: Bet IV (Black+Yellow)

Single-objective variant (payoff only):
  Adm = {Bet III} -- ambiguous bet wins!
  Choice A: Bet II > Bet I -- REVERSED
  Choice B: Bet III > Bet IV -- REVERSED
  Risk aversion does not explain Ellsberg; it reverses it.
```

---

10. What This Example Demonstrates

1. **"Ambiguity aversion" is rational multi-objective decision-making.** The Ellsberg Paradox is not a paradox under MOADT. It arises because EU theory insists on a single prior, a single scalar value function, and a single evaluation function. Relax any of these, and the preference for known-probability bets emerges naturally from the structure of outcome sets under credal uncertainty.
2. **The knowability objective is irreducible.** Section 7 proves that removing the knowability objective *reverses* the Ellsberg pattern. The preference for known probabilities cannot be captured by any concave utility function – it requires a distinct objective. This is not a modeling convenience; it is a structural necessity.
3. **Credal sets and evaluator sets interact nontrivially.** The cautious evaluator favors ambiguous bets on the payoff dimension (via Jensen's inequality), while the credal set generates the spread that makes ambiguous bets vulnerable on knowability. Neither mechanism alone produces the Ellsberg pattern. Their interaction through MOADT's multi-objective outcome sets is what resolves the paradox.

4. **MOADT needs no "ambiguity aversion parameter."** Unlike Maxmin EU, α-MEU, smooth ambiguity models, or variational preferences, MOADT does not introduce a special parameter to capture ambiguity aversion. The phenomenon emerges from the standard MOADT machinery: credal sets, evaluator sets, multiple objectives, and the robust dominance / satisficing protocol. No new axioms are needed.
5. **The protocol terminates with a definite recommendation.** In the full 4-bet problem, MOADT uniquely recommends Bet IV. In each pairwise choice, MOADT uniquely recommends the known-probability bet. There is no deference needed – the structural advantage of known-probability bets is decisive at Layer 2. The theory resolves the paradox completely, without passing the decision back to the principal.

Appendix: Computational Verification

All numerical results in this document were produced by `examples/classic_ellsberg.py` using the MOADT engine (`moadt/_engine.py`). The code is available in the repository and can be run independently to verify every number.

The knowability function used is:

$$\text{knowability}(a, s) = \begin{cases} 1.0 & \text{if bet } a \text{ has a state-independent win probability} \\ 1 - \frac{|b-30|}{30} & \text{otherwise, where } b = \text{number of black balls in state } s \end{cases}$$

The cautious evaluator applies $\sqrt{\cdot}$ to the monetary payoff only; knowability is passed through unchanged.

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MOADT Worked Example 8: The Allais Paradox

A concrete walkthrough of Multi-Objective Admissible Decision Theory applied to the Independence Axiom violation

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Overview

This document applies MOADT — Multi-Objective Admissible Decision Theory — to the Allais Paradox, the most famous demonstration that real human preferences systematically violate Expected Utility theory. Every computation is shown explicitly: the algebraic proof of the EU contradiction, outcome sets under multiple evaluators and priors, all four layers of the choice protocol, and a scalar EU comparison showing the knife-edge impossibility of reproducing Allais preferences with weighted objectives.

The Allais Paradox is not a contrived edge case. It is a clean, minimal demonstration that scalar utility theory cannot capture a pattern of preferences that the overwhelming majority of humans exhibit. The standard response in economics and decision theory is to classify this preference pattern as *irrational* — a bias to be corrected. MOADT offers a different interpretation: the preferences are perfectly rational once you stop forcing them through a one-dimensional bottleneck.

The scenario uses 4 gambles, 100 equiprobable states (encoding the Allais probabilities exactly), 3 objectives, 2 evaluators, and 2 priors. All numerical results were produced by `examples/classic_allais.py` using the MOADT engine (`moadt/_engine.py`) and can be independently verified.

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1. The Scenario

1.1 The Allais Paradox (Allais, 1953)

A decision-maker faces two independent choice problems.

Problem 1:

Gamble	Description
A	\$1,000,000 with certainty
B	89% chance of \$1M, 10% chance of \$5M, 1% chance of \$0

Problem 2:

Gamble	Description
C	11% chance of \$1M, 89% chance of \$0
D	10% chance of \$5M, 90% chance of \$0

The typical human preference pattern is $A \succ B$ and $D \succ C$. People prefer the certainty of \$1M over the risky gamble in Problem 1, but prefer the higher expected value gamble in Problem 2. This is not a niche laboratory finding — it is robust across cultures, stakes, and decades of replication.

1.2 The Independence Axiom Violation

Under Expected Utility with any utility function $u(\cdot)$, normalizing $u(\$0) = 0$:

$A \succ B$ **requires:**

$$u(\$1M) > 0.89 \cdot u(\$1M) + 0.10 \cdot u(\$5M) + 0.01 \cdot u(\$0)$$

$$0.11 \cdot u(\$1M) > 0.10 \cdot u(\$5M)$$

$$\frac{u(\$1M)}{u(\$5M)} > \frac{10}{11} \approx 0.9091$$

$D \succ C$ **requires:**

$$0.10 \cdot u(\$5M) > 0.11 \cdot u(\$1M)$$

$$\frac{u(\$1M)}{u(\$5M)} < \frac{10}{11} \approx 0.9091$$

These two conditions are a direct algebraic contradiction. No utility function — linear, concave, convex, or otherwise — can simultaneously produce $A \succ B$ and $D \succ C$. This is the Allais Paradox.

1.3 MOADT Formulation

Rather than forcing the problem through a scalar utility function, MOADT models it with three genuinely incommensurable objectives:

Objective	Meaning	How computed
f_1: Expected value	Monetary payoff (normalized: \$0 = 0, \$5M = 1)	$\mathbb{E}_P[v(\text{payoff})]$
f_2: Downside protection	Probability of adequate outcome ($\geq$ \$500K)	$\mathbb{E}_P[\mathbf{1}_{\text{payoff}\geq 500K}]$
f_3: Reliability	Probability of nonzero payoff	$\mathbb{E}_P[\mathbf{1}_{\text{payoff}>0}]$

These objectives are genuinely distinct. Expected value measures average return. Downside protection measures the probability of avoiding a bad outcome. Reliability measures the probability of receiving *anything*. A person who values the certainty of \$1M is not exhibiting a "bias" — they are weighting objectives 2 and 3 alongside objective 1.

Evaluators ($|\mathcal{F}| = 2$):

Evaluator	Valuation function	Interpretation
v_{neutral}	$v(x) = x/\$5M$	Risk-neutral: linear in dollars
v_{cautious}	$v(x) = \sqrt{x/\$5M}$	Risk-averse: diminishing marginal returns

Credal set ($|\mathcal{P}| = 2$): We use 100 equiprobable states to encode the Allais probabilities exactly. The exact prior assigns $P(s_i) = 0.01$ uniformly. A perturbed prior shifts 0.002 probability mass per state from the 5 highest-numbered states to the 5 lowest-numbered states, creating a mildly pessimistic variant. Since worst outcomes (e.g., \$0) are assigned to the highest-numbered states, this perturbation slightly increases the probability of bad outcomes.

State encoding: Each gamble's probability distribution is mapped onto the 100 states. For example: - Gamble A: all 100 states yield \$1M - Gamble B: states 1–89 yield \$1M, states 90–99 yield \$5M, state 100 yields \$0 - Gamble C: states 1–11 yield \$1M, states 12–100 yield \$0 - Gamble D: states 1–10 yield \$5M, states 11–100 yield \$0

(Worst outcomes are assigned to the highest-numbered states, so the perturbed prior is genuinely pessimistic.)

---

2. Why Expected Utility Fails

2.1 The Algebraic Contradiction

The proof in Section 1.2 is airtight: no utility function can produce both $A \succ B$ and $D \succ C$. To make this concrete, here are five specific utility functions spanning the full range from risk-neutral to extremely concave:

Utility function	$EU(A)$	$EU(B)$	$A \succ B$?	$EU(C)$	$EU(D)$	$D \succ C$?	Both?
Linear: $u(x) = x/5M$	0.200000	0.278000	No	0.022000	0.100000	Yes	No
Sqrt: $u(x) = \sqrt{x/5M}$	0.447214	0.498020	No	0.049193	0.100000	Yes	No
Log: $u(x) = \ln(1 + x)/\ln(1 + 5M)$	0.895660	0.897138	No	0.098523	0.100000	Yes	No
Power: $u(x) = (x/5M)^{0.3}$	0.617034	0.649160	No	0.067874	0.100000	Yes	No
Extreme: $u(x) = (x/5M)^{0.05}$	0.922681	0.921186	Yes	0.101495	0.100000	No	No

2.2 The Pattern

The table reveals the structural impossibility. As the utility function becomes more concave (from Linear to Extreme), $EU(A)$ eventually exceeds $EU(B)$ — but only by reversing the relationship between C and D. The Extreme utility function achieves $A \succ B$ but forces $C \succ D$. The crossover is inevitable because $A \succ B$ and $D \succ C$ require contradictory constraints on the ratio $u(\$1M)/u(\$5M)$.

No amount of curvature adjustment can thread this needle. The failure is not a calibration problem — it is a theorem. QED.

---

3. Multi-Objective Gamble Statistics

Before running the MOADT protocol, we examine the raw multi-objective statistics for all four gambles:

Gamble	$\mathbb{E}[\$]$	$\mathbb{E}[v]$ neutral	$\mathbb{E}[v]$ sqrt	P(adequate)	Reliability
A (certain \$1M)	\$1,000,000	0.2000	0.4472	1.00	1.00
B (risky mix)	\$1,390,000	0.2780	0.4980	0.99	0.99
C (11% \$1M)	\$110,000	0.0220	0.0492	0.11	0.11
D (10% \$5M)	\$500,000	0.1000	0.1000	0.10	0.10

3.1 Problem 1: The Certainty Premium

B has higher expected value than A under both evaluators ($1.39M vs. $1M; 0.278 vs. 0.200 normalized). But A has **perfect** downside protection (P(adequate) = 1.00 vs. 0.99) and **perfect** reliability (1.00 vs. 0.99). A dominates B on two of three objectives; B dominates A on one. These are genuinely incommensurable — not an "irrational bias."

The 1% gap between 1.00 and 0.99 may look trivially small, but it represents the difference between *certainty* and *uncertainty*. This is a qualitative, not merely quantitative, distinction. Every state under A yields an adequate outcome; under B, one state yields catastrophic loss. The multi-objective framework captures this as a real structural difference.

3.2 Problem 2: The Certainty Dimension Collapses

D has higher expected value than C ($500K vs. $110K; a 4.5x advantage). C has slightly higher downside protection (0.11 vs. 0.10) and reliability (0.11 vs. 0.10). But notice: both gambles are overwhelmingly risky. C gives $0 with probability 0.89; D gives $0 with probability 0.90. Neither gamble offers anything resembling certainty. The downside protection dimension — which drove the structural advantage of A in Problem 1 — is effectively neutralized. D's massive expected value advantage is the salient feature.

3.3 The Structural Asymmetry

This is the key insight: **the certainty dimension creates a genuine multi-objective advantage in Problem 1 that has no analogue in Problem 2.** In Problem 1, gamble A achieves perfection on two objectives. In Problem 2, neither gamble achieves anything close to perfection on any objective. The asymmetry between the two problems is structural, not parametric — it cannot be captured by adjusting a utility function's curvature.

---

4. MOADT Analysis: Problem 1 (A vs. B)

4.1 Outcome Sets

Each action's outcome set $Y(a)$ contains one vector per $(P, \vec{f})$ pair. With $|\mathcal{P}| = 2$ priors and $|\mathcal{F}| = 2$ evaluators, each $Y(a)$ contains 4 vectors in $\mathbb{R}^3$.

Y(**A_certain_1M**)**:**

$(P, \vec{f})$ pair	Expected value	Downside protection	Reliability
$(P_{\text{exact}}, v_{\text{neutral}})$	0.2000	1.0000	1.0000
$(P_{\text{exact}}, v_{\text{cautious}})$	0.4472	1.0000	1.0000
$(P_{\text{perturbed}}, v_{\text{neutral}})$	0.2000	1.0000	1.0000
$(P_{\text{perturbed}}, v_{\text{cautious}})$	0.4472	1.0000	1.0000

A's outcome set is remarkably stable: regardless of evaluator or prior, downside protection and reliability are always exactly 1.0. This is the mathematical signature of certainty — the outcome set has zero spread on objectives 2 and 3.

Y(**B_risky_mix**)**:**

$(P,\vec{f})$ pair	Expected value	Downside protection	Reliability
$(P_{\text{exact}}, v_{\text{neutral}})$	0.2780	0.9900	0.9900
$(P_{\text{exact}}, v_{\text{cautious}})$	0.4980	0.9900	0.9900
$(P_{\text{perturbed}}, v_{\text{neutral}})$	0.2696	0.9880	0.9880
$(P_{\text{perturbed}}, v_{\text{cautious}})$	0.4916	0.9880	0.9880

B's expected value is higher than A's under every $(P,\vec{f})$ pair, but its downside protection and reliability fall below 1.0. The perturbed prior reduces these further (0.988 vs. 0.990), demonstrating sensitivity to probability uncertainty.

4.2 Robust Dominance

We check: does either action robustly dominate the other?

$A \succ_R B$ requires: $\forall \vec{y}_B \in Y(B), \exists \vec{y}_A \in Y(A)$ such that $\vec{y}_A \succ_P \vec{y}_B$.

Take B's best vector $(P_{\text{exact}}, v_{\text{cautious}})$: $(0.498, 0.990, 0.990)$. A's best expected value is 0.4472, which is less than 0.498. So no vector in $Y(A)$ can Pareto-dominate this vector — A always loses on expected value. Therefore $A \not\succ_R B$.

$B \succ_R A$ requires: $\forall \vec{y}_A \in Y(A), \exists \vec{y}_B \in Y(B)$ such that $\vec{y}_B \succ_P \vec{y}_A$.

Take any vector in $Y(A)$: downside protection and reliability are 1.0. No vector in $Y(B)$ reaches 1.0 on these objectives. Therefore $B \not\succ_R A$.

Result: No robust dominance. $\text{Adm}(A) = \{\text{A_certain_1M}, \text{B_risky_mix}\}$.

4.3 Protocol Trace

Layer 1 (Constraints): No binding constraints in this run (constraint_floor = {}).

$$C = \{\text{A_certain_1M}, \text{B_risky_mix}\}$$

$$F = \text{Adm}(C) = \{\text{A_certain_1M}, \text{B_risky_mix}\}$$

Layer 2 (Reference-Point Satisficing): $\vec{r} = (0.15, 0.05, 0.50)$ — modest aspirations reflecting that even a small expected value and minimal downside protection are acceptable, but reliability should be at least 50%.

Both actions satisfy the reference point under all $(P,\vec{f})$ pairs:

$$\text{Sat}(F, \vec{r}) = \{\text{A_certain_1M}, \text{B_risky_mix}\}$$

Layer 3 (Regret-Pareto): For each action and objective, we compute the worst-case regret — the maximum gap between the best alternative and this action, across all $(P,\vec{f})$ pairs:

$$\vec{\rho}(\text{A_certain_1M}) = (0.078,\ 0,\ 0)$$

$$\vec{\rho}(\text{B_risky_mix}) = (0,\ 0.012,\ 0.012)$$

A's only regret is on expected value: B beats it by at most 0.078. A has zero regret on downside protection and reliability — it is never worse than B on these objectives.

B's only regret is on downside protection and reliability: A beats it by at most 0.012 on each. B has zero regret on expected value.

Neither regret vector Pareto-dominates the other: $(0.078, 0, 0)$ vs. $(0, 0.012, 0.012)$. A has lower regret on objectives 2 and 3; B has lower regret on objective 1.

$$R = \{\text{A_certain_1M}, \text{B_risky_mix}\}$$

Layer 4 (Deference): $|R| = 2 > 1$. The protocol defers to the principal:

> Both A and B are admissible. A sacrifices 0.078 in expected value for zero regret on safety dimensions. B sacrifices 0.012 in downside protection and reliability for zero regret on expected value. The theory cannot resolve this tradeoff without a value judgment about how much certainty is worth. This is the principal's decision.

---

5. MOADT Analysis: Problem 2 (C vs. D)

5.1 Outcome Sets

Y(**C_11pct_1M**):

$(P, \vec{f})$ pair	Expected value	Downside protection	Reliability
$(P_{\text{exact}}, v_{\text{neutral}})$	0.0220	0.1100	0.1100
$(P_{\text{exact}}, v_{\text{cautious}})$	0.0492	0.1100	0.1100
$(P_{\text{perturbed}}, v_{\text{neutral}})$	0.0200	0.1000	0.1000
$(P_{\text{perturbed}}, v_{\text{cautious}})$	0.0447	0.1000	0.1000

Y(**D_10pct_5M**):

$(P, \vec{f})$ pair	Expected value	Downside protection	Reliability
$(P_{\text{exact}}, v_{\text{neutral}})$	0.1000	0.1000	0.1000
$(P_{\text{exact}}, v_{\text{cautious}})$	0.1000	0.1000	0.1000
$(P_{\text{perturbed}}, v_{\text{neutral}})$	0.0900	0.0900	0.0900
$(P_{\text{perturbed}}, v_{\text{cautious}})$	0.0900	0.0900	0.0900

Two immediate observations:

1. D's outcome set is identical across both evaluators (neutral and cautious). This is because D's payoff is either \$0 or \$5M. The normalized value of \$5M is 1.0 under both evaluators

($v_{\text{neutral}}(5M) = 1.0$, $v_{\text{cautious}}(5M) = \sqrt{1.0} = 1.0$), and $v(0) = 0$ for both. So the evaluator distinction is irrelevant for D.

2. C has a tiny edge on downside protection and reliability (0.11 vs. 0.10 under exact prior), but D has a massive edge on expected value (0.10 vs. 0.022 under neutral, 0.10 vs. 0.049 under cautious).

5.2 Robust Dominance

Neither action robustly dominates the other. D cannot Pareto-dominate C's vectors because C has higher downside protection and reliability under the exact prior ($0.11 > 0.10$). C cannot Pareto-dominate D because D has much higher expected value.

$$\text{Adm}(A) = \{\text{C_11pct_1M}, \text{D_10pct_5M}\}$$

5.3 Protocol Trace

Layer 1 (Constraints): No binding constraints. Both pass.

$$C = \{\text{C_11pct_1M}, \text{D_10pct_5M}\}, \quad F = \text{Adm}(C) = \{\text{C_11pct_1M}, \text{D_10pct_5M}\}$$

Layer 2 (Reference-Point Satisficing): $\vec{r} = (0.05, 0.00, 0.05)$ — very low aspirations, reflecting that both gambles are overwhelmingly likely to yield \$0.

C's worst vector is $(0.0200, 0.1000, 0.1000)$. The expected value of 0.02 falls below the aspiration of 0.05 on objective 1. **C fails the reference point.**

D's worst vector is $(0.0900, 0.0900, 0.0900)$. All components meet or exceed the aspiration levels. **D passes.**

$$\text{Sat}(F, \vec{r}) = \{\text{D_10pct_5M}\}$$

Layer 3 (Regret-Pareto): With only one action surviving:

$$\vec{\rho}(\text{D_10pct_5M}) = (0,\ 0,\ 0)$$

$$R = \{\text{D_10pct_5M}\}$$

Layer 4 (Deference): $|R| = 1$. **Unique recommendation: D.**

5.4 The Resolution So Far

Without constraints, MOADT produces: - **Problem 1:** Both A and B are admissible; deference to the principal (who can rationally choose A). - **Problem 2:** D is uniquely recommended (C fails satisficing on expected value).

The Allais preference pattern $A \succ B$ and $D \succ C$ is *compatible* with MOADT — the theory does not force $C \succ D$ as a consequence of $A \succ B$. The Independence Axiom has no analogue in MOADT's multi-objective framework.

---

6. MOADT with Constraints

6.1 Problem 1: Downside Floor Eliminates B

We now add a hard constraint: downside protection ≥ 0.5 per state per evaluator. This means every state must yield an adequate outcome ($\geq$ \$500K). This is a non-negotiable floor — not a tradeoff.

Constraint check: - A: Every state gives \$1M $\geq$ \$500K. Adequate indicator = 1.0 in all 100 states. **PASS.** - B: State 100 gives \$0 < \$500K. Adequate indicator = 0.0 in that state. **FAIL.**

The protocol trace:

Layer 1: B fails the constraint.

$$C = \{\text{A_certain_1M}\}, \quad F = \text{Adm}(C) = \{\text{A_certain_1M}\}$$

Layers 2–3: With only one action, A trivially passes satisficing and has zero regret.

$$R = \{\text{A_certain_1M}\}$$

Layer 4: $|R| = 1$. **Unique recommendation: A.**

6.2 Problem 2: Same Floor Eliminates Both

We apply the identical constraint to Problem 2:

Constraint check: - C: 89 states give \$0 < \$500K. Adequate indicator = 0.0 in those states. **FAIL.** - D: 90 states give \$0 < \$500K. Adequate indicator = 0.0 in those states. **FAIL.**

The protocol trace:

Layer 1: Both actions fail. $C = \emptyset$.

$$\text{WARNING: } C = \emptyset \text{ — no action satisfies all constraints!}$$

This is an **error condition**. The constraint is too strict for this decision context — both gambles involve overwhelming ruin risk. The protocol escalates to the principal, who must either relax the constraint or accept the risk.

6.3 The Asymmetry Explained

This is precisely how MOADT dissolves the paradox through constraints:

- In Problem 1, the adequacy constraint provides a clean separation: A passes, B fails. The constraint reflects a genuine qualitative difference between certainty and near-certainty.
- In Problem 2, the same constraint does **not** favor C over D. Both gambles fail, because both involve substantial ruin risk. The constraint that eliminates B (1% ruin) does not force $C \succ D$, because C also has ruin risk (89%).

Under EU, $A \succ B$ logically implies $C \succ D$ (the Independence Axiom). Under MOADT with constraints, "eliminate B because it has ruin risk" does **not** imply "prefer C over D" — because C also has ruin risk. The logical entailment that creates the paradox simply does not exist in the MOADT framework.

---

7. Combined Analysis: All Four Gambles

Running all four gambles through a single MOADT problem reveals the dominance structure:

7.1 Outcome Sets

All outcome sets are as computed in Sections 4.1 and 5.1 above. The key new information is the pairwise dominance analysis.

7.2 Robust Dominance

Dominance pair	Result
A $\succ_R$ C	**Yes** — A dominates C on all objectives under all $(P, \vec{f})$ pairs
A $\succ_R$ D	**Yes** — A dominates D on all objectives under all $(P, \vec{f})$ pairs
B $\succ_R$ C	**Yes** — B dominates C on all objectives under all $(P, \vec{f})$ pairs
B $\succ_R$ D	**Yes** — B dominates D on all objectives under all $(P, \vec{f})$ pairs
A $\succ_R$ B	No
B $\succ_R$ A	No

Both Problem 1 gambles robustly dominate both Problem 2 gambles. This is unsurprising — the Problem 1 gambles have higher expected value, higher downside protection, and higher reliability across the board. Within each problem, the actions remain incomparable.

$$\mathrm{Adm}(A) = \{\text{A_certain_1M}, \text{B_risky_mix}\}$$

C and D are eliminated by robust dominance before the protocol even begins.

7.3 Protocol Trace

Layer 1 (Constraints): No binding constraints. All four pass constraint check, but the feasible set is restricted to the admissible set.

$$F = \mathrm{Adm}(C) = \{\text{A_certain_1M}, \text{B_risky_mix}\}$$

Layer 2 (Satisficing): $\vec{r} = (0.10, 0.00, 0.10)$. Both A and B pass.

$$\mathrm{Sat}(F, \vec{r}) = \{\text{A_certain_1M}, \text{B_risky_mix}\}$$

Layer 3 (Regret-Pareto):

$$\vec{\rho}(\text{A_certain_1M}) = (0.078,\ 0,\ 0)$$
$$\vec{\rho}(\text{B_risky_mix}) = (0,\ 0.012,\ 0.012)$$

Pareto-incomparable. $R = \{\text{A_certain_1M}, \text{B_risky_mix}\}$.

Layer 4 (Deference): $|R| = 2 > 1$. Defer to principal.

7.4 Interpretation

When all four gambles compete simultaneously, the combined analysis confirms the within-problem structure: A and B are the only viable options (C and D are robustly dominated), and the tradeoff between A and B is exactly as described in Section 4. The between-problem comparison is trivial — Problem 1 gambles are simply better than Problem 2 gambles on every dimension.

---

8. Scalar EU Comparison

8.1 The Knife-Edge Problem

Even within the three-objective MOADT framework, one might ask: can we find a weight vector $\vec{w} = (w_1, w_2, w_3)$ that produces $A \succ B$ and $D \succ C$ simultaneously via scalar aggregation $U(a) = \vec{w} \cdot \vec{y}(a)$?

Evaluator: risk-neutral, Prior: exact probabilities:

Weights	A	B	C	D	P1 winner	P2 winner
Pure expected value $(1, 0, 0)$	0.2000	0.2780	0.0220	0.1000	B	D
Pure downside $(0, 1, 0)$	1.0000	0.9900	0.1100	0.1000	A	C
Pure reliability $(0, 0, 1)$	1.0000	0.9900	0.1100	0.1000	A	C
Balanced $(1/3, 1/3, 1/3)$	0.7333	0.7527	0.0807	0.1000	B	D
Value + safety $(0.4, 0.4, 0.2)$	0.6800	0.7052	0.0748	0.1000	B	D

Weights	A	B	C	D	P1 winner	P2 winner
Safety-heavy $(0.2, 0.5, 0.3)$	0.8400	0.8476	0.0924	0.1000	B	D
Value-heavy $(0.7, 0.1, 0.2)$	0.4400	0.4916	0.0484	0.1000	B	D
Knife-edge $(0.1136, 0.4432, 0.4432)$	0.9091	0.9091	0.1000	0.1000	A	C

Evaluator: sqrt-concave, Prior: exact probabilities:

Weights	A	B	C	D	P1 winner	P2 winner
Pure expected value $(1, 0, 0)$	0.4472	0.4980	0.0492	0.1000	B	D
Pure downside $(0, 1, 0)$	1.0000	0.9900	0.1100	0.1000	A	C
Pure reliability $(0, 0, 1)$	1.0000	0.9900	0.1100	0.1000	A	C
Balanced $(1/3, 1/3, 1/3)$	0.8157	0.8260	0.0897	0.1000	B	D
Value + safety $(0.4, 0.4, 0.2)$	0.7789	0.7932	0.0857	0.1000	B	D
Safety-heavy $(0.2, 0.5, 0.3)$	0.8894	0.8916	0.0978	0.1000	B	D
Value-heavy $(0.7, 0.1, 0.2)$	0.6130	0.6456	0.0674	0.1000	B	D
Knife-edge $(0.1136, 0.4432, 0.4432)$	0.9372	0.9341	0.1031	0.1000	A	C

8.2 The Vanishingly Narrow Feasible Region

No weight vector in either table produces the Allais pattern ($A \succ B$ and $D \succ C$). The pattern is always either $B + D$ (expected value dominates) or $A + C$ (safety dominates).

The algebra explains why. Under risk-neutral evaluation with the exact prior:

$$U(A) = 0.20w_1 + 1.00w_2 + 1.00w_3$$
$$U(B) = 0.278w_1 + 0.99w_2 + 0.99w_3$$

$A \geq B$ requires:

$$0.01(w_2 + w_3) \geq 0.078w_1 \quad \Longrightarrow \quad \frac{w_2 + w_3}{w_1} \geq 7.8$$

Meanwhile:

$$U(D) = 0.10w_1 + 0.10w_2 + 0.10w_3$$
$$U(C) = 0.022w_1 + 0.11w_2 + 0.11w_3$$

$D \geq C$ requires:

$$0.078w_1 \geq 0.01(w_2 + w_3) \quad \Longrightarrow \quad \frac{w_2 + w_3}{w_1} \leq 7.8$$

Both thresholds are *exactly* 7.8 — the same coefficients (0.078 and 0.01) appear in both inequalities because the objective-value gaps between the safe and risky options are identical in both problems. The feasible region for strict preferences ($A \succ B$ and $D \succ C$ simultaneously) is therefore *empty*:

$$A \succ B \text{ requires } \frac{w_2 + w_3}{w_1} > 7.8, \qquad D \succ C \text{ requires } \frac{w_2 + w_3}{w_1} < 7.8$$

These are contradictory. The only solution is the exact boundary point $(w_2 + w_3)/w_1 = 7.8$, where $A = B$ and $C = D$ simultaneously (both problems are dead ties). The knife-edge weights $(0.1136, 0.4432, 0.4432)$ sit at this unique point. Any perturbation flips one preference but necessarily flips the other in the opposite direction.

A scalar weighting cannot even weakly "achieve" the Allais pattern — the best it can do is produce simultaneous ties on both problems, which is not a preference at all. **No scalar weighting over three objectives can reproduce the Allais preferences, even approximately.** This is exactly why MOADT's non-scalar approach is necessary: it keeps the objectives separate and never needs to find the "right" weights.

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9. The Mechanism: How MOADT Accommodates the Allais Preferences

MOADT resolves the Allais paradox through three reinforcing mechanisms.

9.1 Multi-Objective Structure (Layer 0)

EU theory forces all considerations into a single scalar $u(x)$. MOADT maintains separate objectives — expected value, downside protection, and reliability — that are genuinely incommensurable.

In Problem 1 (A vs. B): - B beats A on expected value. - A beats B on downside protection (1.00 vs. 0.99) and reliability (1.00 vs. 0.99). - Neither robustly dominates the other. Both are admissible. - The "certainty premium" is captured as a real advantage on 2 of 3 objectives.

In Problem 2 (C vs. D): - D beats C on expected value (0.10 vs. 0.022 neutral; 0.10 vs. 0.049 cautious). - C barely beats D on downside protection (0.11 vs. 0.10) and reliability (0.11 vs. 0.10). - With both gambles being overwhelmingly risky, the certainty dimension collapses. - D's expected

value advantage is massive; C's safety edge is marginal. - D is selected by satisficing; C fails the reference point.

The **structural asymmetry** between problems is the key: in Problem 1, certainty creates a genuine multi-objective advantage on two dimensions. In Problem 2, both gambles are risky, so the dimensions that favored A are effectively neutralized.

9.2 Hard Constraints (Layer 1)

If the agent has a non-negotiable floor on downside protection (e.g., "never risk total ruin"), then: - **Problem 1:** B fails (1% chance of $0), A passes. A is selected. - **Problem 2:** Both fail (both have $0 outcomes). Escalate to principal.

The constraint provides an **absolute veto**, not a tradeoff. This reflects how people actually think about certainty: "guaranteed money" vs. "a chance of nothing" is qualitatively different, not just quantitatively. The constraint that eliminates B does **not** force $C \succ D$, because C also has ruin risk.

9.3 Regret-Pareto Selection (Layer 3)

When both actions survive to Layer 3, minimax regret captures asymmetric opportunity costs: - Choosing A and missing B's upside: regret is modest (expected value gap of 0.078). - Choosing B and getting $0: regret is on the downside and reliability dimensions (gaps of 0.012 each).

In Problem 2, the regret structure is different: D's expected value advantage is so large relative to C's marginal safety edge that satisficing resolves the problem at Layer 2 before regret analysis is needed.

9.4 Why the Independence Axiom Fails in MOADT

The Independence Axiom says: if you prefer A to B, then for any gamble X and any mixing probability α, you should prefer the mixture $\alpha A + (1-\alpha)X$ to $\alpha B + (1-\alpha)X$. The common component X should be irrelevant.

In scalar EU, this axiom is a theorem — it follows from linearity of expectation applied to a single utility function. In MOADT, it is not a theorem because:

1. MOADT does not have a single utility function. It has a vector of objectives.
2. The admissibility criterion (robust dominance) is a partial order, not a total order. Mixing in a common component can change which actions are Pareto-incomparable.
3. The choice protocol layers (constraints, satisficing, regret) are sensitive to the structure of the outcome sets, not just their expected values.

The Allais paradox is only a "paradox" if you accept that a single scalar utility function should capture all human preferences. MOADT rejects this premise.

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10. Summary and What This Demonstrates

```
Problem 1 (A vs B), no constraints:
  Admissible set:      {A_certain_1M, B_risky_mix}
  Satisficing set:     {A_certain_1M, B_risky_mix}
```

```
  Regret-Pareto set:   {A_certain_1M, B_risky_mix}
  Deference needed:    Yes
  -> Both admissible; principal can rationally choose A

Problem 2 (C vs D), no constraints:
  Admissible set:      {C_11pct_1M, D_10pct_5M}
  Satisficing set:     {D_10pct_5M}
  Regret-Pareto set:   {D_10pct_5M}
  Deference needed:    No
  -> D uniquely recommended

Problem 1, downside floor (>= $500K per state):
  Constraint set:      {A_certain_1M}
  -> B eliminated (1% ruin risk). A uniquely selected.

Problem 2, same downside floor:
  Constraint set:      {} (empty)
  -> BOTH fail. Error condition: escalate to principal.
  -> The constraint that eliminates B does NOT force C > D.

Combined (all 4 gambles, no constraints):
  Admissible set:      {A_certain_1M, B_risky_mix}
  Dominance:           A >_R C, A >_R D, B >_R C, B >_R D
  Regret-Pareto set:   {A_certain_1M, B_risky_mix}
  Deference needed:    Yes
```

What This Demonstrates

1. **The Allais "paradox" is not a paradox.** It is a perfectly rational pattern of preferences once you stop forcing multi-dimensional concerns through a one-dimensional bottleneck. People who prefer A to B are not violating rationality — they are giving weight to downside protection and reliability alongside expected value. People who prefer D to C are not contradicting their Problem 1 preference — the certainty dimension that created A's structural advantage simply does not exist in Problem 2.

2. **Scalar utility cannot accommodate the pattern.** No single utility function produces both $A \succ B$ and $D \succ C$. Even scalarizing over three objectives fails: the same coefficients (0.078 and 0.01) appear in both problems, so $A \succ B$ requires $(w_2 + w_3)/w_1 > 7.8$ while $D \succ C$ requires $(w_2 + w_3)/w_1 < 7.8$ — a contradiction. The only solution is the exact tie point where both problems are dead ties. The failure is structural, not parametric.

3. **MOADT's non-scalar structure is essential.** By keeping objectives separate, MOADT allows the dominance structure to differ between problems. A has a genuine multi-objective advantage over B (perfection on 2 of 3 objectives); D has a genuine expected value advantage over C in a context where the safety objectives are neutralized. These are different structural configurations, not contradictory preferences.

4. **Constraints provide qualitative distinctions.** The constraint mechanism captures the qualitative difference between certainty and near-certainty. A constraint that eliminates B (1% ruin risk) does not logically force $C \succ D$ — because C has 89% ruin risk. The logical entailment

that creates the paradox under EU ($A \succ B \Rightarrow C \succ D$) has no analogue in MOADT.

5. **Deference is appropriate.** In Problem 1, MOADT does not force the choice of A — it identifies both A and B as admissible and presents the tradeoff transparently. The theory makes room for the Allais preference without mandating it. A principal who prefers B (accepting 1% ruin risk for higher expected value) is also acting rationally within MOADT.

Appendix: Computational Verification

All numerical results in this document were produced by `examples/classic_allais.py` using the MOADT engine (`moadt/_engine.py`). The code uses the `build_allais_problem_v2` encoding: 100 equiprobable states, 2 evaluators (risk-neutral and sqrt-concave), 2 priors (exact and perturbed), and 3 objectives (expected value, downside protection, reliability). The code is available in the repository and can be run independently to verify every number.

Key implementation details: - States are sorted so that worst outcomes occupy the highest-numbered states, making the perturbed prior genuinely pessimistic. - The downside protection objective uses an indicator function: $\mathbf{1}_{\text{payoff}\geq\$500K}$, so its expectation equals the probability of an adequate outcome. - The reliability objective uses an indicator function: $\mathbf{1}_{\text{payoff}>\$0}$, so its expectation equals the probability of any positive payoff. - Constraints in Section 6 are per-state, per-evaluator checks (not on expected values).

References

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- MOADT (Multi-Objective Admissible Decision Theory) is defined in the companion paper.
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MOADT Worked Example 9: The St. Petersburg Paradox

A concrete walkthrough of Multi-Objective Admissible Decision Theory applied to infinite expected value

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Overview

The St. Petersburg Paradox (Bernoulli, 1738) is perhaps the oldest and most fundamental paradox in decision theory. A fair coin is flipped repeatedly until tails appears; if tails first appears on flip n, the player wins 2^n dollars. The expected value of the game is infinite:

$$E[\text{winnings}] = \sum_{n=1}^{\infty} \frac{1}{2^n} \cdot 2^n = \sum_{n=1}^{\infty} 1 = \infty$$

Standard expected utility theory therefore says a risk-neutral agent should pay *any* finite amount to play. Yet no rational human would pay more than roughly \$20–25. This gap between theory and intuition has generated nearly three centuries of proposed resolutions – diminishing marginal utility, bounded utility functions, finite wealth constraints, risk aversion, prospect theory – each picking a single mechanism and building it into the utility function.

This document applies MOADT to the St. Petersburg Paradox and shows that the paradox dissolves not through any single mechanism but through *structural pluralism*: the interaction of multiple objectives, multiple priors, multiple evaluators, and a layered protocol that filters sequentially. Every computation is shown explicitly. The key finding: MOADT caps the acceptable entry fee at \$22, matching human intuition of \$20–25. All numerical results were produced by `examples/classic_stpetersburg.py` using the MOADT engine.

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1. The Scenario

1.1 The St. Petersburg Game

A casino offers the following game:

1. A fair coin is flipped repeatedly until tails appears.
2. If tails first appears on flip n, you win $\$2^n$.

3. You must pay an entry fee F to play.

The expected winnings are infinite: each flip contributes exactly \$1 to the expected value (probability $1/2^n$ times payoff $2^n = 1$), and there are infinitely many flips. Expected utility theory says: pay any finite fee. Humans say: about \$20–25 at most.

1.2 Truncated Model

We truncate at $N = 20$ flips, with the final state s_{20} absorbing all remaining probability mass. This gives:

- Maximum payoff: $2^{20} = \$1{,}048{,}576$
- Truncated EV (fair coin): \$21.00
- Truncated EV (biased coin, $P_{\text{tails}} = 0.55$): \$9.78

Even truncated, EU theory says pay up to \$21. The paradox remains: a risk-neutral agent should pay \$20 for a game where 75% of the time it wins \$2 or \$4.

1.3 MOADT Formulation

For each entry fee F, we formulate a separate binary MOADT problem:

Component	Specification
Actions	{Play, Dont_Play}
States	20 coin outcome states: s_n = "first tails on flip n"
Objectives ($k = 3$)	(1) Expected net monetary value, (2) Probability of net gain, (3) Downside exposure
Evaluators ($\lvert\mathcal{F}\rvert = 2$)	Risk-neutral, Risk-aware (log compression)
Credal set ($\lvert\mathcal{D}\rvert = 2$)	Fair coin, Biased coin ($P_{\text{tails}} = 0.55$)
Constraint (Layer 1)	Worst-case loss $\leq$ \$20 in any state
Reference point (Layer 2)	EV $\geq$ \$1, $P(\text{gain}) \geq 10\%$, worst case $\geq -\$5$

We sweep over 21 fee levels from \$1 to \$1,000 and find the *crossover point* – the maximum fee at which MOADT still recommends playing.

1.4 Why Three Objectives?

The three objectives are genuinely independent and capture orthogonal dimensions of value:

- **Objective 1 (Expected net value):** Favors playing – the expected payoff exceeds the fee up to $F = \$21$ (truncated) or always (untruncated). This is the dimension that creates the paradox in scalar EU.
- **Objective 2 (Probability of net gain):** Penalizes playing at high fees. At $F = \$15$, you have only a 12.5% chance of coming out ahead. At $F = \$50$, only 3.1%.
- **Objective 3 (Downside exposure):** Measures expected loss magnitude. Per-state values are capped at the baseline (the \$0-net-payoff level), so winning states contribute no upside information — only losing states differentiate actions. The expected downside drops below baseline as soon as any state produces a loss (fee > \$2), and worsens monotonically with fee as losses grow larger and more frequent. This is genuinely independent of objective 1 (which

captures both upside and downside) and objective 2 (which captures only the probability, not the magnitude, of outcomes). The Layer 1 *constraint* on this objective enforces a hard per-state floor on worst-case loss; the *objective value* captures how badly the expectation is dragged down by losing states.

In scalar EU, objective 1 drowns out objectives 2 and 3. In MOADT, all three are maintained as separate dimensions, preserving the tension that the paradox exploits.

1.5 Evaluators and Priors

Evaluators:

- $\vec{f}_{\text{neutral}}$ (risk-neutral): Objective 1 uses normalized net payoffs in $[0, 1]$ across the range $[-998, 1{,}048{,}575]$. Objective 2 (probability of gain) is naturally in $[0, 1]$. Objective 3 uses the same normalization but capped at the baseline: $\min(\text{norm}(\text{net}), \text{baseline})$, so only losing states contribute below-baseline values.
- $\vec{f}_{\text{risk}}$ (risk-aware): Applies logarithmic compression to objective 1 on the upside, while preserving downside linearly. Objective 3 is un-compressed in both evaluators — losses are fully felt regardless of risk attitude. This is Bernoulli's original insight – diminishing marginal value of money – but as *one evaluator among several*, not the sole resolution.

The risk-aware evaluator uses $\log(1 + 99x)/\log(100)$ compression. At the baseline ($0), both evaluators agree: 0.000951. But they diverge for large gains: a $1M payoff normalizes to 0.953719 (risk-neutral) vs. 0.989815 (risk-aware, compressed). The compression reduces the pull of extreme upside without eliminating it.

Priors:

- P_{fair}: Exact fair coin. $P(\text{first tails on flip } n) = (1/2)^n$.
- P_{biased}: Slightly pessimistic coin with $P(\text{tails}) = 0.55$. Long runs of heads (which produce the huge payoffs) are less likely.

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2. Game Structure

The state table shows all 20 states with their probabilities and payoffs:

State	$P(\text{fair})$	$P(\text{biased})$	Payoff	Cumulative $P(\text{fair})$
s_1	0.50000000	0.55000000	$2	0.50000000
s_2	0.25000000	0.24750000	$4	0.75000000
s_3	0.12500000	0.11137500	$8	0.87500000
s_4	0.06250000	0.05011875	$16	0.93750000
s_5	0.03125000	0.02255344	$32	0.96875000
s_6	0.01562500	0.01014905	$64	0.98437500
s_7	0.00781250	0.00456707	$128	0.99218750
s_8	0.00390625	0.00205518	$256	0.99609375
s_9	0.00195312	0.00092483	$512	0.99804688
s_{10}	0.00097656	0.00041617	$1,024	0.99902344
s_{11}	0.00048828	0.00018728	$2,048	0.99951172

State	P(fair)	P(biased)	Payoff	Cumulative P(fair)
s_{12}	0.00024414	0.00008428	\$4,096	0.99975586
s_{13}	0.00012207	0.00003792	\$8,192	0.99987793
s_{14}	0.00006104	0.00001707	\$16,384	0.99993896
s_{15}	0.00003052	0.00000768	\$32,768	0.99996948
s_{16}	0.00001526	0.00000346	\$65,536	0.99998474
s_{17}	0.00000763	0.00000156	\$131,072	0.99999237
s_{18}	0.00000381	0.00000070	\$262,144	0.99999619
s_{19}	0.00000191	0.00000031	\$524,288	0.99999809
s_{20} [abs]	0.00000191	0.00000026	\$1,048,576	1.00000000

Key Observation

75% of the time, you win \$2 or \$4. 93.75% of the time, you win \$16 or less. The huge payoffs (\$1M+) happen with probability less than 0.0001%. The entire expected value of the game is driven by astronomically improbable events. This is the structural feature that makes the paradox: expected value is dominated by tail events that a real decision-maker would never rely on.

The biased coin compresses these tails further. Under P_{biased}, the probability of reaching state s_{20} drops from 0.00000191 to 0.00000026 – a factor of 7 reduction. The truncated EV falls from \$21.00 to \$9.78.

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3. Classical Expected Value Analysis

The following table shows the classical EU analysis for each fee level:

Fee	EV_{fair}	$\text{EV}_{\text{biased}}$	Median	$P(\text{gain})_{\text{fair}}$	Worst Case
\$1	+\$20.00	+\$8.78	+\$1	100.00%	+\$1
\$2	+\$19.00	+\$7.78	\$0	50.00%	\$0
\$3	+\$18.00	+\$6.78	−\$1	50.00%	−\$1
\$4	+\$17.00	+\$5.78	−\$2	25.00%	−\$2
\$5	+\$16.00	+\$4.78	−\$3	25.00%	−\$3
\$8	+\$13.00	+\$1.78	−\$6	12.50%	−\$6
\$10	+\$11.00	−\$0.22	−\$8	12.50%	−\$8
\$12	+\$9.00	−\$2.22	−\$10	12.50%	−\$10
\$15	+\$6.00	−\$5.22	−\$13	12.50%	−\$13
\$18	+\$3.00	−\$8.22	−\$16	6.25%	−\$16
\$20	+\$1.00	−\$10.22	−\$18	6.25%	−\$18
\$22	−\$1.00	−\$12.22	−\$20	6.25%	−\$20
\$25	−\$4.00	−\$15.22	−\$23	6.25%	−\$23
\$30	−\$9.00	−\$20.22	−\$28	6.25%	−\$28
\$40	−\$19.00	−\$30.22	−\$38	3.12%	−\$38
\$50	−\$29.00	−\$40.22	−\$48	3.12%	−\$48

Fee	EV_{fair}	EV_{biased}	Median	$P(\text{gain})_{fair}$	Worst Case
\$75	−\$54.00	−\$65.22	−\$73	1.56%	−\$73
\$100	−\$79.00	−\$90.22	−\$98	1.56%	−\$98
\$200	−\$179.00	−\$190.22	−\$198	0.78%	−\$198
\$500	−\$479.00	−\$490.22	−\$498	0.39%	−\$498
\$1,000	−\$979.00	−\$990.22	−\$998	0.20%	−\$998

The Paradox in Numbers

The absurdity of expected value maximization is starkest at moderate fees:

- **At fee = \$15:** EV = +\$6, but the median outcome is −\$13 and $P(\text{gain})$ is only 12.5%. You have a positive expected value but lose money 87.5% of the time.
- **At fee = \$25:** EV = −\$4 in the truncated game, but *infinite* in the untruncated game. EU theory says "play" even though you lose money 93.75% of the time.
- **At fee = \$100:** EV = −\$79, median = −\$98, $P(\text{gain}) = 1.6\%$. Only in the untruncated game does EU still say "play."

Classical EU says: play at any fee ≤ \$21 (truncated) or ANY fee (untruncated). This is the recommendation that nearly three centuries of decision theory have struggled to escape.

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4. MOADT Analysis

For each fee level, the MOADT protocol runs on the binary decision {Play, Dont_Play} with 20 states, 3 objectives, 2 evaluators, and 2 priors. Each action's outcome set $Y(a)$ contains $|\mathcal{P}| \times |\mathcal{F}| = 4$ vectors in $\mathbb{R}^3$.

Results Across All Fee Levels

Fee	Adm(A)	Constraints	F = Adm(C)	Sat($F, \vec{r}$)	Final R	Verdict
\$1	Play	all pass	Play	Play	Play	**PLAY**
\$2	Play	all pass	Play	Play	Play	**PLAY**
\$3	Play, Dont_Play	all pass	Play, Dont_Play	Play	Play	**PLAY**
\$4	Play, Dont_Play	all pass	Play, Dont_Play	Play	Play	**PLAY**
\$5	Play, Dont_Play	all pass	Play, Dont_Play	Play	Play	**PLAY**
\$8	Play, Dont_Play	all pass	Play, Dont_Play	Play	Play	**PLAY**
\$10	Play, Dont_Play	all pass	Play, Dont_Play	ASF → Play	Play	**PLAY**
\$12	Play, Dont_Play	all pass	Play, Dont_Play	ASF → Play	Play	**PLAY**

Fee	Adm(A)	Constraints	F = Adm(C)	Sat($F, \vec{r}$)	Final R	Verdict
\$15	Play, Dont_Play	all pass	Play, Dont_Play	ASF → Play	Play	**PLAY**
\$18	Play, Dont_Play	all pass	Play, Dont_Play	ASF → Play	Play	**PLAY**
\$20	Play, Dont_Play	all pass	Play, Dont_Play	ASF → Play	Play	**PLAY**
\$22	Play, Dont_Play	all pass	Play, Dont_Play	ASF → Play	Play	**PLAY**
\$25	Play, Dont_Play	Play fail	Dont_Play	ASF → Dont_Play	Dont_Play	**DONT PLAY**
\$30	Play, Dont_Play	Play fail	Dont_Play	ASF → Dont_Play	Dont_Play	**DONT PLAY**
\$40	Play, Dont_Play	Play fail	Dont_Play	ASF → Dont_Play	Dont_Play	**DONT PLAY**
\$50	Play, Dont_Play	Play fail	Dont_Play	ASF → Dont_Play	Dont_Play	**DONT PLAY**
\$75	Play, Dont_Play	Play fail	Dont_Play	ASF → Dont_Play	Dont_Play	**DONT PLAY**
\$100	Play, Dont_Play	Play fail	Dont_Play	ASF → Dont_Play	Dont_Play	**DONT PLAY**
\$200	Play, Dont_Play	Play fail	Dont_Play	ASF → Dont_Play	Dont_Play	**DONT PLAY**
\$500	Play, Dont_Play	Play fail	Dont_Play	ASF → Dont_Play	Dont_Play	**DONT PLAY**
\$1,000	Play, Dont_Play	Play fail	Dont_Play	ASF → Dont_Play	Dont_Play	**DONT PLAY**

The Crossover

- **Last fee where MOADT says PLAY:** \$22
- **First fee where MOADT says DONT PLAY:** \$25
- **Classical EU says:** pay up to \$21 (truncated), infinity (untruncated)
- **Human intuition:** \$20–25

MOADT's maximum recommended fee of \$22 falls squarely within the range of human willingness-to-pay. At the sampled fee points, the crossover is clear: \$22 is a clear PLAY, \$25 is a clear DONT PLAY. (The exact crossover lies somewhere in the \$22–\$25 interval; finer fee sampling would locate it more precisely but would not change the qualitative conclusion.)

Three Regimes

The table reveals three distinct regimes:

1. **Very low fees (\$1–\$2):** Play robustly dominates Dont_Play. At these fees, no state produces a net loss (the minimum payoff is \$2), so Play matches Dont_Play on downside (objective 3) while strictly beating it on expected value and P(gain). Satisficing passes directly.
2. **Low-to-moderate fees (\$3–\$8):** Robust dominance fails — the first-flip state ($\$2 - F < 0$)

drags Play's expected downside below baseline, creating a genuine three-way tension among objectives. But satisficing still selects Play (all three objectives meet aspirations under the better prior-evaluator pairs).

3. **Moderate fees ($10–$22):** Both actions survive dominance. Satisficing fails (P(gain) drops below the 10% aspiration), but the ASF fallback selects Play because its shortfall from aspirations is smaller than Dont_Play's.
4. **High fees ($25+):** Play violates the downside constraint (worst-case loss exceeds $20). Only Dont_Play survives Layer 1. The game is unambiguously not worth playing.

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5. Detailed Traces

5.1 Fee = $5 (Low Fee – Satisficing Resolves)

Outcome sets:

$(P, \vec{f})$ pair	E[net]	P(gain)	Downside
Y(Play):			
$(P_{\text{fair}}, \vec{f}_{\text{neutral}})$	0.000966	0.250000	0.000949
$(P_{\text{fair}}, \vec{f}_{\text{risk}})$	0.001176	0.250000	0.000949
$(P_{\text{biased}}, \vec{f}_{\text{neutral}})$	0.000955	0.202500	0.000949
$(P_{\text{biased}}, \vec{f}_{\text{risk}})$	0.001058	0.202500	0.000949
Y(Dont_Play):			
(all pairs)	0.000951	0.000000	0.000951

Protocol execution:

```
Robust Dominance:   No robust dominance found.
                    Adm(A) = {Play, Dont_Play}

Layer 1 (Constraints): Both pass. C = {Play, Dont_Play}
                       F = Adm(C) = {Play, Dont_Play}

Layer 2 (Satisficing): Sat(F, r) = {Play}
                       Play meets all three aspirations.

Layer 3 (Regret):      rho(Play) = (0, 0, 0)
                       R = {Play}

Layer 4:               |R| = 1 -- unique recommendation: Play
```

At $5, Play beats Dont_Play on objectives 1 and 2 (higher EV and positive P(gain)), but Dont_Play beats Play on objective 3 (downside): Play's expected downside is 0.000949 (dragged below baseline by the 50% chance of losing $3 on the first flip) while Dont_Play's is 0.000951 (baseline). This three-way tension – Play wins on EV and P(gain), Dont_Play wins on downside – means neither robustly dominates the other. But satisficing resolves it: Play meets all three aspirations, so Layer 2 selects Play directly.

5.2 Fee = $10 (Moderate Fee – ASF Resolves)

Outcome sets:

$(P,\vec{f})$ pair	E[net]	P(gain)	Downside
Y(Play):			
$(P_{\text{fair}},\vec{f}_{\text{neutral}})$	0.000961	0.125000	0.000945
$(P_{\text{fair}},\vec{f}_{\text{risk}})$	0.001151	0.125000	0.000945
$(P_{\text{biased}},\vec{f}_{\text{neutral}})$	0.000951	0.091125	0.000945
$(P_{\text{biased}},\vec{f}_{\text{risk}})$	0.001038	0.091125	0.000945
Y(Dont_Play):			
(all pairs)	0.000951	0.000000	0.000951

Protocol execution:

```
Robust Dominance:   No robust dominance found.
                    Adm(A) = {Play, Dont_Play}

Layer 1 (Constraints): Both pass. C = {Play, Dont_Play}
                       F = Adm(C) = {Play, Dont_Play}

Layer 2 (Satisficing): Sat(F, r) = empty
                       Play fails the P(gain) >= 10% aspiration under biased prior
                       (P(gain) = 9.11% < 10%).
                       ASF fallback: ASF(Play) = -0.0089
                                     ASF(Dont_Play) = -0.1000
                       ASF selects Play (best available).

Layer 3:               Skipped (ASF already resolved).

Layer 4:               |R| = 1 -- unique recommendation: Play
```

At $10, neither action robustly dominates — Play loses on downside (objective 3 at 0.000945 vs. baseline 0.000951) while winning on EV and P(gain). Play also *fails* the satisficing aspiration on P(gain): under the biased coin, only 9.11% of outcomes produce a net gain, falling below the 10% threshold. The ASF fallback activates, measuring shortfall from aspirations. Play's worst shortfall is −0.0089 (barely missing the P(gain) target), while Dont_Play's is −0.1000 (missing the P(gain) target completely at 0%). Play wins the ASF comparison handily.

5.3 Fee = $20 (Near the Crossover – ASF Resolves)

Outcome sets:

$(P,\vec{f})$ pair	E[net]	P(gain)	Downside
Y(Play):			
$(P_{\text{fair}},\vec{f}_{\text{neutral}})$	0.000952	0.062500	0.000937
$(P_{\text{fair}},\vec{f}_{\text{risk}})$	0.001123	0.062500	0.000937

$(P,\vec{f})$ pair	E[net]	P(gain)	Downside
$(P_{\text{biased}},\vec{f}_{\text{neutral}})$	0.000941	0.041006	0.000936
$(P_{\text{biased}},\vec{f}_{\text{risk}})$	0.001015	0.041006	0.000936
Y(Dont_Play):			
(all pairs)	0.000951	0.000000	0.000951

Protocol execution:

```
Robust Dominance:   No robust dominance found.
                    Adm(A) = {Play, Dont_Play}

Layer 1 (Constraints): Both pass. C = {Play, Dont_Play}
                       F = Adm(C) = {Play, Dont_Play}

Layer 2 (Satisficing): Sat(F, r) = empty
                       P(gain) = 4.1% under biased prior (far below 10%).
                       ASF fallback: ASF(Play) = -0.0590
                                     ASF(Dont_Play) = -0.1000
                       ASF selects Play.

Layer 3:               Skipped (ASF already resolved).

Layer 4:               |R| = 1 -- unique recommendation: Play
```

At \$20, the tension across all three objectives is fully engaged. Play loses on downside (objective 3 at 0.000936–0.000937 vs. baseline 0.000951) and on EV under the biased prior ($0.000941 < 0.000951$), but wins on P(gain) ($4.1\% > 0\%$) and on EV under the risk-aware evaluator (0.001015–0.001123). Neither action dominates. The ASF shortfall has grown to -0.0590, reflecting that Play is increasingly far from meeting aspirations, but Play still beats Dont_Play's shortfall of -0.1000.

5.4 Fee = \$22 (The Last PLAY – Constraint Boundary)

Outcome sets:

$(P,\vec{f})$ pair	E[net]	P(gain)	Downside
Y(Play):			
$(P_{\text{fair}},\vec{f}_{\text{neutral}})$	0.000950	0.062500	0.000935
$(P_{\text{fair}},\vec{f}_{\text{risk}})$	0.001118	0.062500	0.000935
$(P_{\text{biased}},\vec{f}_{\text{neutral}})$	0.000939	0.041006	0.000934
$(P_{\text{biased}},\vec{f}_{\text{risk}})$	0.001011	0.041006	0.000934
Y(Dont_Play):			
(all pairs)	0.000951	0.000000	0.000951

Protocol execution:

```
Robust Dominance:   No robust dominance found.
```

```
                    Adm(A) = {Play, Dont_Play}

Layer 1 (Constraints): Both pass. C = {Play, Dont_Play}
                       F = Adm(C) = {Play, Dont_Play}

Layer 2 (Satisficing): Sat(F, r) = empty
                       ASF fallback:
                         ASF(Play) = -0.0590
                         ASF(Dont_Play) = -0.1000
                       ASF selects Play.

Layer 3:               Skipped (ASF already resolved).

Layer 4:               |R| = 1 -- unique recommendation: Play
```

At \$22, this is the last fee where MOADT recommends playing. The fee exceeds the truncated EV (\$21), so under the fair coin with risk-neutral evaluation, Play's expected net value (0.000950) falls *below* the baseline (0.000951). Under the biased prior, it is even worse (0.000939). Play's downside (0.000934–0.000935) is well below baseline. Neither action dominates — the three-way tension (Play wins on P(gain), mixed on EV, loses on downside) persists.

Both actions survive to the constraint layer. Both pass (worst-case loss = \$22 − \$2 = \$20, exactly at the limit). Both enter the feasible set. Satisficing fails for both (neither meets P(gain) ≥ 10%). The ASF fallback resolves: Play's shortfall (−0.0590) is smaller than Dont_Play's (−0.1000), so Play is selected. The key: even though Play's P(gain) aspiration shortfall is larger, its *worst* shortfall across all objectives is still less severe than Dont_Play's complete failure on P(gain) (score of 0 vs. aspiration of 0.10).

5.5 Fee = \$25 (The First DONT PLAY – Constraint Eliminates Play)

Outcome sets:

$(P, \vec{f})$ pair	E[net]	P(gain)	Downside
Y(Play):			
$(P_{\text{fair}}, \vec{f}_{\text{neutral}})$	0.000947	0.062500	0.000932
$(P_{\text{fair}}, \vec{f}_{\text{risk}})$	0.001112	0.062500	0.000932
$(P_{\text{biased}}, \vec{f}_{\text{neutral}})$	0.000936	0.041006	0.000932
$(P_{\text{biased}}, \vec{f}_{\text{risk}})$	0.001006	0.041006	0.000932
Y(Dont_Play):			
(all pairs)	0.000951	0.000000	0.000951

Protocol execution:

```
Robust Dominance:   No robust dominance found.
                    Adm(A) = {Play, Dont_Play}

Layer 1 (Constraints): Play FAILS.
                       Worst-case loss = $25 - $2 = $23 > $20 limit.
```

```
                        Dont_Play PASSES.
                        C = {Dont_Play}
                        F = Adm(C) = {Dont_Play}

Layer 2 (Satisficing): Sat(F, r) = empty
                        ASF fallback: ASF(Dont_Play) = -0.1000
                        ASF selects Dont_Play.

Layer 3:                Skipped (ASF already resolved).

Layer 4:                |R| = 1 -- unique recommendation: Dont_Play
```

At \$25, the hard constraint fires. The worst-case net payoff from Play is $\$2 - \$25 = -\$23$, which exceeds the \$20 loss tolerance. Play is eliminated at Layer 1 – no amount of expected upside can compensate for a worst-case loss that the agent cannot tolerate. This is not a utility function adjustment; it is a structural precondition.

5.6 Fee = \$50 (High Fee – Deep into DONT PLAY Territory)

Protocol execution:

```
Robust Dominance:   No robust dominance found.
                    Adm(A) = {Play, Dont_Play}

Layer 1 (Constraints): Play FAILS (worst loss = $48 > $20).
                        F = {Dont_Play}

Layer 2-4:              Dont_Play selected throughout.
                        Unique recommendation: Dont_Play
```

At \$50, the constraint violation is overwhelming (\$48 loss vs. \$20 limit). Play has only a 3.12% chance of gaining and a 1.85% chance under the biased prior. Even the risk-aware evaluator cannot rescue it.

---

6. Mechanism Analysis

Which MOADT Layer Drives the Resolution at Each Fee Level

Fee Range	Resolving Layer	Mechanism
\$1–\$2	Layer 0 (Dominance)	Play robustly dominates: no state produces a loss, so downside = baseline while EV and P(gain) strictly beat Dont_Play.

Fee Range	Resolving Layer	Mechanism
$3–$8	Layer 2 (Satisficing)	Dominance fails — the first-flip loss state drags expected downside below baseline, creating a three-way tension. But Play meets all aspirations, so satisficing selects it directly.
$10–$22	Layer 2 (ASF)	Neither action dominates. Satisficing also fails (P(gain) drops below 10% aspiration). ASF fallback selects Play (smaller shortfall from aspirations than Dont_Play).
$25+	Layer 1 (Constraint)	Play violates the worst-case loss constraint. Eliminated before any further analysis.

Detailed mechanism trace from the script:

```
$    1: Layer 0 (Dominance): Play robustly dominates
$    2: Layer 0 (Dominance): Play robustly dominates
$    3: Layer 2 (Satisficing): excludes Dont_Play
$    4: Layer 2 (Satisficing): excludes Dont_Play
$    5: Layer 2 (Satisficing): excludes Dont_Play
$    8: Layer 2 (Satisficing): excludes Dont_Play
$   10: Layer 2 (ASF): selects Play
$   12: Layer 2 (ASF): selects Play
$   15: Layer 2 (ASF): selects Play
$   18: Layer 2 (ASF): selects Play
$   20: Layer 2 (ASF): selects Play
$   22: Layer 2 (ASF): selects Play
$   25: Layer 1 (Constraint): Play violates downside floor
$   30: Layer 1 (Constraint): Play violates downside floor
$   40: Layer 1 (Constraint): Play violates downside floor
...
$ 1000: Layer 1 (Constraint): Play violates downside floor
```

Why Multiple Mechanisms Matter

No single mechanism resolves the paradox across the full range:

1. **Dominance alone** resolves only fees $1–$2 (where no state produces a loss, so Play wins on all three objectives). From fee $3 onward, the downside objective creates a genuine tension that dominance cannot resolve — both actions remain admissible, and the protocol layers

must do the real work.

2. **Constraints alone** would produce a sharp cutoff at \$22 (fee = loss limit + \$2) but would say nothing about whether to play at \$20 or \$22 – it treats all fees below the cutoff identically.
3. **Satisficing alone** would fail at fees above \$8 (where P(gain) drops below 10%), which is too conservative – most people would still play at \$10 or \$15.
4. **ASF** provides the bridge: it resolves cases where satisficing fails but the game is still worth playing, by measuring *how far* each action falls from aspirations rather than applying a binary pass/fail.

The layered protocol allows each mechanism to operate in its natural range, handing off to the next layer when the current one cannot resolve the decision.

---

7. Comparison with Historical Resolutions

Historical Resolution	Mechanism	MOADT Analog
Bernoulli (1738) – Log utility	Diminishing marginal value of money; $U(x) = \ln(x)$	Risk-aware evaluator $\vec{f}_{\text{risk}}$ with log compression. But it is *one evaluator among two*, not the sole resolution.
Bounded utility (Menger, 1934)	Cap on total utility gain; finite $U_{\max}$ prevents infinite EV	Hard constraints (Layer 1): cap on maximum acceptable *loss*, not maximum utility. Structural, not parametric.
Finite wealth (various)	Cannot bet what you do not have; budget constraint	Downside floor constraint. The agent's loss tolerance is a structural parameter of the decision problem, not a property of the utility function.
Risk aversion (Arrow, Pratt)	Concave utility function; $U'' < 0$	Multi-objective structure: P(gain) and downside exposure as *separate objectives*, not curvature of a single utility function. Risk is preserved as a dimension, not compressed into one.

Historical Resolution	Mechanism	MOADT Analog
Prospect theory (Kahneman & Tversky, 1979)	Probability weighting, loss aversion, reference dependence	Credal set (biased prior models probability distortion) + loss-preserving evaluator (downside stays linear while upside is compressed). Reference point is explicit in Layer 2.

The Key Difference

Each historical resolution picks *one* mechanism and builds it into the utility function. MOADT does not choose – it provides a framework that naturally incorporates elements of *all* of them:

- **Evaluator plurality** captures diminishing marginal value (Bernoulli) without committing to a specific functional form.
- **Constraints** capture wealth bounds (Menger, finite wealth) as structural preconditions rather than utility modifications.
- **Objective plurality** captures risk aversion (Arrow-Pratt) structurally – risk is a separate dimension, not a curvature parameter.
- **Credal sets** capture probability uncertainty (prospect theory's probability weighting) as explicit model uncertainty.

The paradox dissolves through structural pluralism, not a clever fix to the utility function.

---

8. Sensitivity Analysis

Effect of Loss Tolerance on Fee Cap

The Layer 1 constraint sets the maximum acceptable loss in any state. Since the worst state always pays \$2, the constraint translates directly to a fee cap: $F_{\max} = \$2 + \text{loss_limit}$.

Loss Tolerance	Max Fee (Layer 1)	Note
\$5	\$7	Conservative gambler
\$10	\$12	
\$15	\$17	
\$20	\$22	Matches human intuition
\$25	\$27	Beyond truncated EV
\$30	\$32	Beyond truncated EV
\$50	\$52	Beyond truncated EV
\$100	\$102	Beyond truncated EV
\$500	\$502	Beyond truncated EV

With loss tolerance of \$20, MOADT caps the fee at \$22. This is close to human intuition of \$20–25.

Why This Is Not Arbitrary

The constraint level is *not* an arbitrary parameter choice – it represents a genuine structural feature of the decision problem: "How much can the agent afford to lose in a single bet?" Different agents with different wealth levels would set different constraints, producing different fee caps. This is exactly analogous to the empirical finding that wealthier individuals report higher willingness-to-pay for the St. Petersburg game.

The constraint interacts with the multi-objective structure:

- **Low fees** ($< \$3$)**:** Play robustly dominates regardless of constraint level. Layer 0 resolves.
- **Moderate fees** (**\$3–**$F_{\max}$)**:** Neither action dominates, but ASF/satisficing selects Play (positive EV + some P(gain)). The constraint is not binding.
- **High fees** ($> F_{\max}$)**:** The constraint eliminates Play. No further analysis needed.

The transition zone around $F_{\max}$ is where multiple MOADT mechanisms interact to produce the fee cap.

---

9. Summary of Protocol Execution

```
Input: 2 actions x 20 states x 3 objectives x 2 priors x 2 evaluators
       Sweep over 21 fee levels from $1 to $1,000

Fee = $1-$2:   Robust Dominance: Play dominates Dont_Play
               Adm(A) = {Play}
               No state produces a loss, so downside = baseline
               Verdict: PLAY

Fee = $3-$8:   Robust Dominance: Neither dominates
               Adm(A) = {Play, Dont_Play}
               Layer 2: Satisficing selects Play (all aspirations met)
               Verdict: PLAY

Fee = $10-$22: Robust Dominance: Neither dominates
               Adm(A) = {Play, Dont_Play}
               Layer 2: Satisficing fails (P(gain) < 10% aspiration)
                        ASF fallback selects Play
               Verdict: PLAY

Fee = $25+:    Robust Dominance: Neither dominates
               Adm(A) = {Play, Dont_Play}
               Layer 1: Play FAILS (worst loss > $20)
               F = {Dont_Play}
               Verdict: DONT PLAY

MOADT maximum recommended fee: $22
Crossover to refusal: $25
Classical EU maximum: $21 (truncated), infinity (untruncated)
```

```
Human intuition: ~$20-25
```

---

10. What This Example Demonstrates

1. **The multi-objective structure prevents dominance swamping.** In scalar EU, infinite (or large) expected value overwhelms all other concerns – this is the paradox. With 3 independent objectives, Play cannot robustly dominate Dont_Play once the fee exceeds $2, because the expected downside exposure (objective 3) drops below the baseline while expected value remains positive. MOADT preserves the tension rather than collapsing it.
2. **Hard constraints eliminate catastrophic bets.** The per-state floor (max loss $\leq$ $20) means that entry fees above $22 are immediately eliminated at Layer 1. No amount of expected upside can compensate. This is a structural feature of MOADT, not a utility function tweak. The constraint is analogous to bounded utility (Menger) and finite wealth, but operates as a precondition rather than a function shape.
3. **Satisficing filters on multiple aspirations simultaneously.** The reference point requires positive EV AND reasonable P(gain) AND tolerable worst case. As the fee rises, P(gain) drops below the aspiration threshold, filtering out Play even if EV remains positive. This captures the intuition that a game where you lose money 87.5% of the time is not a "good bet" regardless of expected value.
4. **The ASF fallback bridges the gap.** When no action robustly meets aspirations (as happens for fees $10–$22), the achievement scalarizing function measures *how close* each action gets. This allows MOADT to distinguish between "barely misses aspirations" (Play at $10) and "completely fails aspirations" (Dont_Play on P(gain)), avoiding the false conservatism of a binary pass/fail filter.
5. **Structural pluralism replaces single-mechanism fixes.** MOADT incorporates elements of *every* historical resolution – Bernoulli's log utility, bounded utility, finite wealth, risk aversion, prospect theory – as special cases within its framework. The paradox dissolves not because one of these mechanisms is "correct," but because their interaction through the layered protocol produces a recommendation that respects all of them simultaneously.
6. **The framework produces actionable recommendations.** The output is not "it depends" or a philosophical position. It is: "Play at fees up to $22. Do not play at $25 or above." This matches human intuition while being derived from explicit, verifiable computations.

Appendix: Computational Verification

All numerical results in this document were produced by `examples/classic_stpetersburg.py` using the MOADT engine (`moadt/_engine.py`). The code is available in the repository and can be run independently to verify every number.

Key parameters:

- Truncation depth: $N = 20$ flips
- Normalization range: $[-998, 1{,}048{,}575]$

- Baseline (Dont_Play normalized value): 0.000951
- Loss limit: $20 (translates to constraint floor at `normalize_net(-20)`)
- Reference point: $\vec{r} = (\text{normalize_net}(1),\ 0.10,\ \text{normalize_net}(-5))$
- Risk compression: $\log(1 + 99x)/\log(100)$ on upside, linear on downside

---

References

- MOADT (Multi-Objective Admissible Decision Theory) is defined in the companion paper.
- Freeman (2025), "The Scalarization Trap," provides the motivation for why scalar expected utility is structurally problematic for alignment.
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